

Document Number: P0896R4
Date: 2018-11-09
Audience: Library Evolution Working Group,
Library Working Group
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The One Range_s Proposal

(was Merging the Ranges TS)

Three Proposals for Views under the Sky,
Seven for LEWG in their halls of stone,
Nine for the Ranges TS doomed to die,
One for LWG on its dark throne
in the Land of Geneva where the Standards lie.
One Proposal to `ranges::merge` them all, One Proposal to `ranges::find` them,
One Proposal to bring them all and in namespace `ranges` bind them,
In the Land of Geneva where the Standards lie.

With apologies to J.R.R. Tolkien.

Note: ~~t~~This is an ~~early~~final draft. It's known to be both ~~incomplete~~ and ~~in~~correkt, and it has ~~lots-of~~no bad formatting.

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1 Scope

[intro.scope]

“Eventually, all things merge into one, and a river runs through it.”

—*Norman Maclean*

¹ This document proposes to merge the ISO/IEC TS 21425:2017, aka the Ranges TS, into the working draft.

1.1 Revision History

[intro.history]

1.1.1 Revision 4

[intro.history.r4]

- (Throughout) Audit *requires-clauses* for non-primary-expressions. For example, `requires !C<T>` was a valid *requires-clause* per the Concepts TS, but C++20 requires (pun intended) `requires (!C<T>)`.
- (Throughout) Order iterator trait aliases consistently.
- (Throughout) Changed all `T foo {}`; member declarations to `T foo = T();` to avoid [LWG 3149](#).
- [17.4.11](#) Clarify that `t1` and `u1` may or may not be distinct. ADL is also performed for enumeration types. The fallback case is expression-equivalent to “an expression that exchanges the denoted values.”
- [16.3.1](#) Add `__cpp_lib_ranges`
- [19.10.11.1](#) Require nothrow SMFs for no-throw-sentinel; qualify references to `ranges::begin` and `::end`. Refer to quasi-CPOs as “entities” instead of “function templates”, since they likely are not actually function templates (also in [24.3](#)). Correct “explicitly-specified template arguments” to “explicitly-specified template argument lists” and duplicate this note for the identical requirement in [24.3](#). Add notes clarifying that these concepts don’t require that *all* operations of the corresponding iterator concept are non-throwing.
- [19.10.11.3](#) Use *voidify* in `std` algorithms.
- [19.10.11.4](#) Use *voidify* in `std` algorithms.
- [19.10.11.5](#) Use *voidify* in `std` algorithms.
- [19.10.11.6](#) Use *voidify* in `std` algorithms.
- [19.10.11.7](#) Use *voidify* in `std` algorithms.
- [22.1](#) Strike subclauses from Table 73
- [22.3.1](#) Several editorial clarifications from LWG review.
- [22.3.2.1](#) Replace use of `decay_t` with `remove_const_t`.
- [22.3.2.1](#) & [22.3.2.2](#) Strike noise paragraph “XXX is implemented as if:”. Change “names an instantiation of the primary template” to “names a specialization generated from the primary template”.
- [22.3.2.2](#) Change “names an instantiation of the primary template” to “names a specialization generated from the primary template”.
- [22.3.2.3](#) Fix broken wording in P1. Rephrase p2 without “non-program-defined”. Strike unneeded second sentence of P4 (There’s no reason to require that program-defined `iterator_concept` has any relation to a `std` type, or that it has special member functions needed to implement tag dispatch). Rename `BidirectionalIterator` to `BI` in example to avoid confusion with the concept of the same name. Change “program-defined specialization of `iterator_traits`” to “names a specialization generated from the primary template.” *cpp17-forward-iterator* must require an lvalue reference type. Change “names an instantiation of the primary template” to “names a specialization generated from the primary template”. Change “if `incrementable_traits<I>::difference_type` is well-formed...” to “if the *qualified-id* ... is valid and denotes a type.”
- [22.3.3.1](#) Don’t italicize “customization point object”, we’re not defining it here (or in [22.3.3.2](#)). Rephrase in the style of `ranges::swap` ([17.4.11](#)) (and in [22.3.3.2](#)).
- [22.3.3.2](#) Much rephrasing for clarification.

- 22.3.4.1 Strike p1. Clarify p2. Add a note explaining `ITER_TRAITS`. Change “names an instantiation of the primary template” to “names a specialization generated from the primary template”.
- 22.3.4.3 Add a note explaining the weird `const_cast` expressions.
- 22.3.4.8 editorial clarification; missing semicolon
- 22.3.4.11 Add a void cast to avoid (pun intended) ADL comma hijacking
- 22.3.4.13 Clarify p1; strike note and add a `// not defined` comment in the class definition. Harmonize wording with *Advanceable* (23.7.6.2).
- 22.3.6.3 Clarify p1
- 22.3.7.1 Clarify note
- 22.4.3.1 Rephrase all the things.
- 22.4.3.2 Rephrase p1.
- 22.5.1.2 Clarify p2
- 22.5.1.7 Explode p1 into *Constraints*: elements.
- 22.5.1.8 Expand the effects for `iter_move` and `iter_swap` to the intended implementation to avoid confusion over the conditional `noexcept` specifications. Simplify said `noexcepts` by using an lvalue *id-expression* instead of `declval`.
- 22.5.3.1 and 22.5.3.6 Change return type of `operator++` from `decltype(auto)` to `auto`
- 22.5.3.7 Explode p1 into *Constraints*: elements
- 22.5.3.8 Explode p1 into *Constraints*: elements
- 22.5.3.9, 22.5.3.10, 22.5.4.1, 22.5.4.3, 22.5.6.1, and 22.5.6.2 Fix misuses of `ConvertibleTo`.
- 22.5.4 Push hanging paragraphs down into 22.5.4.1.
- 22.5.4.1 Coalesce tiny subclauses. Strike unnecessary specialization of `readable_traits` (given that `iterator_traits` is specialized). Replace nested `difference_type` with a specialization of `incrementable_traits`.
- 22.5.4.4 Move conditional `noexcept` paragraph immediately after declaration of `operator->`. Mark `count()` as `noexcept`.
- 22.5.4.6 Replace `I1` and `S1` with `I` and `S`.
- 22.5.4.7 Get `I2` from `y`, not `I`.
- 22.5.5 Merge [default.sent].
- 22.5.6 Rearrange subclauses / titles / stable names in the style of the working draft. Merge all *Returns*: elements into *Effects*: .
- 22.5.6.1 A counted iterator knows the distance to the *end* of its range. Strike “possibly differing”. Strike unnecessary specialization of `readable_traits` (given that `iterator_traits` is specialized). Rename `cnt` to `length`. Replace nested `difference_type` with a specialization of `incrementable_traits`. Add default member initializers and `default` the default constructor.
- 22.5.6.2 Merge [counted.iter.op.conv].
- 22.5.6.5 Strike unnecessary *Expects*: element from `operator++(int)` requires `ForwardIterator<I>`, `operator+`, and `operator-`.
- 22.5.7.1 Strike “is a placeholder type that”. Strike p1s2.
- Clause 23 Consistently name `View` template parameters `V`, and rename `Range` template parameters from `O` to `R`.
- 23.1 Import [thread.decaycopy], add `noexcept` and `constexpr`, and reference it for uses of `DECAY_COPY` in [range].
- 23.2 Change stable name from “range.syn” to “ranges.syn” to reflect the header name `<ranges>`. Rearrange declarations to clarify namespaces. Add textual descriptions to “link” comments.
- 23.3.2 Clarify exactly which types model `Sentinel` in the note. (Also in 23.3.4, 23.3.6, and 23.3.8.)

- 23.4.1 Change `remove_cvref_t` to `remove_cv_t` since the type of an expression is never a reference type. Extract `disable_sized_range` from 1.2 and 1.3 and turn them into sub-bullets. Simplify 1.3 by naming the expression.
- 23.4.2 Add `()` around expression to clarify. Simplify 1.3 by naming the expression.
- 23.4.3 Remove "return `begin()` if it's a pointer" behavior.
- 23.5.1 Heavy rephrasing.
- 23.5.2 Much rephrasing for clarity. `begin` is *sometimes* required to be equality-preserving. A note is insufficient to cause `begin` and `end` to not generate implicit expression variations. *forwarding-range* is missing the most important semantic requirement! It also needs some explication.
- 23.5.3 Strike extraneous deduction constraint; rephrase.
- 23.5.5 Merge tiny range subclauses into one.
- 23.6.1 Relocate `[range.adaptors.helpers]` here under `[range.utility]`.
- 23.6.2 Require that the template parameter `D` is cv-unqualified. Apply post-completion requirements to `D`. Strike conversion operator in expectation that container constructors are a superior solution. Merge all function specifications into the class synopsis.
- 23.6.3 Reorganize and retitle subclauses. Clarify namespaces in class synopsis. Cleanup exposition-only concept definitions.
- 23.6.3.1 Use constructor delegation to save some paragraphs.
- 23.7.2 Change stable name from `[range.adaptor.semiregular_wrapper]`; editorial rephrasing.
- 23.7.3 Change stable name from `[range.adaptor.all]`.
- 23.7.3.1 Strike excessive conditional `noexcept`. Merge `[range.view.ref.ops]` into `[range.view.ref]`. Apply the LWG 2993 P/R to *ref-view*'s constructor, so *ref-view*<const R> isn't constructible from rvalues. Inline silly one-liner "Effects: Equivalent to" members into the class. Non-member `begin` and `end` take their argument by-value.
- 23.7.4 Change stable names, combine subclauses. Merge `end` overloads with deduced return type, and specify directly in the class body. Add missing precondition to `begin`.
- 23.7.4.3 Changing the value of an element in a filter view is ok, changing it to something the predicate would reject is UB. Negate the result of the predicate in `operator--`.
- 23.7.5 Rearrange subclauses. Inline effects of functions with deduced return types into class bodies.
- 23.7.5.2 Constraints should use `RegularInvocable` instead of `Invocable` (unlike the algorithm, the view can visit each element multiple times).
- 23.7.5.3 Fix definition of `Base`.
- 23.7.6 Use `W` as the name of `WeaklyIncrementable` template parameters.
- 23.7.6.2 Harmonize wording for *Advanceable* with `RandomAccessIterator`. Inline `size` into the class body since it has a deduced return type.
- 23.7.6.3 Use `i.value_` consistently in friends.
- 23.7.7.1 Change "the first `N` elements" to "at most `N` elements".
- 23.7.8.3 Pass an lvalue to `ranges::end` even when *inner-range* is an xvalue.
- 23.7.10 Remove extraneous `single_view` deduction guide.
- 23.7.11.2 Ensure that `remove_reference_t<R>::size()` is a constant expression before evaluating it in *tiny-range*.
- 23.7.11.5 Rename `zero_` to `incremented_`. Fix specification of `iter_swap`.
- 23.7.12 Properly require that `F` is convertible to `iter_difference_t<T>`. Force conversion of `F` to `T`'s difference type before adding to `E`. Don't force conversion of `F` to `T`'s difference type to construct `counted_iterator`; the conversion happens implicitly.
- 23.7.13.3 Clarify p1.1.
- 23.7.14.2 `reverse_iterator` CTAD breaks for `reverse_iterators`; use `make_reverse_iterator` instead.

- 24.3 Editorial clarification.
- 24.6.1 `last + M` should be `first + M` in `copy_n`'s “Returns:”.
- 24.7.3.3 and 24.7.3.4 Strike extraneous requirement.
- 24.7.4 Move effect from *Returns:* to *Effects:* .
- C.5.9 Add Annex C wording for incompatibilities introduced by forbidding explicit template arguments for algorithms in 24.3/14.

1.1.2 Revision 3

[intro.history.r3]

- (Throughout) Use `R` (`r`) instead of `Rng` (`rng`) for `Range` template (function) parameter names.
- (Throughout) In addition to changing the cross references, replace “is a contiguous iterator” in container requirements with “models `ContiguousIterator`”.
- 15.3.18 Use `string_view` instead of `char*` in example.
- 15.5.1.3 Rephrase final sentence
- 17.4.11 Define semantics of the swap concepts completely in the `ranges::swap` customization point, which moves here from [utility]
- 22.2 Replace bogus `-> auto&&` deduction constraints with a legit requirement that the expression has a referenceable type.
- 22.3.3.2 Vastly simplify conditional `noexcept` for the exposition-only *iter-exchange-move*.
- 23.6.3 Merge `subrange` deduction guides with identical parameter-declaration-clauses that otherwise conflict per [temp.deduct.guide]/3.
- 24.2 Simplify algorithm declarations with the form:

```
template<ForwardIterator I, [...]>
    requires Permutable<I> && [...]
[...]
```

to the equivalent (since `Permutable` subsumes `ForwardIterator`):

```
template<Permutable I, [...]>
    requires [...]
[...]
```

which favors smaller declarations over consistency in form between iterator-sentinel and range overloads of algorithms that require `Permutable`.

1.1.3 Revision 2

[intro.history.r2]

- Merge P0789R3.
- Merge P1033R1.
- Reformulate non-member operators of `common_iterator` and `counted_iterator` as members or hidden friends.
- Merge P1037R0.
- Merge P0970R1: Drop `dangling` per LEWG request, and make calls that would have returned a `dangling` iterator ill-formed instead by redefining `safe_iterator_t<R>` to be ill-formed when iterators from `R` may dangle.
- Merge P0944R0.
- Drop `tagged` and related machinery. Algorithm `foo` that did return a `tagged tuple` or `pair` now instead returns a named type `foo_result` with public data members whose names are the same as the previous set of tag names. Exceptions:
 - The single-range `transform` overload returns `unary_transform_result`.
 - The dual-range `transform` overload returns `binary_transform_result`.
- LEWG was disturbed by the use of `enable_if` to define `difference_type` and `value_type`; use `requires` clauses instead.
- Per LEWG direction, rename header `<range>` to `<ranges>` to agree with the namespace name `std::ranges`.

- Remove inappropriate usage of `value_type_t` in the insert iterators: design intent of `value_type_t` is to be only an associated type trait for `Readable`, and the `Container` type parameter of the insert iterators is not `Readable`.
- Use `remove_cvref_t` where appropriate.
- Restore the design intent that neither `Writable` types nor non-`Readable` Iterator types are required to have an equality-preserving `*` operator.
- Require semantics for the `Constructible` requirements of `IndirectlyMovableStorable` and `IndirectlyCopyableStorable`.
- Declare `constexpr` the algorithms that are so declared in the working draft.
- `constexpr` some `move_sentinel` members that we apparently missed in P0579.

1.1.4 Revision 1

[intro.history.r1]

- Remove section [std2.numerics] which is incorporated into P0898.
- Do not propose `ranges::exchange`: it is not used in the Ranges TS.
- Rewrite nearly everything to merge into `std::ranges`¹ rather than into `std2`:
 - Occurrences of “std2.” in stable names are either removed, or replaced with “range” when the name resulting from removal would conflict with an existing stable name.
- Incorporate the `std2::swap` customization point from P0898R0 as `ranges::swap`. (This was necessarily dropped from P0898R1.) Perform the necessary surgery on the `Swappable` concept from P0898R1 to restore the intended design that uses the renamed customization point.

1) `std::two` was another popular suggestion.

2 General Principles

[intro]

2.1 Goals

[intro.goals]

- ¹ The primary goal of this proposal is to deliver high-quality, constrained generic Standard Library components at the same time that the language gets support for such components.

2.2 Rationale

[intro.rationale]

- ¹ The best, and arguably only practical way to achieve the goal stated above is by incorporating the Ranges TS into the working paper. The sooner we can agree on what we want “**Iterator**” and “**Range**” to mean going forward (for instance), and the sooner users are able to rely on them, the sooner we can start building and delivering functionality on top of those fundamental abstractions. (For example, see “P0789: Range Adaptors and Utilities” ([4]).)
- ² The cost of not delivering such a set of Standard Library concepts and algorithms is that users will either do without or create a babel of mutually incompatible concepts and algorithms, often without the rigour followed by the Ranges TS. The experience of the authors and implementors of the Ranges TS is that getting concept definitions and algorithm constraints right is *hard*. The Standard Library should save its users from needless heartache.

2.3 Risks

[intro.risks]

- ¹ Shipping constrained components from the Ranges TS in the C++20 timeframe is not without risk. As of the time of writing (February 1, 2018), no major Standard Library vendor has shipped an implementation of the Ranges TS. Two of the three major compiler vendors have not even shipped an implementation of the concepts language feature. Arguably, we have not yet gotten the usage experience for which all Technical Specifications are intended.
- ² On the other hand, the components of Ranges TS have been vetted very thoroughly by the range-v3 ([3]) project, on which the Ranges TS is based. There is no part of the Ranges TS – concepts included – that has not seen extensive use via range-v3. (The concepts in range-v3 are emulated with high fidelity through the use of generalized SFINAE for expressions.) As an Open Source project, usage statistics are hard to come by, but the following may be indicative:
 - (2.1) — The range-v3 GitHub project has over 1,400 stars, over 120 watchers, and 145 forks.
 - (2.2) — It is cloned on average about 6,000 times a month.
 - (2.3) — A GitHub search, restricted to C++ files, for the string “**range/v3**” (a path component of all of range-v3’s header files), turns up over 7,000 hits.
- ³ Lacking true concepts, range-v3 cannot emulate concept-based function overloading, or the sorts of constraints-checking short-circuit evaluation required by true concepts. For that reason, the authors of the Ranges TS have created a reference implementation: CMCSTL2 ([1]) using true concepts. To this reference implementation, the authors ported all of range-v3’s tests. These exposed only a handful of concepts-specific bugs in the components of the Ranges TS (and a great many more bugs in compilers). Those improvements were back-ported to range-v3 where they have been thoroughly vetted over the past 2 years.
- ⁴ In short, concern about lack of implementation experience should not be a reason to withhold this important Standard Library advance from users.

2.4 Methodology

[intro.methodology]

- ¹ The contents of the Ranges TS, Clause 7 (“Concepts library”) are proposed for namespace `std` by P0898, “Standard Library Concepts” ([2]). Additionally, P0898 proposes the `identity` function object (ISO/IEC TS 21425:2017 §[func.identity]) and the `common_reference` type trait (ISO/IEC TS 21425:2017 §[meta.trans.other]) for namespace `std`. The changes proposed by the Ranges TS to `common_type` are merged into the working paper (also by P0898). The “`invoke`” function and the “`swappable`” type traits (e.g., `is_swappable_with`) already exist in the text of the working paper, so they are omitted here.
- ² The salient, high-level features of this proposal are as follows:
 - (2.1) — The remaining library components in the Ranges TS are proposed for namespace `::std::ranges`.

- (2.2) — The text of the Ranges TS is rebased on the latest working draft.
- (2.3) — Structurally, this paper proposes to specify each piece of `std::ranges` alongside the content of `std` from the same header. Since some Ranges TS components reuse names that previously had meaning in the C++ Standard, we sometimes rename old content to avoid name collisions.
- (2.4) — The content of headers from the Ranges TS with the same base name as a standard header are merged into that standard header. For example, the content of `<experimental/ranges/iterator>` will be merged into `<iterator>`. The new header `<experimental/ranges/range>` will be added under the name `<ranges>`.

2.5 Style of presentation

[intro.style]

- ¹ The remainder of this document is a technical specification in the form of editorial instructions directing that changes be made to the text of the C++ working draft. The formatting of the text suggests the origin of each portion of the wording.

Existing wording from the C++ working draft - included to provide context - is presented without decoration.

Entire clauses / subclauses / paragraphs incorporated from the Ranges TS are presented in a distinct cyan color.

In-line additions of wording from the Ranges TS to the C++ working draft are presented in cyan with underline.

~~In-line bits of wording that the Ranges TS strikes from the C++ working draft are presented in red with strike-through.~~

Wording to be added which is original to this document appears in gold with underline.

~~Wording which this document strikes is presented in magenta with strikethrough. (Hopefully context makes it clear whether the wording is currently in the C++ working draft, or wording that is not being added from the Ranges TS.)~~

Ideally, these formatting conventions make it clear which wording comes from which document in this three-way merge.

15 Library introduction

[library]

[...]

15.1 General

[library.general]

[...]

[Editor's note: Insert a new row in Table 17 for the ranges library:]

Table 17 — Library categories

Clause	Category
Clause [language.support]	Language support library
Clause 17	Concepts library
Clause [diagnostics]	Diagnostics library
Clause 19	General utilities library
Clause 20	Strings library
Clause [localization]	Localization library
Clause 21	Containers library
Clause 22	Iterators library
Clause 23	Ranges library
Clause 24	Algorithms library
Clause 25	Numerics library
Clause [input.output]	Input/output library
Clause [re]	Regular expressions library
Clause [atomics]	Atomic operations library
Clause [thread]	Thread support library

[...]

- ⁹ The containers ([Clause 21](#)), iterators ([Clause 22](#)), [ranges](#) ([Clause 23](#)), and algorithms ([Clause 24](#)) libraries provide a C++ program with access to a subset of the most widely used algorithms and data structures.

15.3 Definitions

[definitions]

[...]

15.3.18

[defns.projection]

projection

⟨function object argument⟩ transformation that an algorithm applies before inspecting the values of elements

[*Example:*

```
std::pair<int, std::string_view> pairs[] = {{2, "foo"}, {1, "bar"}, {0, "baz"}};
std::ranges::sort(pairs, std::ranges::less<>{}, [](auto const& p) { return p.first; });
```

sorts the pairs in increasing order of their first members:

```
{0, "baz"}, {1, "bar"}, {2, "foo"}
```

— *end example*

[...]

15.5 Library-wide requirements

[requirements]

[...]

15.5.1.2 Headers

[headers]

[...]

Table 18 — C++ library headers

<algorithm>	<forward_list>	<new>	<string_view>
<any>	<fstream>	<numeric>	<stringstream>
<array>	<functional>	<optional>	<syncstream>
<atomic>	<future>	<ostream>	<system_error>
<bit>	<initializer_list>	<queue>	<thread>
<bitset>	<iomanip>	<random>	<tuple>
<charconv>	<ios>	<ranges>	<typeindex>
<chrono>	<iosfwd>	<ratio>	<typeinfo>
<codecvt>	<iostream>	<regex>	<type_traits>
<compare>	<istream>	<scoped_allocator>	<unordered_map>
<complex>	<iterator>	<set>	<unordered_set>
<concepts>	<limits>	<shared_mutex>	<utility>
<condition_variable>	<list>		<valarray>
<contract>	<locale>	<sstream>	<variant>
<deque>	<map>	<stack>	<vector>
<exception>	<memory>	<stdexcept>	<version>
<execution>	<memory_resource>	<streambuf>	
<filesystem>	<mutex>	<string>	

15.5.1.3 Cpp17Allocator requirements

[allocator.requirements]

[...]

- ⁵ An allocator type `X` shall ~~satisfy~~meet the *Cpp17CopyConstructible* requirements (Table [copyconstructible]). The `X::pointer`, `X::const_pointer`, `X::void_pointer`, and `X::const_void_pointer` types shall ~~satisfy~~meet the *Cpp17NullablePointer* requirements (Table [nullablepointer]). No constructor, comparison function, copy operation, move operation, or swap operation on these pointer types shall exit via an exception. `X::pointer` and `X::const_pointer` shall also ~~satisfy~~meet the requirements for a ~~random access iterator~~Cpp17RandomAccessIterator (22.3.5.6) and ~~of a contiguous iterator (22.3.1)~~the additional requirement that, when `a` and `(a + n)` are dereferenceable pointer values for some integral value `n`,

```
addressof(*(a + n)) == addressof(*a) + n
```

is true.

16 Language support library

[language.support]

[...]

16.3 Implementation properties

[support.limits]

16.3.1 General

[support.limits.general]

[...]

Table 35 — Standard library feature-test macros

Macro name	Value	Header(s)
[...]	[...]	[...]
<code>__cpp_lib_ranges</code>	TBD	<code><algorithm></code> <code><functional></code> <code><iterator></code> <code><memory></code> <code><ranges></code>
[...]	[...]	[...]

[...]

17 Concepts library

[concepts]

17.3 Header <concepts> synopsis

[concepts.syn]

```

namespace std {
    [...]

    // [concept.assignable], concept Assignable
    template<class LHS, class RHS>
        concept Assignable = see below;

    // 17.4.11, concept Swappable
    namespace ranges {
        inline namespace unspecified {
            inline constexpr unspecified swap = unspecified;
        }
    }

    template<class T>
        concept Swappable = see below;
    template<class T, class U>
        concept SwappableWith = see below;

    [...]
}

```

17.4 Language-related concepts

[concepts.lang]

17.4.11 Concept Swappable

[concept.swappable]

- ¹ Let **t1** and **t2** be equality-preserving expressions that denote distinct equal objects of type **T**, and let **u1** and **u2** similarly denote distinct equal objects of type **U**. [Note: **t1** and **u1** can denote distinct objects, or the same object. — end note] An operation *exchanges the values* denoted by **t1** and **u1** if and only if the operation modifies neither **t2** nor **u2** and:
 - (1.1) — If **T** and **U** are the same type, the result of the operation is that **t1** equals **u2** and **u1** equals **t2**.
 - (1.2) — If **T** and **U** are different types that model `CommonReference<const T&, const U&>`, the result of the operation is that `C(t1)` equals `C(u2)` and `C(u1)` equals `C(t2)` where `C` is `common_reference_t<const T&, const U&>`.
- ² The name `ranges::swap` denotes a customization point object ([customization.point.object]). The expression `ranges::swap(E1, E2)` for some subexpressions **E1** and **E2** is expression-equivalent to:
 - (2.1) — `(void)swap(E1, E2)`², if **E1** or **E2** has class or enumeration type ([basic.compound]) and that expression is valid, with overload resolution performed in a context that includes the declarations


```

template<class T>
    void swap(T&, T&) = delete;
template<class T, size_t N>
    void swap(T(&)[N], T(&)[N]) = delete;
                    
```

 and does not include a declaration of `ranges::swap`. If the function selected by overload resolution does not exchange the values denoted by **E1** and **E2**, the program is ill-formed with no diagnostic required.
 - (2.2) — Otherwise, `(void)ranges::swap_ranges(E1, E2)` if **E1** and **E2** are lvalues of array types ([basic.compound]) with equal extent and `ranges::swap(*E1, *E2)` is a valid expression, except that `noexcept(ranges::swap(E1, E2))` is equal to `noexcept(ranges::swap(*E1, *E2))`.
 - (2.3) — Otherwise, if **E1** and **E2** are lvalues of the same type **T** that models `MoveConstructible<T>` and `Assignable<T&, T>`, an expression that exchanges the denoted values. `ranges::swap(E1, E2)` is a constant expression if

2) The name `swap` is used here unqualified.

- (2.3.1) — `T` is a literal type ([[basic.types](#)]),
- (2.3.2) — both `E1 = std::move(E2)` and `E2 = std::move(E1)` are constant subexpressions ([[defs.const.subexpr](#)]), and
- (2.3.3) — the full-expressions of the initializers in the declarations

```
T t1(std::move(E1));
T t2(std::move(E2));
```

are constant subexpressions.

`noexcept(ranges::swap(E1, E2))` is equal to `is_nothrow_move_constructible_v<T> && is_nothrow_move_assignable_v<T>`.

- (2.4) — Otherwise, `ranges::swap(E1, E2)` is ill-formed. [*Note: This case can result in substitution failure when `ranges::swap(E1, E2)` appears in the immediate context of a template instantiation. — end note*]
- ³ [*Note: Whenever `ranges::swap(E1, E2)` is a valid expression, it exchanges the values denoted by `E1` and `E2` and has type `void`. — end note*]

```
template<class T>
concept Swappable = is_swappable_v<T>;
concept Swappable = requires(T& a, T& b) { ranges::swap(a, b); };
```

- ⁴ Let `a1` and `a2` denote distinct equal objects of type `T`, and let `b1` and `b2` similarly denote distinct equal objects of type `T`. `Swappable<T>` is satisfied only if after evaluating either `swap(a1, b1)` or `swap(b1, a1)` in the context described below, `a1` is equal to `b2` and `b1` is equal to `a2`.

- ⁵ The context in which `swap(a1, b1)` or `swap(b1, a1)` is evaluated shall ensure that a binary non-member function named `swap` is selected via overload resolution ([[over.match](#)]) on a candidate set that includes:

- (5.1) — the two `swap` function templates defined in `<utility>` ([[utility](#)]) and
- (5.2) — the lookup set produced by argument-dependent lookup ([[basic.lookup.argdep](#)]).

```
template<class T, class U>
concept SwappableWith =
is_swappable_with_v<T, T> && is_swappable_with_v<U, U> &&
CommonReference<const remove_reference_t<T>&, const remove_reference_t<U>&> &&
is_swappable_with_v<T, U> && is_swappable_with_v<U, T>;
requires(T&& t, U&& u) {
    ranges::swap(std::forward<T>(t), std::forward<T>(t));
    ranges::swap(std::forward<U>(u), std::forward<U>(u));
    ranges::swap(std::forward<T>(t), std::forward<U>(u));
    ranges::swap(std::forward<U>(u), std::forward<T>(t));
};
```

- ⁶ Let:

- (6.1) — `t1` and `t2` denote distinct equal objects of type `remove_cvref_t<T>`,
- (6.2) — `Et` be an expression that denotes `t1` such that `decltype((Et))` is `T`,
- (6.3) — `u1` and `u2` similarly denote distinct equal objects of type `remove_cvref_t<U>`,
- (6.4) — `Eu` be an expression that denotes `u1` such that `decltype((Eu))` is `U`, and
- (6.5) — `C` be

```
common_reference_t<const remove_reference_t<T>&, const remove_reference_t<U>&>
```

`SwappableWith<T, U>` is satisfied only if after evaluating either `swap(Et, Eu)` or `swap(Eu, Et)` in the context described above, `C(t1)` is equal to `C(u2)` and `C(u1)` is equal to `C(t2)`.

- ⁷ The context in which `swap(Et, Eu)` or `swap(Eu, Et)` is evaluated shall ensure that a binary non-member function named `swap` is selected via overload resolution ([[over.match](#)]) on a candidate set that includes:

- (7.1) — the two `swap` function templates defined in `<utility>` ([[utility](#)]) and

(7.2) — the lookup set produced by argument-dependent lookup ([basic.lookup.argdep]).

⁸ [Note: The semantics of the Swappable and SwappableWith concepts are fully defined by the `ranges::swap` customization point. — end note]

⁹ [Example: User code can ensure that the evaluation of `swap` calls is performed in an appropriate context under the various conditions as follows:

```
#include <cassert>
#include <concepts>
#include <utility>

namespace ranges = std::ranges;

template<class T, std::SwappableWith<T> U>
void value_swap(T&& t, U&& u) {
    using std::swap;
    ranges::swap(std::forward<T>(t), std::forward<U>(u)); // OK: uses ‘swappable with’ conditions
                                                         // for rvalues and lvalues
}

template<std::Swappable T>
void lv_swap(T& t1, T& t2) {
    using std::swap;
    ranges::swap(t1, t2); // OK: uses swappable conditions for
                          // lvalues of type T
}

namespace N {
    struct A { int m; };
    struct Proxy { A* a; };
    Proxy proxy(A& a) { return Proxy{ &a }; }

    void swap(A& x, Proxy p) {
        std::ranges::swap(x.m, p.a->m); // OK: uses context equivalent to swappable
                                           // conditions for fundamental types
    }
    void swap(Proxy p, A& x) { swap(x, p); } // satisfy symmetry constraint requirement
}

int main() {
    int i = 1, j = 2;
    lv_swap(i, j);
    assert(i == 2 && j == 1);

    N::A a1 = { 5 }, a2 = { -5 };
    value_swap(a1, proxy(a2));
    assert(a1.m == -5 && a2.m == 5);
}
```

— end example]

19 General utilities library

[utilities]

[...]

19.5 Tuples

[tuple]

[...]

19.5.3 Class template tuple

[tuple.tuple]

[...]

19.5.3.6 Tuple helper classes

[tuple.helper]

[...]

6 In addition to being available via inclusion of the `<tuple>` header, the three templates are available when ~~either~~any of the headers `<array>`, ~~or~~ `<ranges>`, or `<utility>` are included.

[...]

8 In addition to being available via inclusion of the `<tuple>` header, the three templates are available when ~~either~~any of the headers `<array>`, ~~or~~ `<ranges>`, or `<utility>` are included.

[...]

19.10 Memory

[memory]

19.10.2 Header `<memory>` synopsis

[memory.syn]

[...]

```
namespace std {
    [...]

    // [default allocator], the default allocator
    template<class T> class allocator;
    template<class T, class U>
        bool operator==(const allocator<T>&, const allocator<U>&) noexcept;
    template<class T, class U>
        bool operator!=(const allocator<T>&, const allocator<U>&) noexcept;

    // 19.10.11, specialized algorithms
    // 19.10.11.1, special memory concepts
    template<class I>
        concept no-throw-input-iterator = see below; // exposition only

    template<class I>
        concept no-throw-forward-iterator = see below; // exposition only

    template<class S, class I>
        concept no-throw-sentinel = see below; // exposition only

    template<class R>
        concept no-throw-input-range = see below; // exposition only

    template<class R>
        concept no-throw-forward-range = see below; // exposition only

    template<class T>
        constexpr T* addressof(T& r) noexcept;
    template<class T>
        const T* addressof(const T&&) = delete;
    template<class ForwardIterator>
        void uninitialized_default_construct(ForwardIterator first, ForwardIterator last);
```

```

template<class ExecutionPolicy, class ForwardIterator>
    void uninitialized_default_construct(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
                                       ForwardIterator first, ForwardIterator last);
template<class ForwardIterator, class Size>
    ForwardIterator uninitialized_default_construct_n(ForwardIterator first, Size n);
template<class ExecutionPolicy, class ForwardIterator, class Size>
    ForwardIterator uninitialized_default_construct_n(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
                                                   ForwardIterator first, Size n);

namespace ranges {
    template<no-throw-forward-iterator I, no-throw-sentinel<I> S>
        requires DefaultConstructible<iter_value_t<I>>>
        I uninitialized_default_construct(I first, S last);
    template<no-throw-forward-range R>
        requires DefaultConstructible<iter_value_t<iterator_t<R>>>
        safe_iterator_t<R> uninitialized_default_construct(R&& r);

    template<no-throw-forward-iterator I>
        requires DefaultConstructible<iter_value_t<I>>>
        I uninitialized_default_construct_n(I first, iter_difference_t<I> n);
}

template<class ForwardIterator>
    void uninitialized_value_construct(ForwardIterator first, ForwardIterator last);
template<class ExecutionPolicy, class ForwardIterator>
    void uninitialized_value_construct(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
                                       ForwardIterator first, ForwardIterator last);
template<class ForwardIterator, class Size>
    ForwardIterator uninitialized_value_construct_n(ForwardIterator first, Size n);
template<class ExecutionPolicy, class ForwardIterator, class Size>
    ForwardIterator uninitialized_value_construct_n(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
                                                   ForwardIterator first, Size n);

namespace ranges {
    template<no-throw-forward-iterator I, no-throw-sentinel<I> S>
        requires DefaultConstructible<iter_value_t<I>>>
        I uninitialized_value_construct(I first, S last);
    template<no-throw-forward-range R>
        requires DefaultConstructible<iter_value_t<iterator_t<R>>>
        safe_iterator_t<R> uninitialized_value_construct(R&& r);

    template<no-throw-forward-iterator I>
        requires DefaultConstructible<iter_value_t<I>>>
        I uninitialized_value_construct_n(I first, iter_difference_t<I> n);
}

template<class InputIterator, class ForwardIterator>
    ForwardIterator uninitialized_copy(InputIterator first, InputIterator last,
                                     ForwardIterator result);
template<class ExecutionPolicy, class InputIterator, class ForwardIterator>
    ForwardIterator uninitialized_copy(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
                                     InputIterator first, InputIterator last,
                                     ForwardIterator result);
template<class InputIterator, class Size, class ForwardIterator>
    ForwardIterator uninitialized_copy_n(InputIterator first, Size n,
                                       ForwardIterator result);
template<class ExecutionPolicy, class InputIterator, class Size, class ForwardIterator>
    ForwardIterator uninitialized_copy_n(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
                                       InputIterator first, Size n,
                                       ForwardIterator result);

namespace ranges {
    template<class I, class O>
        using uninitialized_copy_result = copy_result<I, O>;
}

```

```

template<InputIterator I, Sentinel<I> S1, no-throw-forward-iterator O, no-throw-sentinel<O> S2>
    requires Constructible<iter_value_t<O>, iter_reference_t<I>>
    uninitialized_copy_result<I, O>
    uninitialized_copy(I ifirst, S1 ilast, O ofirst, S2 olast);
template<InputRange IR, no-throw-forward-range OR>
    requires Constructible<iter_value_t<iterator_t<OR>>, iter_reference_t<iterator_t<IR>>>
    uninitialized_copy_result<safe_iterator_t<IR>, safe_iterator_t<OR>>
    uninitialized_copy(IR&& input_range, OR&& output_range);

template<class I, class O>
    using uninitialized_copy_n_result = uninitialized_copy_result<I, O>;
template<InputIterator I, no-throw-forward-iterator O, no-throw-sentinel<O> S>
    requires Constructible<iter_value_t<O>, iter_reference_t<I>>
    uninitialized_copy_n_result<I, O>
    uninitialized_copy_n(I ifirst, iter_difference_t<I> n, O ofirst, S olast);
}

template<class InputIterator, class ForwardIterator>
    ForwardIterator uninitialized_move(InputIterator first, InputIterator last,
                                     ForwardIterator result);
template<class ExecutionPolicy, class InputIterator, class ForwardIterator>
    ForwardIterator uninitialized_move(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
                                     InputIterator first, InputIterator last,
                                     ForwardIterator result);
template<class InputIterator, class Size, class ForwardIterator>
    pair<InputIterator, ForwardIterator> uninitialized_move_n(InputIterator first, Size n,
                                                            ForwardIterator result);
template<class ExecutionPolicy, class InputIterator, class Size, class ForwardIterator>
    pair<InputIterator, ForwardIterator> uninitialized_move_n(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
                                                            InputIterator first, Size n,
                                                            ForwardIterator result);

namespace ranges {
    template<class I, class O>
        using uninitialized_move_result = uninitialized_copy_result<I, O>;
    template<InputIterator I, Sentinel<I> S1, no-throw-forward-iterator O, no-throw-sentinel<O> S2>
        requires Constructible<iter_value_t<O>, iter_rvalue_reference_t<I>>
        uninitialized_move_result<I, O>
        uninitialized_move(I ifirst, S1 ilast, O ofirst, S2 olast);
    template<InputRange IR, no-throw-forward-range OR>
        requires Constructible<iter_value_t<iterator_t<OR>>, iter_rvalue_reference_t<iterator_t<IR>>>
        uninitialized_move_result<safe_iterator_t<IR>, safe_iterator_t<OR>>
        uninitialized_move(IR&& input_range, OR&& output_range);

    template<class I, class O>
        using uninitialized_move_n_result = uninitialized_copy_result<I, O>;
    template<InputIterator I, no-throw-forward-iterator O, no-throw-sentinel<O> S>
        requires Constructible<iter_value_t<O>, iter_rvalue_reference_t<I>>
        uninitialized_move_n_result<I, O>
        uninitialized_move_n(I ifirst, iter_difference_t<I> n, O ofirst, S olast);
}

template<class ForwardIterator, class T>
    void uninitialized_fill(ForwardIterator first, ForwardIterator last, const T& x);
template<class ExecutionPolicy, class ForwardIterator, class T>
    void uninitialized_fill(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
                           ForwardIterator first, ForwardIterator last, const T& x);
template<class ForwardIterator, class Size, class T>
    ForwardIterator uninitialized_fill_n(ForwardIterator first, Size n, const T& x);
template<class ExecutionPolicy, class ForwardIterator, class Size, class T>
    ForwardIterator uninitialized_fill_n(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
                                       ForwardIterator first, Size n, const T& x);

```

```

namespace ranges {
    template<no-throw-forward-iterator I, no-throw-sentinel<I> S, class T>
        requires Constructible<iter_value_t<I>, const T&>
        I uninitialized_fill(I first, S last, const T& x);
    template<no-throw-forward-range R, class T>
        requires Constructible<iter_value_t<iterator_t<R>>, const T&>
        safe_iterator_t<R> uninitialized_fill(R&& r, const T& x);

    template<no-throw-forward-iterator I, class T>
        requires Constructible<iter_value_t<I>, const T&>
        I uninitialized_fill_n(I first, iter_difference_t<I> n, const T& x);
}

template<class T>
    void destroy_at(T* location);
template<class ForwardIterator>
    void destroy(ForwardIterator first, ForwardIterator last);
template<class ExecutionPolicy, class ForwardIterator>
    void destroy(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
        ForwardIterator first, ForwardIterator last);
template<class ForwardIterator, class Size>
    ForwardIterator destroy_n(ForwardIterator first, Size n);
template<class ExecutionPolicy, class ForwardIterator, class Size>
    ForwardIterator destroy_n(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
        ForwardIterator first, Size n);

namespace ranges {
    template<Destructible T>
        void destroy_at(T* location) noexcept;

    template<no-throw-input-iterator I, no-throw-sentinel<I> S>
        requires Destructible<iter_value_t<I>>
        I destroy(I first, S last) noexcept;
    template<no-throw-input-range R>
        requires Destructible<iter_value_t<iterator_t<R>>
        safe_iterator_t<R> destroy(R&& r) noexcept;

    template<no-throw-input-iterator I>
        requires Destructible<iter_value_t<I>>
        I destroy_n(I first, iter_difference_t<I> n) noexcept;
}

[...]
}
[...]
```

19.10.11 Specialized algorithms

[specialized.algorithms]

- ¹ Throughout this subclause, the names of template parameters are used to express type requirements for those algorithms defined directly in namespace std.
- (1.1) — If an algorithm's template parameter is named `InputIterator`, the template argument shall ~~satisfy~~meet the *Cpp17InputIterator* requirements (22.3.5.2).
- (1.2) — If an algorithm's template parameter is named `ForwardIterator`, the template argument shall ~~satisfy~~meet the *Cpp17ForwardIterator* requirements (22.3.5.4), and is required to have the property that no exceptions are thrown from increment, assignment, comparison, or indirection through valid iterators.

~~Unless otherwise specified, if an exception is thrown in the following algorithms there are no effects.~~

- ² Unless otherwise specified, if an exception is thrown in the following algorithms there are no effects.
- ³ In the description of the algorithms, operator `-` is used for some of the iterator categories for which it need not be defined. In these cases, the value of the expression `b - a` is the number of increments of `a` needed to make `bool(a == b)` be true.

⁴ The following additional requirements apply for those algorithms defined in namespace `std::ranges`:

- (4.1) — The entities defined in the `std::ranges` namespace in this subclause are not found by argument-dependent name lookup ([[basic.lookup.argdep](#)]). When found by unqualified ([[basic.lookup.unqual](#)]) name lookup for the *postfix-expression* in a function call ([[expr.call](#)]), they inhibit argument-dependent name lookup.
- (4.2) — Overloads of algorithms that take `Range` arguments ([23.5.2](#)) behave as if they are implemented by calling `ranges::begin` and `ranges::end` on the `Range(s)` and dispatching to the overload that takes separate iterator and sentinel arguments.
- (4.3) — The number and order of deducible template parameters for algorithm declarations is unspecified, except where explicitly stated otherwise. [Note: Consequently, these algorithms may not be called with explicitly-specified template argument lists. — end note]

⁵ [Note: When invoked on ranges of potentially overlapping subobjects ([[intro.object](#)]), the algorithms specified in this subclause [19.10.11](#) result in undefined behavior. — end note]

⁶ Some algorithms defined in this clause make use of the exposition-only function *voidify*:

```
template<class T>
void* voidify(T& ptr) noexcept {
    return const_cast<void*>(static_cast<const volatile void*>(addressof(ptr)));
}
```

19.10.11.1 Special memory concepts

[[special.mem.concepts](#)]

¹ Some algorithms in this subclause are constrained with the following exposition-only concepts:

```
template<class I>
concept no_throw_input_iterator = // exposition only
    InputIterator<I> &&
    is_lvalue_reference_v<iter_reference_t<I>> &&
    Same<remove_cvref_t<iter_reference_t<I>>, iter_value_t<I>>;
```

² No exceptions are thrown from increment, copy construction, move construction, copy assignment, move assignment, or indirection through valid iterators.

³ [Note: This concept allows some `InputIterator` ([22.3.4.9](#)) operations to throw exceptions. — end note]

```
template<class S, class I>
concept no_throw_sentinel = Sentinel<S, I>; // exposition only
```

⁴ No exceptions are thrown from copy construction, move construction, copy assignment, move assignment, or comparisons between valid values of type `I` and `S`.

⁵ [Note: This concept allows some `Sentinel` ([22.3.4.7](#)) operations to throw exceptions. — end note]

```
template<class R>
concept no_throw_input_range = // exposition only
    Range<R> &&
    no_throw_input_iterator<iterator_t<R>> &&
    no_throw_sentinel<sentinel_t<R>, iterator_t<R>>;
```

⁶ No exceptions are thrown from calls to `ranges::begin` and `ranges::end` on an object of type `R`.

```
template<class I>
concept no_throw_forward_iterator = // exposition only
    no_throw_input_iterator<I> &&
    ForwardIterator<I> &&
    no_throw_sentinel<I, I>;
```

⁷ [Note: This concept allows some `ForwardIterator` ([22.3.4.11](#)) operations to throw exceptions. — end note]

```
template<class R>
concept no_throw_forward_range = // exposition only
    no_throw_input_range<R> &&
    no_throw_forward_iterator<iterator_t<R>>;
```

19.10.11.2 **addressof** [specialized.addressof]
 [...]

19.10.11.3 **uninitialized_default_construct** [uninitialized.construct.default]

```
template<class ForwardIterator>
void uninitialized_default_construct(ForwardIterator first, ForwardIterator last);
```

1 *Effects:* Equivalent to:

```
    for (; first != last; ++first)
        ::new (static_cast<void*>(addressof voidify(*first)))
            typename iterator_traits<ForwardIterator>::value_type;
```

```
namespace ranges {
    template<no-throw-forward-iterator I, no-throw-sentinel<I> S>
        requires DefaultConstructible<iter_value_t<I>>
        I uninitialized_default_construct(I first, S last);
    template<no-throw-forward-range R>
        requires DefaultConstructible<iter_value_t<iterator_t<R>>>
        safe_iterator_t<R> uninitialized_default_construct(R&& r);
}
```

2 *Effects:* Equivalent to:

```
    for (; first != last; ++first)
        ::new (voidify(*first)) remove_reference_t<iter_reference_t<I>>;
    return first;
```

```
template<class ForwardIterator, class Size>
ForwardIterator uninitialized_default_construct_n(ForwardIterator first, Size n);
```

3 *Effects:* Equivalent to:

```
    for (; n > 0; (void)++first, --n)
        ::new (static_cast<void*>(addressof voidify(*first)))
            typename iterator_traits<ForwardIterator>::value_type;
    return first;
```

```
namespace ranges {
    template<no-throw-forward-iterator I>
        requires DefaultConstructible<iter_value_t<I>>
        I uninitialized_default_construct_n(I first, iter_difference_t<I> n);
}
```

4 *Effects:* Equivalent to:

```
    return uninitialized_default_construct(counted_iterator(first, n),
                                          default_sentinel).base();
```

19.10.11.4 **uninitialized_value_construct** [uninitialized.construct.value]

```
template<class ForwardIterator>
void uninitialized_value_construct(ForwardIterator first, ForwardIterator last);
```

1 *Effects:* Equivalent to:

```
    for (; first != last; ++first)
        ::new (static_cast<void*>(addressof voidify(*first)))
            typename iterator_traits<ForwardIterator>::value_type();
```

```
namespace ranges {
    template<no-throw-forward-iterator I, no-throw-sentinel<I> S>
        requires DefaultConstructible<iter_value_t<I>>
        I uninitialized_value_construct(I first, S last);
    template<no-throw-forward-range R>
        requires DefaultConstructible<iter_value_t<iterator_t<R>>>
        safe_iterator_t<R> uninitialized_value_construct(R&& r);
}
```

2 *Effects:* Equivalent to:

```

    for (; first != last; ++first)
        ::new (voidify(*first)) remove_reference_t<iter_reference_t<I>>>();
    return first;

```

```

template<class ForwardIterator, class Size>
ForwardIterator uninitialized_value_construct_n(ForwardIterator first, Size n);

```

3 *Effects:* Equivalent to:

```

    for (; n > 0; (void)++first, --n)
        ::new (static_cast<void*>(addressof voidify(*first)))
            typename iterator_traits<ForwardIterator>::value_type();
    return first;

```

```

namespace ranges {
    template<no-throw-forward-iterator I>
        requires DefaultConstructible<iter_value_t<I>>
        I uninitialized_value_construct_n(I first, iter_difference_t<I> n);
}

```

4 *Effects:* Equivalent to:

```

    return uninitialized_value_construct(counted_iterator(first, n),
        default_sentinel).base();

```

19.10.11.5 uninitialized_copy

[uninitialized.copy]

```

template<class InputIterator, class ForwardIterator>
ForwardIterator uninitialized_copy(InputIterator first, InputIterator last,
    ForwardIterator result);

```

1 *Expects:* [result, (last - first)) shall not overlap with [first, last).

2 *Effects:* As-if-by Equivalent to:

```

    for (; first != last; ++result, (void) ++first)
        ::new (static_cast<void*>(addressof voidify(*result)))
            typename iterator_traits<ForwardIterator>::value_type(*first);

```

3 *Returns:* result.

```

namespace ranges {
    template<InputIterator I, Sentinel<I> S1, no-throw-forward-iterator O, no-throw-sentinel<O> S2>
        requires Constructible<iter_value_t<O>, iter_reference_t<I>>
        uninitialized_copy_result<I, O>
            uninitialized_copy(I ifirst, S1 ilast, O ofirst, S2 olast);
    template<InputRange IR, no-throw-forward-range OR>
        requires Constructible<iter_value_t<iterator_t<OR>>, iter_reference_t<iterator_t<IR>>>
        uninitialized_copy_result<safe_iterator_t<IR>, safe_iterator_t<OR>>
            uninitialized_copy(IR&& input_range, OR&& output_range);
}

```

4 *Expects:* [ofirst, olast) shall not overlap with [ifirst, ilast).

5 *Effects:* Equivalent to:

```

    for (; ifirst != ilast && ofirst != olast; ++ofirst, (void)++ifirst) {
        ::new (voidify(*ofirst)) remove_reference_t<iter_reference_t<O>>(*ifirst);
    }
    return {ifirst, ofirst};

```

```

template<class InputIterator, class Size, class ForwardIterator>
ForwardIterator uninitialized_copy_n(InputIterator first, Size n, ForwardIterator result);

```

6 *Expects:* [result, n) shall not overlap with [first, n).

7 *Effects:* As-if-by Equivalent to:

```

    for (; n > 0; ++result, (void) ++first, --n) {
        ::new (static_cast<void*>(addressof voidify(*result)))
            typename iterator_traits<ForwardIterator>::value_type(*first);
    }

```

8 *Returns:* result.

```
namespace ranges {
    template<InputIterator I, no-throw-forward-iterator O, no-throw-sentinel<O> S>
        requires Constructible<iter_value_t<O>, iter_reference_t<I>>
        uninitialized_copy_n_result<I, O>
        uninitialized_copy_n(I ifirst, iter_difference_t<I> n, O ofirst, S olast);
}
```

9 *Expects:* [ofirst, olast) shall not overlap with [ifirst, n).

10 *Effects:* Equivalent to:

```
    auto t = uninitialized_copy(counted_iterator(ifirst, n),
                                default_sentinel, ofirst, olast);
    return {t.in.base(), t.out};
```

19.10.11.6 uninitialized_move

[uninitialized.move]

```
template<class InputIterator, class ForwardIterator>
    ForwardIterator uninitialized_move(InputIterator first, InputIterator last,
                                      ForwardIterator result);
```

1 *Expects:* [result, (last - first)) shall not overlap with [first, last).

2 *Effects:* Equivalent to:

```
    for (; first != last; (void)++result, ++first)
        ::new (static_cast<void*>(addressof voidify(*result)))
            typename iterator_traits<ForwardIterator>::value_type(std::move(*first));
    return result;
```

3 *Remarks:* If an exception is thrown, some objects in the range [first, last) are left in a valid but unspecified state.

```
namespace ranges {
    template<InputIterator I, Sentinel<I> S1, no-throw-forward-iterator O, no-throw-sentinel<O> S2>
        requires Constructible<iter_value_t<O>, iter_rvalue_reference_t<I>>
        uninitialized_move_result<I, O>
        uninitialized_move(I ifirst, S1 ilast, O ofirst, S2 olast);
    template<InputRange IR, no-throw-forward-range OR>
        requires Constructible<iter_value_t<iterator_t<OR>>, iter_rvalue_reference_t<iterator_t<IR>>>
        uninitialized_move_result<safe_iterator_t<IR>, safe_iterator_t<OR>>
        uninitialized_move(IR&& input_range, OR&& output_range);
}
```

4 *Expects:* [ofirst, olast) shall not overlap with [ifirst, ilast).

5 *Effects:* Equivalent to:

```
    for (; ifirst != ilast && ofirst != olast; ++ofirst, (void)++ifirst) {
        ::new (voidify(*ofirst)) remove_reference_t<iter_reference_t<O>>(ranges::iter_move(ifirst));
    }
    return {ifirst, ofirst};
```

6 [Note: If an exception is thrown, some objects in the range [first, last) are left in a valid, but unspecified state. — end note]

```
template<class InputIterator, class Size, class ForwardIterator>
    pair<InputIterator, ForwardIterator>
    uninitialized_move_n(InputIterator first, Size n, ForwardIterator result);
```

7 *Expects:* [result, n) shall not overlap with [first, n).

8 *Effects:* Equivalent to:

```
    for (; n > 0; ++result, (void) ++first, --n)
        ::new (static_cast<void*>(addressof voidify(*result)))
            typename iterator_traits<ForwardIterator>::value_type(std::move(*first));
    return {first, result};
```

9 *Remarks:* If an exception is thrown, some objects in the range [first, std::next(first, n)] are left in a valid but unspecified state.

```

namespace ranges {
    template<InputIterator I, no-throw-forward-iterator O, no-throw-sentinel<O> S>
        requires Constructible<iter_value_t<O>, iter_rvalue_reference_t<I>>
        uninitialized_move_n_result<I, O>
        uninitialized_move_n(I ifirst, iter_difference_t<I> n, O ofirst, S olast);
}

```

10 Expects: [ofirst, olast) shall not overlap with [ifirst, n).

11 Effects: Equivalent to:

```

    auto t = uninitialized_move(counted_iterator(ifirst, n),
                                default_sentinel, ofirst, olast);
    return {t.in.base(), t.out};

```

12 [Note: If an exception is thrown, some objects in the range [first, n) are left in a valid but unspecified state. — end note]

19.10.11.7 uninitialized_fill

[uninitialized.fill]

```

template<class ForwardIterator, class T>
void uninitialized_fill(ForwardIterator first, ForwardIterator last, const T& x);

```

1 Effects: ~~As if by~~ Equivalent to:

```

    for (; first != last; ++first)
        ::new (static_cast<void*>(&voidify(*first)))
            typename iterator_traits<ForwardIterator>::value_type(x);

```

```

namespace ranges {
    template<no-throw-forward-iterator I, no-throw-sentinel<I> S, class T>
        requires Constructible<iter_value_t<I>, const T&>
        I uninitialized_fill(I first, S last, const T& x);
    template<no-throw-forward-range R, class T>
        requires Constructible<iter_value_t<iterator_t<R>>, const T&>
        safe_iterator_t<R> uninitialized_fill(R&& r, const T& x);
}

```

Effects: Equivalent to:

```

    for (; first != last; ++first) {
        ::new (voidify(*first)) remove_reference_t<iter_reference_t<I>>(x);
    }
    return first;

```

```

template<class ForwardIterator, class Size, class T>
ForwardIterator uninitialized_fill_n(ForwardIterator first, Size n, const T& x);

```

2 Effects: ~~As if by~~ Equivalent to:

```

    for (; n--; ++first)
        ::new (static_cast<void*>(&voidify(*first)))
            typename iterator_traits<ForwardIterator>::value_type(x);
    return first;

```

```

namespace ranges {
    template<no-throw-forward-iterator I, class T>
        requires Constructible<iter_value_t<I>, const T&>
        I uninitialized_fill_n(I first, iter_difference_t<I> n, const T& x);
}

```

3 Effects: Equivalent to:

```

    return uninitialized_fill(counted_iterator(first, n), default_sentinel, x).base();

```

19.10.11.8 destroy

[specialized.destroy]

```

template<class T>
void destroy_at(T* location);

```

```
namespace ranges {
    template<Destructible T>
        void destroy_at(T* location) noexcept;
}
```

1 *Effects:*

(1.1) — If T is an array type, equivalent to `destroy(begin(*location), end(*location))`.

(1.2) — Otherwise, equivalent to `location->~T()`~~;~~;

```
template<class ForwardIterator>
    void destroy(ForwardIterator first, ForwardIterator last);
```

2 *Effects:* Equivalent to:

```
    for (; first!=last; ++first)
        destroy_at(addressof(*first));
```

```
namespace ranges {
    template<no-throw-input-iterator I, no-throw-sentinel<I> S>
        requires Destructible<iter_value_t<I>>
        I destroy(I first, S last) noexcept;
    template<no-throw-input-range R>
        requires Destructible<iter_value_t<iterator_t<R>>>
        safe_iterator_t<R> destroy(R&& r) noexcept;
}
```

3 *Effects:* Equivalent to:

```
    for (; first != last; ++first)
        destroy_at(addressof(*first));
    return first;
```

```
template<class ForwardIterator, class Size>
    ForwardIterator destroy_n(ForwardIterator first, Size n);
```

4 *Effects:* Equivalent to:

```
    for (; n > 0; (void)++first, --n)
        destroy_at(addressof(*first));
    return first;
```

```
namespace ranges {
    template<no-throw-input-iterator I>
        requires Destructible<iter_value_t<I>>
        I destroy_n(I first, iter_difference_t<I> n) noexcept;
}
```

5 *Effects:* Equivalent to:

```
    return destroy(counted_iterator(first, n), default_sentinel).base();
```

[...]

19.14 Function Objects

[function.objects]

19.14.1 Header <functional> synopsis

[functional.syn]

[...]

```
template<class T>
    inline constexpr bool is_bind_expression_v = is_bind_expression<T>::value;
template<class T>
    inline constexpr int is_placeholder_v = is_placeholder<T>::value;

namespace ranges {
    // 19.14.8, comparisons
    template<class T = void>
        requires see below
        struct equal_to;
```

```

    template<class T = void>
        requires see below
    struct not_equal_to;

    template<class T = void>
        requires see below
    struct greater;

    template<class T = void>
        requires see below
    struct less;

    template<class T = void>
        requires see below
    struct greater_equal;

    template<class T = void>
        requires see below
    struct less_equal;

    template<> struct equal_to<void>;
    template<> struct not_equal_to<void>;
    template<> struct greater<void>;
    template<> struct less<void>;
    template<> struct greater_equal<void>;
    template<> struct less_equal<void>;
}
}
[...]
```

19.14.7 Comparisons

[comparisons]

[...]

19.14.8 Comparisons (ranges)

[range.comparisons]

- ¹ In this subclause, *BUILTIN_PTR_CMP*(*T*, *op*, *U*) for types *T* and *U* and where *op* is an equality ([*expr.eq*]) or relational operator ([*expr.rel*]) is a boolean constant expression. *BUILTIN_PTR_CMP*(*T*, *op*, *U*) is *true* if and only if *op* in the expression *declval*<*T*>() *op* *declval*<*U*>() resolves to a built-in operator comparing pointers.
- ² There is an implementation-defined strict total ordering over all pointer values of a given type. This total ordering is consistent with the partial order imposed by the builtin operators <, >, <=, and >=.

```

template<class T = void>
    requires EqualityComparable<T> || Same<T, void> || BUILTIN_PTR_CMP(const T&, ==, const T&)
struct equal_to {
    constexpr bool operator()(const T& x, const T& y) const;
};
```

- ³ operator() has effects equivalent to: return ranges::equal_to<>{}(x, y);

```

template<class T = void>
    requires EqualityComparable<T> || Same<T, void> || BUILTIN_PTR_CMP(const T&, ==, const T&)
struct not_equal_to {
    constexpr bool operator()(const T& x, const T& y) const;
};
```

- ⁴ operator() has effects equivalent to: return !ranges::equal_to<>{}(x, y);

```

template<class T = void>
    requires StrictTotallyOrdered<T> || Same<T, void> || BUILTIN_PTR_CMP(const T&, <, const T&)
struct greater {
    constexpr bool operator()(const T& x, const T& y) const;
};
```

- ⁵ operator() has effects equivalent to: return ranges::less<>{}(y, x);

```

template<class T = void>
    requires StrictTotallyOrdered<T> || Same<T, void> || BUILTIN_PTR_CMP(const T&, <, const T&)
struct less {
    constexpr bool operator()(const T& x, const T& y) const;
};
6     operator() has effects equivalent to: return ranges::less<>{}(x, y);

template<class T = void>
    requires StrictTotallyOrdered<T> || Same<T, void> || BUILTIN_PTR_CMP(const T&, <, const T&)
struct greater_equal {
    constexpr bool operator()(const T& x, const T& y) const;
};
7     operator() has effects equivalent to: return !ranges::less<>{}(x, y);

template<class T = void>
    requires StrictTotallyOrdered<T> || Same<T, void> || BUILTIN_PTR_CMP(const T&, <, const T&)
struct less_equal {
    constexpr bool operator()(const T& x, const T& y) const;
};
8     operator() has effects equivalent to: return !ranges::less<>{}(y, x);

template<> struct equal_to<void> {
    template<class T, class U>
        requires EqualityComparableWith<T, U> || BUILTIN_PTR_CMP(T, ==, U)
        constexpr bool operator()(T&& t, U&& u) const;

    using is_transparent = unspecified;
};
9     Expects: If the expression std::forward<T>(t) == std::forward<U>(u) results in a call to a built-in
operator == comparing pointers of type P, the conversion sequences from both T and U to P shall be
equality-preserving ([concepts.equality]).
10    Effects:
(10.1)    — If the expression std::forward<T>(t) == std::forward<U>(u) results in a call to a built-in
operator == comparing pointers of type P: returns false if either (the converted value of) t
precedes u or u precedes t in the implementation-defined strict total order over pointers of type P
and otherwise true.
(10.2)    — Otherwise, equivalent to: return std::forward<T>(t) == std::forward<U>(u);

template<> struct not_equal_to<void> {
    template<class T, class U>
        requires EqualityComparableWith<T, U> || BUILTIN_PTR_CMP(T, ==, U)
        constexpr bool operator()(T&& t, U&& u) const;

    using is_transparent = unspecified;
};
11    operator() has effects equivalent to:
        return !ranges::equal_to<>{}(std::forward<T>(t), std::forward<U>(u));

template<> struct greater<void> {
    template<class T, class U>
        requires StrictTotallyOrderedWith<T, U> || BUILTIN_PTR_CMP(U, <, T)
        constexpr bool operator()(T&& t, U&& u) const;

    using is_transparent = unspecified;
};
12    operator() has effects equivalent to:
        return ranges::less<>{}(std::forward<U>(u), std::forward<T>(t));

```

```

template<> struct less<void> {
    template<class T, class U>
        requires StrictTotallyOrderedWith<T, U> || BUILTIN_PTR_CMP(T, <, U)
        constexpr bool operator()(T&& t, U&& u) const;

    using is_transparent = unspecified;
};

```

13 *Expects:* If the expression `std::forward<T>(t) < std::forward<U>(u)` results in a call to a built-in operator `<` comparing pointers of type `P`, the conversion sequences from both `T` and `U` to `P` shall be equality-preserving ([[concepts.equality](#)]). For any expressions `ET` and `EU` such that `decltype((ET))` is `T` and `decltype((EU))` is `U`, exactly one of `ranges::less<>{}(ET, EU)`, `ranges::less<>{}(EU, ET)`, or `ranges::equal_to<>{}(ET, EU)` shall be true.

14 *Effects:*

(14.1) — If the expression `std::forward<T>(t) < std::forward<U>(u)` results in a call to a built-in operator `<` comparing pointers of type `P`: returns `true` if (the converted value of) `t` precedes `u` in the implementation-defined strict total order over pointers of type `P` and otherwise `false`.

(14.2) — Otherwise, equivalent to: `return std::forward<T>(t) < std::forward<U>(u);`

```

template<> struct greater_equal<void> {
    template<class T, class U>
        requires StrictTotallyOrderedWith<T, U> || BUILTIN_PTR_CMP(T, <, U)
        constexpr bool operator()(T&& t, U&& u) const;

    using is_transparent = unspecified;
};

```

15 `operator()` has effects equivalent to:

```
return !ranges::less<>{}(std::forward<T>(t), std::forward<U>(u));
```

```

template<> struct less_equal<void> {
    template<class T, class U>
        requires StrictTotallyOrderedWith<T, U> || BUILTIN_PTR_CMP(U, <, T)
        constexpr bool operator()(T&& t, U&& u) const;

    using is_transparent = unspecified;
};

```

16 `operator()` has effects equivalent to:

```
return !ranges::less<>{}(std::forward<U>(u), std::forward<T>(t));
```

19.14.9 Logical operations

[logical.operations]

[...]

20 Strings library

[strings]

[...]

20.4 String view classes

[string.view]

[...]

20.4.2 Class template `basic_string_view`

[string.view.template]

```
template<class charT, class traits = char_traits<charT>>
class basic_string_view {
public:
    [...]
```

```
// 20.4.2.2, iterator support
```

```
constexpr const_iterator begin() const noexcept;
constexpr const_iterator end() const noexcept;
constexpr const_iterator cbegin() const noexcept;
constexpr const_iterator cend() const noexcept;
constexpr const_reverse_iterator rbegin() const noexcept;
constexpr const_reverse_iterator rend() const noexcept;
constexpr const_reverse_iterator crbegin() const noexcept;
constexpr const_reverse_iterator crend() const noexcept;
```

```
friend constexpr const_iterator begin(basic_string_view sv) noexcept { return sv.begin(); }
friend constexpr const_iterator end(basic_string_view sv) noexcept { return sv.end(); }
```

```
// [string.view.capacity], capacity
```

```
constexpr size_type size() const noexcept;
constexpr size_type length() const noexcept;
constexpr size_type max_size() const noexcept;
[[nodiscard]] constexpr bool empty() const noexcept;
```

```
    [...]
```

```
};
```

[...]

20.4.2.2 Iterator support

[string.view.iterators]

```
using const_iterator = implementation-defined;
```

- ¹ A type that meets the requirements of a constant ~~random-access iterator~~ [Cpp17RandomAccessIterator \(22.3.5.6\)](#) and ~~of a contiguous iterator (22.3.4.14)~~ [models ContiguousIterator \(22.3.4.14\)](#), whose `value_type` is the template parameter `charT`.

- ² [...]

[...]

21 Containers library

[containers]

[...]

21.2 Container requirements

[container.requirements]

21.2.1 General container requirements

[container.requirements.general]

[...]

- ¹³ A *contiguous container* is a container ~~that supports random access iterators (22.3.5.6) and~~ whose member types `iterator` and `const_iterator` ~~are contiguous iterators (22.3.1)~~ meet the *Cpp17RandomAccessIterator* requirements (22.3.5.6) and model *ContiguousIterator* (22.3.4.14).

[...]

21.7 Views

[views]

[...]

21.7.3 Class template `span`

[views.span]

21.7.3.1 Overview

[span.overview]

[...]

- ² The iterator types `span::iterator` and `span::const_iterator` ~~are random access iterators (22.3.5.6), contiguous iterators (22.3.1), and~~ each model *ContiguousIterator* (22.3.4.14), meet the *Cpp17RandomAccessIterator* requirements (22.3.5.6), and meet the requirements for `constexpr` iterators (22.3.1). All requirements on container iterators (21.2) apply to `span::iterator` and `span::const_iterator` as well.
- ³ All member functions of `span` have constant time complexity.

```
namespace std {
    template<class ElementType, ptrdiff_t Extent = dynamic_extent>
    class span {
    public:
        [...]

        // [span.iterators], iterator support
        constexpr iterator begin() const noexcept;
        constexpr iterator end() const noexcept;
        constexpr const_iterator cbegin() const noexcept;
        constexpr const_iterator cend() const noexcept;
        constexpr reverse_iterator rbegin() const noexcept;
        constexpr reverse_iterator rend() const noexcept;
        constexpr const_reverse_iterator crbegin() const noexcept;
        constexpr const_reverse_iterator crend() const noexcept;

        friend constexpr iterator begin(span s) noexcept { return s.begin(); }
        friend constexpr iterator end(span s) noexcept { return s.end(); }
    };
};
```

[...]

};

}

[...]

22 Iterators library

[iterators]

22.1 General

[iterators.general]

- ¹ This Clause describes components that C++ programs may use to perform iterations over containers (Clause 21), streams ([[iostream.format](#)]), ~~and~~ stream buffers ([[stream.buffers](#)]), ~~and other ranges~~ (Clause 23).
- ² The following subclauses describe iterator requirements, and components for iterator primitives, predefined iterators, and stream iterators, as summarized in Table 73.

Table 73 — Iterators library summary

Subclause	Header(s)
22.3 R Iterator requirements	<iterator>
22.4 Iterator primitives	<iterator>
22.5 Predefined iterators	
22.6 Stream iterators	

[Editor's note: Move [iterator.synopsis] here immediately after [iterators.general] and modify as follows:]

22.2 Header <iterator> synopsis

[iterator.synopsis]

```
#include <concepts>

namespace std {
    template<class T> using with-reference // exposition only
        = T&;
    template<class T> concept can-reference // exposition only
        = requires { typename with-reference<T>; };
    template<class T> concept dereferenceable // exposition only
        = requires(T& t) {
            *t; // not required to be equality-preserving
            requires can-reference<decltype(*t)>;
        };

    // 22.3.2, associated types
    // 22.3.2.1, incrementable traits
    template<class> struct incrementable_traits;
    template<class T>
        using iter_difference_t = see below;

    // 22.3.2.2, readable traits
    template<class> struct readable_traits;
    template<class T>
        using iter_value_t = see below;

    // 22.4, primitives22.3.2.3, Iterator traits
    template<class IteratorI> struct iterator_traits;
    template<class T> struct iterator_traits<T*>;

    template<dereferenceable T>
        using iter_reference_t = decltype(*declval<T&>());

    namespace ranges {
        // 22.3.3, customization points
        inline namespace unspecified {
            // 22.3.3.1, iter_move
            inline constexpr unspecified iter_move = unspecified;
        }
    }
}
```

```

    // 22.3.3.2, iter_swap
    inline constexpr unspecified iter_swap = unspecified;
}

template<dereferenceable T>
requires requires(T& t) {
    ranges::iter_move(t);
    requires can-reference<decltype(ranges::iter_move(t))>;
}
using iter_rvalue_reference_t
    = decltype(ranges::iter_move(declval<T&>()));

// 22.3.4, iterator concepts
// 22.3.4.2, Readable
template<class In>
    concept Readable = see below;

template<Readable T>
    using iter_common_reference_t =
        common_reference_t<iter_reference_t<T>, iter_value_t<T>&&>;

// 22.3.4.3, Writable
template<class Out, class T>
    concept Writable = see below;

// 22.3.4.4, WeaklyIncrementable
template<class I>
    concept WeaklyIncrementable = see below;

// 22.3.4.5, Incrementable
template<class I>
    concept Incrementable = see below;

// 22.3.4.6, Iterator
template<class I>
    concept Iterator = see below;

// 22.3.4.5, Sentinel
template<class S, class I>
    concept Sentinel = see below;

// 22.3.4.8, SizedSentinel
template<class S, class I>
    inline constexpr bool disable_sized_sentinel = false;

template<class S, class I>
    concept SizedSentinel = see below;

// 22.3.4.9, InputIterator
template<class I>
    concept InputIterator = see below;

// 22.3.4.10, OutputIterator
template<class I, class T>
    concept OutputIterator = see below;

// 22.3.4.11, ForwardIterator
template<class I>
    concept ForwardIterator = see below;

// 22.3.4.12, BidirectionalIterator
template<class I>
    concept BidirectionalIterator = see below;

```

```

// 22.3.4.13, RandomAccessIterator
template<class I>
    concept RandomAccessIterator = see below;

// 22.3.4.14, ContiguousIterator
template<class I>
    concept ContiguousIterator = see below;

// 22.3.6, indirect callable requirements
// 22.3.6.2, indirect callables
template<class F, class I>
    concept IndirectUnaryInvocable = see below;

template<class F, class I>
    concept IndirectRegularUnaryInvocable = see below;

template<class F, class I>
    concept IndirectUnaryPredicate = see below;

template<class F, class I1, class I2 = I1>
    concept IndirectRelation = see below;

template<class F, class I1, class I2 = I1>
    concept IndirectStrictWeakOrder = see below;

template<class F, class... Is>
    requires (Readable<Is> && ...) && Invocable<F, iter_reference_t<Is>...>
    using indirect_result_t = invoke_result_t<F, iter_reference_t<Is>...>;

// 22.3.6.3, projected
template<Readable I, IndirectRegularUnaryInvocable<I> Proj>
    struct projected;

template<WeaklyIncrementable I, class Proj>
    struct incrementable_traits<projected<I, Proj>>;

// 22.3.7, common algorithm requirements
// 22.3.7.2 IndirectlyMovable
template<class In, class Out>
    concept IndirectlyMovable = see below;

template<class In, class Out>
    concept IndirectlyMovableStorable = see below;

// 22.3.7.3 IndirectlyCopyable
template<class In, class Out>
    concept IndirectlyCopyable = see below;

template<class In, class Out>
    concept IndirectlyCopyableStorable = see below;

// 22.3.7.4 IndirectlySwappable
template<class I1, class I2 = I1>
    concept IndirectlySwappable = see below;

// 22.3.7.5 IndirectlyComparable
template<class I1, class I2, class R, class P1 = identity, class P2 = identity>
    concept IndirectlyComparable = see below;

// 22.3.7.6 Permutable
template<class I>
    concept Permutable = see below;

```

```

// 22.3.7.7 Mergeable
template<class I1, class I2, class Out,
        class R = ranges::less<>, class P1 = identity, class P2 = identity>
    concept Mergeable = see below;

template<class I, class R = ranges::less<>, class P = identity>
    concept Sortable = see below;

// 22.4, primitives
// 22.4.1, iterator tags
struct input_iterator_tag { };
struct output_iterator_tag { };
struct forward_iterator_tag: public input_iterator_tag { };
struct bidirectional_iterator_tag: public forward_iterator_tag { };
struct random_access_iterator_tag: public bidirectional_iterator_tag { };
struct contiguous_iterator_tag: public random_access_iterator_tag { };

// 22.4.2, iterator operations
template<class InputIterator, class Distance>
    constexpr void
        advance(InputIterator& i, Distance n);
template<class InputIterator>
    constexpr typename iterator_traits<InputIterator>::difference_type
        distance(InputIterator first, InputIterator last);
template<class InputIterator>
    constexpr InputIterator
        next(InputIterator x,
            typename iterator_traits<InputIterator>::difference_type n = 1);
template<class BidirectionalIterator>
    constexpr BidirectionalIterator
        prev(BidirectionalIterator x,
            typename iterator_traits<BidirectionalIterator>::difference_type n = 1);

// 22.4.3, range iterator operations
namespace ranges {
    // 22.4.3.1, ranges::advance
    template<Iterator I>
        constexpr void advance(I& i, iter_difference_t<I> n);
    template<Iterator I, Sentinel<I> S>
        constexpr void advance(I& i, S bound);
    template<Iterator I, Sentinel<I> S>
        constexpr iter_difference_t<I> advance(I& i, iter_difference_t<I> n, S bound);

    // 22.4.3.2, ranges::distance
    template<Iterator I, Sentinel<I> S>
        constexpr iter_difference_t<I> distance(I first, S last);
    template<Range R>
        constexpr iter_difference_t<iterator_t<R>> distance(R&& r);

    // 22.4.3.3, ranges::next
    template<Iterator I>
        constexpr I next(I x);
    template<Iterator I>
        constexpr I next(I x, iter_difference_t<I> n);
    template<Iterator I, Sentinel<I> S>
        constexpr I next(I x, S bound);
    template<Iterator I, Sentinel<I> S>
        constexpr I next(I x, iter_difference_t<I> n, S bound);

    // 22.4.3.4, ranges::prev
    template<BidirectionalIterator I>
        constexpr I prev(I x);
    template<BidirectionalIterator I>
        constexpr I prev(I x, iter_difference_t<I> n);

```

```

    template<BidirectionalIterator I>
        constexpr I prev(I x, iter_difference_t<I> n, I bound);
}

// 22.5, predefined iterators and sentinels
// 22.5.1, reverse iterators
template<class Iterator> class reverse_iterator;

template<class Iterator1, class Iterator2>
    constexpr bool operator==(
        const reverse_iterator<Iterator1>& x,
        const reverse_iterator<Iterator2>& y);
template<class Iterator1, class Iterator2>
    constexpr bool operator!=(
        const reverse_iterator<Iterator1>& x,
        const reverse_iterator<Iterator2>& y);
template<class Iterator1, class Iterator2>
    constexpr bool operator<(
        const reverse_iterator<Iterator1>& x,
        const reverse_iterator<Iterator2>& y);
template<class Iterator1, class Iterator2>
    constexpr bool operator>(
        const reverse_iterator<Iterator1>& x,
        const reverse_iterator<Iterator2>& y);
template<class Iterator1, class Iterator2>
    constexpr bool operator<=(
        const reverse_iterator<Iterator1>& x,
        const reverse_iterator<Iterator2>& y);
template<class Iterator1, class Iterator2>
    constexpr bool operator>=(
        const reverse_iterator<Iterator1>& x,
        const reverse_iterator<Iterator2>& y);

template<class Iterator1, class Iterator2>
    constexpr auto operator-(
        const reverse_iterator<Iterator1>& x,
        const reverse_iterator<Iterator2>& y) -> decltype(y.base() - x.base());
template<class Iterator>
    constexpr reverse_iterator<Iterator>
        operator+(
            typename reverse_iterator<Iterator>::difference_type n,
            const reverse_iterator<Iterator>& x);

template<class Iterator>
    constexpr reverse_iterator<Iterator> make_reverse_iterator(Iterator i);

// 22.5.2, insert iterators
template<class Container> class back_insert_iterator;
template<class Container>
    back_insert_iterator<Container> back_inserter(Container& x);

template<class Container> class front_insert_iterator;
template<class Container>
    front_insert_iterator<Container> front_inserter(Container& x);

template<class Container> class insert_iterator;
template<class Container>
    insert_iterator<Container> inserter(Container& x, typename Container::iteratoriterator_t<Container> i);

// 22.5.3, move iterators and sentinels
template<class Iterator> class move_iterator;

```

```

template<class Iterator1, class Iterator2>
constexpr bool operator==(
    const move_iterator<Iterator1>& x, const move_iterator<Iterator2>& y);
template<class Iterator1, class Iterator2>
constexpr bool operator!=(
    const move_iterator<Iterator1>& x, const move_iterator<Iterator2>& y);
template<class Iterator1, class Iterator2>
constexpr bool operator<(
    const move_iterator<Iterator1>& x, const move_iterator<Iterator2>& y);
template<class Iterator1, class Iterator2>
constexpr bool operator<=(
    const move_iterator<Iterator1>& x, const move_iterator<Iterator2>& y);
template<class Iterator1, class Iterator2>
constexpr bool operator>(
    const move_iterator<Iterator1>& x, const move_iterator<Iterator2>& y);
template<class Iterator1, class Iterator2>
constexpr bool operator>=(
    const move_iterator<Iterator1>& x, const move_iterator<Iterator2>& y);

template<class Iterator1, class Iterator2>
constexpr auto operator-(
    const move_iterator<Iterator1>& x,
    const move_iterator<Iterator2>& y) -> decltype(x.base() - y.base());
template<class Iterator>
constexpr move_iterator<Iterator> operator+(
    typename move_iterator<Iterator>::difference_type n, const move_iterator<Iterator>& x);
template<class Iterator>
constexpr move_iterator<Iterator> make_move_iterator(Iterator i);
template<Semiregular S> class move_sentinel;

// 22.5.4, common iterators
template<Iterator I, Sentinel<I> S>
requires (!Same<I, S>)
class common_iterator;

template<class I, class S>
struct incrementable_traits<common_iterator<I, S>>;

template<InputIterator I, class S>
struct iterator_traits<common_iterator<I, S>>;

// 22.5.5, default sentinels
struct default_sentinel_t;
inline constexpr default_sentinel_t default_sentinel{};

// 22.5.6, counted iterators
template<Iterator I> class counted_iterator;

template<class I>
struct incrementable_traits<counted_iterator<I>>;

template<InputIterator I>
struct iterator_traits<counted_iterator<I>>;

// 22.5.7, unreachable sentinels
struct unreachable_sentinel_t;
inline constexpr unreachable_sentinel_t unreachable_sentinel{};

// 22.6, stream iterators
[...]
```

```

}
```

22.3 Iterator requirements

[iterator.requirements]

22.3.1 In general

[iterator.requirements.general]

- ¹ Iterators are a generalization of pointers that allow a C++ program to work with different data structures ([for example](#), containers [and ranges](#)) in a uniform manner. To be able to construct template algorithms that work correctly and efficiently on different types of data structures, the library formalizes not just the interfaces but also the semantics and complexity assumptions of iterators. An input iterator *i* supports the expression `*i`, resulting in a value of some object type *T*, called the *value type* of the iterator. An output iterator *i* has a non-empty set of types that are *writable* to the iterator; for each such type *T*, the expression `*i = o` is valid where *o* is a value of type *T*. ~~An iterator *i* for which the expression `(*i).m` is well-defined supports the expression `i->m` with the same semantics as `(*i).m`.~~ For every iterator type *X* ~~for which equality is defined~~, there is a corresponding signed integer type called the *difference type* of the iterator.
- ² Since iterators are an abstraction of pointers, their semantics [isare](#) a generalization of most of the semantics of pointers in C++. This ensures that every function template that takes iterators works as well with regular pointers. This document defines [fivesix](#) categories of iterators, according to the operations defined on them: *input iterators*, *output iterators*, *forward iterators*, *bidirectional iterators*, *random access iterators*, and [contiguous iterators](#), as shown in Table 74.

Table 74 — Relations among iterator categories

Contiguous	→	Random Access	→	Bidirectional	→	Forward	→	Input	→	Output
----------------------------	---	---------------	---	---------------	---	---------	---	-------	---	--------

- ³ The six categories of iterators correspond to the iterator concepts [InputIterator](#) (22.3.4.9), [OutputIterator](#) (22.3.4.10), [ForwardIterator](#) (22.3.4.11), [BidirectionalIterator](#) (22.3.4.12), [RandomAccessIterator](#) (22.3.4.13), and [ContiguousIterator](#) (22.3.4.14), respectively. The generic term *iterator* refers to any type that models the [Iterator](#) concept (22.3.4.6).
- ⁴ Forward iterators satisfy all the requirements of input iterators and can be used whenever an input iterator is specified; Bidirectional iterators also satisfy all the requirements of forward iterators and can be used whenever a forward iterator is specified; Random access iterators also satisfy all the requirements of bidirectional iterators and can be used whenever a bidirectional iterator is specified; [Contiguous iterators also satisfy all the requirements of random access iterators and can be used whenever a random access iterator is specified.](#)
- ⁵ Iterators that further satisfy the requirements of output iterators are called *mutable iterators*. Nonmutable iterators are referred to as *constant iterators*.
- ⁶ In addition to the requirements in this subclause, the nested *typedef-names* specified in 22.3.2.3 shall be provided for the iterator type. [Note: Either the iterator type must provide the *typedef-names* directly (in which case `iterator_traits` picks them up automatically), or an `iterator_traits` specialization must provide them. — end note]
- ⁷ Iterators that further satisfy the requirement that, for integral values *n* and dereferenceable iterator values *a* and `(a + n)`, `*(a + n)` is equivalent to `*(addressof(*a) + n)`, are called *contiguous iterators*. [Note: For example, the type “pointer to `int`” is a contiguous iterator, but `reverse_iterator<int*>` is not. For a valid iterator range `[a,b)` with dereferenceable *a*, the corresponding range denoted by pointers is `[addressof(*a),addressof(*a) + (b - a))`; *b* might not be dereferenceable. — end note]
- ⁸ Just as a regular pointer to an array guarantees that there is a pointer value pointing past the last element of the array, so for any iterator type there is an iterator value that points past the last element of a corresponding sequence. These values are called *past-the-end* values. Values of an iterator *i* for which the expression `*i` is defined are called *dereferenceable*. The library never assumes that past-the-end values are dereferenceable. Iterators can also have singular values that are not associated with any sequence. [Example: After the declaration of an uninitialized pointer *x* (as with `int* x;`), *x* must always be assumed to have a singular value of a pointer. — end example] Results of most expressions are undefined for singular values; the only exceptions are destroying an iterator that holds a singular value, the assignment of a non-singular value to an iterator that holds a singular value, and, for iterators that satisfy the *Cpp17DefaultConstructible* requirements, using a value-initialized iterator as the source of a copy or move operation. [Note: This guarantee is not offered for default-initialization, although the distinction only matters for types with trivial default constructors such as pointers or aggregates holding pointers. — end note] In these cases the singular value is overwritten the same way as any other value. Dereferenceable values are always non-singular.

- ⁹ An iterator *j* is called *reachable* from an iterator *i* if and only if there is a finite sequence of applications of the expression `++i` that makes `i == j`. If *j* is reachable from *i*, they refer to elements of the same sequence.
- ¹⁰ Most of the library's algorithmic templates that operate on data structures have interfaces that use ranges. A *range* is a pair of iterators that designate the beginning and end of the computation. A range `[i, i)` is an empty range; in general, a range `[i, j)` refers to the elements in the data structure starting with the element pointed to by *i* and up to but not including the element pointed to by *j*. Range `[i, j)` is valid if and only if *j* is reachable from *i*. The result of the application of functions in the library to invalid ranges is undefined.
- ¹¹ Most of the library's algorithmic templates that operate on data structures have interfaces that use ranges. A *range* is an iterator and a *sentinel* that designate the beginning and end of the computation, or an iterator and a count that designate the beginning and the number of elements to which the computation is to be applied.³
- ¹² An iterator and a sentinel denoting a range are comparable. A range `[i, s)` is empty if `i == s`; otherwise, `[i, s)` refers to the elements in the data structure starting with the element pointed to by *i* and up to but not including the element, if any, pointed to by the first iterator *j* such that `j == s`.
- ¹³ A sentinel *s* is called *reachable* from an iterator *i* if and only if there is a finite sequence of applications of the expression `++i` that makes `i == s`. If *s* is reachable from *i*, `[i, s)` denotes a valid range.
- ¹⁴ A counted range `[i, n)` is empty if `n == 0`; otherwise, `[i, n)` refers to the *n* elements in the data structure starting with the element pointed to by *i* and up to but not including the element, if any, pointed to by the result of *n* applications of `++i`. A counted range `[i, n)` is valid if and only if `n == 0`; or *n* is positive, *i* is dereferenceable, and `[++i, --n)` is valid.
- ¹⁵ The result of the application of library functions to invalid ranges is undefined.
- ¹⁶ All the categories of iterators require only those functions that are realizable for a given category in constant time (amortized). Therefore, requirement tables [and concept definitions](#) for the iterators do not ~~have a~~ [specify](#) complexity ~~column~~.
- ¹⁷ Destruction of an [non-forward](#) iterator may invalidate pointers and references previously obtained from that iterator.
- ¹⁸ An *invalid* iterator is an iterator that may be singular.⁴
- ¹⁹ Iterators are called *constexpr iterators* if all operations provided to satisfy iterator category operations are constexpr functions, except for
- (19.1) — `swap`,
- (19.2) — a pseudo-destructor call (`[expr.pseudo]`), and
- (19.3) — the construction of an iterator with a singular value.

[Note: For example, the types “pointer to `int`” and `reverse_iterator<int*>` are constexpr iterators. — end note]

- ²⁰ In the following sections, *a* and *b* denote values of type *X* or `const X`, *difference_type* and *reference* refer to the types `iterator_traits<X>::difference_type` and `iterator_traits<X>::reference`, respectively, *n* denotes a value of *difference_type*, *u*, *tmp*, and *m* denote identifiers, *r* denotes a value of *X*&, *t* denotes a value of value type *T*, *o* denotes a value of some type that is writable to the output iterator. [Note: For an iterator type *X* there must be an instantiation of `iterator_traits<X>` (22.3.2.3). — end note]

22.3.2 Associated types

[iterator.assoc.types]

22.3.2.1 Incrementable traits

[incrementable.traits]

- ¹ To implement algorithms only in terms of incrementable types, it is often necessary to determine the difference type that corresponds to a particular incrementable type. Accordingly, it is required that if *WI* is the name of a type that models the `WeaklyIncrementable` concept (22.3.4.4), the type

```
iter_difference_t<WI>
```

³) The sentinel denoting the end of a range may have the same type as the iterator denoting the beginning of the range, or a different type.

⁴) This definition applies to pointers, since pointers are iterators. The effect of dereferencing an iterator that has been invalidated is undefined.

be defined as the incrementable type's difference type.

```
namespace std {
    template<class> struct incrementable_traits { };

    template<class T>
        requires is_object_v<T>
    struct incrementable_traits<T*> {
        using difference_type = ptrdiff_t;
    };

    template<class I>
    struct incrementable_traits<const I>
        : incrementable_traits<remove_const_t<I>> { };

    template<class T>
        requires requires { typename T::difference_type; }
    struct incrementable_traits<T> {
        using difference_type = typename T::difference_type;
    };

    template<class T>
        requires (!requires { typename T::difference_type; }) &&
        requires(const T& a, const T& b) { { a - b } -> Integral; }
    struct incrementable_traits<T> {
        using difference_type = make_signed_t<decltype(declval<T>() - declval<T>())>;
    };

    template<class T>
        using iter_difference_t = see below;
}
```

- ² If `iterator_traits<I>` names a specialization generated from the primary template, then `iter_difference_t<I>` denotes `incrementable_traits<I>::difference_type`; otherwise, it denotes `iterator_traits<I>::difference_type`.
- ³ Users may specialize `incrementable_traits` on program-defined types.

22.3.2.2 Readable traits

[readable.traits]

- ¹ To implement algorithms only in terms of readable types, it is often necessary to determine the value type that corresponds to a particular readable type. Accordingly, it is required that if `R` is the name of a type that models the Readable concept (22.3.4.2), the type

```
iter_value_t<R>
```

be defined as the readable type's value type.

```
template<class> struct cond-value-type { }; // exposition only
template<class T>
    requires is_object_v<T>
struct cond-value-type {
    using value_type = remove_cv_t<T>;
};

template<class> struct readable_traits { };

template<class T>
struct readable_traits<T*>
    : cond-value-type<T> { };

template<class I>
    requires is_array_v<I>
struct readable_traits<I> {
    using value_type = remove_cv_t<remove_extent_t<I>>;
};
```

```

template<class I>
struct readable_traits<const I>
    : readable_traits<remove_const_t<I>> { };

template<class T>
    requires requires { typename T::value_type; }
struct readable_traits<T>
    : cond_value_type<typename T::value_type> { };

template<class T>
    requires requires { typename T::element_type; }
struct readable_traits<T>
    : cond_value_type<typename T::element_type> { };

template<class T> using iter_value_t = see below;

```

- ² If `iterator_traits<I>` names a specialization generated from the primary template, then `iter_value_t<I>` denotes `readable_traits<I>::value_type`; otherwise, it denotes `iterator_traits<I>::value_type`.
- ³ Class template `readable_traits` may be specialized on program-defined types.
- ⁴ [Note: Some legacy output iterators define a nested type named `value_type` that is an alias for `void`. These types are not `Readable` and have no associated value types. — end note]
- ⁵ [Note: Smart pointers like `shared_ptr<int>` are `Readable` and have an associated value type, but a smart pointer like `shared_ptr<void>` is not `Readable` and has no associated value type. — end note]

[Editor's note: Relocate [iterator.traits] here and modify as follows:]

22.3.2.3 Iterator traits

[iterator.traits]

- ¹ To implement algorithms only in terms of iterators, it is ~~often~~^{sometimes} necessary to determine the ~~value and difference types~~^{iterator category} that corresponds to a particular iterator type. Accordingly, it is required that if `IteratorI` is the type of an iterator, the types

```

iterator_traits<Iterator>::difference_type
iterator_traits<Iterator>::value_type
iterator_traits<IteratorI>::iterator_category

```

be defined as the iterator's ~~difference type, value type and~~ iterator category, ~~respectively~~. In addition, the types

```

iterator_traits<Iterator>::reference
iterator_traits<IteratorI>::pointer
iterator_traits<I>::reference

```

shall be defined as the iterator's pointer and reference ~~and pointer~~ types; that is, for an iterator object a of class type, the same type as ~~the type of *a and a->~~ `decltype(a.operator->())` and `decltype(*a)`, respectively. The type `iterator_traits<I>::pointer` shall be void for an iterator of class type I that does not support `operator->`. Additionally, i~~n~~ the case of an output iterator, the types

```

iterator_traits<I>::value_type
iterator_traits<IteratorI>::difference_type
iterator_traits<Iterator>::value_type
iterator_traits<IteratorI>::reference
iterator_traits<Iterator>::pointer

```

may be defined as `void`.

- ² The definitions in this subclause make use of the following exposition-only concepts:

```

template<class I>
concept cpp17-iterator =
    Copyable<I> && requires(I i) {
        *i; requires can-reference<decltype(*i)>;
        { ++i } -> Same<I>&;
        *i++; requires can-reference<decltype(*i++)>;
    };

```

```

template<class I>
concept cpp17-input-iterator =
    cpp17-iterator<I> && EqualityComparable<I> && requires(I i) {
        typename incrementable_traits<I>::difference_type;
        typename readable_traits<I>::value_type;
        typename common_reference_t<iter_reference_t<I> &&,
            typename readable_traits<I>::value_type &>;

        *i++;
        typename common_reference_t<decltype(*i++) &&,
            typename readable_traits<I>::value_type &>;
        requires SignedIntegral<typename incrementable_traits<I>::difference_type>;
    };

template<class I>
concept cpp17-forward-iterator =
    cpp17-input-iterator<I> && Constructible<I> &&
    is_lvalue_reference_v<iter_reference_t<I>> &&
    Same<remove_cvref_t<iter_reference_t<I>>, typename readable_traits<I>::value_type> &&
    requires(I i) {
        { i++ } -> const I&;
        requires Same<iter_reference_t<I>, decltype(*i++)>;
    };

template<class I>
concept cpp17-bidirectional-iterator =
    cpp17-forward-iterator<I> && requires(I i) {
        { --i } -> Same<I>&;
        { i-- } -> const I&;
        *i--; requires Same<iter_reference_t<I>, decltype(*i--)>;
    };

template<class I>
concept cpp17-random-access-iterator =
    cpp17-bidirectional-iterator<I> && StrictTotallyOrdered<I> &&
    requires(I i, typename incrementable_traits<I>::difference_type n) {
        { i += n } -> Same<I>&;
        { i -= n } -> Same<I>&;
        i + n; requires Same<I, decltype(i + n)>;
        n + i; requires Same<I, decltype(n + i)>;
        i - n; requires Same<I, decltype(i - n)>;
        i - i; requires Same<decltype(n), decltype(i - i)>;
        { i[n] } -> iter_reference_t<I>;
    };

```

³ The members of a specialization `iterator_traits<I>` generated from the `iterator_traits` primary template are computed as follows:

- (3.1) — If **IteratorI** has valid (`[temp.deduct]`) member types `difference_type`, `value_type`, **pointer**, **reference**, and `iterator_category`, then `iterator_traits<IteratorI>` ~~shall have~~has the following **as** publicly accessible members:

```

using iterator_category = typename I::iterator_category;
using difference_type = typename Iterator::difference_type;
using value_type         = typename IteratorI::value_type;
using difference_type     = typename I::difference_type;
using pointer            = typename Iterator::pointersee below;
using reference          = typename IteratorI::reference;
using iterator_category = typename Iterator::iterator_category;

```

If the *qualified-id* `I::pointer` is valid and denotes a type, then `iterator_traits<I>::pointer` names that type; otherwise, it names `void`.

- (3.2) — Otherwise, if `I` satisfies the exposition-only concept `cpp17-input-iterator`, then `iterator_traits<I>` has the following publicly accessible members:

```

using iterator_category = see below;
using value_type         = typename readable_traits<I>::value_type;

```

```

        using difference_type    = typename incrementable_traits<I>::difference_type;
        using pointer            = see below;
        using reference          = see below;

```

(3.2.1) — If the *qualified-id* `I::pointer` is valid and denotes a type, `pointer` names that type. Otherwise, if `decltype(declval<I&>().operator->())` is well-formed, then `pointer` names that type. Otherwise, `pointer` names `void`.

(3.2.2) — If the *qualified-id* `I::reference` is valid and denotes a type, `reference` names that type. Otherwise, `reference` names `iter_reference_t<I>`.

(3.2.3) — If the *qualified-id* `I::iterator_category` is valid and denotes a type, `iterator_category` names that type. Otherwise, if `I` satisfies *cpp17-random-access-iterator*, `iterator_category` names `random_access_iterator_tag`. Otherwise, if `I` satisfies *cpp17-bidirectional-iterator*, `iterator_category` names `bidirectional_iterator_tag`. Otherwise, if `I` satisfies *cpp17-forward-iterator*, `iterator_category` names `forward_iterator_tag`. Otherwise, `iterator_category` names `input_iterator_tag`.

(3.3) — Otherwise, if `I` satisfies the exposition-only concept *cpp17-iterator*, then `iterator_traits<I>` has the following publicly accessible members:

```

        using iterator_category = output_iterator_tag;
        using value_type        = void;
        using difference_type    = see below;
        using pointer            = void;
        using reference          = void;

```

If the *qualified-id* `incrementable_traits<I>::difference_type` is valid and denotes a type, then `difference_type` names that type; otherwise, it names `void`.

(3.4) — Otherwise, `iterator_traits<IteratorI>` ~~shall have~~has no members by any of the above names.

4 Explicit or partial specializations of `iterator_traits` may have a member type `iterator_concept` that is used to indicate conformance to the iterator concepts (22.3.4).

5 ~~It~~`iterator_traits` is specialized for pointers as

```

namespace std {
    template<class T>
        requires is_object_v<T>
    struct iterator_traits<T*> {
        using iterator_concept    = contiguous_iterator_tag;
        using iterator_category    = random_access_iterator_tag;
        using value_type          = remove_cv_t<T>;
        using difference_type      = ptrdiff_t;
        using value_type          = remove_cv_t<T>;
        using pointer              = T*;
        using reference            = T&;
        using iterator_category    = random_access_iterator_tag;
    };
}

```

6 [Example: To implement a generic `reverse` function, a C++ program can do the following:

```

template<class BidirectionalIteratorI>
void reverse(BidirectionalIteratorI first, BidirectionalIteratorI last) {
    typename iterator_traits<BidirectionalIteratorI>::difference_type n =
        distance(first, last);
    --n;
    while(n > 0) {
        typename iterator_traits<BidirectionalIteratorI>::value_type
            tmp = *first;
        *first++ = *--last;
        *last = tmp;
        n -= 2;
    }
}

```

— end example]

22.3.3 Customization points

[iterator.cust]

22.3.3.1 iter_move

[iterator.cust.move]

- ¹ The name `iter_move` denotes a customization point object ([customization.point.object]). The expression `ranges::iter_move(E)` for some subexpression `E` is expression-equivalent to the following:

- (1.1) — `iter_move(E)`, if that expression is valid, with overload resolution performed in a context that does not include a declaration of `ranges::iter_move`.
- (1.2) — Otherwise, if the expression `*E` is well-formed:
 - (1.2.1) — if `*E` is an lvalue, `std::move(*E)`;
 - (1.2.2) — otherwise, `*E`.
- (1.3) — Otherwise, `ranges::iter_move(E)` is ill-formed. [Note: This case can result in substitution failure when `ranges::iter_move(E)` appears in the immediate context of a template instantiation. — end note]

- ² If `ranges::iter_move(E)` is not equal to `*E`, the program is ill-formed with no diagnostic required.

22.3.3.2 iter_swap

[iterator.cust.swap]

- ¹ The name `iter_swap` denotes a customization point object ([customization.point.object]) that exchanges the values (17.4.11) denoted by its arguments.
- ² Let *iter-exchange-move* be the exposition-only function:

```
template<class X, class Y>
constexpr iter_value_t<remove_reference_t<X>> iter-exchange-move(X&& x, Y&& y)
    noexcept(noexcept(iter_value_t<remove_reference_t<X>>(iter_move(x))) &&
             noexcept(*x = iter_move(y)));
```

- ³ *Effects:* Equivalent to:

```
iter_value_t<remove_reference_t<X>> old_value(iter_move(x));
*x = iter_move(y);
return old_value;
```

- ⁴ The expression `ranges::iter_swap(E1, E2)` for some subexpressions `E1` and `E2` is expression-equivalent to the following:

- (4.1) — `(void)iter_swap(E1, E2)`, if that expression is valid, with overload resolution performed in a context that includes the declaration

```
template<class I1, class I2>
void iter_swap(I1, I2) = delete;
```

and does not include a declaration of `ranges::iter_swap`. If the function selected by overload resolution does not exchange the values denoted by `E1` and `E2`, the program is ill-formed with no diagnostic required.

- (4.2) — Otherwise, if the types of `E1` and `E2` each model `Readable`, and if the reference types of `E1` and `E2` model `SwappableWith` (17.4.11), then `ranges::swap(*E1, *E2)`.
- (4.3) — Otherwise, if the types `T1` and `T2` of `E1` and `E2` model `IndirectlyMovableStorable<T1, T2>` and `IndirectlyMovableStorable<T2, T1>`, then `(void)(*E1 = iter-exchange-move(E2, E1))`, except that `E1` is evaluated only once.
- (4.4) — Otherwise, `ranges::iter_swap(E1, E2)` is ill-formed. [Note: This case can result in substitution failure when `ranges::iter_swap(E1, E2)` appears in the immediate context of a template instantiation. — end note]

22.3.4 Iterator concepts

[iterator.concepts]

22.3.4.1 General

[iterator.concepts.general]

- ¹ For a type `I`, let `ITER_TRAITS(I)` denote the type `I` if `iterator_traits<I>` names a specialization generated from the primary template. Otherwise, `ITER_TRAITS(I)` denotes `iterator_traits<I>`.

- (1.1) — If the qualified-id `ITER_TRAITS(I)::iterator_concept` is valid and names a type, then `ITER_CONCEPT(I)` denotes that type.

- (1.2) — Otherwise, if the *qualified-id* `ITER_TRAITS(I)::iterator_category` is valid and names a type, then `ITER_CONCEPT(I)` denotes that type.
- (1.3) — Otherwise, if `iterator_traits<I>` names a specialization generated from the primary template, then `ITER_CONCEPT(I)` denotes `random_access_iterator_tag`.
- (1.4) — Otherwise, `ITER_CONCEPT(I)` does not denote a type.

² [Note: `ITER_TRAITS` enables independent syntactic determination of an iterator's category and concept. — end note] [Example:

```
struct I {
    using value_type = int;
    using difference_type = int;

    int operator*() const;
    I& operator++();
    I operator++(int);
    I& operator--();
    I operator--(int);

    bool operator==(I) const;
    bool operator!=(I) const;
};
```

`iterator_traits<I>::iterator_category` denotes `input_iterator_tag`, and `ITER_CONCEPT(I)` denotes `random_access_iterator_tag`. — end example]

22.3.4.2 Concept Readable

[iterator.concept.readable]

- ¹ Types that are readable by applying `operator*` model the `Readable` concept, including pointers, smart pointers, and iterators.

```
template<class In>
concept Readable =
    requires {
        typename iter_value_t<In>;
        typename iter_reference_t<In>;
        typename iter_rvalue_reference_t<In>;
    } &&
    CommonReference<iter_reference_t<In>&&, iter_value_t<In>&> &&
    CommonReference<iter_reference_t<In>&&, iter_rvalue_reference_t<In>&&> &&
    CommonReference<iter_rvalue_reference_t<In>&&, const iter_value_t<In>&>;
```

- ² Given a value `i` of type `I`, `I` models `Readable` only if the expression `*i` (which is indirectly required to be valid via the exposition-only *dereferenceable* concept (22.2)) is equality-preserving.

22.3.4.3 Concept Writable

[iterator.concept.writable]

- ¹ The `Writable` concept specifies the requirements for writing a value into an iterator's referenced object.

```
template<class Out, class T>
concept Writable =
    requires(Out&& o, T&& t) {
        *o = std::forward<T>(t); // not required to be equality-preserving
        *std::forward<Out>(o) = std::forward<T>(t); // not required to be equality-preserving
        const_cast<const iter_reference_t<Out>&&>(*o) =
            std::forward<T>(t); // not required to be equality-preserving
        const_cast<const iter_reference_t<Out>&&>(*std::forward<Out>(o)) =
            std::forward<T>(t); // not required to be equality-preserving
    };
};
```

- ² Let `E` be an expression such that `decltype(E)` is `T`, and let `o` be a dereferenceable object of type `Out`. `Out` and `T` model `Writable<Out, T>` only if

- (2.1) — If `Out` and `T` model `Readable<Out> && Same<iter_value_t<Out>, decay_t<T>>`, then `*o` after any above assignment is equal to the value of `E` before the assignment.
- ³ After evaluating any above assignment expression, `o` is not required to be dereferenceable.
- ⁴ If `E` is an xvalue ([basic.lval]), the resulting state of the object it denotes is valid but unspecified ([lib.types.movedfrom]).

- ⁵ [Note: The only valid use of an `operator*` is on the left side of the assignment statement. Assignment through the same value of the writable type happens only once. — *end note*]
- ⁶ [Note: `Writable` has the awkward `const_cast` expressions to reject iterators with prvalue non-proxy reference types that permit rvalue assignment but do not also permit `const` rvalue assignment. Consequently, an iterator type `I` that returns `std::string` by value does not model `Writable<I, std::string>`. — *end note*]

22.3.4.4 Concept `WeaklyIncrementable`

[`iterator.concept.weaklyincrementable`]

- ¹ The `WeaklyIncrementable` concept specifies the requirements on types that can be incremented with the pre- and post-increment operators. The increment operations are not required to be equality-preserving, nor is the type required to be `EqualityComparable`.

```
template<class I>
concept WeaklyIncrementable =
    Semiregular<I> &&
    requires(I i) {
        typename iter_difference_t<I>;
        requires SignedIntegral<iter_difference_t<I>>;
        { ++i } -> Same<I>&; // not required to be equality-preserving
        i++; // not required to be equality-preserving
    };
```

- ² Let `i` be an object of type `I`. When `i` is in the domain of both pre- and post-increment, `i` is said to be *incrementable*. `I` models `WeaklyIncrementable<I>` only if
- (2.1) — The expressions `++i` and `i++` have the same domain.
- (2.2) — If `i` is incrementable, then both `++i` and `i++` advance `i` to the next element.
- (2.3) — If `i` is incrementable, then `addressof(++i)` is equal to `addressof(i)`.
- ³ [Note: For `WeaklyIncrementable` types, `a` equals `b` does not imply that `++a` equals `++b`. (Equality does not guarantee the substitution property or referential transparency.) Algorithms on weakly incrementable types should never attempt to pass through the same incrementable value twice. They should be single-pass algorithms. These algorithms can be used with istreams as the source of the input data through the `istream_iterator` class template. — *end note*]

22.3.4.5 Concept `Incrementable`

[`iterator.concept.incrementable`]

- ¹ The `Incrementable` concept specifies requirements on types that can be incremented with the pre- and post-increment operators. The increment operations are required to be equality-preserving, and the type is required to be `EqualityComparable`. [Note: This supersedes the annotations on the increment expressions in the definition of `WeaklyIncrementable`. — *end note*]

```
template<class I>
concept Incrementable =
    Regular<I> &&
    WeaklyIncrementable<I> &&
    requires(I i) {
        i++; requires Same<decltype(i++), I>;
    };
```

- ² Let `a` and `b` be incrementable objects of type `I`. `I` models `Incrementable` only if
- (2.1) — If `bool(a == b)` then `bool(a++ == b)`.
- (2.2) — If `bool(a == b)` then `bool(((void)a++, a) == ++b)`.
- ³ [Note: The requirement that `a` equals `b` implies `++a` equals `++b` (which is not true for weakly incrementable types) allows the use of multi-pass one-directional algorithms with types that model `Incrementable`. — *end note*]

22.3.4.6 Concept `Iterator`

[`iterator.concept.iterator`]

- ¹ The `Iterator` concept forms the basis of the iterator concept taxonomy; every iterator models `Iterator`. This concept specifies operations for dereferencing and incrementing an iterator. Most algorithms will require additional operations to compare iterators with sentinels (22.3.4.7), to read (22.3.4.9) or write (22.3.4.10) values, or to provide a richer set of iterator movements (22.3.4.11, 22.3.4.12, 22.3.4.13).

```
template<class I>
concept Iterator =
    requires(I i) {
        *i; requires can-reference<decltype(*i)>;
    } &&
    WeaklyIncrementable<I>;
```

22.3.4.7 Concept Sentinel

[iterator.concept.sentinel]

- ¹ The **Sentinel** concept specifies the relationship between an **Iterator** type and a **Semiregular** type whose values denote a range.

```
template<class S, class I>
concept Sentinel =
    Semiregular<S> &&
    Iterator<I> &&
    weakly-equality-comparable-with<S, I>; // See [concept.equalitycomparable]
```

- ² Let *s* and *i* be values of type *S* and *I* such that [*i*, *s*) denotes a range. Types *S* and *I* model **Sentinel**<*S*, *I*> only if
- (2.1) — *i* == *s* is well-defined.
- (2.2) — If `bool(i != s)` then *i* is dereferenceable and [*++i*, *s*) denotes a range.
- ³ The domain of == is not static. Given an iterator *i* and sentinel *s* such that [*i*, *s*) denotes a range and *i* != *s*, *i* and *s* are not required to continue to denote a range after incrementing any other iterator equal to *i*. Consequently, *i* == *s* is no longer required to be well-defined.

22.3.4.8 Concept SizedSentinel

[iterator.concept.sizedsentinel]

- ¹ The **SizedSentinel** concept specifies requirements on an **Iterator** and a **Sentinel** that allow the use of the - operator to compute the distance between them in constant time.

```
template<class S, class I>
concept SizedSentinel =
    Sentinel<S, I> &&
    !disable_sized_sentinel<remove_cv_t<S>, remove_cv_t<I>> &&
    requires(const I& i, const S& s) {
        s - i; requires Same<decltype(s - i), iter_difference_t<I>>;
        i - s; requires Same<decltype(i - s), iter_difference_t<I>>;
    };
```

- ² Let *i* be an iterator of type *I*, and *s* a sentinel of type *S* such that [*i*, *s*) denotes a range. Let *N* be the smallest number of applications of *++i* necessary to make `bool(i == s)` be true. *S* and *I* model **SizedSentinel**<*S*, *I*> only if
- (2.1) — If *N* is representable by `iter_difference_t<I>`, then *s* - *i* is well-defined and equals *N*.
- (2.2) — If -*N* is representable by `iter_difference_t<I>`, then *i* - *s* is well-defined and equals -*N*.
- ³ [Note: `disable_sized_sentinel` allows use of sentinels and iterators with the library that satisfy but do not in fact model **SizedSentinel**. — end note]
- ⁴ [Example: The **SizedSentinel** concept is modeled by pairs of **RandomAccessIterators** (22.3.4.13) and by counted iterators and their sentinels (22.5.6.1). — end example]

22.3.4.9 Concept InputIterator

[iterator.concept.input]

- ¹ The **InputIterator** concept defines requirements for a type whose referenced values can be read (from the requirement for **Readable** (22.3.4.2)) and which can be both pre- and post-incremented. [Note: Unlike the *Cpp17InputIterator* requirements (22.3.5.2), the **InputIterator** concept does not need equality comparison since iterators are typically compared to sentinels. — end note]

```
template<class I>
concept InputIterator =
    Iterator<I> &&
    Readable<I> &&
    requires { typename ITER_CONCEPT(I); } &&
    DerivedFrom<ITER_CONCEPT(I), input_iterator_tag>;
```

22.3.4.10 Concept OutputIterator**[iterator.concept.output]**

- ¹ The `OutputIterator` concept defines requirements for a type that can be used to write values (from the requirement for `Writable` (22.3.4.3)) and which can be both pre- and post-incremented. [Note: Output iterators are not required to model `EqualityComparable`. — end note]

```
template<class I, class T>
concept OutputIterator =
    Iterator<I> &&
    Writable<I, T> &&
    requires(I i, T&& t) {
        *i++ = std::forward<T>(t); // not required to be equality-preserving
    };

```

- ² Let `E` be an expression such that `decltype((E))` is `T`, and let `i` be a dereferenceable object of type `I`. `I` and `T` model `OutputIterator<I, T>` only if `*i++ = E`; has effects equivalent to:

```
*i = E;
++i;

```

- ³ [Note: Algorithms on output iterators should never attempt to pass through the same iterator twice. They should be single-pass algorithms. — end note]

22.3.4.11 Concept ForwardIterator**[iterator.concept.forward]**

- ¹ The `ForwardIterator` concept adds equality comparison and the multi-pass guarantee, specified below.

```
template<class I>
concept ForwardIterator =
    InputIterator<I> &&
    DerivedFrom<ITER_CONCEPT(I), forward_iterator_tag> &&
    Incrementable<I> &&
    Sentinel<I, I>;

```

- ² The domain of `==` for forward iterators is that of iterators over the same underlying sequence. However, value-initialized iterators of the same type may be compared and shall compare equal to other value-initialized iterators of the same type. [Note: Value-initialized iterators behave as if they refer past the end of the same empty sequence. — end note]

- ³ Pointers and references obtained from a forward iterator into a range `[i, s)` shall remain valid while `[i, s)` continues to denote a range.

- ⁴ Two dereferenceable iterators `a` and `b` of type `X` offer the *multi-pass guarantee* if:

(4.1) — `a == b` implies `++a == ++b` and

(4.2) — The expression `((void) [] (X x){++x;}(a), *a)` is equivalent to the expression `*a`.

- ⁵ [Note: The requirement that `a == b` implies `++a == ++b` and the removal of the restrictions on the number of assignments through a mutable iterator (which applies to output iterators) allow the use of multi-pass one-directional algorithms with forward iterators. — end note]

22.3.4.12 Concept BidirectionalIterator**[iterator.concept.bidirectional]**

- ¹ The `BidirectionalIterator` concept adds the ability to move an iterator backward as well as forward.

```
template<class I>
concept BidirectionalIterator =
    ForwardIterator<I> &&
    DerivedFrom<ITER_CONCEPT(I), bidirectional_iterator_tag> &&
    requires(I i) {
        { --i } -> Same<I>&;
        i--; requires Same<decltype(i--), I>;
    };

```

- ² A bidirectional iterator `r` is decrementable if and only if there exists some `q` such that `++q == r`. Decrementable iterators `r` shall be in the domain of the expressions `--r` and `r--`.

- ³ Let `a` and `b` be equal objects of type `I`. `I` models `BidirectionalIterator` only if:

(3.1) — If `a` and `b` are decrementable, then all of the following are true:

(3.1.1) — `addressof(--a) == addressof(a)`

- (3.1.2) — `bool(a-- == b)`
- (3.1.3) — after evaluating both `a--` and `--b` `bool(a == b)` is still `true`
- (3.1.4) — `bool(++(--a) == b)`
- (3.2) — If `a` and `b` are incrementable, then `bool(--(++a) == b)`.

22.3.4.13 Concept `RandomAccessIterator`

[`iterator.concept.random.access`]

- ¹ The `RandomAccessIterator` concept adds support for constant-time advancement with `+=`, `+`, `-=`, and `-`, as well as the computation of distance in constant time with `-`. Random access iterators also support array notation via subscripting.

```
template<class I>
concept RandomAccessIterator =
    BidirectionalIterator<I> &&
    DerivedFrom<ITER_CONCEPT(I), random_access_iterator_tag> &&
    StrictTotallyOrdered<I> &&
    SizedSentinel<I, I> &&
    requires(I i, const I j, const iter_difference_t<I> n) {
        { i += n } -> Same<I>&&;
        j + n; requires Same<decltype(j + n), I>;
        n + j; requires Same<decltype(n + j), I>;
        { i -= n } -> Same<I>&&;
        j - n; requires Same<decltype(j - n), I>;
        j[n]; requires Same<decltype(j[n]), iter_reference_t<I>>;
    };

```

- ² Let `a` and `b` be valid iterators of type `I` such that `b` is reachable from `a` after `n` applications of `++a`, let `D` be `iter_difference_t<I>`, and let `n` denote a value of type `D`. `I` models `RandomAccessIterator` only if

- (2.1) — `(a += n)` is equal to `b`.
- (2.2) — `addressof(a += n)` is equal to `addressof(a)`.
- (2.3) — `(a + n)` is equal to `(a += n)`.
- (2.4) — For any two positive values `x` and `y` of type `D`, if `(a + D(x + y))` is valid, then `(a + D(x + y))` is equal to `((a + x) + y)`.
- (2.5) — `(a + D(0))` is equal to `a`.
- (2.6) — If `(a + D(n - 1))` is valid, then `(a + n)` is equal to `++(a + D(n - 1))`.
- (2.7) — `(b += -n)` is equal to `a`.
- (2.8) — `(b -= n)` is equal to `a`.
- (2.9) — `addressof(b -= n)` is equal to `addressof(b)`.
- (2.10) — `(b - n)` is equal to `(b -= n)`.
- (2.11) — If `b` is dereferenceable, then `a[n]` is valid and is equal to `*b`.
- (2.12) — `bool(a <= b)` is `true`.

22.3.4.14 Concept `ContiguousIterator`

[`iterator.concept.contiguous`]

- ¹ The `ContiguousIterator` concept provides a guarantee that the denoted elements are stored contiguously in memory.

```
template<class I>
concept ContiguousIterator =
    RandomAccessIterator<I> &&
    DerivedFrom<ITER_CONCEPT(I), contiguous_iterator_tag> &&
    is_lvalue_reference_v<iter_reference_t<I>> &&
    Same<iter_value_t<I>, remove_cvref_t<iter_reference_t<I>>>;

```

- ² Let `a` and `b` be dereferenceable iterators of type `I` such that `b` is reachable from `a`, and let `D` be `iter_difference_t<I>`. `I` models `ContiguousIterator` only if `addressof(*(a + D(b - a)))` is equal to `addressof(*a + D(b - a))`.

22.3.5 C++17 iterator requirements [iterator.cpp17]

- ¹ In the following sections, **a** and **b** denote values of type **X** or **const X**, **difference_type** and **reference** refer to the types `iterator_traits<X>::difference_type` and `iterator_traits<X>::reference`, respectively, **n** denotes a value of **difference_type**, **u**, **tmp**, and **m** denote identifiers, **r** denotes a value of **X&**, **t** denotes a value of value type **T**, **o** denotes a value of some type that is writable to the output iterator. *[Note: For an iterator type **X** there must be an instantiation of `iterator_traits<X>` (22.3.2.3). — end note]*

[Editor's note: Relocate [iterator.iterators] here:]

22.3.5.1 Cpp17Iterator [iterator.iterators]

[...]

[Editor's note: Relocate [input.iterators] here:]

22.3.5.2 Input iterators [input.iterators]

[...]

[Editor's note: Relocate [output.iterators] here:]

22.3.5.3 Output iterators [output.iterators]

[...]

[Editor's note: Relocate [forward.iterators] here:]

22.3.5.4 Forward iterators [forward.iterators]

[...]

[Editor's note: Relocate [bidirectional.iterators] here:]

22.3.5.5 Bidirectional iterators [bidirectional.iterators]

[...]

[Editor's note: Relocate [random.access.iterators] here:]

22.3.5.6 Random access iterators [random.access.iterators]

[...]

22.3.6 Indirect callable requirements [indirectcallable]

22.3.6.1 General [indirectcallable.general]

- ¹ There are several concepts that group requirements of algorithms that take callable objects (**[func.require]**) as arguments.

22.3.6.2 Indirect callables [indirectcallable.indirectinvocable]

- ¹ The indirect callable concepts are used to constrain those algorithms that accept callable objects (**[func.def]**) as arguments.

```
namespace std {
    template<class F, class I>
        concept IndirectUnaryInvocable =
            Readable<I> &&
            CopyConstructible<F> &&
            Invocable<F&, iter_value_t<I>&> &&
            Invocable<F&, iter_reference_t<I>>> &&
            Invocable<F&, iter_common_reference_t<I>>> &&
            CommonReference<
                invoke_result_t<F&, iter_value_t<I>&>,
                invoke_result_t<F&, iter_reference_t<I>>>>;
```

```

template<class F, class I>
concept IndirectRegularUnaryInvocable =
    Readable<I> &&
    CopyConstructible<F> &&
    RegularInvocable<F&, iter_value_t<I>&> &&
    RegularInvocable<F&, iter_reference_t<I>>> &&
    RegularInvocable<F&, iter_common_reference_t<I>>> &&
    CommonReference<
        invoke_result_t<F&, iter_value_t<I>&>,
        invoke_result_t<F&, iter_reference_t<I>>>>;

template<class F, class I>
concept IndirectUnaryPredicate =
    Readable<I> &&
    CopyConstructible<F> &&
    Predicate<F&, iter_value_t<I>&> &&
    Predicate<F&, iter_reference_t<I>>> &&
    Predicate<F&, iter_common_reference_t<I>>>;

template<class F, class I1, class I2 = I1>
concept IndirectRelation =
    Readable<I1> && Readable<I2> &&
    CopyConstructible<F> &&
    Relation<F&, iter_value_t<I1>&, iter_value_t<I2>&> &&
    Relation<F&, iter_value_t<I1>&, iter_reference_t<I2>>> &&
    Relation<F&, iter_reference_t<I1>, iter_value_t<I2>&> &&
    Relation<F&, iter_reference_t<I1>, iter_reference_t<I2>>> &&
    Relation<F&, iter_common_reference_t<I1>, iter_common_reference_t<I2>>>;

template<class F, class I1, class I2 = I1>
concept IndirectStrictWeakOrder =
    Readable<I1> && Readable<I2> &&
    CopyConstructible<F> &&
    StrictWeakOrder<F&, iter_value_t<I1>&, iter_value_t<I2>&> &&
    StrictWeakOrder<F&, iter_value_t<I1>&, iter_reference_t<I2>>> &&
    StrictWeakOrder<F&, iter_reference_t<I1>, iter_value_t<I2>&> &&
    StrictWeakOrder<F&, iter_reference_t<I1>, iter_reference_t<I2>>> &&
    StrictWeakOrder<F&, iter_common_reference_t<I1>, iter_common_reference_t<I2>>>;
}

```

22.3.6.3 Class template projected

[projected]

- ¹ Class template projected is used to constrain algorithms that accept callable objects and projections (15.3.18). It combines a Readable type I and a callable object type Proj into a new Readable type whose reference type is the result of applying Proj to the iter_reference_t of I.

```

namespace std {
    template<Readable I, IndirectRegularUnaryInvocable<I> Proj>
    struct projected {
        using value_type = remove_cvref_t<indirect_result_t<Proj&, I>>;
        indirect_result_t<Proj&, I> operator*() const; // not defined
    };

    template<WeaklyIncrementable I, class Proj>
    struct incrementable_traits<projected<I, Proj>> {
        using difference_type = iter_difference_t<I>;
    };
}

```

22.3.7 Common algorithm requirements

[common.alg.req]

22.3.7.1 General

[common.alg.req.general]

- ¹ There are several additional iterator concepts that are commonly applied to families of algorithms. These group together iterator requirements of algorithm families. There are three relational concepts that specify how

element values are transferred between `Readable` and `Writable` types: `IndirectlyMovable`, `IndirectlyCopyable`, and `IndirectlySwappable`. There are three relational concepts for rearrangements: `Permutable`, `Mergeable`, and `Sortable`. There is one relational concept for comparing values from different sequences: `IndirectlyComparable`.

- ² [Note: The `ranges::less<>` function object type used in the concepts below imposes constraints on the concepts' arguments in addition to those that appear in the concepts' bodies (19.14.8). — end note]

22.3.7.2 Concept `IndirectlyMovable` [common.alg.req.indirectlymovable]

- ¹ The `IndirectlyMovable` concept specifies the relationship between a `Readable` type and a `Writable` type between which values may be moved.

```
template<class In, class Out>
concept IndirectlyMovable =
    Readable<In> &&
    Writable<Out, iter_rvalue_reference_t<In>>;
```

- ² The `IndirectlyMovableStorable` concept augments `IndirectlyMovable` with additional requirements enabling the transfer to be performed through an intermediate object of the `Readable` type's value type.

```
template<class In, class Out>
concept IndirectlyMovableStorable =
    IndirectlyMovable<In, Out> &&
    Writable<Out, iter_value_t<In>> &&
    Movable<iter_value_t<In>> &&
    Constructible<iter_value_t<In>, iter_rvalue_reference_t<In>> &&
    Assignable<iter_value_t<In>&, iter_rvalue_reference_t<In>>;
```

- ³ Let `i` be a dereferenceable value of type `In`. `In` and `Out` model `IndirectlyMovableStorable<In, Out>` only if after the initialization of the object `obj` in

```
iter_value_t<In> obj(ranges::iter_move(i));
```

`obj` is equal to the value previously denoted by `*i`. If `iter_rvalue_reference_t<In>` is an rvalue reference type, the resulting state of the value denoted by `*i` is valid but unspecified ([lib.types.movedfrom]).

22.3.7.3 Concept `IndirectlyCopyable` [common.alg.req.indirectlycopyable]

- ¹ The `IndirectlyCopyable` concept specifies the relationship between a `Readable` type and a `Writable` type between which values may be copied.

```
template<class In, class Out>
concept IndirectlyCopyable =
    Readable<In> &&
    Writable<Out, iter_reference_t<In>>;
```

- ² The `IndirectlyCopyableStorable` concept augments `IndirectlyCopyable` with additional requirements enabling the transfer to be performed through an intermediate object of the `Readable` type's value type. It also requires the capability to make copies of values.

```
template<class In, class Out>
concept IndirectlyCopyableStorable =
    IndirectlyCopyable<In, Out> &&
    Writable<Out, const iter_value_t<In>&> &&
    Copyable<iter_value_t<In>> &&
    Constructible<iter_value_t<In>, iter_reference_t<In>> &&
    Assignable<iter_value_t<In>&, iter_reference_t<In>>;
```

- ³ Let `i` be a dereferenceable value of type `In`. `In` and `Out` model `IndirectlyCopyableStorable<In, Out>` only if after the initialization of the object `obj` in

```
iter_value_t<In> obj(*i);
```

`obj` is equal to the value previously denoted by `*i`. If `iter_reference_t<In>` is an rvalue reference type, the resulting state of the value denoted by `*i` is valid but unspecified ([lib.types.movedfrom]).

22.3.7.4 Concept `IndirectlySwappable` [common.alg.req.indirectlyswappable]

- ¹ The `IndirectlySwappable` concept specifies a swappable relationship between the values referenced by two `Readable` types.

```
template<class I1, class I2 = I1>
concept IndirectlySwappable =
    Readable<I1> && Readable<I2> &&
    requires(I1& i1, I2& i2) {
        ranges::iter_swap(i1, i1);
        ranges::iter_swap(i2, i2);
        ranges::iter_swap(i1, i2);
        ranges::iter_swap(i2, i1);
    };

```

22.3.7.5 Concept IndirectlyComparable [common.alg.req.indirectlycomparable]

- ¹ The `IndirectlyComparable` concept specifies the common requirements of algorithms that compare values from two different sequences.

```
template<class I1, class I2, class R, class P1 = identity,
        class P2 = identity>
concept IndirectlyComparable =
    IndirectRelation<R, projected<I1, P1>, projected<I2, P2>>;

```

22.3.7.6 Concept Permutable [common.alg.req.permutable]

- ¹ The `Permutable` concept specifies the common requirements of algorithms that reorder elements in place by moving or swapping them.

```
template<class I>
concept Permutable =
    ForwardIterator<I> &&
    IndirectlyMovableStorable<I, I> &&
    IndirectlySwappable<I, I>;

```

22.3.7.7 Concept Mergeable [common.alg.req.mergeable]

- ¹ The `Mergeable` concept specifies the requirements of algorithms that merge sorted sequences into an output sequence by copying elements.

```
template<class I1, class I2, class Out, class R = ranges::less<>,
        class P1 = identity, class P2 = identity>
concept Mergeable =
    InputIterator<I1> &&
    InputIterator<I2> &&
    WeaklyIncrementable<Out> &&
    IndirectlyCopyable<I1, Out> &&
    IndirectlyCopyable<I2, Out> &&
    IndirectStrictWeakOrder<R, projected<I1, P1>, projected<I2, P2>>;

```

22.3.7.8 Concept Sortable [common.alg.req.sortable]

- ¹ The `Sortable` concept specifies the common requirements of algorithms that permute sequences into ordered sequences (e.g., `sort`).

```
template<class I, class R = ranges::less<>, class P = identity>
concept Sortable =
    Permutable<I> &&
    IndirectStrictWeakOrder<R, projected<I, P>>;

```

22.4 Iterator primitives [iterator.primitives]

- ¹ To simplify the task of defining iterators, the library provides several classes and functions:

22.4.1 Standard iterator tags [std.iterator.tags]

- ¹ It is often desirable for a function template specialization to find out what is the most specific category of its iterator argument, so that the function can select the most efficient algorithm at compile time. To facilitate this, the library introduces *category tag* classes which are used as compile time tags for algorithm selection. They are: `output_iterator_tag`, `input_iterator_tag`, ~~`output_iterator_tag`~~, `forward_iterator_tag`, `bidirectional_iterator_tag`, ~~and~~ `random_access_iterator_tag`, ~~and~~ `contiguous_iterator_tag`. For every iterator of type ~~`IteratorI`~~, `iterator_traits<IteratorI>::iterator_category` shall be defined to be a category tag that describes the iterator's behavior. Additionally, `iterator_traits<I>::iterator_concept` may be used to indicate conformance to the iterator concepts (22.3.4).

```

namespace std {
    struct output_iterator_tag { };
    struct input_iterator_tag { };
    struct output_iterator_tag { };

    struct forward_iterator_tag: public input_iterator_tag { };
    struct bidirectional_iterator_tag: public forward_iterator_tag { };
    struct random_access_iterator_tag: public bidirectional_iterator_tag { };
    struct contiguous_iterator_tag: random_access_iterator_tag { };
}

```

[...]

22.4.2 Iterator operations

[iterator.operations]

[...]

22.4.3 Range iterator operations

[range.iterator.operations]

- ¹ The library includes the operations `ranges::advance`, `ranges::distance`, `ranges::next`, and `ranges::prev` to manipulate iterators. These operations adapt to the set of operators provided by each iterator category to provide the most efficient implementation possible for a concrete iterator type. [Example: `ranges::advance` uses the `+` operator to move a `RandomAccessIterator` forward `n` steps in constant time. For an iterator type that does not model `RandomAccessIterator`, `ranges::advance` instead performs `n` individual increments with the `++` operator. — end example]
- ² The function templates defined in this subclause are not found by argument-dependent name lookup ([basic.lookup.argdep]). When found by unqualified ([basic.lookup.unqual]) name lookup for the *postfix-expression* in a function call ([expr.call]), they inhibit argument-dependent name lookup.

[Example:

```

void foo() {
    using namespace std::ranges;
    std::vector<int> vec{1,2,3};
    distance(begin(vec), end(vec)); // #1
}

```

The function call expression at #1 invokes `std::ranges::distance`, not `std::distance`, despite that (a) the iterator type returned from `begin(vec)` and `end(vec)` may be associated with namespace `std` and (b) `std::distance` is more specialized ([temp.func.order]) than `std::ranges::distance` since the former requires its first two parameters to have the same type. — end example]

- ³ The number and order of deducible template parameters for the function templates defined in this subclause is unspecified, except where explicitly stated otherwise.

22.4.3.1 `ranges::advance`

[range.iterator.operations.advance]

```

template<Iterator I>
constexpr void advance(I& i, iter_difference_t<I> n);

```

- ¹ *Expects:* `n` shall be negative only if `I` models `BidirectionalIterator`.

- ² *Effects:*

- (2.1) — If `I` models `RandomAccessIterator`, equivalent to `i += n`.
- (2.2) — Otherwise, if `n` is non-negative, increments `i` by `n`.
- (2.3) — Otherwise, decrements `i` by `-n`.

```

template<Iterator I, Sentinel<I> S>
constexpr void advance(I& i, S bound);

```

- ³ *Expects:* `[i, bound)` shall denote a range.

- ⁴ *Effects:*

- (4.1) — If `I` and `S` model `Assignable<I&, S>`, equivalent to `i = std::move(bound)`.
- (4.2) — Otherwise, if `S` and `I` model `SizedSentinel<S, I>`, equivalent to `ranges::advance(i, bound - i)`.
- (4.3) — Otherwise, while `bool(i != bound)` is true, increments `i`.

```
template<Iterator I, Sentinel<I> S>
constexpr iter_difference_t<I> advance(I& i, iter_difference_t<I> n, S bound);
```

5 *Expects:* If $n > 0$, $[i, \text{bound})$ shall denote a range. If $n == 0$, $[i, \text{bound})$ or $[\text{bound}, i)$ shall denote a range. If $n < 0$, $[\text{bound}, i)$ shall denote a range, I shall model `BidirectionalIterator`, and I and S shall model `Same<I, S>`.

6 *Effects:*

(6.1) — If S and I model `SizedSentinel<S, I>`:

(6.1.1) — If $|n| \geq |\text{bound} - i|$, equivalent to `ranges::advance(i, bound)`.

(6.1.2) — Otherwise, equivalent to `ranges::advance(i, n)`.

(6.2) — Otherwise,

(6.2.1) — if n is non-negative, while `bool(i != bound)` is `true`, increments i but at most n times.

(6.2.2) — Otherwise, while `bool(i != bound)` is `true`, decrements i but at most $-n$ times.

7 *Returns:* $n - M$, where M is the difference between the ending and starting positions of i .

22.4.3.2 `ranges::distance`

[range.iterator.operations.distance]

```
template<Iterator I, Sentinel<I> S>
constexpr iter_difference_t<I> distance(I first, S last);
```

1 *Expects:* $[\text{first}, \text{last})$ shall denote a range, or $[\text{last}, \text{first})$ shall denote a range and S and I shall model `Same<S, I> && SizedSentinel<S, I>`.

2 *Effects:* If S and I model `SizedSentinel<S, I>`, returns `(last - first)`; otherwise, returns the number of increments needed to get from `first` to `last`.

```
template<Range R>
constexpr iter_difference_t<iterator_t<R>> distance(R&& r);
```

3 *Effects:* If R models `SizedRange`, equivalent to:

```
return ranges::size(r); // 23.4.1
```

Otherwise, equivalent to:

```
return ranges::distance(ranges::begin(r), ranges::end(r)); // 23.3
```

22.4.3.3 `ranges::next`

[range.iterator.operations.next]

```
template<Iterator I>
constexpr I next(I x);
```

1 *Effects:* Equivalent to: `++x; return x;`

```
template<Iterator I>
constexpr I next(I x, iter_difference_t<I> n);
```

2 *Effects:* Equivalent to: `ranges::advance(x, n); return x;`

```
template<Iterator I, Sentinel<I> S>
constexpr I next(I x, S bound);
```

3 *Effects:* Equivalent to: `ranges::advance(x, bound); return x;`

```
template<Iterator I, Sentinel<I> S>
constexpr I next(I x, iter_difference_t<I> n, S bound);
```

4 *Effects:* Equivalent to: `ranges::advance(x, n, bound); return x;`

22.4.3.4 `ranges::prev`

[range.iterator.operations.prev]

```
template<BidirectionalIterator I>
constexpr I prev(I x);
```

1 *Effects:* Equivalent to: `--x; return x;`

```

template<BidirectionalIterator I>
constexpr I prev(I x, iter_difference_t<I> n);
2     Effects: Equivalent to: ranges::advance(x, -n); return x;

template<BidirectionalIterator I>
constexpr I prev(I x, iter_difference_t<I> n, I bound);
3     Effects: Equivalent to: ranges::advance(x, -n, bound); return x;

```

22.5 Iterator adaptors

[predef.iterators]

22.5.1 Reverse iterators

[reverse.iterators]

- ¹ Class template `reverse_iterator` is an iterator adaptor that iterates from the end of the sequence defined by its underlying iterator to the beginning of that sequence. ~~The fundamental relation between a reverse iterator and its corresponding iterator `i` is established by the identity: `&*(reverse_iterator(i)) == &(i-1)`.~~

22.5.1.1 Class template `reverse_iterator`

[reverse.iterator]

```

namespace std {
    template<class Iterator>
    class reverse_iterator {
    public:
        using iterator_type      = Iterator;
        using iterator_concept   = see below;
        using iterator_category = typename iterator_traits<Iterator>::iterator_category;
        using iterator_category   = see below;
        using value_type          = typename iterator_traits<Iterator>::value_type;
        using value_type          = iter_value_t<Iterator>;
        using difference_type     = typename iterator_traits<Iterator>::difference_type;
        using difference_type     = iter_difference_t<Iterator>;
        using pointer            = typename iterator_traits<Iterator>::pointer;
        using reference           = typename iterator_traits<Iterator>::reference;
        using reference           = iter_reference_t<Iterator>;

        constexpr reverse_iterator();
        constexpr explicit reverse_iterator(Iterator x);
        template<class U> constexpr reverse_iterator(const reverse_iterator<U>& u);
        template<class U> constexpr reverse_iterator& operator=(const reverse_iterator<U>& u);

        constexpr Iterator base() const;      // explicit
        constexpr reference operator*() const;
        constexpr pointer  operator->() const requires see below;

        [...]

        constexpr reverse_iterator& operator+=(difference_type n);
        constexpr unspecified operator[] (difference_type n) const;

        friend constexpr iter_rvalue_reference_t<Iterator> iter_move(const reverse_iterator& i)
            noexcept(see below);
        template<IndirectlySwappable<Iterator> Iterator2>
            friend constexpr void iter_swap(const reverse_iterator& x,
                                           const reverse_iterator<Iterator2>& y)
                noexcept(see below);

    protected:
        Iterator current;
    };

    [...]

    template<class Iterator>
    constexpr reverse_iterator<Iterator> make_reverse_iterator(Iterator i);

```

```

template<class Iterator1, class Iterator2>
    requires (!SizedSentinel<Iterator1, Iterator2>)
    inline constexpr bool disable_sized_sentinel<reverse_iterator<Iterator1>,
                                                    reverse_iterator<Iterator2>> = true;
}

```

- ¹ The member *typedef-name* `iterator_concept` denotes `random_access_iterator_tag` if `Iterator` models `RandomAccessIterator`, and `bidirectional_iterator_tag` otherwise.
- ² The member *typedef-name* `iterator_category` denotes `random_access_iterator_tag` if `iterator_traits<Iterator>::iterator_category` models `DerivedFrom<random_access_iterator_tag>`, and `iterator_traits<Iterator>::iterator_category` otherwise.

22.5.1.2 `reverse_iterator` requirements [reverse.iter.requirements]

- ¹ The template parameter `Iterator` shall ~~satisfy all~~ *either meet* the requirements of a *Cpp17BidirectionalIterator* (22.3.5.5) *or model BidirectionalIterator* (22.3.4.12).
- ² Additionally, `Iterator` shall ~~satisfy~~ *either meet* the requirements of a *Cpp17RandomAccessIterator* (22.3.5.6) *or model RandomAccessIterator* (22.3.4.13) if *the definitions of* any of the members
 - (2.1) — `operator+`, `operator-`, `operator+=`, `operator-=` (22.5.1.6), *or*
 - (2.2) — `operator[]` (22.5.1.5),
 or the non-member operators (22.5.1.7)
 - (2.3) — `operator<`, `operator>`, `operator<=`, `operator>=`, `operator-`, or `operator+` (22.5.1.8)
 are ~~referenced in a way that requires instantiation~~ *instantiated* ([temp.inst]).

[...]

22.5.1.5 `reverse_iterator` element access [reverse.iter.elem]

[...]

[Editor's note: This change incorporates the PR of LWG 1052:]

```

constexpr pointer operator->() const requires is_pointer_v<Iterator>
    || requires(const Iterator i) { i.operator->(); };

```

- ² ~~Returns: `addressof(operator*())`.~~ *Effects:*
 - (2.1) — If `Iterator` is a pointer type, equivalent to: `return prev(current);`
 - (2.2) — Otherwise, equivalent to: `return prev(current).operator->();`
- [...]

22.5.1.6 `reverse_iterator` navigation [reverse.iter.nav]

[...]

22.5.1.7 `reverse_iterator` comparisons [reverse.iter.cmp]

```

template<class Iterator1, class Iterator2>
constexpr bool operator==(
    const reverse_iterator<Iterator1>& x,
    const reverse_iterator<Iterator2>& y);

```

- ¹ *Constraints:* The expression `x.current == y.current` shall be valid and convertible to `bool`.
- ² *Returns:* `x.current == y.current`.

```

template<class Iterator1, class Iterator2>
constexpr bool operator!=(
    const reverse_iterator<Iterator1>& x,
    const reverse_iterator<Iterator2>& y);

```

- ³ *Constraints:* The expression `x.current != y.current` shall be valid and convertible to `bool`.
- ⁴ *Returns:* `x.current != y.current`.

```
template<class Iterator1, class Iterator2>
constexpr bool operator<(
    const reverse_iterator<Iterator1>& x,
    const reverse_iterator<Iterator2>& y);
```

5 *Constraints:* The expression `x.current > y.current` shall be valid and convertible to `bool`.

6 *Returns:* `x.current > y.current`.

```
template<class Iterator1, class Iterator2>
constexpr bool operator>(
    const reverse_iterator<Iterator1>& x,
    const reverse_iterator<Iterator2>& y);
```

7 *Constraints:* The expression `x.current < y.current` shall be valid and convertible to `bool`.

8 *Returns:* `x.current < y.current`.

```
template<class Iterator1, class Iterator2>
constexpr bool operator<=(
    const reverse_iterator<Iterator1>& x,
    const reverse_iterator<Iterator2>& y);
```

9 *Constraints:* The expression `x.current >= y.current` shall be valid and convertible to `bool`.

10 *Returns:* `x.current >= y.current`.

```
template<class Iterator1, class Iterator2>
constexpr bool operator>=(
    const reverse_iterator<Iterator1>& x,
    const reverse_iterator<Iterator2>& y);
```

11 *Constraints:* The expression `x.current <= y.current` shall be valid and convertible to `bool`.

12 *Returns:* `x.current <= y.current`.

22.5.1.8 Non-member functions

[reverse.iter.nonmember]

[...]

2 *Returns:* `reverse_iterator<Iterator> (x.current - n)`.

```
friend constexpr iter_rvalue_reference_t<Iterator> iter_move(const reverse_iterator& i)
noexcept(see below);
```

3 *Effects:* Equivalent to:

```
    auto tmp = i.current;
    return ranges::iter_move(--tmp);
```

4 *Remarks:* The expression in `noexcept` is equivalent to:

```
    is_nothrow_copy_constructible_v<Iterator> &&
    noexcept(ranges::iter_move(--declval<Iterator>()))
```

```
template<IndirectlySwappable<Iterator> Iterator2>
friend constexpr void iter_swap(const reverse_iterator& x, const reverse_iterator<Iterator2>& y)
noexcept(see below);
```

5 *Effects:* Equivalent to:

```
    auto xtmp = x.current;
    auto ytmp = y.current;
    ranges::iter_swap(--xtmp, --ytmp);
```

6 *Remarks:* The expression in `noexcept` is equivalent to:

```
    is_nothrow_copy_constructible_v<Iterator> &&
    is_nothrow_copy_constructible_v<Iterator2> &&
    noexcept(ranges::iter_swap(--declval<Iterator>(), --declval<Iterator2>()))
```

```
template<class Iterator>
constexpr reverse_iterator<Iterator> make_reverse_iterator(Iterator i);
```

[...]

22.5.2 Insert iterators

[insert.iterators]

[...]

22.5.2.1 Class template back_insert_iterator

[back.insert.iterator]

```

namespace std {
    template<class Container>
    class back_insert_iterator {
    protected:
        Container* container = nullptr;

    public:
        using iterator_category = output_iterator_tag;
        using value_type        = void;
        using difference_type    = void_ptrdiff_t;
        using pointer            = void;
        using reference          = void;
        using container_type     = Container;

        constexpr back_insert_iterator() noexcept = default;
        explicit back_insert_iterator(Container& x);
        back_insert_iterator& operator=(const typename Container::value_type& value);
        back_insert_iterator& operator=(typename Container::value_type&& value);

        back_insert_iterator& operator*();
        back_insert_iterator& operator++();
        back_insert_iterator operator++(int);
    };

    template<class Container>
        back_insert_iterator<Container> back_inserter(Container& x);
}

```

[...]

22.5.2.2 Class template front_insert_iterator

[front.insert.iterator]

```

namespace std {
    template<class Container>
    class front_insert_iterator {
    protected:
        Container* container = nullptr;

    public:
        using iterator_category = output_iterator_tag;
        using value_type        = void;
        using difference_type    = void_ptrdiff_t;
        using pointer            = void;
        using reference          = void;
        using container_type     = Container;

        constexpr front_insert_iterator() noexcept = default;
        explicit front_insert_iterator(Container& x);
        front_insert_iterator& operator=(const typename Container::value_type& value);
        front_insert_iterator& operator=(typename Container::value_type&& value);

        front_insert_iterator& operator*();
        front_insert_iterator& operator++();
        front_insert_iterator operator++(int);
    };

    template<class Container>
        front_insert_iterator<Container> front_inserter(Container& x);
}

```

[...]

22.5.2.3 Class template insert_iterator

[insert.iterator]

```

namespace std {
    template<class Container>
    class insert_iterator {
    protected:
        Container* container = nullptr;
        typename Container::iterator iterator_t<Container> iter =
            iterator_t<Container>();
    public:
        using iterator_category = output_iterator_tag;
        using value_type        = void;
        using difference_type    = voidptrdiff_t;
        using pointer            = void;
        using reference          = void;
        using container_type     = Container;

        insert_iterator() = default;
        insert_iterator(Container& x, typename Container::iterator iterator_t<Container> i);
        insert_iterator& operator=(const typename Container::value_type& value);
        insert_iterator& operator=(typename Container::value_type&& value);

        insert_iterator& operator*();
        insert_iterator& operator++();
        insert_iterator& operator++(int);
    };

    template<class Container>
    insert_iterator<Container> inserter(Container& x, typename Container::iterator iterator_t<Container> i);
}

```

22.5.2.3.1 insert_iterator operations

[insert.iter.ops]

```
insert_iterator(Container& x, typename Container::iterator iterator_t<Container> i);
```

[...]

22.5.2.3.2 inserter

[inserter]

```

template<class Container>
insert_iterator<Container> inserter(Container& x, typename Container::iterator iterator_t<Container> i);

```

¹ *Returns:* insert_iterator<Container>(x, i).

[Editor's note: Note the change to the title of [move.iterators].]

22.5.3 Move iterators and sentinels

[move.iterators]

[...]

22.5.3.1 Class template move_iterator

[move.iterator]

```

namespace std {
    template<class Iterator>
    class move_iterator {
    public:
        using iterator_type      = Iterator;
        using iterator_concept   = input_iterator_tag;
        using iterator_category  = typename iterator_traits<Iterator>::iterator_categorysee below;
        using value_type         = typename iterator_traits<Iterator>::value_typeiter_value_t<Iterator>;
        using difference_type    = typename iterator_traits<Iterator>::difference_typeiter_difference_t<Iterator>;
        using pointer            = Iterator;
        using reference          = see belowiter_rvalue_reference_t<Iterator>;

        constexpr move_iterator();
        constexpr explicit move_iterator(Iterator i);

        [...]
    };
}

```

```
constexpr move_iterator& operator++();
constexpr move_iteratorauto operator++(int);
constexpr move_iterator& operator--();
```

[...]

```
constexpr move_iterator operator-(difference_type n) const;
constexpr move_iterator& operator--(difference_type n);
constexpr unspecifiedreference operator[](difference_type n) const;
```

```
template<Sentinel<Iterator> S>
    friend constexpr bool operator==(
        const move_iterator& x, const move_sentinel<S>& y);
template<Sentinel<Iterator> S>
    friend constexpr bool operator==(
        const move_sentinel<S>& x, const move_iterator& y);
template<Sentinel<Iterator> S>
    friend constexpr bool operator!=(
        const move_iterator& x, const move_sentinel<S>& y);
template<Sentinel<Iterator> S>
    friend constexpr bool operator!=(
        const move_sentinel<S>& x, const move_iterator& y);

template<SizedSentinel<Iterator> S>
    friend constexpr iter_difference_t<Iterator> operator-(
        const move_sentinel<S>& x, const move_iterator& y);
template<SizedSentinel<Iterator> S>
    friend constexpr iter_difference_t<Iterator> operator-(
        const move_iterator& x, const move_sentinel<S>& y);

friend constexpr iter_rvalue_reference_t<Iterator> iter_move(const move_iterator& i)
    noexcept(noexcept(ranges::iter_move(i.current)));
template<IndirectlySwappable<Iterator> Iterator2>
    friend constexpr void iter_swap(const move_iterator& x, const move_iterator<Iterator2>& y)
        noexcept(noexcept(ranges::iter_swap(x.current, y.current)));
```

```
private:
    Iterator current;    // exposition only
};
```

[...]

```
template<class Iterator>
    constexpr move_iterator<Iterator> operator+(
        typename move_iterator<Iterator>::difference_typeiter_difference_t<Iterator> n,
        const move_iterator<Iterator>& x);
template<class Iterator>
    constexpr move_iterator<Iterator> make_move_iterator(Iterator i);
}
```

- ¹ Let R denote `iterator_traits<Iterator>::reference`. If `is_reference_v<R>` is true, the template specialization `move_iterator<Iterator>` shall define the nested type named `reference` as a synonym for `remove_reference_t<R>&&`, otherwise as a synonym for R .
- ² The member `typedef-name` `iterator_category` denotes `random_access_iterator_tag` if `iterator_traits<Iterator>::iterator_category` models `DerivedFrom<random_access_iterator_tag>`, and `iterator_traits<Iterator>::iterator_category` otherwise.

22.5.3.2 move_iterator requirements

[move.iter.requirements]

- ¹ The template parameter `Iterator` shall ~~satisfy either meet~~ the *Cpp17InputIterator* requirements (22.3.5.2) ~~or model InputIterator~~ (22.3.4.9). Additionally, if any of the bidirectional ~~or random access~~ traversal functions are instantiated, the template parameter shall ~~satisfy either meet~~ the *Cpp17BidirectionalIterator* requirements (22.3.5.5) ~~or model BidirectionalIterator~~ (22.3.4.12) ~~Cpp17RandomAccessIterator requirements~~ (22.3.5.6), ~~respectively~~. If any of the random access traversal functions are instantiated, the template

parameter shall either meet the *Cpp17RandomAccessIterator* requirements (22.3.5.6) or model *RandomAccessIterator* (22.3.4.13).

[...]

22.5.3.5 move_iterator element access

[move.iter.elem]

constexpr reference operator*() const;

1 ~~Returns: static_cast<reference>(*current).~~

Effects: Equivalent to: return ranges::iter_move(current);

constexpr pointer operator->() const;

2 ~~Returns: current.~~

constexpr ~~unspecified~~reference operator[](difference_type n) const;

3 ~~Returns: std::move(current[n]).~~

Effects: Equivalent to: ranges::iter_move(current + n);

22.5.3.6 move_iterator navigation

[move.iter.nav]

[...]

constexpr ~~move_iterator~~auto operator++(int);

3 ~~Effects: As if by~~If Iterator models ForwardIterator, equivalent to:

move_iterator tmp = *this;

++current;

return tmp;

Otherwise, equivalent to ++current.

[...]

22.5.3.7 move_iterator comparisons

[move.iter.op.comp]

template<class Iterator1, class Iterator2>

constexpr bool operator==(const move_iterator<Iterator1>& x, const move_iterator<Iterator2>& y);

template<Sentinel<Iterator> S>

friend constexpr bool operator==(const move_iterator& x, const move_sentinel<S>& y);

template<Sentinel<Iterator> S>

friend constexpr bool operator==(const move_sentinel<S>& x, const move_iterator& y);

1 Constraints: The expression x.base() == y.base() shall be valid and convertible to bool.

2 ~~Returns: x.base() == y.base().~~

template<class Iterator1, class Iterator2>

constexpr bool operator!=(const move_iterator<Iterator1>& x, const move_iterator<Iterator2>& y);

template<Sentinel<Iterator> S>

friend constexpr bool operator!=(const move_iterator& x, const move_sentinel<S>& y);

template<Sentinel<Iterator> S>

friend constexpr bool operator!=(const move_sentinel<S>& x, const move_iterator& y);

3 Constraints: The expression x.base() == y.base() shall be valid and convertible to bool.

4 ~~Returns: !(x == y).~~

template<class Iterator1, class Iterator2>

constexpr bool operator<(const move_iterator<Iterator1>& x, const move_iterator<Iterator2>& y);

5 Constraints: The expression x.base() < y.base() shall be valid and convertible to bool.

6 ~~Returns: x.base() < y.base().~~

template<class Iterator1, class Iterator2>

constexpr bool operator>(const move_iterator<Iterator1>& x, const move_iterator<Iterator2>& y);

7 Constraints: The expression y.base() < x.base() shall be valid and convertible to bool.

8 ~~Returns: y < x.~~

```
template<class Iterator1, class Iterator2>
constexpr bool operator<=(const move_iterator<Iterator1>& x, const move_iterator<Iterator2>& y);
```

9 *Constraints:* The expression `y.base() < x.base()` shall be valid and convertible to `bool`.

10 *Returns:* `!(y < x)`.

```
template<class Iterator1, class Iterator2>
constexpr bool operator>=(const move_iterator<Iterator1>& x, const move_iterator<Iterator2>& y);
```

11 *Constraints:* The expression `x.base() < y.base()` shall be valid and convertible to `bool`.

12 *Returns:* `!(x < y)`.

22.5.3.8 move_iterator non-member functions

[move.iter.nonmember]

```
template<class Iterator1, class Iterator2>
constexpr auto operator-(
    const move_iterator<Iterator1>& x,
    const move_iterator<Iterator2>& y) -> decltype(x.base() - y.base());
template<Sentinel<Iterator> S>
friend constexpr iter_difference_t<Iterator> operator-(
    const move_sentinel<S>& x, const move_iterator<S>& y);
template<Sentinel<Iterator> S>
friend constexpr iter_difference_t<Iterator> operator-(
    const move_iterator<S>& x, const move_sentinel<S>& y);
```

1 *Returns:* `x.base() - y.base()`.

```
template<class Iterator>
constexpr move_iterator<Iterator> operator+(
    typename move_iterator<Iterator>::difference_type iter_difference_t<Iterator> n,
    const move_iterator<Iterator>& x);
```

2 *Constraints:* The expression `x + n` shall be valid and have type `Iterator`.

3 *Returns:* `x + n`.

```
friend constexpr iter_rvalue_reference_t<Iterator> iter_move(const move_iterator& i)
noexcept(noexcept(ranges::iter_move(i.current)));
```

4 *Effects:* Equivalent to: `return ranges::iter_move(i.current);`

```
template<IndirectlySwappable<Iterator> Iterator2>
friend constexpr void iter_swap(const move_iterator& x, const move_iterator<Iterator2>& y)
noexcept(noexcept(ranges::iter_swap(x.current, y.current)));
```

5 *Effects:* Equivalent to: `ranges::iter_swap(x.current, y.current)`.

```
template<class Iterator>
constexpr move_iterator<Iterator> make_move_iterator(Iterator i);
```

6 *Returns:* `move_iterator<Iterator>(i)`.

22.5.3.9 Class template move_sentinel

[move.sentinel]

1 Class template `move_sentinel` is a sentinel adaptor useful for denoting ranges together with `move_iterator`. When an input iterator type `I` and sentinel type `S` model `Sentinel<S, I>`, `move_sentinel<S>` and `move_iterator<I>` model `Sentinel<move_sentinel<S>, move_iterator<I>>` as well.

2 [*Example:* A `move_if` algorithm is easily implemented with `copy_if` using `move_iterator` and `move_sentinel`:

```
template<InputIterator I, Sentinel<I> S, WeaklyIncrementable O,
        IndirectUnaryPredicate<I> Pred>
requires IndirectlyMovable<I, O>
void move_if(I first, S last, O out, Pred pred) {
    std::ranges::copy_if(move_iterator<I>{first}, move_sentinel<S>{last}, out, pred);
}
```

— end example]

```

namespace std {
    template<Semiregular S>
    class move_sentinel {
    public:
        constexpr move_sentinel();
        constexpr explicit move_sentinel(S s);
        template<class S2>
            requires ConvertibleTo<const S2&, S>
            constexpr move_sentinel(const move_sentinel<S2>& s);
        template<class S2>
            requires Assignable<S&, const S2&>
            constexpr move_sentinel& operator=(const move_sentinel<S2>& s);

        constexpr S base() const;

    private:
        S last; // exposition only
    };
}

```

22.5.3.10 move_sentinel operations

[move.sent.ops]

```
constexpr move_sentinel();
```

- ¹ *Effects:* Value-initializes last. If `is_trivially_default_constructible_v<S>` is true, then this constructor is a `constexpr` constructor.

```
constexpr explicit move_sentinel(S s);
```

- ² *Effects:* Initializes last with `std::move(s)`.

```

template<class S2>
    requires ConvertibleTo<const S2&, S>
    constexpr move_sentinel(const move_sentinel<S2>& s);

```

- ³ *Effects:* Initializes last with `s.last`.

```

template<class S2>
    requires Assignable<S&, const S2&>
    constexpr move_sentinel& operator=(const move_sentinel<S2>& s);

```

- ⁴ *Effects:* Equivalent to: `last = s.last; return *this;`

22.5.4 Common iterators

[iterators.common]

22.5.4.1 Class template common_iterator

[common.iterator]

- ¹ Class template `common_iterator` is an iterator/sentinel adaptor that is capable of representing a non-common range of elements (where the types of the iterator and sentinel differ) as a common range (where they are the same). It does this by holding either an iterator or a sentinel, and implementing the equality comparison operators appropriately.
- ² [Note: The `common_iterator` type is useful for interfacing with legacy code that expects the begin and end of a range to have the same type. — end note]
- ³ [Example:

```

template<class ForwardIterator>
void fun(ForwardIterator begin, ForwardIterator end);

list<int> s;
// populate the list s
using CI = common_iterator<counted_iterator<list<int>::iterator>, default_sentinel_t>;
// call fun on a range of 10 ints
fun(CI(counted_iterator(s.begin(), 10)), CI(default_sentinel));
— end example]

```

```

namespace std {
    template<Iterator I, Sentinel<I> S>
        requires (!Same<I, S>)
        class common_iterator {
        public:
            constexpr common_iterator() = default;
            constexpr common_iterator(I i);
            constexpr common_iterator(S s);
            template<class I2, class S2>
                requires ConvertibleTo<const I2&, I> && ConvertibleTo<const S2&, S>
                constexpr common_iterator(const common_iterator<I2, S2>& x);

            template<class I2, class S2>
                requires ConvertibleTo<const I2&, I> && ConvertibleTo<const S2&, S> &&
                    Assignable<I&, const I2&> && Assignable<S&, const S2&>
                common_iterator& operator=(const common_iterator<I2, S2>& x);

            decltype(auto) operator*();
            decltype(auto) operator*() const
                requires dereferenceable<const I>;
            decltype(auto) operator->() const
                requires see below;

            common_iterator& operator++();
            decltype(auto) operator++(int);

            template<class I2, Sentinel<I> S2>
                requires Sentinel<S, I2>
            friend bool operator==(
                const common_iterator& x, const common_iterator<I2, S2>& y);
            template<class I2, Sentinel<I> S2>
                requires Sentinel<S, I2> && EqualityComparableWith<I, I2>
            friend bool operator==(
                const common_iterator& x, const common_iterator<I2, S2>& y);
            template<class I2, Sentinel<I> S2>
                requires Sentinel<S, I2>
            friend bool operator!=(
                const common_iterator& x, const common_iterator<I2, S2>& y);

            template<SizedSentinel<I> I2, SizedSentinel<I> S2>
                requires SizedSentinel<S, I2>
            friend iter_difference_t<I2> operator-(
                const common_iterator& x, const common_iterator<I2, S2>& y);

            friend iter_rvalue_reference_t<I> iter_move(const common_iterator& i)
                noexcept(noexcept(ranges::iter_move(declval<const I&>())))
                requires InputIterator<I>;
            template<IndirectlySwappable<I> I2, class S2>
                friend void iter_swap(const common_iterator& x, const common_iterator<I2, S2>& y)
                    noexcept(noexcept(ranges::iter_swap(declval<const I&>(), declval<const I2&>())));

        private:
            variant<I, S> v_; // exposition only
        };

        template<class I, class S>
        struct incrementable_traits<common_iterator<I, S>> {
            using difference_type = iter_difference_t<I>;
        };

        template<InputIterator I, class S>
        struct iterator_traits<common_iterator<I, S>> {
            using iterator_concept = see below;
            using iterator_category = see below;

```

```

        using value_type = iter_value_t<I>;
        using difference_type = iter_difference_t<I>;
        using pointer = see below;
        using reference = iter_reference_t<I>;
    };
}

```

22.5.4.2 Associated types

[common.iter.types]

¹ The nested *typedef-names* of the specialization of `iterator_traits` for `common_iterator<I, S>` are defined as follows.

- (1.1) — `iterator_concept` denotes `forward_iterator_tag` if `I` models `ForwardIterator`; otherwise it denotes `input_iterator_tag`.
- (1.2) — Let `C` denote the type `iterator_traits<I>::iterator_category`. If `C` models `DerivedFrom<forward_iterator_tag>`, `iterator_category` denotes `forward_iterator_tag`. Otherwise, `iterator_category` denotes `input_iterator_tag`.
- (1.3) — If the expression `a.operator->()` is well-formed, where `a` is an lvalue of type `const common_iterator<I, S>`, then `pointer` denotes the type of that expression. Otherwise, `pointer` denotes `void`.

22.5.4.3 Constructors and conversions

[common.iter.const]

```
constexpr common_iterator(I i);
```

¹ *Effects*: Initializes `v_` as if by `v_{in_place_type<I>, std::move(i)}`.

```
constexpr common_iterator(S s);
```

² *Effects*: Initializes `v_` as if by `v_{in_place_type<S>, std::move(s)}`.

```
template<class I2, class S2>
requires ConvertibleTo<const I2&, I> && ConvertibleTo<const S2&, S>
constexpr common_iterator(const common_iterator<I2, S2>& x);
```

³ *Expects*: `x.v_.valueless_by_exception()` is false.

⁴ *Effects*: Initializes `v_` as if by `v_{in_place_index<i>, get<i>(x.v_)}`, where `i` is `x.v_.index()`.

```
template<class I2, class S2>
requires ConvertibleTo<const I2&, I> && ConvertibleTo<const S2&, S> &&
    Assignable<I&, const I2&> && Assignable<S&, const S2&>
common_iterator& operator=(const common_iterator<I2, S2>& x);
```

⁵ *Expects*: `x.v_.valueless_by_exception()` is false.

⁶ *Effects*: Equivalent to:

(6.1) — If `v_.index() == x.v_.index()`, then `get<i>(v_) = get<i>(x.v_)`.

(6.2) — Otherwise, `v_.emplace<i>(get<i>(x.v_))`.

where `i` is `x.v_.index()`.

⁷ *Returns*: `*this`

22.5.4.4 Accessors

[common.iter.access]

```
decltype(auto) operator*();
decltype(auto) operator*() const
requires dereferenceable<const I>;
```

¹ *Expects*: `holds_alternative<I>(v_)`.

² *Effects*: Equivalent to: `return *get<I>(v_)`;

```
decltype(auto) operator->() const
requires see below;
```

³ The expression in the *requires* clause is equivalent to:

```
Readable<const I> &&
    (requires(const I& i) { i.operator->(); } ||
```

```
is_reference_v<iter_reference_t<I>> ||
Constructible<iter_value_t<I>, iter_reference_t<I>>>)
```

4 *Expects:* holds_alternative<I>(v_).

5 *Effects:*

(5.1) — If I is a pointer type or if the expression `get<I>(v_).operator->()` is well-formed, equivalent to: `return get<I>(v_);`

(5.2) — Otherwise, if `iter_reference_t<I>` is a reference type, equivalent to:

```
auto&& tmp = *get<I>(v_);
return addressof(tmp);
```

(5.3) — Otherwise, equivalent to: `return proxy(*get<I>(v_));` where *proxy* is the exposition-only class:

```
class proxy {
    iter_value_t<I> keep_;
    proxy(iter_reference_t<I>&& x)
        : keep_(std::move(x)) {}
public:
    const iter_value_t<I>* operator->() const {
        return addressof(keep_);
    }
};
```

22.5.4.5 Navigation

[common.iter.nav]

```
common_iterator& operator++();
```

1 *Expects:* holds_alternative<I>(v_).

2 *Effects:* Equivalent to `++get<I>(v_)`.

3 *Returns:* `*this`.

```
decltype(auto) operator++(int);
```

4 *Expects:* holds_alternative<I>(v_).

5 *Effects:* If I models ForwardIterator, equivalent to:

```
common_iterator tmp = *this;
++*this;
return tmp;
```

Otherwise, equivalent to: `return get<I>(v_)+;`

22.5.4.6 Comparisons

[common.iter.cmp]

```
template<class I2, Sentinel<I> S2>
requires Sentinel<S, I2>
friend bool operator==(
    const common_iterator& x, const common_iterator<I2, S2>& y);
```

1 *Expects:* `x.v_.valueless_by_exception()` and `y.v_.valueless_by_exception()` are each false.

2 *Returns:* true if *i* == *j*, and otherwise `get<i>(x.v_) == get<j>(y.v_)`, where *i* is `x.v_.index()` and *j* is `y.v_.index()`.

```
template<class I2, Sentinel<I> S2>
requires Sentinel<S, I2> && EqualityComparableWith<I, I2>
friend bool operator==(
    const common_iterator& x, const common_iterator<I2, S2>& y);
```

3 *Expects:* `x.v_.valueless_by_exception()` and `y.v_.valueless_by_exception()` are each false.

4 *Returns:* true if *i* and *j* are each 1, and otherwise `get<i>(x.v_) == get<j>(y.v_)`, where *i* is `x.v_.index()` and *j* is `y.v_.index()`.

```
template<class I2, Sentinel<I> S2>
    requires Sentinel<S, I2>
friend bool operator!=(
    const common_iterator& x, const common_iterator<I2, S2>& y);
```

5 *Effects:* Equivalent to: `return !(x == y);`

```
template<SizedSentinel<I> I2, SizedSentinel<I> S2>
    requires SizedSentinel<S, I2>
friend iter_difference_t<I2> operator-(
    const common_iterator& x, const common_iterator<I2, S2>& y);
```

6 *Expects:* `x.v_.valueless_by_exception()` and `y.v_.valueless_by_exception()` are each false.

7 *Returns:* 0 if *i* and *j* are each 1, and otherwise `get<i>(x.v_) - get<j>(y.v_)`, where *i* is `x.v_.index()` and *j* is `y.v_.index()`.

22.5.4.7 Customization

[common.iter.cust]

```
friend iter_rvalue_reference_t<I> iter_move(const common_iterator& i)
    noexcept(noexcept(ranges::iter_move(declval<const I&>())))
    requires InputIterator<I>;
```

1 *Expects:* `holds_alternative<I>(v_)`.

2 *Effects:* Equivalent to: `return ranges::iter_move(get<I>(i.v_));`

```
template<IndirectlySwappable<I> I2, class S2>
    friend void iter_swap(const common_iterator& x, const common_iterator<I2, S2>& y)
        noexcept(noexcept(ranges::iter_swap(declval<const I&>(), declval<const I2&>())));
```

3 *Expects:* `holds_alternative<I>(x.v_)` and `holds_alternative<I2>(y.v_)` are each true.

4 *Effects:* Equivalent to `ranges::iter_swap(get<I>(x.v_), get<I2>(y.v_))`.

22.5.5 Default sentinels

[default.sentinel]

```
namespace std {
    struct default_sentinel_t { };
}
```

1 Class `default_sentinel_t` is an empty type used to denote the end of a range. It can be used together with iterator types that know the bound of their range (e.g., `counted_iterator` (22.5.6.1)).

22.5.6 Counted iterators

[iterators.counted]

22.5.6.1 Class template `counted_iterator`

[counted.iterator]

1 Class template `counted_iterator` is an iterator adaptor with the same behavior as the underlying iterator except that it keeps track of the distance to the end of its range. It can be used together with `default_sentinel` in calls to generic algorithms to operate on a range of *N* elements starting at a given position without needing to know the end position a priori.

[Editor's note: The following example incorporates the PR for [stl2#554](#):]

2 [Example:

```
list<string> s;
// populate the list s with at least 10 strings
vector<string> v;
// copies 10 strings into v:
ranges::copy(counted_iterator(s.begin(), 10), default_sentinel, back_inserter(v));
```

— end example]

3 Two values `i1` and `i2` of types `counted_iterator<I1>` and `counted_iterator<I2>` refer to elements of the same sequence if and only if `next(i1.base(), i1.count())` and `next(i2.base(), i2.count())` refer to the same (possibly past-the-end) element.

```

namespace std {
    template<Iterator I>
    class counted_iterator {
    public:
        using iterator_type = I;

        constexpr counted_iterator() = default;
        constexpr counted_iterator(I x, iter_difference_t<I> n);
        template<class I2>
            requires ConvertibleTo<const I2&, I>
            constexpr counted_iterator(const counted_iterator<I2>& x);

        template<class I2>
            requires Assignable<I&, const I2&>
            constexpr counted_iterator& operator=(const counted_iterator<I2>& x);

        constexpr I base() const;
        constexpr iter_difference_t<I> count() const noexcept;
        constexpr decltype(auto) operator*();
        constexpr decltype(auto) operator*() const
            requires dereferenceable<const I>;

        constexpr counted_iterator& operator++();
        decltype(auto) operator++(int);
        constexpr counted_iterator operator++(int)
            requires ForwardIterator<I>;
        constexpr counted_iterator& operator--();
        requires BidirectionalIterator<I>;
        constexpr counted_iterator operator--(int)
            requires BidirectionalIterator<I>;

        constexpr counted_iterator operator+(iter_difference_t<I> n) const
            requires RandomAccessIterator<I>;
        friend constexpr counted_iterator operator+(
            iter_difference_t<I> n, const counted_iterator& x)
            requires RandomAccessIterator<I>;
        constexpr counted_iterator& operator+=(iter_difference_t<I> n)
            requires RandomAccessIterator<I>;

        constexpr counted_iterator operator-(iter_difference_t<I> n) const
            requires RandomAccessIterator<I>;
        template<Common<I> I2>
            friend constexpr iter_difference_t<I2> operator-(
                const counted_iterator& x, const counted_iterator<I2>& y);
        friend constexpr iter_difference_t<I> operator-(
            const counted_iterator& x, default_sentinel_t);
        friend constexpr iter_difference_t<I> operator-(
            default_sentinel_t, const counted_iterator& y);
        constexpr counted_iterator& operator-=(iter_difference_t<I> n)
            requires RandomAccessIterator<I>;

        constexpr decltype(auto) operator[](iter_difference_t<I> n) const
            requires RandomAccessIterator<I>;

        template<Common<I> I2>
            friend constexpr bool operator==(
                const counted_iterator& x, const counted_iterator<I2>& y);
        friend constexpr bool operator==(
            const counted_iterator& x, default_sentinel_t);
        friend constexpr bool operator==(
            default_sentinel_t, const counted_iterator& x);
    };

```

```

template<Common<I> I2>
    friend constexpr bool operator!=(
        const counted_iterator& x, const counted_iterator<I2>& y);
friend constexpr bool operator!=(
    const counted_iterator& x, default_sentinel_t y);
friend constexpr bool operator!=(
    default_sentinel_t x, const counted_iterator& y);

template<Common<I> I2>
    friend constexpr bool operator<(
        const counted_iterator& x, const counted_iterator<I2>& y);
template<Common<I> I2>
    friend constexpr bool operator>(
        const counted_iterator& x, const counted_iterator<I2>& y);
template<Common<I> I2>
    friend constexpr bool operator<=(
        const counted_iterator& x, const counted_iterator<I2>& y);
template<Common<I> I2>
    friend constexpr bool operator>=(
        const counted_iterator& x, const counted_iterator<I2>& y);

friend constexpr iter_rvalue_reference_t<I> iter_move(const counted_iterator& i)
    noexcept(noexcept(ranges::iter_move(i.current)))
    requires InputIterator<I>;
template<IndirectlySwappable<I> I2>
    friend constexpr void iter_swap(const counted_iterator& x, const counted_iterator<I2>& y)
        noexcept(noexcept(ranges::iter_swap(x.current, y.current)));

private:
    I current = I(); // exposition only
    iter_difference_t<I> length = 0; // exposition only
};

template<class I>
struct incrementable_traits<counted_iterator<I>> {
    using difference_type = iter_difference_t<I>;
};

template<InputIterator I>
struct iterator_traits<counted_iterator<I>> : iterator_traits<I> {
    using pointer = void;
};
}

```

22.5.6.2 Constructors and conversions

[counted.iter.const]

```
constexpr counted_iterator(I i, iter_difference_t<I> n);
```

1 *Expects:* $n \geq 0$.

2 *Effects:* Initializes `current` with `i` and `length` with `n`.

```

template<class I2>
    requires ConvertibleTo<const I2&, I>
    constexpr counted_iterator(const counted_iterator<I2>& x);

```

3 *Effects:* Initializes `current` with `x.current` and `length` with `x.length`.

```

template<class I2>
    requires Assignable<I&, const I2>
    constexpr counted_iterator& operator=(const counted_iterator<I2>& x);

```

4 *Effects:* Assigns `x.current` to `current` and `x.length` to `length`.

5 *Returns:* `*this`.

22.5.6.3 Accessors

[counted.iter.access]

```
constexpr I base() const;
1     Effects: Equivalent to: return current;

constexpr iter_difference_t<I> count() const noexcept;
2     Effects: Equivalent to: return length;
```

22.5.6.4 Element access

[counted.iter.elem]

```
constexpr decltype(auto) operator*();
constexpr decltype(auto) operator*() const
    requires dereferenceable<const I>;
1     Effects: Equivalent to: return *current;

constexpr decltype(auto) operator[](iter_difference_t<I> n) const
    requires RandomAccessIterator<I>;
2     Expects: n < length.
3     Effects: Equivalent to: return current[n];
```

22.5.6.5 Navigation

[counted.iter.nav]

```
constexpr counted_iterator& operator++();
1     Expects: length > 0.
2     Effects: Equivalent to:
        ++current;
        --length;
        return *this;

decltype(auto) operator++(int);
3     Expects: length > 0.
4     Effects: Equivalent to:
        --length;
        try { return current++; }
        catch(...) { ++length; throw; }

constexpr counted_iterator operator++(int)
    requires ForwardIterator<I>;
5     Effects: Equivalent to:
        counted_iterator tmp = *this;
        ++*this;
        return tmp;

constexpr counted_iterator& operator--();
    requires BidirectionalIterator<I>
6     Effects: Equivalent to:
        --current;
        ++length;
        return *this;

constexpr counted_iterator operator--(int)
    requires BidirectionalIterator<I>;
7     Effects: Equivalent to:
        counted_iterator tmp = *this;
        --*this;
        return tmp;
```

```

constexpr counted_iterator operator+(iter_difference_t<I> n) const
    requires RandomAccessIterator<I>;
8     Effects: Equivalent to: return counted_iterator(current + n, length - n);

friend constexpr counted_iterator operator+(
    iter_difference_t<I> n, const counted_iterator& x)
    requires RandomAccessIterator<I>;
9     Effects: Equivalent to: return x + n;

constexpr counted_iterator& operator+=(iter_difference_t<I> n)
    requires RandomAccessIterator<I>;
10    Expects: n <= length.
11    Effects: Equivalent to:
        current += n;
        length -= n;
        return *this;

constexpr counted_iterator operator-(iter_difference_t<I> n) const
    requires RandomAccessIterator<I>;
12    Effects: Equivalent to: return counted_iterator(current - n, length + n);

template<Common<I> I2>
    friend constexpr iter_difference_t<I2> operator-(
        const counted_iterator& x, const counted_iterator<I2>& y);
13    Expects: x and y shall refer to elements of the same sequence (22.5.6.1).
14    Effects: Equivalent to: return y.length - x.length;

friend constexpr iter_difference_t<I> operator-(
    const counted_iterator& x, default_sentinel_t);
15    Effects: Equivalent to: return -x.length;

friend constexpr iter_difference_t<I> operator-(
    default_sentinel_t, const counted_iterator& y);
16    Effects: Equivalent to: return y.length;

constexpr counted_iterator& operator--=(iter_difference_t<I> n)
    requires RandomAccessIterator<I>;
17    Expects: -n <= length.
18    Effects: Equivalent to:
        current -= n;
        length += n;
        return *this;

```

22.5.6.6 Comparisons

[counted.iter.cmp]

```

template<Common<I> I2>
    friend constexpr bool operator==(
        const counted_iterator& x, const counted_iterator<I2>& y);
1     Expects: x and y shall refer to elements of the same sequence (22.5.6.1).
2     Effects: Equivalent to: return x.length == y.length;

friend constexpr bool operator==(
    const counted_iterator& x, default_sentinel_t);
friend constexpr bool operator==(
    default_sentinel_t, const counted_iterator& x);
3     Effects: Equivalent to: return x.length == 0;

```

```

template<Common<I> I2>
    friend constexpr bool operator!=(
        const counted_iterator& x, const counted_iterator<I2>& y);
friend constexpr bool operator!=(
    const counted_iterator& x, default_sentinel_t y);
friend constexpr bool operator!=(
    default_sentinel_t x, const counted_iterator& y);
4     Effects: Equivalent to: return !(x == y);

template<Common<I> I2>
    friend constexpr bool operator<(
        const counted_iterator& x, const counted_iterator<I2>& y);
5     Expects: x and y shall refer to elements of the same sequence (22.5.6.1).
6     Effects: Equivalent to: return y.length < x.length;
7     [Note: The argument order in the Effects element is reversed because length counts down, not up.
    — end note]

template<Common<I> I2>
    friend constexpr bool operator>(
        const counted_iterator& x, const counted_iterator<I2>& y);
8     Effects: Equivalent to: return y < x;

template<Common<I> I2>
    friend constexpr bool operator<=(
        const counted_iterator& x, const counted_iterator<I2>& y);
9     Effects: Equivalent to: return !(y < x);

template<Common<I> I2>
    friend constexpr bool operator>=(
        const counted_iterator& x, const counted_iterator<I2>& y);
10    Effects: Equivalent to: return !(x < y);

```

22.5.6.7 Customizations

[counted.iter.cust]

```

friend constexpr iter_rvalue_reference_t<I> iter_move(const counted_iterator& i)
    noexcept(noexcept(ranges::iter_move(i.current)))
    requires InputIterator<I>;
1     Effects: Equivalent to: return ranges::iter_move(i.current);

template<IndirectlySwappable<I> I2>
    friend constexpr void iter_swap(const counted_iterator& x, const counted_iterator<I2>& y)
        noexcept(noexcept(ranges::iter_swap(x.current, y.current)));
2     Effects: Equivalent to ranges::iter_swap(x.current, y.current).

```

22.5.7 Unreachable sentinel

[unreachable.sentinel]

22.5.7.1 Class unreachable_sentinel_t

[unreachable.sentinel]

[Editor’s note: This wording integrates the PR for [stl2#507](#))]

- 1 Class `unreachable_sentinel_t` can be used with any `WeaklyIncrementable` type to denote the “upper bound” of an open interval.
- 2 [*Example:*

```

char* p;
// set p to point to a character buffer containing newlines
char* nl = find(p, unreachable_sentinel, '\n');

```

Provided a newline character really exists in the buffer, the use of `unreachable_sentinel` above potentially makes the call to `find` more efficient since the loop test against the sentinel does not require a conditional branch. — end example]

```

namespace std {
    struct unreachable_sentinel_t {
        template<WeaklyIncrementable I>
            friend constexpr bool operator==(unreachable_sentinel_t, const I&) noexcept;
        template<WeaklyIncrementable I>
            friend constexpr bool operator==(const I&, unreachable_sentinel_t) noexcept;
        template<WeaklyIncrementable I>
            friend constexpr bool operator!=(unreachable_sentinel_t, const I&) noexcept;
        template<WeaklyIncrementable I>
            friend constexpr bool operator!=(const I&, unreachable_sentinel_t) noexcept;
    };
}

```

22.5.7.2 Comparisons

[unreachable.sentinel.cmp]

```

template<WeaklyIncrementable I>
    friend constexpr bool operator==(unreachable_sentinel_t, const I&) noexcept;
template<WeaklyIncrementable I>
    friend constexpr bool operator==(const I&, unreachable_sentinel_t) noexcept;

```

¹ *Returns:* false.

```

template<WeaklyIncrementable I>
    friend constexpr bool operator!=(unreachable_sentinel_t, const I&) noexcept;
template<WeaklyIncrementable I>
    friend constexpr bool operator!=(const I&, unreachable_sentinel_t) noexcept;

```

² *Returns:* true.

22.6 Stream iterators

[stream.iterators]

[...]

22.6.1 Class template istream_iterator

[istream.iterator]

[...]

```

namespace std {
    template<class T, class charT = char, class traits = char_traits<charT>,
             class Distance = ptrdiff_t>
    class istream_iterator {
    public:
        [...]

        constexpr istream_iterator();
        constexpr istream_iterator(default_sentinel_t);
        istream_iterator(istream_type& s);

        [...]

        istream_iterator operator++(int);

        friend bool operator==(const istream_iterator& i, default_sentinel_t);
        friend bool operator==(default_sentinel_t, const istream_iterator& i);
        friend bool operator!=(const istream_iterator& x, default_sentinel_t y);
        friend bool operator!=(default_sentinel_t x, const istream_iterator& y);

    private:
        [...]
    };

    [...]
}

```

22.6.1.1 istream_iterator constructors and destructor

[istream.iterator.cons]

```
constexpr istream_iterator();
```

```
constexpr istream_iterator(default_sentinel_t);
```

- 1 *Effects:* Constructs the end-of-stream iterator. If `is_trivially_default_constructible_v<T>` is true, then ~~this constructor is at~~ these constructors are constexpr constructors.

[Editor's note: If P0738 "I Stream, You Stream, We All Stream for `istream_iterator`" lands at the same time as this proposal, the recommended merge for this sentence is "If `is_trivially_default_constructible_v<T>` is true, then these constructors are constexpr constructors."]

- 2 *Ensures:* `in_stream == 0`.

[...]

22.6.1.2 istream_iterator operations

[istream.iterator.ops]

[...]

```
template<class T, class charT, class traits, class Distance>
    bool operator==(const istream_iterator<T,charT,traits,Distance>& x,
                    const istream_iterator<T,charT,traits,Distance>& y);
```

- 8 *Returns:* `x.in_stream == y.in_stream`.

```
friend bool operator==(default_sentinel_t, const istream_iterator& i);
friend bool operator==(const istream_iterator& i, default_sentinel_t);
```

- 9 *Returns:* `!i.in_stream`.

```
template<class T, class charT, class traits, class Distance>
    bool operator!=(const istream_iterator<T,charT,traits,Distance>& x,
                    const istream_iterator<T,charT,traits,Distance>& y);
friend bool operator!=(default_sentinel_t x, const istream_iterator& y);
friend bool operator!=(const istream_iterator& x, default_sentinel_t y);
```

- 10 *Returns:* `!(x == y)`

22.6.2 Class template ostream_iterator

[ostream.iterator]

[...]

- 2 `ostream_iterator` is defined as:

```
namespace std {
    template<class T, class charT = char, class traits = char_traits<charT>>
    class ostream_iterator {
    public:
        using iterator_category = output_iterator_tag;
        using value_type        = void;
        using difference_type    = void_ptrdiff_t;
        using pointer            = void;
        using reference           = void;
        using char_type          = charT;
        using traits_type        = traits;
        using ostream_type       = basic_ostream<charT,traits>;

        constexpr ostream_iterator() noexcept = default;
        ostream_iterator(ostream_type& s);

        [...]

    private:
        basic_ostream<charT,traits>* out_stream = nullptr; // exposition only
        const charT* delim = nullptr;                // exposition only
    };
}
```

[...]

22.6.3 Class template istreambuf_iterator

[istreambuf.iterator]

[...]

```

namespace std {
    template<class charT, class traits = char_traits<charT>>
    class istreambuf_iterator {
    public:
        [...]

        constexpr istreambuf_iterator() noexcept;
        constexpr istreambuf_iterator(default_sentinel_t) noexcept;
        istreambuf_iterator(const istreambuf_iterator&) noexcept = default;

        [...]

        bool equal(const istreambuf_iterator& b) const;

        friend bool operator==(default_sentinel_t s, const istreambuf_iterator& i);
        friend bool operator==(const istreambuf_iterator& i, default_sentinel_t s);
        friend bool operator!=(default_sentinel_t a, const istreambuf_iterator& b);
        friend bool operator!=(const istreambuf_iterator& a, default_sentinel_t b);

    private:
        streambuf_type* sbuf_;                // exposition only
    };

    [...]
}

```

22.6.3.2 istreambuf_iterator constructors

[istreambuf.iterator.cons]

[...]

```

constexpr istreambuf_iterator() noexcept;
constexpr istreambuf_iterator(default_sentinel_t) noexcept;

```

2 *Effects:* Initializes sbuf_ with nullptr.

[...]

22.6.3.3 istreambuf_iterator operations

[istreambuf.iterator.ops]

[...]

```

template<class charT, class traits>
    bool operator==(const istreambuf_iterator<charT,traits>& a,
                    const istreambuf_iterator<charT,traits>& b);

```

6 *Returns:* a.equal(b).

```

friend bool operator==(default_sentinel_t s, const istreambuf_iterator& i);
friend bool operator==(const istreambuf_iterator& i, default_sentinel_t s);

```

7 *Returns:* i.equal(s).

```

template<class charT, class traits>
    bool operator!=(const istreambuf_iterator<charT,traits>& a,
                    const istreambuf_iterator<charT,traits>& b);
friend bool operator!=(default_sentinel_t a, const istreambuf_iterator& b);
friend bool operator!=(const istreambuf_iterator& a, default_sentinel_t b);

```

8 *Returns:* ~~a.equal(b)~~!(a == b).

22.6.4 Class template ostreambuf_iterator

[ostreambuf.iterator]

```

namespace std {
    template<class charT, class traits = char_traits<charT>>
    class ostreambuf_iterator {
    public:
        using iterator_category = output_iterator_tag;
        using value_type        = void;

```

```

using difference_type    = voidptrdiff_t;
using pointer            = void;
using reference           = void;
using char_type          = charT;
using traits_type        = traits;
using streambuf_type     = basic_streambuf<charT,traits>;
using ostream_type      = basic_ostream<charT,traits>;

constexpr ostreambuf_iterator() noexcept = default;

[...]

private:
    streambuf_type* sbuf_ = nullptr;    // exposition only
};
}

```

[Editor's note: Add a new clause between [iterators] and [algorithms] with the following content:]

23 Ranges library

[range]

23.1 General

[range.general]

- ¹ This clause describes components for dealing with ranges of elements.
- ² The following subclauses describe range and view requirements, and components for range primitives as summarized in Table 75.

Table 75 — Ranges library summary

Subclause	Header(s)
23.3 Range access	<ranges>
23.4 Range primitives	
23.5 Requirements	
23.6 Range utilities	
23.7 Range adaptors	

- ³ Several places in this clause use the expression *DECAY_COPY(x)* with semantics as defined in [31.2.6](#).

23.2 Header <ranges> synopsis

[ranges.syn]

```
#include <initializer_list>
#include <iterator>

namespace std::ranges {
    inline namespace unspecified {
        // 23.3, range access
        inline constexpr unspecified begin = unspecified;
        inline constexpr unspecified end = unspecified;
        inline constexpr unspecified cbegin = unspecified;
        inline constexpr unspecified cend = unspecified;
        inline constexpr unspecified rbegin = unspecified;
        inline constexpr unspecified rend = unspecified;
        inline constexpr unspecified crbegin = unspecified;
        inline constexpr unspecified crend = unspecified;

        // 23.4, range primitives
        inline constexpr unspecified size = unspecified;
        inline constexpr unspecified empty = unspecified;
        inline constexpr unspecified data = unspecified;
        inline constexpr unspecified cdata = unspecified;
    }

    // 23.5.2, Range
    template<class T>
        using iterator_t = decltype(ranges::begin(declval<T>()));

    template<class T>
        using sentinel_t = decltype(ranges::end(declval<T>()));

    template<forwarding-range R>
        using safe_iterator_t = iterator_t<R>;

    template<class T>
        concept Range = see below;

    // 23.5.3, SizedRange
    template<class>
        inline constexpr bool disable_sized_range = false;
```

```

template<class T>
    concept SizedRange = see below;

// 23.5.4, View
template<class T>
    inline constexpr bool enable_view = see below;

struct view_base { };

template<class T>
    concept View = see below;

// 23.5.5, common range refinements
template<class R, class T>
    concept OutputRange = see below;

template<class T>
    concept InputRange = see below;

template<class T>
    concept ForwardRange = see below;

template<class T>
    concept BidirectionalRange = see below;

template<class T>
    concept RandomAccessRange = see below;

template<class T>
    concept ContiguousRange = see below;

template<class T>
    concept CommonRange = see below;

template<class T>
    concept ViewableRange = see below;

// 23.6.2, class template view_interface
template<class D>
    requires is_class_v<D>
class view_interface;

// 23.6.3, sub-ranges
enum class subrange_kind : bool { unsized, sized };

template<Iterator I, Sentinel<I> S = I, subrange_kind K = see below>
    requires (K == subrange_kind::sized || !SizedSentinel<S, I>)
class subrange;

template<forwarding-range R>
    using safe_subrange_t = subrange<iterator_t<R>>;

// 23.7.3, all view
namespace view { inline constexpr unspecified all = unspecified; }

template<ViewableRange R>
    using all_view = decltype(view::all(declval<R>()));

// 23.7.4, filter view
template<InputRange V, IndirectUnaryPredicate<iterator_t<V>> Pred>
    requires View<V>
class filter_view;

namespace view { inline constexpr unspecified filter = unspecified; }

```

```

// 23.7.5, transform view
template<InputRange V, CopyConstructible F>
    requires View<V> && is_object_v<F> &&
        RegularInvocable<F&, iter_reference_t<iterator_t<V>>>
class transform_view;

namespace view { inline constexpr unspecified transform = unspecified; }

// 23.7.6, iota view
template<WeaklyIncrementable W, Semiregular Bound = unreachable_sentinel_t>
    requires weakly-equality-comparable-with<W, Bound>
class iota_view;

namespace view { inline constexpr unspecified iota = unspecified; }

// 23.7.7, take view
template<View> class take_view;

namespace view { inline constexpr unspecified take = unspecified; }

// 23.7.8, join view
template<InputRange V>
    requires View<V> && InputRange<iter_reference_t<iterator_t<V>>> &&
        (is_reference_v<iter_reference_t<iterator_t<V>>> ||
         View<iter_value_t<iterator_t<V>>>)
class join_view;

namespace view { inline constexpr unspecified join = unspecified; }

// 23.7.9, empty view
template<class T>
    requires is_object_v<T>
class empty_view;

namespace view {
    template<class T>
        inline constexpr empty_view<T> empty {};
}

// 23.7.10, single view
template<CopyConstructible T>
    requires is_object_v<T>
class single_view;

namespace view { inline constexpr unspecified single = unspecified; }

// 23.7.11, split view
template<class R>
    concept tiny-range = see below; // exposition only

template<InputRange V, ForwardRange Pattern>
    requires View<V> && View<Pattern> &&
        IndirectlyComparable<iterator_t<V>, iterator_t<Pattern>, ranges::equal_to<>> &&
        (ForwardRange<V> || tiny-range<Pattern>)
class split_view;

namespace view { inline constexpr unspecified split = unspecified; }

// 23.7.12, counted view
namespace view { inline constexpr unspecified counted = unspecified; }

```

```

// 23.7.13, common view
template<View V>
    requires (!CommonRange<V>)
class common_view;

namespace view { inline constexpr unspecified common = unspecified; }

// 23.7.14, reverse view
template<View V>
    requires BidirectionalRange<V>
class reverse_view;

namespace view { inline constexpr unspecified reverse = unspecified; }
}

namespace std {
    namespace view = ranges::view;

    template<class I, class S, ranges::subrange_kind K>
    struct tuple_size<ranges::subrange<I, S, K>>
        : integral_constant<size_t, 2> {};
    template<class I, class S, ranges::subrange_kind K>
    struct tuple_element<0, ranges::subrange<I, S, K>> {
        using type = I;
    };
    template<class I, class S, ranges::subrange_kind K>
    struct tuple_element<1, ranges::subrange<I, S, K>> {
        using type = S;
    };
}

```

23.3 Range access

[range.access]

[Editor's note: This wording integrates the PR for [stl2#547](#).]

- ¹ In addition to being available via inclusion of the `<ranges>` header, the customization point objects in 23.3 are available when `<iterator>` is included.

23.3.1 `ranges::begin`

[range.access.begin]

- ¹ The name `ranges::begin` denotes a customization point object ([customization.point.object]). The expression `ranges::begin(E)` for some subexpression `E` is expression-equivalent to:

- (1.1) — `E + 0` if `E` is an lvalue of array type ([basic.compound]).
- (1.2) — Otherwise, if `E` is an lvalue, `DECAY_COPY(E.begin())` if it is a valid expression and its type `I` models `Iterator`.
- (1.3) — Otherwise, `DECAY_COPY(begin(E))` if it is a valid expression and its type `I` models `Iterator` with overload resolution performed in a context that includes the declarations:

```

template<class T> void begin(T&&) = delete;
template<class T> void begin(initializer_list<T>&&) = delete;

```

and does not include a declaration of `ranges::begin`.

- (1.4) — Otherwise, `ranges::begin(E)` is ill-formed. [Note: This case can result in substitution failure when `ranges::begin(E)` appears in the immediate context of a template instantiation. — end note]
- ² [Note: Whenever `ranges::begin(E)` is a valid expression, its type models `Iterator`. — end note]

23.3.2 `ranges::end`

[range.access.end]

- ¹ The name `ranges::end` denotes a customization point object ([customization.point.object]). The expression `ranges::end(E)` for some subexpression `E` is expression-equivalent to:

- (1.1) — `E + extent_v<T>` if `E` is an lvalue of array type ([basic.compound]) `T`.

- (1.2) — Otherwise, if `E` is an lvalue, `DECAY_COPY(E.end())` if it is a valid expression and its type `S` models `Sentinel<decltype(ranges::begin(E))>`.
- (1.3) — Otherwise, `DECAY_COPY(end(E))` if it is a valid expression and its type `S` models `Sentinel<decltype(ranges::begin(E))>` with overload resolution performed in a context that includes the declarations:

```
template<class T> void end(T&&) = delete;
template<class T> void end(initializer_list<T>&&) = delete;
```

and does not include a declaration of `ranges::end`.

- (1.4) — Otherwise, `ranges::end(E)` is ill-formed. [Note: This case can result in substitution failure when `ranges::end(E)` appears in the immediate context of a template instantiation. — end note]
- 2 [Note: Whenever `ranges::end(E)` is a valid expression, the types `S` and `I` of `ranges::end(E)` and `ranges::begin(E)` model `Sentinel<S, I>`. — end note]

23.3.3 `ranges::cbegin` [range.access.cbegin]

- 1 The name `ranges::cbegin` denotes a customization point object ([customization.point.object]). The expression `ranges::cbegin(E)` for some subexpression `E` of type `T` is expression-equivalent to:
 - (1.1) — `ranges::begin(static_cast<const T&>(E))` if `E` is an lvalue.
 - (1.2) — Otherwise, `ranges::begin(static_cast<const T&&>(E))`.
- 2 [Note: Whenever `ranges::cbegin(E)` is a valid expression, its type models `Iterator`. — end note]

23.3.4 `ranges::cend` [range.access.cend]

- 1 The name `ranges::cend` denotes a customization point object ([customization.point.object]). The expression `ranges::cend(E)` for some subexpression `E` of type `T` is expression-equivalent to:
 - (1.1) — `ranges::end(static_cast<const T&>(E))` if `E` is an lvalue.
 - (1.2) — Otherwise, `ranges::end(static_cast<const T&&>(E))`.
- 2 [Note: Whenever `ranges::cend(E)` is a valid expression, the types `S` and `I` of `ranges::cend(E)` and `ranges::cbegin(E)` model `Sentinel<S, I>`. — end note]

23.3.5 `ranges::rbegin` [range.access.rbegin]

- 1 The name `ranges::rbegin` denotes a customization point object ([customization.point.object]). The expression `ranges::rbegin(E)` for some subexpression `E` is expression-equivalent to:
 - (1.1) — If `E` is an lvalue, `DECAY_COPY(E.rbegin())` if it is a valid expression and its type `I` models `Iterator`.
 - (1.2) — Otherwise, `DECAY_COPY(rbegin(E))` if it is a valid expression and its type `I` models `Iterator` with overload resolution performed in a context that includes the declaration:


```
template<class T> void rbegin(T&&) = delete;
```

and does not include a declaration of `ranges::rbegin`.

- (1.3) — Otherwise, `make_reverse_iterator(ranges::end(E))` if both `ranges::begin(E)` and `ranges::end(E)` are valid expressions of the same type `I` which models `BidirectionalIterator` (22.3.4.12).
- (1.4) — Otherwise, `ranges::rbegin(E)` is ill-formed. [Note: This case can result in substitution failure when `ranges::rbegin(E)` appears in the immediate context of a template instantiation. — end note]
- 2 [Note: Whenever `ranges::rbegin(E)` is a valid expression, its type models `Iterator`. — end note]

23.3.6 `ranges::rend` [range.access.rend]

- 1 The name `ranges::rend` denotes a customization point object ([customization.point.object]). The expression `ranges::rend(E)` for some subexpression `E` is expression-equivalent to:
 - (1.1) — If `E` is an lvalue, `DECAY_COPY(E.rend())` if it is a valid expression and its type `S` models `Sentinel<decltype(ranges::rbegin(E))>`.
 - (1.2) — Otherwise, `DECAY_COPY(rend(E))` if it is a valid expression and its type `S` models `Sentinel<decltype(ranges::rbegin(E))>` with overload resolution performed in a context that includes the declaration:


```
template<class T> void rend(T&&) = delete;
```

and does not include a declaration of `ranges::rend`.

- (1.3) — Otherwise, `make_reverse_iterator(ranges::begin(E))` if both `ranges::begin(E)` and `ranges::end(E)` are valid expressions of the same type `I` which models `BidirectionalIterator` (22.3.4.12).
- (1.4) — Otherwise, `ranges::rend(E)` is ill-formed. [Note: This case can result in substitution failure when `ranges::rend(E)` appears in the immediate context of a template instantiation. — end note]
- ² [Note: Whenever `ranges::rend(E)` is a valid expression, the types `S` and `I` of `ranges::rend(E)` and `ranges::rbegin(E)` model `Sentinel<S, I>`. — end note]

23.3.7 `ranges::crbegin`

[range.access.crbegin]

- ¹ The name `ranges::crbegin` denotes a customization point object ([customization.point.object]). The expression `ranges::crbegin(E)` for some subexpression `E` of type `T` is expression-equivalent to:
 - (1.1) — `ranges::rbegin(static_cast<const T&>(E))` if `E` is an lvalue.
 - (1.2) — Otherwise, `ranges::rbegin(static_cast<const T&&>(E))`.
- ² [Note: Whenever `ranges::crbegin(E)` is a valid expression, its type models `Iterator`. — end note]

23.3.8 `ranges::crend`

[range.access.crend]

- ¹ The name `ranges::crend` denotes a customization point object ([customization.point.object]). The expression `ranges::crend(E)` for some subexpression `E` of type `T` is expression-equivalent to:
 - (1.1) — `ranges::rend(static_cast<const T&>(E))` if `E` is an lvalue.
 - (1.2) — Otherwise, `ranges::rend(static_cast<const T&&>(E))`.
- ² [Note: Whenever `ranges::crend(E)` is a valid expression, the types `S` and `I` of `ranges::crend(E)` and `ranges::crbegin(E)` model `Sentinel<S, I>`. — end note]

23.4 Range primitives

[range.primitives]

- ¹ In addition to being available via inclusion of the `<ranges>` header, the customization point objects in 23.4 are available when `<iterator>` is included.

23.4.1 `size`

[range.primitives.size]

- ¹ The name `size` denotes a customization point object ([customization.point.object]). The expression `ranges::size(E)` for some subexpression `E` with type `T` is expression-equivalent to:
 - (1.1) — `DECAY_COPY(extent_v<T>)` if `T` is an array type ([basic.compound]).
 - (1.2) — Otherwise, if `disable_sized_range<remove_cv_t<T>>` (23.5.3) is false:
 - (1.2.1) — `DECAY_COPY(E.size())` if it is a valid expression and its type `I` models `Integral`.
 - (1.2.2) — Otherwise, `DECAY_COPY(size(E))` if it is a valid expression and its type `I` models `Integral` with overload resolution performed in a context that includes the declaration:

```
template<class T> void size(T&&) = delete;
```

and does not include a declaration of `ranges::size`.

- (1.3) — Otherwise, `(ranges::end(E) - ranges::begin(E))` if it is a valid expression and the types `I` and `S` of `ranges::begin(E)` and `ranges::end(E)` model `SizedSentinel<S, I>` (22.3.4.8) and `ForwardIterator<I>`. However, `E` is evaluated only once.
- (1.4) — Otherwise, `ranges::size(E)` is ill-formed. [Note: This case can result in substitution failure when `ranges::size(E)` appears in the immediate context of a template instantiation. — end note]
- ² [Note: Whenever `ranges::size(E)` is a valid expression, its type models `Integral`. — end note]

23.4.2 `empty`

[range.primitives.empty]

- ¹ The name `empty` denotes a customization point object ([customization.point.object]). The expression `ranges::empty(E)` for some subexpression `E` is expression-equivalent to:
 - (1.1) — `bool((E).empty())` if it is a valid expression.
 - (1.2) — Otherwise, `(ranges::size(E) == 0)` if it is a valid expression.

- (1.3) — Otherwise, EQ, where EQ is `bool(ranges::begin(E) == ranges::end(E))` except that E is only evaluated once, if EQ is a valid expression and the type of `ranges::begin(E)` models `ForwardIterator`.
- (1.4) — Otherwise, `ranges::empty(E)` is ill-formed. [Note: This case can result in substitution failure when `ranges::empty(E)` appears in the immediate context of a template instantiation. — end note]
- ² [Note: Whenever `ranges::empty(E)` is a valid expression, it has type `bool`. — end note]

23.4.3 data

[range.primitives.data]

- ¹ The name `data` denotes a customization point object ([customization.point.object]). The expression `ranges::data(E)` for some subexpression E is expression-equivalent to:
 - (1.1) — If E is an lvalue, `DECAY_COPY(E.data())` if it is a valid expression of pointer to object type.
 - (1.2) — Otherwise, if `ranges::begin(E)` is a valid expression whose type models `ContiguousIterator`,

`ranges::begin(E) == ranges::end(E) ? nullptr : addressof(*ranges::begin(E))`

except that E is evaluated only once.
- (1.3) — Otherwise, `ranges::data(E)` is ill-formed. [Note: This case can result in substitution failure when `ranges::data(E)` appears in the immediate context of a template instantiation. — end note]
- ² [Note: Whenever `ranges::data(E)` is a valid expression, it has pointer to object type. — end note]

23.4.4 cdata

[range.primitives.cdata]

- ¹ The name `cdata` denotes a customization point object ([customization.point.object]). The expression `ranges::cdata(E)` for some subexpression E of type T is expression-equivalent to:
 - (1.1) — `ranges::data(static_cast<const T&>(E))` if E is an lvalue.
 - (1.2) — Otherwise, `ranges::data(static_cast<const T&&>(E))`.
- ² [Note: Whenever `ranges::cdata(E)` is a valid expression, it has pointer to object type. — end note]

23.5 Range requirements

[range.requirements]

23.5.1 General

[range.requirements.general]

- ¹ Ranges are an abstraction that allow a C++ program to operate on elements of data structures uniformly. Calling `ranges::begin` on a range returns an object whose type models `Iterator` (22.3.4.6). Calling `ranges::end` on a range returns an object whose type S, together with the type I of the object returned by `ranges::begin`, models `Sentinel<S, I>`. The library formalizes the interfaces, semantics, and complexity of ranges to enable algorithms and range adaptors that work efficiently on different types of sequences.
- ² The `Range` concept requires that `ranges::begin` and `ranges::end` return an iterator and a sentinel, respectively. The `SizedRange` concept refines `Range` with the requirement that the number of elements in the range can be determined in constant time using the `ranges::size` function. The `View` concept specifies requirements on a `Range` type with constant-time copy and assign operations.
- ³ Several refinements of `Range` group requirements that arise frequently in concepts and algorithms. Common ranges are ranges for which `ranges::begin` and `ranges::end` return objects of the same type. Random access ranges are ranges for which `ranges::begin` returns a type that models `RandomAccessIterator` (22.3.4.13). (Contiguous, bidirectional, forward, input, and output ranges are defined similarly.) Viewable ranges can be converted to views.

23.5.2 Ranges

[range.range]

- ¹ The `Range` concept defines the requirements of a type that allows iteration over its elements by providing an iterator and sentinel that denote the elements of the range.

```
template<class T>
concept range_impl = // exposition only
    requires(T&& t) {
        ranges::begin(std::forward<T>(t)); // sometimes equality-preserving (see below)
        ranges::end(std::forward<T>(t));
    };

```

```
template<class T>
    concept Range = range-impl<T&&>;
```

```
template<class T>
    concept forwarding-range = // exposition only
    Range<T> && range-impl<T>;
```

- 2 The required expressions `ranges::begin(std::forward<T>(t))` and `ranges::end(std::forward<T>(t))` of the *range-impl* concept do not require implicit expression variations ([[concepts.equality](#)]).
- 3 Given an expression `E` such that `decltype((E))` is `T`, `T` models *range-impl* only if
- (3.1) — `[ranges::begin(E), ranges::end(E))` denotes a range (22.3.1),
- (3.2) — both `ranges::begin(E)` and `ranges::end(E)` are amortized constant time and non-modifying, and
- (3.3) — if the type of `ranges::begin(E)` models `ForwardIterator`, `ranges::begin(E)` is equality-preserving.
- 4 [Note: Equality preservation of both `ranges::begin` and `ranges::end` enables passing a `Range` whose iterator type models `ForwardIterator` to multiple algorithms and making multiple passes over the range by repeated calls to `ranges::begin` and `ranges::end`. Since `ranges::begin` is not required to be equality-preserving when the return type does not model `ForwardIterator`, repeated calls might not return equal values or might not be well-defined; `ranges::begin` should be called at most once for such a range. — end note]
- 5 Given an expression `E` such that `decltype((E))` is `T` and an lvalue `t` that denotes the same object as `E`, `T` models *forwarding-range* only if
- (5.1) — `ranges::begin(E)` and `ranges::begin(t)` are expression-equivalent,
- (5.2) — `ranges::end(E)` and `ranges::end(t)` are expression-equivalent, and
- (5.3) — the validity of iterators obtained from the object denoted by `E` is not tied to the lifetime of that object.
- 6 [Note: Since the validity of iterators is not tied to the lifetime of an object whose type models *forwarding-range*, a function can accept arguments of such a type by value and return iterators obtained from it without danger of dangling. — end note]
- 7 [Example: Specializations of class template `subrange` (23.6.3) model *forwarding-range*. `subrange` provides non-member rvalue overloads of `begin` and `end` with the same semantics as its member lvalue overloads, and `subrange`’s iterators - since they are “borrowed” from some other range - do not have validity tied to the lifetime of a `subrange` object. — end example]

23.5.3 Sized ranges

[range.sized]

- 1 The `SizedRange` concept specifies the requirements of a `Range` type that knows its size in constant time with the `size` function.

```
template<class T>
    concept SizedRange =
        Range<T> &&
        !disable_sized_range<remove_cvref_t<T>> &&
        requires(T& t) { ranges::size(t); };
```

- 2 Given an lvalue `t` of type `remove_reference_t<T>`, `T` models `SizedRange` only if
- (2.1) — `ranges::size(t)` is $\mathcal{O}(1)$, does not modify `t`, and is equal to `ranges::distance(t)`, and
- (2.2) — if `iterator_t<T>` models `ForwardIterator`, `ranges::size(t)` is well-defined regardless of the evaluation of `ranges::begin(t)`. [Note: `ranges::size(t)` is otherwise not required to be well-defined after evaluating `ranges::begin(t)`. For example, `ranges::size(t)` might be well-defined for a `SizedRange` whose iterator type does not model `ForwardIterator` only if evaluated before the first call to `ranges::begin(t)`. — end note]
- 3 [Note: The complexity requirement for the evaluation of `ranges::size` is non-amortized, unlike the case for the complexity of the evaluations of `ranges::begin` and `ranges::end` in the `Range` concept. — end note]

- 4 [Note: `disable_sized_range` allows use of range types with the library that satisfy but do not in fact model `SizedRange`. — end note]

23.5.4 Views

[range.view]

- 1 The `View` concept specifies the requirements of a `Range` type that has constant time copy, move, and assignment operators; that is, the cost of these operations is not proportional to the number of elements in the `View`.

- 2 [Example: Examples of `Views` are:

- (2.1) — A `Range` type that wraps a pair of iterators.
- (2.2) — A `Range` type that holds its elements by `shared_ptr` and shares ownership with all its copies.
- (2.3) — A `Range` type that generates its elements on demand.

Most containers (Clause 21) are not views since copying the container copies the elements, which cannot be done in constant time. — end example]

```
template<class T>
    inline constexpr bool enable_view = see below;
```

```
template<class T>
    concept View =
        Range<T> && Semiregular<T> && enable_view<T>;
```

- 3 Since the difference between `Range` and `View` is largely semantic, the two are differentiated with the help of `enable_view`.

- 4 For a type `T`, the default value of `enable_view<T>` is:

- (4.1) — If `DerivedFrom<T, view_base>` is true, true.
- (4.2) — Otherwise, if `T` is a specialization of class template `initializer_list` ([support.initlist]), `set` ([set]), `multiset` ([multiset]), `unordered_set` ([unord.set]), `unordered_multiset` ([unord.multiset]), or `match_results` ([re.results]), false.
- (4.3) — Otherwise, if both `T` and `const T` model `Range` and `iter_reference_t<iterator_t<T>>` is not the same type as `iter_reference_t<iterator_t<const T>>`, false. [Note: Deep const-ness implies element ownership, whereas shallow const-ness implies reference semantics. — end note]
- (4.4) — Otherwise, true.

- 5 Pursuant to [namespace.std], users may specialize `enable_view` to true for types which model `View`, and false for types which do not.

23.5.5 Common range refinements

[range.refinements]

- 1 The `OutputRange` concept specifies requirements of a `Range` type for which `ranges::begin` returns a model of `OutputIterator` (22.3.4.10). `InputRange`, `ForwardRange`, `BidirectionalRange`, and `RandomAccessRange` are defined similarly.

```
template<class R, class T>
    concept OutputRange =
        Range<R> && OutputIterator<iterator_t<R>, T>;
```

```
template<class T>
    concept InputRange =
        Range<T> && InputIterator<iterator_t<T>>;
```

```
template<class T>
    concept ForwardRange =
        InputRange<T> && ForwardIterator<iterator_t<T>>;
```

```
template<class T>
    concept BidirectionalRange =
        ForwardRange<T> && BidirectionalIterator<iterator_t<T>>;
```

```
template<class T>
concept RandomAccessRange =
    BidirectionalRange<T> && RandomAccessIterator<iterator_t<T>>;
```

- ² ContiguousRange additionally requires that the `ranges::data` customization point (23.4.3) is usable with the range.

```
template<class T>
concept ContiguousRange =
    RandomAccessRange<T> && ContiguousIterator<iterator_t<T>> &&
    requires(T& t) {
        ranges::data(t);
        requires Same<decltype(ranges::data(t)), add_pointer_t<iter_reference_t<iterator_t<T>>>>;
    };
```

- ³ The CommonRange concept specifies requirements of a Range type for which `ranges::begin` and `ranges::end` return objects of the same type. [Example: The standard containers ([containers]) model CommonRange. — end example]

```
template<class T>
concept CommonRange =
    Range<T> && Same<iterator_t<T>, sentinel_t<T>>;
```

- ⁴ The ViewableRange concept specifies the requirements of a Range type that can be converted to a View safely.

```
template<class T>
concept ViewableRange =
    Range<T> && (forwarding-range<T> || View<decay_t<T>>);
```

23.6 Range utilities

[range.utility]

- ¹ The components in this subclause are general utilities for representing and manipulating ranges.

23.6.1 Helper concepts

[range.utility.helpers]

- ¹ Many of the types in this subclause are specified in terms of the following exposition-only concepts:

```
template<class R>
concept simple-view =
    View<R> && Range<const R> &&
    Same<iterator_t<R>, iterator_t<const R>> &&
    Same<sentinel_t<R>, sentinel_t<const R>>;

template<InputIterator I>
concept has-arrow =
    is_pointer_v<I> || requires(I i) { i.operator->(); };

template<class T, class U>
concept not-same-as =
    !Same<remove_cvref_t<T>, remove_cvref_t<U>>;
```

23.6.2 View interface

[view.interface]

- ¹ The class template `view_interface` is a helper for defining View-like types that offer a container-like interface. It is parameterized with the type that inherits from it.

```
namespace std::ranges {
    template<Range R>
    struct range-common-iterator-impl { // exposition only
        using type = common_iterator<iterator_t<R>, sentinel_t<R>>;
    };
    template<CommonRange R>
    struct range-common-iterator-impl<R> { // exposition only
        using type = iterator_t<R>;
    };
    template<Range R>
    using range-common-iterator = // exposition only
        typename range-common-iterator-impl<R>::type;
```

```

template<class D>
    requires is_class_v<D> && Same<D, remove_cv_t<D>>
class view_interface : public view_base {
private:
    constexpr D& derived() noexcept { // exposition only
        return static_cast<D&>(*this);
    }
    constexpr const D& derived() const noexcept { // exposition only
        return static_cast<const D&>(*this);
    }
public:
    constexpr bool empty() requires ForwardRange<D> {
        return ranges::begin(derived()) == ranges::end(derived());
    }
    constexpr bool empty() const requires ForwardRange<const D> {
        return ranges::begin(derived()) == ranges::end(derived());
    }

    constexpr explicit operator bool()
        requires requires { ranges::empty(derived()); } {
        return !ranges::empty(derived());
    }
    constexpr explicit operator bool() const
        requires requires { ranges::empty(derived()); } {
        return !ranges::empty(derived());
    }

    constexpr auto data() requires ContiguousIterator<iterator_t<D>> {
        return ranges::empty(derived()) ? nullptr : addressof(*ranges::begin(derived()));
    }
    constexpr auto data() const
        requires Range<const D> && ContiguousIterator<iterator_t<const D>> {
        return ranges::empty(derived()) ? nullptr : addressof(*ranges::begin(derived()));
    }

    constexpr auto size() requires ForwardRange<D> &&
        SizedSentinel<sentinel_t<D>, iterator_t<D>> {
        return ranges::end(derived()) - ranges::begin(derived());
    }
    constexpr auto size() const requires ForwardRange<const D> &&
        SizedSentinel<sentinel_t<const D>, iterator_t<const D>> {
        return ranges::end(derived()) - ranges::begin(derived());
    }

    constexpr decltype(auto) front() requires ForwardRange<D>;
    constexpr decltype(auto) front() const requires ForwardRange<const D>;

    constexpr decltype(auto) back() requires BidirectionalRange<D> && CommonRange<D>;
    constexpr decltype(auto) back() const
        requires BidirectionalRange<const D> && CommonRange<const D>;

    template<RandomAccessRange R = D>
        constexpr decltype(auto) operator[] (iter_difference_t<iterator_t<R>> n) {
            return ranges::begin(derived())[n];
        }
    template<RandomAccessRange R = const D>
        constexpr decltype(auto) operator[] (iter_difference_t<iterator_t<R>> n) const {
            return ranges::begin(derived())[n];
        }
};
}

```

- ² The template parameter `D` for `view_interface` may be an incomplete type. Before any member of the resulting specialization of `view_interface` other than special member functions is referenced, `D` shall be complete, and model both `DerivedFrom<view_interface<D>>` and `View`.

23.6.2.1 Members

[view.interface.members]

```
constexpr decltype(auto) front() requires ForwardRange<D>;
constexpr decltype(auto) front() const requires ForwardRange<const D>;

1   Expects: !empty().
2   Effects: Equivalent to: return *ranges::begin(derived());

constexpr decltype(auto) back() requires BidirectionalRange<D> && CommonRange<D>;
constexpr decltype(auto) back() const
    requires BidirectionalRange<const D> && CommonRange<const D>;

3   Expects: !empty().
4   Effects: Equivalent to: return *ranges::prev(ranges::end(derived()));
```

23.6.3 Sub-ranges

[range.subrange]

- ¹ The `subrange` class template combines together an iterator and a sentinel into a single object that models the `View` concept. Additionally, it models the `SizedRange` concept when the final template parameter is `subrange_kind::sized`.

```
namespace std::ranges {
    template<class T>
        concept pair-like = // exposition only
            !is_reference_v<T> && requires(T t) {
                typename tuple_size<T>::type; // ensures tuple_size<T> is complete
                requires DerivedFrom<tuple_size<T>, integral_constant<size_t, 2>>;
                typename tuple_element_t<0, remove_const_t<T>>;
                typename tuple_element_t<1, remove_const_t<T>>;
                { get<0>(t) } -> const tuple_element_t<0, T>&;
                { get<1>(t) } -> const tuple_element_t<1, T>&;
            };

    template<class T, class U, class V>
        concept pair-like-convertible-to = // exposition only
            !Range<T> && pair-like<remove_reference_t<T>> &&
            requires(T&& t) {
                get<0>(std::forward<T>(t));
                requires ConvertibleTo<decltype(get<0>(std::forward<T>(t))), U>;
                get<1>(std::forward<T>(t));
                requires ConvertibleTo<decltype(get<1>(std::forward<T>(t))), V>;
            };

    template<class T, class U, class V>
        concept pair-like-convertible-from = // exposition only
            !Range<T> && pair-like<T> && Constructible<T, U, V>;

    template<class T>
        concept iterator-sentinel-pair = // exposition only
            !Range<T> && pair-like<T> &&
            Sentinel<tuple_element_t<1, T>, tuple_element_t<0, T>>;

    template<Iterator I, Sentinel<I> S = I, subrange_kind K =
        SizedSentinel<S, I> ? subrange_kind::sized : subrange_kind::unsized>
        requires (K == subrange_kind::sized || !SizedSentinel<S, I>)
        class subrange : public view_interface<subrange<I, S, K>> {
        private:
            static constexpr bool StoreSize = // exposition only
                K == subrange_kind::sized && !SizedSentinel<S, I>;
            I begin_ = I(); // exposition only
            S end_ = S(); // exposition only
```

```

    iter_difference_t<I> size_ = 0; // exposition only; present only when StoreSize is true

public:
    subrange() = default;

    constexpr subrange(I i, S s) requires (!StoreSize);

    constexpr subrange(I i, S s, iter_difference_t<I> n)
        requires (K == subrange_kind::sized);

    template<not-same-as<subrange> R>
        requires forwarding-range<R> &&
            ConvertibleTo<iterator_t<R>, I> && ConvertibleTo<sentinel_t<R>, S>
    constexpr subrange(R&& r) requires (!StoreSize) || SizedRange<R>;

    template<forwarding-range R>
        requires ConvertibleTo<iterator_t<R>, I> && ConvertibleTo<sentinel_t<R>, S>
    constexpr subrange(R&& r, iter_difference_t<I> n)
        requires (K == subrange_kind::sized)
        : subrange{ranges::begin(r), ranges::end(r), n}
    {}

    template<not-same-as<subrange> PairLike>
        requires pair-like-convertible-to<PairLike, I, S>
    constexpr subrange(PairLike&& r) requires (!StoreSize)
        : subrange{std::get<0>(std::forward<PairLike>(r)),
                    std::get<1>(std::forward<PairLike>(r))}
    {}

    template<pair-like-convertible-to<I, S> PairLike>
    constexpr subrange(PairLike&& r, iter_difference_t<I> n)
        requires (K == subrange_kind::sized)
        : subrange{std::get<0>(std::forward<PairLike>(r)),
                    std::get<1>(std::forward<PairLike>(r)), n}
    {}

    template<not-same-as<subrange> PairLike>
        requires pair-like-convertible-from<PairLike, const I&, const S&>
    constexpr operator PairLike() const;

    constexpr I begin() const;
    constexpr S end() const;

    constexpr bool empty() const;
    constexpr iter_difference_t<I> size() const
        requires (K == subrange_kind::sized);

    [[nodiscard]] constexpr subrange next(iter_difference_t<I> n = 1) const;
    [[nodiscard]] constexpr subrange prev(iter_difference_t<I> n = 1) const
        requires BidirectionalIterator<I>;
    constexpr subrange& advance(iter_difference_t<I> n);

    friend constexpr I begin(subrange&& r) { return r.begin(); }
    friend constexpr S end(subrange&& r) { return r.end(); }
};

template<Iterator I, Sentinel<I> S>
    subrange(I, S, iter_difference_t<I>) -> subrange<I, S, subrange_kind::sized>;

template<iterator-sentinel-pair P>
    subrange(P) -> subrange<tuple_element_t<0, P>, tuple_element_t<1, P>>;

```

```

template<iterator-sentinel-pair P>
    subrange(P, iter_difference_t<tuple_element_t<0, P>>) ->
        subrange<tuple_element_t<0, P>, tuple_element_t<1, P>, subrange_kind::sized>;

template<forwarding-range R>
    subrange(R&&) ->
        subrange<iterator_t<R>, sentinel_t<R>,
            (SizedRange<R> || SizedSentinel<sentinel_t<R>, iterator_t<R>>)
                ? subrange_kind::sized : subrange_kind::unsized>;

template<forwarding-range R>
    subrange(R&&, iter_difference_t<iterator_t<R>>) ->
        subrange<iterator_t<R>, sentinel_t<R>, subrange_kind::sized>;

template<size_t N, class I, class S, subrange_kind K>
    requires (N < 2)
    constexpr auto get(const subrange<I, S, K>& r);
}

namespace std {
    using ranges::get;
}

```

23.6.3.1 Constructors and conversions

[range.subrange.ctor]

```
constexpr subrange(I i, S s) requires (!StoreSize);
```

1 *Effects:* Initializes `begin_` with `i` and `end_` with `s`.

```
constexpr subrange(I i, S s, iter_difference_t<I> n)
    requires (K == subrange_kind::sized);
```

2 *Expects:* `n == ranges::distance(i, s)`.

3 *Effects:* Initializes `begin_` with `i` and `end_` with `s`. If `StoreSize` is `true`, initializes `size_` with `n`.

4 [Note: Accepting the length of the range and storing it to later return from `size()` enables `subrange` to model `SizedRange` even when it stores an iterator and sentinel that do not model `SizedSentinel`. — end note]

```
template<not-same-as<subrange> R>
    requires forwarding-range<R> &&
        ConvertibleTo<iterator_t<R>, I> && ConvertibleTo<sentinel_t<R>, S>
    constexpr subrange(R&& r) requires (!StoreSize) || SizedRange<R>;
```

5 *Effects:* Equivalent to:

(5.1) — If `StoreSize` is `true`, `subrange{ranges::begin(r), ranges::end(r), ranges::size(r)}`.

(5.2) — Otherwise, `subrange{ranges::begin(r), ranges::end(r)}`.

```
template<not-same-as<subrange> PairLike>
    requires pair-like-convertible-from<PairLike, const I&, const S&>
    constexpr operator PairLike() const;
```

6 *Effects:* Equivalent to: `return PairLike(begin_, end_);`

23.6.3.2 Accessors

[range.subrange.access]

```
constexpr I begin() const;
```

1 *Effects:* Equivalent to: `return begin_;`

```
constexpr S end() const;
```

2 *Effects:* Equivalent to: `return end_;`

```
constexpr bool empty() const;
```

3 *Effects:* Equivalent to: `return begin_ == end_;`

```
constexpr iter_difference_t<I> size() const
    requires (K == subrange_kind::sized);

4     Effects:
(4.1)     — If StoreSize is true, equivalent to: return size_;
(4.2)     — Otherwise, equivalent to: return end_ - begin_;

[[nodiscard]] constexpr subrange next(iter_difference_t<I> n = 1) const;

5     Effects: Equivalent to:
        auto tmp = *this;
        tmp.advance(n);
        return tmp;

6     [Note: If I does not model ForwardIterator, next can invalidate *this. — end note]

[[nodiscard]] constexpr subrange prev(iter_difference_t<I> n = 1) const
    requires BidirectionalIterator<I>;

7     Effects: Equivalent to:
        auto tmp = *this;
        tmp.advance(-n);
        return tmp;

constexpr subrange& advance(iter_difference_t<I> n);

8     Effects: Equivalent to:
(8.1)     — If StoreSize is true,
            size_ -= n - ranges::advance(begin_, n, end_);
            return *this;
(8.2)     — Otherwise,
            ranges::advance(begin_, n, end_);
            return *this;

template<size_t N, class I, class S, subrange_kind K>
    requires (N < 2)
constexpr auto get(const subrange<I, S, K>& r);

9     Effects: Equivalent to:
        if constexpr (N == 0)
            return r.begin();
        else
            return r.end();
```

23.7 Range adaptors

[range.adaptors]

- 1 This subclause defines *range adaptors*, which are utilities that transform a **Range** into a **View** with custom behaviors. These adaptors can be chained to create pipelines of range transformations that evaluate lazily as the resulting view is iterated.
- 2 Range adaptors are declared in namespace `std::ranges::view`.
- 3 The bitwise or operator is overloaded for the purpose of creating adaptor chain pipelines. The adaptors also support function call syntax with equivalent semantics.
- 4 *[Example:*

```
vector<int> ints{0,1,2,3,4,5};
auto even = [](int i){ return 0 == i % 2; };
auto square = [](int i) { return i * i; };
for (int i : ints | view::filter(even) | view::transform(square)) {
    cout << i << ' '; // prints: 0 4 16
}
assert(ranges::equal(ints | view::filter(even), view::filter(ints, even)));
```

— end example]

23.7.1 Range adaptor objects

[range.adaptor.object]

- ¹ A *range adaptor closure object* is a unary function object that accepts a `ViewableRange` argument and returns a `View`. For a range adaptor closure object `C` and an expression `R` such that `decltype(R)` models `ViewableRange`, the following expressions are equivalent and yield a `View`:

```
C(R)
R | C
```

Given an additional range adaptor closure object `D`, the expression `C | D` is well-formed and produces another range adaptor closure object such that the following two expressions are equivalent:

```
R | C | D
R | (C | D)
```

- ² A *range adaptor object* is a customization point object ([[customization.point.object](#)]) that accepts a `ViewableRange` as its first argument and returns a `View`.
- ³ If a range adaptor object accepts only one argument, then it is a range adaptor closure object.
- ⁴ If a range adaptor object accepts more than one argument, then the following expressions are equivalent:

```
adaptor(range, args...)
adaptor(args...)(range)
range | adaptor(args...)
```

In this case, `adaptor(args...)` is a range adaptor closure object.

23.7.2 Semiregular wrapper

[range.semi.wrap]

- ¹ Many of the types in this subclause are specified in terms of an exposition-only class template `semiregular`. `semiregular<T>` behaves exactly like `optional<T>` with the following differences:
- (1.1) — `semiregular<T>` constrains its type parameter `T` with `CopyConstructible<T> && is_object_v<T>`.
- (1.2) — If `T` models `DefaultConstructible`, the default constructor of `semiregular<T>` is equivalent to:
- ```
constexpr semiregular() noexcept(is_nothrow_default_constructible_v<T>)
 : semiregular{in_place}
{ }
```
- (1.3) — If `Assignable<T&, const T&>` is not satisfied, the copy assignment operator is equivalent to:
- ```
semiregular& operator=(const semiregular& that)
    noexcept(is_nothrow_copy_constructible_v<T>)
{
    if (that) emplace(*that);
    else reset();
    return *this;
}
```
- (1.4) — If `Assignable<T&, T>` is not satisfied, the move assignment operator is equivalent to:
- ```
semiregular& operator=(semiregular&& that)
 noexcept(is_nothrow_move_constructible_v<T>)
{
 if (that) emplace(std::move(*that));
 else reset();
 return *this;
}
```

### 23.7.3 All view

[range.all]

- <sup>1</sup> `view::all` returns a `View` that includes all elements of its `Range` argument.
- <sup>2</sup> The name `view::all` denotes a range adaptor object (23.7.1). For some subexpression `E`, the expression `view::all(E)` is expression-equivalent to:
- (2.1) — `DECAY_COPY(E)` if the decayed type of `E` models `View`.
- (2.2) — Otherwise, `ref-view{E}` if that expression is well-formed, where `ref-view` is the exposition-only `View` specified below.
- (2.3) — Otherwise, `subrange{E}`.

**23.7.3.1 *ref-view*****[range.view.ref]**

```

namespace std::ranges {
 template<Range R>
 requires is_object_v<R>
 class ref-view : public view_interface<ref-view<R>> {
 private:
 R* r_ = nullptr; // exposition only
 public:
 constexpr ref-view() noexcept = default;

 template<not-same-as<ref-view> T>
 requires see below
 constexpr ref-view(T&& t);

 constexpr R& base() const { return *r_; }

 constexpr iterator_t<R> begin() const { return ranges::begin(*r_); }
 constexpr sentinel_t<R> end() const { return ranges::end(*r_); }

 constexpr bool empty() const
 requires requires { ranges::empty(*r_); }
 { return ranges::empty(*r_); }

 constexpr auto size() const requires SizedRange<R>
 { return ranges::size(*r_); }

 constexpr auto data() const requires ContiguousRange<R>
 { return ranges::data(*r_); }

 friend constexpr iterator_t<R> begin(ref-view r)
 { return r.begin(); }

 friend constexpr sentinel_t<R> end(ref-view r)
 { return r.end(); }
 };
}

template<not-same-as<ref-view> T>
 requires see below
constexpr ref-view(T&& t);

```

<sup>1</sup> *Remarks:* Let *FUN* denote the exposition-only functions

```

void FUN(R&);
void FUN(R&&) = delete;

```

The expression in the *requires-clause* is equivalent to

```
ConvertibleTo<T, R&> && requires { FUN(declval<T>()); }
```

<sup>2</sup> *Effects:* Initializes *r\_* with `addressof(static_cast<R&>(std::forward<T>(t)))`.

**23.7.4 Filter view****[range.filter]****23.7.4.1 Overview****[range.filter.overview]**

<sup>1</sup> *filter\_view* presents a *View* of an underlying sequence without the elements that fail to satisfy a predicate.

<sup>2</sup> [*Example:*

```

vector<int> is{ 0, 1, 2, 3, 4, 5, 6 };
filter_view evens{is, [](int i) { return 0 == i % 2; }};
for (int i : evens)
 cout << i << ' '; // prints: 0 2 4 6

```

— end example]

## 23.7.4.2 Class template filter\_view

[range.filter.view]

```

namespace std::ranges {
 template<InputRange V, IndirectUnaryPredicate<iterator_t<V>> Pred>
 requires View<V> && is_object_v<Pred>
 class filter_view : public view_interface<filter_view<V, Pred>> {
 private:
 V base_ = V(); // exposition only
 semiregular<Pred> pred_; // exposition only

 class iterator; // exposition only
 class sentinel; // exposition only

 public:
 filter_view() = default;
 constexpr filter_view(V base, Pred pred);
 template<InputRange R>
 requires ViewableRange<R> && Constructible<V, all_view<R>>
 constexpr filter_view(R&& r, Pred pred);

 constexpr V base() const;

 constexpr iterator begin();
 constexpr auto end() {
 if constexpr (CommonRange<V>)
 return iterator{*this, ranges::end(base_)};
 else
 return sentinel{*this};
 }
 };

 template<class R, class Pred>
 filter_view(R&&, Pred) -> filter_view<all_view<R>, Pred>;
}

```

```
constexpr filter_view(V base, Pred pred);
```

1 *Effects:* Initializes `base_` with `std::move(base)` and initializes `pred_` with `std::move(pred)`.

```

template<InputRange R>
 requires ViewableRange<R> && Constructible<V, all_view<R>>
constexpr filter_view(R&& r, Pred pred);

```

2 *Effects:* Initializes `base_` with `view::all(std::forward<R>(r))` and initializes `pred_` with `std::move(pred)`.

```
constexpr V base() const;
```

3 *Effects:* Equivalent to: `return base_;`

```
constexpr iterator begin();
```

4 *Expects:* `pred_.has_value()`.

5 *Returns:* `{*this, ranges::find_if(base_, ref(*pred_))}`.

6 *Remarks:* In order to provide the amortized constant time complexity required by the **Range** concept, this function caches the result within the `filter_view` for use on subsequent calls.

## 23.7.4.3 Class filter\_view::iterator

[range.filter.iterator]

```

namespace std::ranges {
 template<class V, class Pred>
 class filter_view<V, Pred>::iterator {
 private:
 iterator_t<V> current_ = iterator_t<V>(); // exposition only
 filter_view* parent_ = nullptr; // exposition only
 public:
 using iterator_concept = see below;
 using iterator_category = see below;
 };
}

```

```

using value_type = iter_value_t<iterator_t<V>>;
using difference_type = iter_difference_t<iterator_t<V>>;

iterator() = default;
constexpr iterator(filter_view& parent, iterator_t<V> current);

constexpr iterator_t<V> base() const;
constexpr iter_reference_t<iterator_t<V>> operator*() const;
constexpr iterator_t<V> operator->() const
 requires has_arrow<iterator_t<V>>;

constexpr iterator& operator++();
constexpr void operator++(int);
constexpr iterator operator++(int) requires ForwardRange<V>;

constexpr iterator& operator--() requires BidirectionalRange<V>;
constexpr iterator operator--(int) requires BidirectionalRange<V>;

friend constexpr bool operator==(const iterator& x, const iterator& y)
 requires EqualityComparable<iterator_t<V>>;
friend constexpr bool operator!=(const iterator& x, const iterator& y)
 requires EqualityComparable<iterator_t<V>>;

friend constexpr iter_rvalue_reference_t<iterator_t<V>> iter_move(const iterator& i)
 noexcept(noexcept(ranges::iter_move(i.current_)));
friend constexpr void iter_swap(const iterator& x, const iterator& y)
 noexcept(noexcept(ranges::iter_swap(x.current_, y.current_)))
 requires IndirectlySwappable<iterator_t<V>>;
};
}

```

<sup>1</sup> Modification of the element a `filter_view::iterator` denotes is permitted, but results in undefined behavior if the resulting value does not satisfy the filter predicate.

<sup>2</sup> `iterator::iterator_concept` is defined as follows:

- (2.1) — If `V` models `BidirectionalRange`, then `iterator_concept` denotes `bidirectional_iterator_tag`.
- (2.2) — Otherwise, if `V` models `ForwardRange`, then `iterator_concept` denotes `forward_iterator_tag`.
- (2.3) — Otherwise, `iterator_concept` denotes `input_iterator_tag`.

<sup>3</sup> `iterator::iterator_category` is defined as follows:

- (3.1) — Let `C` denote the type `iterator_traits<iterator_t<V>>::iterator_category`.
- (3.2) — If `C` models `DerivedFrom<bidirectional_iterator_tag>`, then `iterator_category` denotes `bidirectional_iterator_tag`.
- (3.3) — Otherwise, if `C` models `DerivedFrom<forward_iterator_tag>`, then `iterator_category` denotes `forward_iterator_tag`.
- (3.4) — Otherwise, `iterator_category` denotes `input_iterator_tag`.

```
constexpr iterator(filter_view& parent, iterator_t<V> current);
```

<sup>4</sup> *Effects:* Initializes `current_` with `current` and `parent_` with `addressof(parent)`.

```
constexpr iterator_t<V> base() const;
```

<sup>5</sup> *Effects:* Equivalent to: `return current_;`

```
constexpr iter_reference_t<iterator_t<V>> operator*() const;
```

<sup>6</sup> *Effects:* Equivalent to: `return *current_;`

```
constexpr iterator_t<V> operator->() const
 requires has_arrow<iterator_t<V>>;
```

<sup>7</sup> *Effects:* Equivalent to: `return current_;`

```

constexpr iterator& operator++();
8 Effects: Equivalent to:
 current_ = ranges::find_if(++current_, ranges::end(parent_->base_), ref(*parent_->pred_));
 return *this;

constexpr void operator++(int);
9 Effects: Equivalent to ++*this.

constexpr iterator operator++(int) requires ForwardRange<V>;
10 Effects: Equivalent to:
 auto tmp = *this;
 ++*this;
 return tmp;

constexpr iterator& operator--() requires BidirectionalRange<V>;
11 Effects: Equivalent to:
 do
 --current_;
 while (!invoke(*parent_->pred_, *current_));
 return *this;

constexpr iterator operator--(int) requires BidirectionalRange<V>;
12 Effects: Equivalent to:
 auto tmp = *this;
 --*this;
 return tmp;

friend constexpr bool operator==(const iterator& x, const iterator& y)
 requires EqualityComparable<iterator_t<V>>;
13 Effects: Equivalent to: return x.current_ == y.current_;

friend constexpr bool operator!=(const iterator& x, const iterator& y)
 requires EqualityComparable<iterator_t<V>>;
14 Effects: Equivalent to: return !(x == y);

friend constexpr iter_rvalue_reference_t<iterator_t<V>> iter_move(const iterator& i)
 noexcept(noexcept(ranges::iter_move(i.current_)));
15 Effects: Equivalent to: return ranges::iter_move(i.current_);

friend constexpr void iter_swap(const iterator& x, const iterator& y)
 noexcept(noexcept(ranges::iter_swap(x.current_, y.current_)))
 requires IndirectlySwappable<iterator_t<V>>;
16 Effects: Equivalent to ranges::iter_swap(x.current_, y.current_).

```

#### 23.7.4.4 Class filter\_view::sentinel

[range.filter.sentinel]

```

namespace std::ranges {
 template<class V, class Pred>
 class filter_view<V, Pred>::sentinel {
 private:
 sentinel_t<V> end_ = sentinel_t<V>(); // exposition only
 public:
 sentinel() = default;
 constexpr explicit sentinel(filter_view& parent);

 constexpr sentinel_t<V> base() const;

 friend constexpr bool operator==(const iterator& x, const sentinel& y);
 friend constexpr bool operator==(const sentinel& x, const iterator& y);
 friend constexpr bool operator!=(const iterator& x, const sentinel& y);

```

```

 friend constexpr bool operator!=(const sentinel& x, const iterator& y);
};
}

constexpr explicit sentinel(filter_view& parent);
1 Effects: Initializes end_ with ranges::end(parent).

constexpr sentinel_t<V> base() const;
2 Effects: Equivalent to: return end_;

friend constexpr bool operator==(const iterator& x, const sentinel& y);
3 Effects: Equivalent to: return x.current_ == y.end_;

friend constexpr bool operator==(const sentinel& x, const iterator& y);
4 Effects: Equivalent to: return y == x;

friend constexpr bool operator!=(const iterator& x, const sentinel& y);
5 Effects: Equivalent to: return !(x == y);

friend constexpr bool operator!=(const sentinel& x, const iterator& y);
6 Effects: Equivalent to: return !(y == x);

```

#### 23.7.4.5 view::filter

[range.filter.adaptor]

- 1 The name `view::filter` denotes a range adaptor object (23.7.1). For some subexpressions *E* and *P*, the expression `view::filter(E, P)` is expression-equivalent to `filter_view{E, P}`.

### 23.7.5 Transform view

[range.transform]

#### 23.7.5.1 Overview

[range.transform.overview]

- 1 `transform_view` presents a View of an underlying sequence after applying a transformation function to each element.
- 2 *[Example:*

```

vector<int> is{ 0, 1, 2, 3, 4 };
transform_view squares{is, [](int i) { return i * i; }};
for (int i : squares)
 cout << i << ' '; // prints: 0 1 4 9 16

```

— end example]

#### 23.7.5.2 Class template transform\_view

[range.transform.view]

```

namespace std::ranges {
 template<InputRange V, CopyConstructible F>
 requires View<V> && is_object_v<F> &&
 RegularInvocable<F&, iter_reference_t<iterator_t<V>>>
 class transform_view : public view_interface<transform_view<V, F>> {
 private:
 template<bool> struct iterator; // exposition only
 template<bool> struct sentinel; // exposition only

 V base_ = V(); // exposition only
 semiregular<F> fun_; // exposition only

 public:
 transform_view() = default;
 constexpr transform_view(V base, F fun);
 template<InputRange R>
 requires ViewableRange<R> && Constructible<V, all_view<R>>
 constexpr transform_view(R&& r, F fun);

 constexpr V base() const;

```

```

constexpr iterator<false> begin();
constexpr iterator<true> begin() const requires Range<const V> &&
 RegularInvocable<const F&, iter_reference_t<iterator_t<const V>>>;

constexpr sentinel<false> end();
constexpr iterator<false> end() requires CommonRange<V>;
constexpr sentinel<true> end() const requires Range<const V> &&
 RegularInvocable<const F&, iter_reference_t<iterator_t<const V>>>;
constexpr iterator<true> end() const requires CommonRange<const V> &&
 RegularInvocable<const F&, iter_reference_t<iterator_t<const V>>>;

constexpr auto size() requires SizedRange<V> { return ranges::size(base_); }
constexpr auto size() const requires SizedRange<const V>
{ return ranges::size(base_); }
};

```

```

template<class R, class F>
 transform_view(R&&, F) -> transform_view<all_view<R>, F>;
}

```

```
constexpr transform_view(V base, F fun);
```

1     *Effects:* Initializes `base_` with `std::move(base)` and `fun_` with `std::move(fun)`.

```

template<InputRange R>
 requires ViewableRange<R> && Constructible<V, all_view<R>>
constexpr transform_view(R&& r, F fun);

```

2     *Effects:* Initializes `base_` with `view::all(std::forward<R>(r))` and `fun_` with `std::move(fun)`.

```
constexpr V base() const;
```

3     *Effects:* Equivalent to: `return base_;`

```
constexpr iterator<false> begin();
```

4     *Effects:* Equivalent to:

```
 return iterator<false>{*this, ranges::begin(base_)};
```

```

constexpr iterator<true> begin() const requires Range<const V> &&
 RegularInvocable<const F&, iter_reference_t<iterator_t<const V>>>;

```

5     *Effects:* Equivalent to:

```
 return iterator<true>{*this, ranges::begin(base_)};
```

```
constexpr sentinel<false> end();
```

6     *Effects:* Equivalent to:

```
 return sentinel<false>{ranges::end(base_)};
```

```
constexpr iterator<false> end() requires CommonRange<V>;
```

7     *Effects:* Equivalent to:

```
 return iterator<false>{*this, ranges::end(base_)};
```

```

constexpr sentinel<true> end() const requires Range<const V> &&
 RegularInvocable<const F&, iter_reference_t<iterator_t<const V>>>;

```

8     *Effects:* Equivalent to:

```
 return sentinel<true>{ranges::end(base_)};
```

```

constexpr iterator<true> end() const requires CommonRange<const V> &&
 RegularInvocable<const F&, iter_reference_t<iterator_t<const V>>>;

```

9     *Effects:* Equivalent to:

```
 return iterator<true>{*this, ranges::end(base_)};
```

## 23.7.5.3 Class template transform\_view::iterator

[range.transform.iterator]

```

namespace std::ranges {
 template<class V, class F>
 template<bool Const>
 class transform_view<V, F>::iterator { // exposition only
 private:
 using Parent = // exposition only
 conditional_t<Const, const transform_view, transform_view>;
 using Base = // exposition only
 conditional_t<Const, const V, V>;
 iterator_t<Base> current_ = iterator_t<Base>(); // exposition only
 Parent* parent_ = nullptr; // exposition only
 public:
 using iterator_concept = see below;
 using iterator_category = see below;
 using value_type =
 remove_cvref_t<invoke_result_t<F&, iter_reference_t<iterator_t<Base>>>>;
 using difference_type = iter_difference_t<iterator_t<Base>>;

 iterator() = default;
 constexpr iterator(Parent& parent, iterator_t<Base> current);
 constexpr iterator(iterator_t<Base> i)
 requires Const && ConvertibleTo<iterator_t<V>, iterator_t<Base>>;

 constexpr iterator_t<Base> base() const;
 constexpr decltype(auto) operator*() const
 { return invoke(*parent_>fun_, *current_); }

 constexpr iterator& operator++();
 constexpr void operator++(int);
 constexpr iterator operator++(int) requires ForwardRange<Base>;

 constexpr iterator& operator--() requires BidirectionalRange<Base>;
 constexpr iterator operator--(int) requires BidirectionalRange<Base>;

 constexpr iterator& operator+=(difference_type n)
 requires RandomAccessRange<Base>;
 constexpr iterator& operator-=(difference_type n)
 requires RandomAccessRange<Base>;
 constexpr decltype(auto) operator[](difference_type n) const
 requires RandomAccessRange<Base>
 { return invoke(*parent_>fun_, current_[n]); }

 friend constexpr bool operator==(const iterator& x, const iterator& y)
 requires EqualityComparable<iterator_t<Base>>;
 friend constexpr bool operator!=(const iterator& x, const iterator& y)
 requires EqualityComparable<iterator_t<Base>>;

 friend constexpr bool operator<(const iterator& x, const iterator& y)
 requires RandomAccessRange<Base>;
 friend constexpr bool operator>(const iterator& x, const iterator& y)
 requires RandomAccessRange<Base>;
 friend constexpr bool operator<=(const iterator& x, const iterator& y)
 requires RandomAccessRange<Base>;
 friend constexpr bool operator>=(const iterator& x, const iterator& y)
 requires RandomAccessRange<Base>;

 friend constexpr iterator operator+(iterator i, difference_type n)
 requires RandomAccessRange<Base>;
 friend constexpr iterator operator+(difference_type n, iterator i)
 requires RandomAccessRange<Base>;

 friend constexpr iterator operator-(iterator i, difference_type n)
 requires RandomAccessRange<Base>;

```

```

friend constexpr difference_type operator-(const iterator& x, const iterator& y)
 requires RandomAccessRange<Base>;

friend constexpr decltype(auto) iter_move(const iterator& i)
 noexcept(noexcept(invoke(*i.parent_->fun_, *i.current_)))
{
 if constexpr (is_lvalue_reference_v<decltype(*i)>)
 return std::move(*i);
 else
 return *i;
}

friend constexpr void iter_swap(const iterator& x, const iterator& y)
 noexcept(noexcept(ranges::iter_swap(x.current_, y.current_)))
 requires IndirectlySwappable<iterator_t<Base>>;
};
}

```

<sup>1</sup> `iterator::iterator_concept` is defined as follows:

- (1.1) — If `V` models `RandomAccessRange`, then `iterator_concept` denotes `random_access_iterator_tag`.
- (1.2) — Otherwise, if `V` models `BidirectionalRange`, then `iterator_concept` denotes `bidirectional_iterator_tag`.
- (1.3) — Otherwise, if `V` models `ForwardRange`, then `iterator_concept` denotes `forward_iterator_tag`.
- (1.4) — Otherwise, `iterator_concept` denotes `input_iterator_tag`.

<sup>2</sup> Let `C` denote the type `iterator_traits<iterator_t<Base>>::iterator_category`. If `C` models `DerivedFrom<contiguous_iterator_tag>`, then `iterator_category` denotes `random_access_iterator_tag`; otherwise, `iterator_category` denotes `C`.

```
constexpr iterator(Parent& parent, iterator_t<Base> current);
```

<sup>3</sup> *Effects:* Initializes `current_` with `current` and `parent_` with `addressof(parent)`.

```
constexpr iterator(iterator<!Const> i)
 requires Const && ConvertibleTo<iterator_t<V>, iterator_t<Base>>;
```

<sup>4</sup> *Effects:* Initializes `current_` with `std::move(i.current_)` and `parent_` with `i.parent_`.

```
constexpr iterator_t<Base> base() const;
```

<sup>5</sup> *Effects:* Equivalent to: `return current_;`

```
constexpr iterator& operator++();
```

<sup>6</sup> *Effects:* Equivalent to:

```
++current_;
return *this;
```

```
constexpr void operator++(int);
```

<sup>7</sup> *Effects:* Equivalent to `++current_`.

```
constexpr iterator operator++(int) requires ForwardRange<Base>;
```

<sup>8</sup> *Effects:* Equivalent to:

```
auto tmp = *this;
++*this;
return tmp;
```

```
constexpr iterator& operator--() requires BidirectionalRange<Base>;
```

<sup>9</sup> *Effects:* Equivalent to:

```
--current_;
return *this;
```

```

constexpr iterator operator--(int) requires BidirectionalRange<Base>;
10 Effects: Equivalent to:
 auto tmp = *this;
 --*this;
 return tmp;

constexpr iterator& operator+=(difference_type n)
 requires RandomAccessRange<Base>;
11 Effects: Equivalent to:
 current_ += n;
 return *this;

constexpr iterator& operator-=(difference_type n)
 requires RandomAccessRange<Base>;
12 Effects: Equivalent to:
 current_ -= n;
 return *this;

friend constexpr bool operator==(const iterator& x, const iterator& y)
 requires EqualityComparable<iterator_t<Base>>;
13 Effects: Equivalent to: return x.current_ == y.current_;

friend constexpr bool operator!=(const iterator& x, const iterator& y)
 requires EqualityComparable<iterator_t<Base>>;
14 Effects: Equivalent to: return !(x == y);

friend constexpr bool operator<(const iterator& x, const iterator& y)
 requires RandomAccessRange<Base>;
15 Effects: Equivalent to: return x.current_ < y.current_;

friend constexpr bool operator>(const iterator& x, const iterator& y)
 requires RandomAccessRange<Base>;
16 Effects: Equivalent to: return y < x;

friend constexpr bool operator<=(const iterator& x, const iterator& y)
 requires RandomAccessRange<Base>;
17 Effects: Equivalent to: return !(y < x);

friend constexpr bool operator>=(const iterator& x, const iterator& y)
 requires RandomAccessRange<Base>;
18 Effects: Equivalent to: return !(x < y);

friend constexpr iterator operator+(iterator i, difference_type n)
 requires RandomAccessRange<Base>;
friend constexpr iterator operator+(difference_type n, iterator i)
 requires RandomAccessRange<Base>;
19 Effects: Equivalent to: return iterator{*i.parent_, i.current_ + n};

friend constexpr iterator operator-(iterator i, difference_type n)
 requires RandomAccessRange<Base>;
20 Effects: Equivalent to: return iterator{*i.parent_, i.current_ - n};

friend constexpr difference_type operator-(const iterator& x, const iterator& y)
 requires RandomAccessRange<Base>;
21 Effects: Equivalent to: return x.current_ - y.current_;

friend constexpr void iter_swap(const iterator& x, const iterator& y)
 noexcept(noexcept(ranges::iter_swap(x.current_, y.current_)))

```

requires IndirectlySwappable<iterator\_t<Base>>;

22 *Effects:* Equivalent to `ranges::iter_swap(x.current_, y.current_)`.

#### 23.7.5.4 Class template `transform_view::sentinel`

[range.transform.sentinel]

```
namespace std::ranges {
 template<class V, class F>
 template<bool Const>
 class transform_view<V, F>::sentinel { // exposition only
 private:
 using Parent = // exposition only
 conditional_t<Const, const transform_view, transform_view>;
 using Base = conditional_t<Const, const V, V>; // exposition only
 sentinel_t<Base> end_ = sentinel_t<Base>(); // exposition only
 public:
 sentinel() = default;
 constexpr explicit sentinel(sentinel_t<Base> end);
 constexpr sentinel(sentinel_t<Base> i)
 requires Const && ConvertibleTo<sentinel_t<V>, sentinel_t<Base>>;

 constexpr sentinel_t<Base> base() const;

 friend constexpr bool operator==(const iterator<Const>& x, const sentinel& y);
 friend constexpr bool operator==(const sentinel& x, const iterator<Const>& y);
 friend constexpr bool operator!=(const iterator<Const>& x, const sentinel& y);
 friend constexpr bool operator!=(const sentinel& x, const iterator<Const>& y);

 friend constexpr iter_difference_t<iterator_t<Base>>
 operator-(const iterator<Const>& x, const sentinel& y)
 requires SizedSentinel<sentinel_t<Base>, iterator_t<Base>>;
 friend constexpr iter_difference_t<iterator_t<Base>>
 operator-(const sentinel& y, const iterator<Const>& x)
 requires SizedSentinel<sentinel_t<Base>, iterator_t<Base>>;
 };
}
```

constexpr explicit sentinel(sentinel\_t<Base> end);

1 *Effects:* Initializes `end_` with `end`.

constexpr sentinel(sentinel\_t<Base> i)  
requires Const && ConvertibleTo<sentinel\_t<V>, sentinel\_t<Base>>;

2 *Effects:* Initializes `end_` with `std::move(i.end_)`.

constexpr sentinel\_t<Base> base() const;

3 *Effects:* Equivalent to: return `end_`;

friend constexpr bool operator==(const iterator<Const>& x, const sentinel& y);

4 *Effects:* Equivalent to: return `x.current_ == y.end_`;

friend constexpr bool operator==(const sentinel& x, const iterator<Const>& y);

5 *Effects:* Equivalent to: return `y == x`;

friend constexpr bool operator!=(const iterator<Const>& x, const sentinel& y);

6 *Effects:* Equivalent to: return `!(x == y)`;

friend constexpr bool operator!=(const sentinel& x, const iterator<Const>& y);

7 *Effects:* Equivalent to: return `!(y == x)`;

friend constexpr iter\_difference\_t<iterator\_t<Base>>  
operator-(const iterator<Const>& x, const sentinel& y)  
requires SizedSentinel<sentinel\_t<Base>, iterator\_t<Base>>;

8 *Effects:* Equivalent to: return `x.current_ - y.end_`;

```
friend constexpr iter_difference_t<iterator_t<Base>>
operator-(const sentinel& y, const iterator<Const>& x)
requires SizedSentinel<sentinel_t<Base>, iterator_t<Base>>;
```

9 *Effects:* Equivalent to: return x.end\_ - y.current\_;

### 23.7.5.5 view::transform [range.transform.adaptor]

1 The name `view::transform` denotes a range adaptor object (23.7.1). For some subexpressions E and F, the expression `view::transform(E, F)` is expression-equivalent to `transform_view{E, F}`.

## 23.7.6 Iota view [range.iota]

### 23.7.6.1 Overview [range.iota.overview]

1 `iota_view` generates a sequence of elements by repeatedly incrementing an initial value.

2 *[Example:*

```
for (int i : iota_view{1, 10})
 cout << i << ' '; // prints: 1 2 3 4 5 6 7 8 9
```

*—end example]*

### 23.7.6.2 Class template `iota_view` [range.iota.view]

```
namespace std::ranges {
 template<class I>
 concept Decrementable = // exposition only
 see below;
 template<class I>
 concept Advanceable = // exposition only
 see below;

 template<WeaklyIncrementable W, Semiregular Bound = unreachable_sentinel_t>
 requires weakly-equality-comparable-with<W, Bound>
 class iota_view : public view_interface<iota_view<W, Bound>> {
 private:
 struct iterator; // exposition only
 struct sentinel; // exposition only
 W value_ = W(); // exposition only
 Bound bound_ = Bound(); // exposition only
 public:
 iota_view() = default;
 constexpr explicit iota_view(W value);
 constexpr iota_view(type_identity_t<W> value,
 type_identity_t<Bound> bound);

 constexpr iterator begin() const;
 constexpr sentinel end() const;
 constexpr iterator end() const requires Same<W, Bound>;

 constexpr auto size() const
 requires (Same<W, Bound> && Advanceable<W>) ||
 (Integral<W> && Integral<Bound>) ||
 SizedSentinel<Bound, W>
 { return bound_ - value_; }
 };

 template<class W, class Bound>
 requires
 (!Integral<W> || !Integral<Bound> || is_signed_v<W> == is_signed_v<Bound>)
 iota_view(W, Bound) -> iota_view<W, Bound>;
}
```

1 The exposition-only *Decrementable* concept is equivalent to:

```

template<class I>
concept Decrementable =
 Incrementable<I> && requires(I i) {
 { --i } -> Same<I>&;
 i--; requires Same<I, decltype(i--)>;
 };

```

2 When an object is in the domain of both pre- and post-decrement, the object is said to be *decrementable*.

3 Let *a* and *b* be equal objects of type *I*. *I* models *Decrementable* only if

(3.1) — If *a* and *b* are decrementable, then the following are all true:

(3.1.1) — `addressof(--a) == addressof(a)`

(3.1.2) — `bool(a-- == b)`

(3.1.3) — `bool(((void)a--, a) == --b)`

(3.1.4) — `bool(++(--a) == b)`.

(3.2) — If *a* and *b* are incrementable, then `bool(--(++a) == b)`.

4 The exposition-only *Advanceable* concept is equivalent to:

```

template<class I>
concept Advanceable =
 Decrementable<I> && StrictTotallyOrdered<I> &&
 requires(I i, const I j, const iter_difference_t<I> n) {
 { i += n } -> Same<I>&;
 { i -= n } -> Same<I>&;
 j + n; requires Same<I, decltype(j + n)>;
 n + j; requires Same<I, decltype(n + j)>;
 j - n; requires Same<I, decltype(j - n)>;
 j - j; requires Same<iter_difference_t<I>, decltype(j - j)>;
 };

```

Let *a* and *b* be objects of type *I* such that *b* is reachable from *a* after *n* applications of `++a`, for some value *n* of type `iter_difference_t<I>`, and let *D* be `iter_difference_t<I>`. *I* models *Advanceable* only if

(4.1) — `(a += n)` is equal to *b*.

(4.2) — `addressof(a += n)` is equal to `addressof(a)`.

(4.3) — `(a + n)` is equal to `(a += n)`.

(4.4) — For any two positive values *x* and *y* of type *D*, if `(a + D(x + y))` is well-defined, then `(a + D(x + y))` is equal to `((a + x) + y)`.

(4.5) — `(a + D(0))` is equal to *a*.

(4.6) — If `(a + D(n - 1))` is well-defined, then `(a + n)` is equal to `++(a + D(n - 1))`.

(4.7) — `(b += -n)` is equal to *a*.

(4.8) — `(b -= n)` is equal to *a*.

(4.9) — `addressof(b -= n)` is equal to `addressof(b)`.

(4.10) — `(b - n)` is equal to `(b -= n)`.

(4.11) — `(b - a)` is equal to *n*.

(4.12) — `(a - b)` is equal to `-n`.

(4.13) — `bool(a <= b)` is true.

```

constexpr explicit iota_view(W value);

```

5 *Expects:* *Bound* denotes `unreachable_sentinel_t` or `Bound()` is reachable from *value*.

6 *Effects:* Initializes *value\_* with *value*.

```

constexpr iota_view(type_identity_t<W> value, type_identity_t<Bound> bound);

```

7 *Expects:* *Bound* denotes `unreachable_sentinel_t` or *bound* is reachable from *value*.

8 *Effects:* Initializes *value\_* with *value* and *bound\_* with *bound*.

```
constexpr iterator begin() const;
9 Effects: Equivalent to: return iterator{value_};

constexpr sentinel end() const;
10 Effects: Equivalent to: return sentinel{bound_};

constexpr iterator end() const requires Same<W, Bound>;
11 Effects: Equivalent to: return iterator{bound_};
```

### 23.7.6.3 Class `iota_view::iterator`

[range.iota.iterator]

```
namespace std::ranges {
 template<class W, class Bound>
 struct iota_view<W, Bound>::iterator {
 private:
 W value_ = W(); // exposition only
 public:
 using iterator_category = see below;
 using value_type = W;
 using difference_type = iter_difference_t<W>;

 iterator() = default;
 constexpr explicit iterator(W value);

 constexpr W operator*() const noexcept(is_nothrow_copy_constructible_v<W>);

 constexpr iterator& operator++();
 constexpr void operator++(int);
 constexpr iterator operator++(int) requires Incrementable<W>;

 constexpr iterator& operator--() requires Decrementable<W>;
 constexpr iterator operator--(int) requires Decrementable<W>;

 constexpr iterator& operator+=(difference_type n)
 requires Advanceable<W>;
 constexpr iterator& operator-=(difference_type n)
 requires Advanceable<W>;
 constexpr W operator[](difference_type n) const
 requires Advanceable<W>;

 friend constexpr bool operator==(const iterator& x, const iterator& y)
 requires EqualityComparable<W>;
 friend constexpr bool operator!=(const iterator& x, const iterator& y)
 requires EqualityComparable<W>;

 friend constexpr bool operator<(const iterator& x, const iterator& y)
 requires StrictTotallyOrdered<W>;
 friend constexpr bool operator>(const iterator& x, const iterator& y)
 requires StrictTotallyOrdered<W>;
 friend constexpr bool operator<=(const iterator& x, const iterator& y)
 requires StrictTotallyOrdered<W>;
 friend constexpr bool operator>=(const iterator& x, const iterator& y)
 requires StrictTotallyOrdered<W>;

 friend constexpr iterator operator+(iterator i, difference_type n)
 requires Advanceable<W>;
 friend constexpr iterator operator+(difference_type n, iterator i)
 requires Advanceable<W>;

 friend constexpr iterator operator-(iterator i, difference_type n)
 requires Advanceable<W>;
 friend constexpr difference_type operator-(const iterator& x, const iterator& y)
 requires Advanceable<W>;
```

```
};
}
```

1 `iterator::iterator_category` is defined as follows:

- (1.1) — If `W` models *Advanceable*, then `iterator_category` is `random_access_iterator_tag`.
- (1.2) — Otherwise, if `W` models *Decrementable*, then `iterator_category` is `bidirectional_iterator_tag`.
- (1.3) — Otherwise, if `W` models *Incrementable*, then `iterator_category` is `forward_iterator_tag`.
- (1.4) — Otherwise, `iterator_category` is `input_iterator_tag`.

2 [Note: Overloads for `iter_move` and `iter_swap` are omitted intentionally. — end note]

```
constexpr explicit iterator(W value);
```

3 *Effects:* Initializes `value_` with `value`.

```
constexpr W operator*() const noexcept(is_nothrow_copy_constructible_v<W>);
```

4 *Effects:* Equivalent to: `return value_;`

5 [Note: The `noexcept` clause is needed by the default `iter_move` implementation. — end note]

```
constexpr iterator& operator++();
```

6 *Effects:* Equivalent to:

```
++value_;
return *this;
```

```
constexpr void operator++(int);
```

7 *Effects:* Equivalent to `++*this`.

```
constexpr iterator operator++(int) requires Incrementable<W>;
```

8 *Effects:* Equivalent to:

```
auto tmp = *this;
++*this;
return tmp;
```

```
constexpr iterator& operator--() requires Decrementable<W>;
```

9 *Effects:* Equivalent to:

```
--value_;
return *this;
```

```
constexpr iterator operator--(int) requires Decrementable<W>;
```

10 *Effects:* Equivalent to:

```
auto tmp = *this;
--*this;
return tmp;
```

```
constexpr iterator& operator+=(difference_type n)
requires Advanceable<W>;
```

11 *Effects:* Equivalent to:

```
value_ += n;
return *this;
```

```
constexpr iterator& operator-=(difference_type n)
requires Advanceable<W>;
```

12 *Effects:* Equivalent to:

```
value_ -= n;
return *this;
```

```

constexpr W operator[](difference_type n) const
 requires Advanceable<W>;
13 Effects: Equivalent to: return value_ + n;

friend constexpr bool operator==(const iterator& x, const iterator& y)
 requires EqualityComparable<W>;
14 Effects: Equivalent to: return x.value_ == y.value_;

friend constexpr bool operator!=(const iterator& x, const iterator& y)
 requires EqualityComparable<W>;
15 Effects: Equivalent to: return !(x == y);

friend constexpr bool operator<(const iterator& x, const iterator& y)
 requires StrictTotallyOrdered<W>;
16 Effects: Equivalent to: return x.value_ < y.value_;

friend constexpr bool operator>(const iterator& x, const iterator& y)
 requires StrictTotallyOrdered<W>;
17 Effects: Equivalent to: return y < x;

friend constexpr bool operator<=(const iterator& x, const iterator& y)
 requires StrictTotallyOrdered<W>;
18 Effects: Equivalent to: return !(y < x);

friend constexpr bool operator>=(const iterator& x, const iterator& y)
 requires StrictTotallyOrdered<W>;
19 Effects: Equivalent to: return !(x < y);

friend constexpr iterator operator+(iterator i, difference_type n)
 requires Advanceable<W>;
20 Effects: Equivalent to: return iterator{i.value_ + n};

friend constexpr iterator operator+(difference_type n, iterator i)
 requires Advanceable<W>;
21 Effects: Equivalent to: return i + n;

friend constexpr iterator operator-(iterator i, difference_type n)
 requires Advanceable<W>;
22 Effects: Equivalent to: return i + -n;

friend constexpr difference_type operator-(const iterator& x, const iterator& y)
 requires Advanceable<W>;
23 Effects: Equivalent to: return x.value_ - y.value_;

```

#### 23.7.6.4 Class `iota_view::sentinel`

[range.iota.sentinel]

```

namespace std::ranges {
 template<class W, class Bound>
 struct iota_view<W, Bound>::sentinel {
 private:
 Bound bound_ = Bound(); // exposition only
 public:
 sentinel() = default;
 constexpr explicit sentinel(Bound bound);

 friend constexpr bool operator==(const iterator& x, const sentinel& y);
 friend constexpr bool operator==(const sentinel& x, const iterator& y);
 friend constexpr bool operator!=(const iterator& x, const sentinel& y);
 friend constexpr bool operator!=(const sentinel& x, const iterator& y);
 };
}

```

```
constexpr explicit sentinel(Bound bound);
1 Effects: Initializes bound_ with bound.

friend constexpr bool operator==(const iterator& x, const sentinel& y);
2 Effects: Equivalent to: return x.value_ == y.bound_;

friend constexpr bool operator==(const sentinel& x, const iterator& y);
3 Effects: Equivalent to: return y == x;

friend constexpr bool operator!=(const iterator& x, const sentinel& y);
4 Effects: Equivalent to: return !(x == y);

friend constexpr bool operator!=(const sentinel& x, const iterator& y);
5 Effects: Equivalent to: return !(y == x);
```

### 23.7.6.5 view::iota [range.iota.adaptor]

1 The name `view::iota` denotes a customization point object ([[customization.point.object](#)]). For some subexpressions `E` and `F`, the expressions `view::iota(E)` and `view::iota(E, F)` are expression-equivalent to `iota_view{E}` and `iota_view{E, F}`, respectively.

## 23.7.7 Take view [range.take]

### 23.7.7.1 Overview [range.take.overview]

1 `take_view` produces a `View` of the first  $N$  elements from another `View`, or all the elements if the adapted `View` contains fewer than  $N$ .

2 [*Example:*

```
vector<int> is{0,1,2,3,4,5,6,7,8,9};
take_view few{is, 5};
for (int i : few)
 cout << i << ' '; // prints: 0 1 2 3 4
```

— *end example*]

### 23.7.7.2 Class template `take_view` [range.take.view]

```
namespace std::ranges {
 template<View V>
 class take_view : public view_interface<take_view<V>> {
 private:
 V base_ = V(); // exposition only
 iter_difference_t<iterator_t<V>> count_ = 0; // exposition only
 template<bool> struct sentinel; // exposition only
 public:
 take_view() = default;
 constexpr take_view(V base, iter_difference_t<iterator_t<V>> count);
 template<ViewableRange R>
 requires Constructible<V, all_view<R>>
 constexpr take_view(R&& r, iter_difference_t<iterator_t<V>> count);

 constexpr V base() const;

 constexpr auto begin() requires (!simple-view<V>) {
 if constexpr (SizedRange<V>) {
 if constexpr (RandomAccessRange<V>)
 return ranges::begin(base_);
 else
 return counted_iterator{ranges::begin(base_), size()};
 } else
 return counted_iterator{ranges::begin(base_), count_};
 }
 }
```

```

constexpr auto begin() const requires Range<const V> {
 if constexpr (SizedRange<const V>) {
 if constexpr (RandomAccessRange<const V>)
 return ranges::begin(base_);
 else
 return counted_iterator{ranges::begin(base_), size()};
 } else
 return counted_iterator{ranges::begin(base_), count_};
}

constexpr auto end() requires (!simple-view<V>) {
 if constexpr (SizedRange<V>) {
 if constexpr (RandomAccessRange<V>)
 return ranges::begin(base_) + size();
 else
 return default_sentinel;
 } else
 return sentinel<false>{ranges::end(base_)};
}

constexpr auto end() const requires Range<const V> {
 if constexpr (SizedRange<const V>) {
 if constexpr (RandomAccessRange<const V>)
 return ranges::begin(base_) + size();
 else
 return default_sentinel;
 } else
 return sentinel<true>{ranges::end(base_)};
}

constexpr auto size() requires SizedRange<V> {
 auto n = ranges::size(base_);
 return ranges::min(n, static_cast<decltype(n)>(count_));
}

constexpr auto size() const requires SizedRange<const V> {
 auto n = ranges::size(base_);
 return ranges::min(n, static_cast<decltype(n)>(count_));
}
};

template<Range R>
 take_view(R&&, iter_difference_t<iterator_t<R>>)
 -> take_view<all_view<R>>;
}

```

```
constexpr take_view(V base, iter_difference_t<iterator_t<V>> count);
```

<sup>1</sup> *Effects:* Initializes `base_` with `std::move(base)` and `count_` with `count`.

```

template<ViewableRange R>
 requires Constructible<V, all_view<R>>
constexpr take_view(R&& r, iter_difference_t<iterator_t<V>> count);

```

<sup>2</sup> *Effects:* Initializes `base_` with `view::all(std::forward<R>(r))` and `count_` with `count`.

```
constexpr V base() const;
```

<sup>3</sup> *Effects:* Equivalent to: `return base_;`

### 23.7.7.3 Class template `take_view::sentinel`

[range.take.sentinel]

```

namespace std::ranges {
 template<class V>
 template<bool Const>
 class take_view<V>::sentinel { // exposition only
 private:

```

```

 using Base = conditional_t<Const, const V, V>; // exposition only
 using CI = counted_iterator<iterator_t<Base>>; // exposition only
 sentinel_t<Base> end_ = sentinel_t<Base>(); // exposition only
public:
 sentinel() = default;
 constexpr explicit sentinel(sentinel_t<Base> end);
 constexpr sentinel(sentinel_t<!Const> s)
 requires Const && ConvertibleTo<sentinel_t<V>, sentinel_t<Base>>;

 constexpr sentinel_t<Base> base() const;

 friend constexpr bool operator==(const sentinel& x, const CI& y);
 friend constexpr bool operator==(const CI& y, const sentinel& x);
 friend constexpr bool operator!=(const sentinel& x, const CI& y);
 friend constexpr bool operator!=(const CI& y, const sentinel& x);
};
}

```

```
constexpr explicit sentinel(sentinel_t<Base> end);
```

1 *Effects:* Initializes `end_` with `end`.

```
constexpr sentinel(sentinel_t<!Const> s)
 requires Const && ConvertibleTo<sentinel_t<V>, sentinel_t<Base>>;
```

2 *Effects:* Initializes `end_` with `std::move(s.end_)`.

```
constexpr sentinel_t<Base> base() const;
```

3 *Effects:* Equivalent to: `return end_;`

```
friend constexpr bool operator==(const sentinel& x, const CI& y);
friend constexpr bool operator==(const CI& y, const sentinel& x);
```

4 *Effects:* Equivalent to: `return y.count() == 0 || y.base() == x.end_;`

```
friend constexpr bool operator!=(const sentinel& x, const CI& y);
friend constexpr bool operator!=(const CI& y, const sentinel& x);
```

5 *Effects:* Equivalent to: `return !(x == y);`

#### 23.7.7.4 `view::take`

[range.take.adaptor]

1 The name `view::take` denotes a range adaptor object (23.7.1). For some subexpressions `E` and `F`, the expression `view::take(E, F)` is expression-equivalent to `take_view{E, F}`.

### 23.7.8 Join view

[range.join]

#### 23.7.8.1 Overview

[range.join.overview]

1 `join_view` flattens a `View` of ranges into a `View`.

2 [Example:

```
vector<string> ss{"hello", " ", "world", "!"};
join_view greeting{ss};
for (char ch : greeting)
 cout << ch; // prints: hello world!
```

— end example]

#### 23.7.8.2 Class template `join_view`

[range.join.view]

```
namespace std::ranges {
 template<InputRange V>
 requires View<V> && InputRange<iter_reference_t<iterator_t<V>>> &&
 (is_reference_v<iter_reference_t<iterator_t<V>>> ||
 View<iter_value_t<iterator_t<V>>>)
 class join_view : public view_interface<join_view<V>> {
 private:
 using InnerRng = // exposition only
 iter_reference_t<iterator_t<V>>>;

```

```

template<bool Const>
 struct iterator; // exposition only
template<bool Const>
 struct sentinel; // exposition only

V base_ = V(); // exposition only
all_view<InnerRng> inner_ = // exposition only, present only when !is_reference_v<InnerRng>
 all_view<InnerRng>();

public:
 join_view() = default;
 constexpr explicit join_view(V base);

template<InputRange R>
 requires ViewableRange<R> && Constructible<V, all_view<R>>
 constexpr explicit join_view(R&& r);

constexpr auto begin() {
 return iterator<simple-view<V>>{*this, ranges::begin(base_)};
}

constexpr auto begin() const requires InputRange<const V> &&
 is_reference_v<iter_reference_t<iterator_t<const V>>> {
 return iterator<true>{*this, ranges::begin(base_)};
}

constexpr auto end() {
 if constexpr (ForwardRange<V> &&
 is_reference_v<InnerRng> && ForwardRange<InnerRng> &&
 CommonRange<V> && CommonRange<InnerRng>)
 return iterator<simple-view<V>>{*this, ranges::end(base_)};
 else
 return sentinel<simple-view<V>>{*this};
}

constexpr auto end() const requires InputRange<const V> &&
 is_reference_v<iter_reference_t<iterator_t<const V>>> {
 if constexpr (ForwardRange<const V> &&
 is_reference_v<iter_reference_t<iterator_t<const V>>> &&
 ForwardRange<iter_reference_t<iterator_t<const V>>> &&
 CommonRange<const V> &&
 CommonRange<iter_reference_t<iterator_t<const V>>>)
 return iterator<true>{*this, ranges::end(base_)};
 else
 return sentinel<true>{*this};
}
};

template<class R>
 explicit join_view(R&&) -> join_view<all_view<R>>;
}

```

```
constexpr explicit join_view(V base);
```

<sup>1</sup> *Effects:* Initializes `base_` with `std::move(base)`.

```

template<InputRange R>
 requires ViewableRange<R> && Constructible<V, all_view<R>>
 constexpr explicit join_view(R&& r);

```

<sup>2</sup> *Effects:* Initializes `base_` with `view::all(std::forward<R>(r))`.

### 23.7.8.3 Class template `join_view::iterator`

[range.join.iterator]

<sup>1</sup>

```

namespace std::ranges {
template<class V>
template<bool Const>
struct join_view<V>::iterator { // exposition only
private:
 using Parent = // exposition only
 conditional_t<Const, const join_view, join_view>;
 using Base = conditional_t<Const, const V, V>; // exposition only

 static constexpr bool ref_is_glvalue = // exposition only
 is_reference_v<iter_reference_t<iterator_t<Base>>>;

 iterator_t<Base> outer_ = iterator_t<Base>(); // exposition only
 iterator_t<iter_reference_t<iterator_t<Base>>> inner_ = // exposition only
 iterator_t<iter_reference_t<iterator_t<Base>>>();
 Parent* parent_ = nullptr; // exposition only

 constexpr void satisfy(); // exposition only
public:
 using iterator_concept = see below;
 using iterator_category = see below;
 using value_type =
 iter_value_t<iterator_t<iter_reference_t<iterator_t<Base>>>>;
 using difference_type = see below;

 iterator() = default;
 constexpr iterator(Parent& parent, iterator_t<V> outer);
 constexpr iterator(iterator<!Const> i) requires Const &&
 ConvertibleTo<iterator_t<V>, iterator_t<Base>> &&
 ConvertibleTo<iterator_t<InnerRng>,
 iterator_t<iter_reference_t<iterator_t<Base>>>>;

 constexpr decltype(auto) operator*() const { return *inner_; }

 constexpr iterator_t<Base> operator->() const
 requires has_arrow<iterator_t<Base>>;

 constexpr iterator& operator++();
 constexpr void operator++(int);
 constexpr iterator operator++(int)
 requires ref_is_glvalue && ForwardRange<Base> &&
 ForwardRange<iter_reference_t<iterator_t<Base>>>;

 constexpr iterator& operator--()
 requires ref_is_glvalue && BidirectionalRange<Base> &&
 BidirectionalRange<iter_reference_t<iterator_t<Base>>>;

 constexpr iterator operator--(int)
 requires ref_is_glvalue && BidirectionalRange<Base> &&
 BidirectionalRange<iter_reference_t<iterator_t<Base>>>;

 friend constexpr bool operator==(const iterator& x, const iterator& y)
 requires ref_is_glvalue && EqualityComparable<iterator_t<Base>> &&
 EqualityComparable<iterator_t<iter_reference_t<iterator_t<Base>>>>;

 friend constexpr bool operator!=(const iterator& x, const iterator& y)
 requires ref_is_glvalue && EqualityComparable<iterator_t<Base>> &&
 EqualityComparable<iterator_t<iter_reference_t<iterator_t<Base>>>>;

 friend constexpr decltype(auto) iter_move(const iterator& i)
 noexcept(noexcept(ranges::iter_move(i.inner_))) {
 return ranges::iter_move(i.inner_);
 }
}

```

```

 friend constexpr void iter_swap(const iterator& x, const iterator& y)
 noexcept(noexcept(ranges::iter_swap(x.inner_, y.inner_)));
 };
}

```

<sup>2</sup> `iterator::iterator_concept` is defined as follows:

- (2.1) — If `ref_is_glvalue` is true,
  - (2.1.1) — If `Base` and `iter_reference_t<iterator_t<Base>>` each model `BidirectionalRange`, then `iterator_concept` denotes `bidirectional_iterator_tag`.
  - (2.1.2) — Otherwise, if `Base` and `iter_reference_t<iterator_t<Base>>` each model `ForwardRange`, then `iterator_concept` denotes `forward_iterator_tag`.
- (2.2) — Otherwise, `iterator_concept` denotes `input_iterator_tag`.

<sup>3</sup> `iterator::iterator_category` is defined as follows:

- (3.1) — Let *OUTERC* denote `iterator_traits<iterator_t<Base>>::iterator_category`, and let *INNERC* denote `iterator_traits<iterator_t<iter_reference_t<iterator_t<Base>>>::iterator_category`.
- (3.2) — If `ref_is_glvalue` is true,
  - (3.2.1) — If *OUTERC* and *INNERC* each model `DerivedFrom<bidirectional_iterator_tag>`, `iterator_category` denotes `bidirectional_iterator_tag`.
  - (3.2.2) — Otherwise, if *OUTERC* and *INNERC* each model `DerivedFrom<forward_iterator_tag>`, `iterator_category` denotes `forward_iterator_tag`.
- (3.3) — Otherwise, `iterator_category` denotes `input_iterator_tag`.

<sup>4</sup> `iterator::difference_type` denotes the type:

```

common_type_t<
 iter_difference_t<iterator_t<Base>>,
 iter_difference_t<iterator_t<iter_reference_t<iterator_t<Base>>>>

```

<sup>5</sup> `join_view` iterators use the `satisfy` function to skip over empty inner ranges.

```
constexpr void satisfy(); // exposition only
```

<sup>6</sup> *Effects:* Equivalent to:

```

auto update_inner = [this](iter_reference_t<iterator_t<Base>> x) -> decltype(auto) {
 if constexpr (ref_is_glvalue) // x is a reference
 return (x); // (x) is an lvalue
 else
 return (parent_->inner_ = view::all(x));
};

for (; outer_ != ranges::end(parent_->base_); ++outer_) {
 auto& inner = update_inner(*outer_);
 inner_ = ranges::begin(inner);
 if (inner_ != ranges::end(inner))
 return;
}
if constexpr (ref_is_glvalue)
 inner_ = iterator_t<iter_reference_t<iterator_t<Base>>>();

```

```
constexpr iterator(Parent& parent, iterator_t<V> outer)
```

<sup>7</sup> *Effects:* Initializes `outer_` with `outer` and `parent_` with `addressof(parent)`; then calls `satisfy()`.

```

constexpr iterator(iterator<!Const> i) requires Const &&
 ConvertibleTo<iterator_t<V>, iterator_t<Base>> &&
 ConvertibleTo<iterator_t<InnerRng>,
 iterator_t<iter_reference_t<iterator_t<Base>>>>;

```

<sup>8</sup> *Effects:* Initializes `outer_` with `std::move(i.outer_)`, `inner_` with `std::move(i.inner_)`, and `parent_` with `i.parent_`.

```

constexpr iterator_t<Base> operator->() const
 requires has_arrow<iterator_t<Base>>;

9 Effects: Equivalent to return inner_;

constexpr iterator& operator++();

10 Let inner-range be:
(10.1) — If ref_is_glvalue is true, *outer_.
(10.2) — Otherwise, parent_->inner_.

11 Effects: Equivalent to:
 auto&& inner_rng = inner-range;
 if (++inner_ == ranges::end(inner_rng)) {
 ++outer_;
 satisfy();
 }
 return *this;

constexpr void operator++(int);

12 Effects: Equivalent to: ++*this.

constexpr iterator operator++(int)
 requires ref_is_glvalue && ForwardRange<Base> &&
 ForwardRange<iter_reference_t<iterator_t<Base>>>;

13 Effects: Equivalent to:
 auto tmp = *this;
 ++*this;
 return tmp;

constexpr iterator& operator--()
 requires ref_is_glvalue && BidirectionalRange<Base> &&
 BidirectionalRange<iter_reference_t<iterator_t<Base>>>;

14 Effects: Equivalent to:
 if (outer_ == ranges::end(parent_->base_))
 inner_ = ranges::end(*--outer_);
 while (inner_ == ranges::begin(*outer_))
 inner_ = ranges::end(*--outer_);
 --inner_;
 return *this;

constexpr iterator operator--(int)
 requires ref_is_glvalue && BidirectionalRange<Base> &&
 BidirectionalRange<iter_reference_t<iterator_t<Base>>>;

15 Effects: Equivalent to:
 auto tmp = *this;
 --*this;
 return tmp;

friend constexpr bool operator==(const iterator& x, const iterator& y)
 requires ref_is_glvalue && EqualityComparable<iterator_t<Base>> &&
 EqualityComparable<iterator_t<iter_reference_t<iterator_t<Base>>>>;

16 Effects: Equivalent to: return x.outer_ == y.outer_ && x.inner_ == y.inner_;

friend constexpr bool operator!=(const iterator& x, const iterator& y)
 requires ref_is_glvalue && EqualityComparable<iterator_t<Base>> &&
 EqualityComparable<iterator_t<iter_reference_t<iterator_t<Base>>>>;

17 Effects: Equivalent to: return !(x == y);

friend constexpr void iter_swap(const iterator& x, const iterator& y)

```

```
noexcept(noexcept(ranges::iter_swap(x.inner_, y.inner_)));
```

18 *Effects:* Equivalent to: `return ranges::iter_swap(x.inner_, y.inner_);`

#### 23.7.8.4 Class template `join_view::sentinel`

[range.join.sentinel]

```
namespace std::ranges {
 template<class V>
 template<bool Const>
 struct join_view<V>::sentinel { // exposition only
 private:
 using Parent = // exposition only
 conditional_t<Const, const join_view, join_view>;
 using Base = conditional_t<Const, const V, V>; // exposition only
 sentinel_t<Base> end_ = sentinel_t<Base>(); // exposition only
 public:
 sentinel() = default;

 constexpr explicit sentinel(Parent& parent);
 constexpr sentinel(sentinel<!Const> s) requires Const &&
 ConvertibleTo<sentinel_t<V>, sentinel_t<Base>>;

 friend constexpr bool operator==(const iterator<Const>& x, const sentinel& y);
 friend constexpr bool operator==(const sentinel& x, const iterator<Const>& y);
 friend constexpr bool operator!=(const iterator<Const>& x, const sentinel& y);
 friend constexpr bool operator!=(const sentinel& x, const iterator<Const>& y);
 };
}
```

```
constexpr explicit sentinel(Parent& parent);
```

1 *Effects:* Initializes `end_` with `ranges::end(parent.base_)`.

```
constexpr sentinel(sentinel<!Const> s) requires Const &&
 ConvertibleTo<sentinel_t<V>, sentinel_t<Base>>;
```

2 *Effects:* Initializes `end_` with `std::move(s.end_)`.

```
friend constexpr bool operator==(const iterator<Const>& x, const sentinel& y);
```

3 *Effects:* Equivalent to: `return x.outer_ == y.end_;`

```
friend constexpr bool operator==(const sentinel& x, const iterator<Const>& y);
```

4 *Effects:* Equivalent to: `return y == x;`

```
friend constexpr bool operator!=(const iterator<Const>& x, const sentinel& y);
```

5 *Effects:* Equivalent to: `return !(x == y);`

```
friend constexpr bool operator!=(const sentinel& x, const iterator<Const>& y);
```

6 *Effects:* Equivalent to: `return !(y == x);`

#### 23.7.8.5 `view::join`

[range.join.adaptor]

1 The name `view::join` denotes a range adaptor object (23.7.1). For some subexpression *E*, the expression `view::join(E)` is expression-equivalent to `join_view{E}`.

### 23.7.9 Empty view

[range.empty]

#### 23.7.9.1 Overview

[range.empty.overview]

1 `empty_view` produces a `View` of no elements of a particular type.

2 [Example:

```
empty_view<int> e;
static_assert(ranges::empty(e));
static_assert(0 == e.size());
```

— end example]

**23.7.9.2 Class template `empty_view`****[range.empty.view]**

```

namespace std::ranges {
 template<class T>
 requires is_object_v<T>
 class empty_view : public view_interface<empty_view<T>> {
 public:
 static constexpr T* begin() noexcept { return nullptr; }
 static constexpr T* end() noexcept { return nullptr; }
 static constexpr T* data() noexcept { return nullptr; }
 static constexpr ptrdiff_t size() noexcept { return 0; }
 static constexpr bool empty() noexcept { return true; }

 friend constexpr T* begin(empty_view) noexcept { return nullptr; }
 friend constexpr T* end(empty_view) noexcept { return nullptr; }
 };
}

```

**23.7.10 Single view****[range.single]****23.7.10.1 Overview****[range.single.overview]**

<sup>1</sup> `single_view` produces a `View` that contains exactly one element of a specified value.

<sup>2</sup> [Example:

```

single_view s{4};
for (int i : s)
 cout << i; // prints 4

```

— end example]

**23.7.10.2 Class template `single_view`****[range.single.view]**

```

namespace std::ranges {
 template<CopyConstructible T>
 requires is_object_v<T>
 class single_view : public view_interface<single_view<T>> {
 private:
 semiregular<T> value_; // exposition only
 public:
 single_view() = default;
 constexpr explicit single_view(const T& t);
 constexpr explicit single_view(T&& t);
 template<class... Args>
 requires Constructible<T, Args...>
 constexpr single_view(in_place_t, Args&&... args);

 constexpr T* begin() noexcept;
 constexpr const T* begin() const noexcept;
 constexpr T* end() noexcept;
 constexpr const T* end() const noexcept;
 static constexpr ptrdiff_t size() noexcept;
 constexpr T* data() noexcept;
 constexpr const T* data() const noexcept;
 };
}

```

```
constexpr explicit single_view(const T& t);
```

<sup>1</sup> *Effects:* Initializes `value_` with `t`.

```
constexpr explicit single_view(T&& t);
```

<sup>2</sup> *Effects:* Initializes `value_` with `std::move(t)`.

```

template<class... Args>
constexpr single_view(in_place_t, Args&&... args);

```

<sup>3</sup> *Effects:* Initializes `value_` as if by `value_{in_place, std::forward<Args>(args)...}`.

```
constexpr T* begin() noexcept;
constexpr const T* begin() const noexcept;
4 Effects: Equivalent to: return data();

constexpr T* end() noexcept;
constexpr const T* end() const noexcept;
5 Effects: Equivalent to: return data() + 1;

static constexpr ptrdiff_t size() noexcept;
6 Effects: Equivalent to: return 1;

constexpr T* data() noexcept;
constexpr const T* data() const noexcept;
7 Effects: Equivalent to: return value_.operator->();
```

### 23.7.10.3 view::single [range.single.adaptor]

- <sup>1</sup> The name `view::single` denotes a customization point object ([[customization.point.object](#)]). For some subexpression `E`, the expression `view::single(E)` is expression-equivalent to `single_view{E}`.

## 23.7.11 Split view [range.split]

### 23.7.11.1 Overview [range.split.overview]

- <sup>1</sup> `split_view` takes a `View` and a delimiter, and splits the `View` into subranges on the delimiter. The delimiter can be a single element or a `View` of elements.

- <sup>2</sup> [Example:

```
string str{"the quick brown fox"};
split_view sentence{str, ' '};
for (auto word : sentence) {
 for (char ch : word)
 cout << ch;
 cout << ' ';
}
// The above prints: the*quick*brown*fox*
```

— end example]

### 23.7.11.2 Class template split\_view [range.split.view]

```
namespace std::ranges {
 template<auto> struct require-constant; // exposition only

 template<class R>
 concept tiny-range = // exposition only
 SizedRange<R> &&
 requires { typename require-constant<remove_reference_t<R>::size()>; } &&
 (remove_reference_t<R>::size() <= 1);

 template<InputRange V, ForwardRange Pattern>
 requires View<V> && View<Pattern> &&
 IndirectlyComparable<iterator_t<V>, iterator_t<Pattern>, ranges::equal_to<>> &&
 (ForwardRange<V> || tiny-range<Pattern>)
 class split_view : public view_interface<split_view<V, Pattern>> {
 private:
 V base_ = V(); // exposition only
 Pattern pattern_ = Pattern(); // exposition only
 iterator_t<V> current_ = iterator_t<V>(); // exposition only, present only if !ForwardRange<V>
 template<bool> struct outer_iterator; // exposition only
 template<bool> struct inner_iterator; // exposition only
 public:
 split_view() = default;
 constexpr split_view(V base, Pattern pattern);
```

```

template<InputRange R, ForwardRange P>
 requires
 Constructible<V, all_view<R>> &&
 Constructible<Pattern, all_view<P>>
constexpr split_view(R&& r, P&& p);

template<InputRange R>
 requires
 Constructible<V, all_view<R>> &&
 Constructible<Pattern, single_view<iter_value_t<iterator_t<R>>>>
constexpr split_view(R&& r, iter_value_t<iterator_t<R>> e);

constexpr auto begin() {
 if constexpr (ForwardRange<V>)
 return outer_iterator<simple-view<V>>{*this, ranges::begin(base_)};
 else {
 current_ = ranges::begin(base_);
 return outer_iterator<false>{*this};
 }
}

constexpr auto begin() const
 requires ForwardRange<V> && ForwardRange<const V> {
 return outer_iterator<true>{*this, ranges::begin(base_)};
}

constexpr auto end()
 requires ForwardRange<V> && CommonRange<V> {
 return outer_iterator<simple-view<V>>{*this, ranges::end(base_)};
}

constexpr auto end() const {
 if constexpr (ForwardRange<V> && ForwardRange<const V> && CommonRange<const V>)
 return outer_iterator<true>{*this, ranges::end(base_)};
 else
 return default_sentinel;
}
};

template<class R, class P>
 split_view(R&&, P&&) -> split_view<all_view<R>, all_view<P>>;

template<InputRange R>
 split_view(R&&, iter_value_t<iterator_t<R>>)
 -> split_view<all_view<R>, single_view<iter_value_t<iterator_t<R>>>>;
}

```

```
constexpr split_view(V base, Pattern pattern);
```

- <sup>1</sup> *Effects:* Initializes `base_` with `std::move(base)`, and `pattern_` with `std::move(pattern)`.

```

template<InputRange R, ForwardRange P>
 requires
 Constructible<V, all_view<R>> &&
 Constructible<Pattern, all_view<P>>
constexpr split_view(R&& r, P&& p);

```

- <sup>2</sup> *Effects:* Initializes `base_` with `view::all(std::forward<R>(r))` and `pattern_` with `view::all(std::forward<P>(p))`.

```

template<InputRange R>
 requires
 Constructible<V, all_view<R>> &&
 Constructible<Pattern, single_view<iter_value_t<iterator_t<R>>>>

```

```
constexpr split_view(R&& r, iter_value_t<iterator_t<R>> e);
```

- 3 *Effects:* Initializes `base_` with `view::all(std::forward<R>(r))` and `pattern_` with `single_view{std::move(e)}`.

### 23.7.11.3 Class template `split_view::outer_iterator`

[range.split.outer]

```
namespace std::ranges {
 template<class V, class Pattern>
 template<bool Const>
 struct split_view<V, Pattern>::outer_iterator { // exposition only
 private:
 using Parent = // exposition only
 conditional_t<Const, const split_view, split_view>;
 using Base = // exposition only
 conditional_t<Const, const V, V>;
 Parent* parent_ = nullptr; // exposition only
 iterator_t<Base> current_ = // exposition only, present only if V models ForwardRange
 iterator_t<Base>();
 public:
 using iterator_concept =
 conditional_t<ForwardRange<Base>, forward_iterator_tag, input_iterator_tag>;
 using iterator_category = input_iterator_tag;
 struct value_type; // See 23.7.11.4
 using difference_type = iter_difference_t<iterator_t<Base>>;

 outer_iterator() = default;
 constexpr explicit outer_iterator(Parent& parent)
 requires (!ForwardRange<Base>);
 constexpr outer_iterator(Parent& parent, iterator_t<Base> current)
 requires ForwardRange<Base>;
 constexpr outer_iterator(outer_iterator<!Const> i) requires Const &&
 ConvertibleTo<iterator_t<V>, iterator_t<const V>>;

 constexpr value_type operator*() const;

 constexpr outer_iterator& operator++();
 constexpr decltype(auto) operator++(int) {
 if constexpr (ForwardRange<Base>) {
 auto tmp = *this;
 ++*this;
 return tmp;
 } else
 ++*this;
 }

 friend constexpr bool operator==(const outer_iterator& x, const outer_iterator& y)
 requires ForwardRange<Base>;
 friend constexpr bool operator!=(const outer_iterator& x, const outer_iterator& y)
 requires ForwardRange<Base>;

 friend constexpr bool operator==(const outer_iterator& x, default_sentinel_t);
 friend constexpr bool operator==(default_sentinel_t, const outer_iterator& x);
 friend constexpr bool operator!=(const outer_iterator& x, default_sentinel_t y);
 friend constexpr bool operator!=(default_sentinel_t y, const outer_iterator& x);
 };
}
```

- 1 Many of the following specifications refer to the notional member *current* of `outer_iterator`. *current* is equivalent to `current_` if `V` models `ForwardRange`, and `parent_ -> current_` otherwise.

```
constexpr explicit outer_iterator(Parent& parent)
 requires (!ForwardRange<Base>);
```

- 2 *Effects:* Initializes `parent_` with `addressof(parent)`.

```

constexpr outer_iterator(Parent& parent, iterator_t<Base> current)
 requires ForwardRange<Base>;

3 Effects: Initializes parent_ with addressof(parent) and current_ with current.

constexpr outer_iterator(outer_iterator<!Const> i) requires Const &&
 ConvertibleTo<iterator_t<V>, iterator_t<const V>>;

4 Effects: Initializes parent_ with i.parent_ and current_ with std::move(i.current_).

constexpr value_type operator*() const;

5 Effects: Equivalent to: return value_type{*this};

constexpr outer_iterator& operator++();

6 Effects: Equivalent to:
 const auto end = ranges::end(parent_>base_);
 if (current == end) return *this;
 const auto [pbegin, pend] = subrange{parent_>pattern_};
 if (pbegin == pend) ++current;
 else {
 do {
 const auto [b, p] = ranges::mismatch(current, end, pbegin, pend);
 if (p == pend) {
 current = b; // The pattern matched; skip it
 break;
 }
 } while (++current != end);
 }
 return *this;

friend constexpr bool operator==(const outer_iterator& x, const outer_iterator& y)
 requires ForwardRange<Base>;

7 Effects: Equivalent to: return x.current_ == y.current_;

friend constexpr bool operator!=(const outer_iterator& x, const outer_iterator& y)
 requires ForwardRange<Base>;

8 Effects: Equivalent to: return !(x == y);

friend constexpr bool operator==(const outer_iterator& x, default_sentinel_t);
friend constexpr bool operator==(default_sentinel_t, const outer_iterator& x);

9 Effects: Equivalent to: return x.current == ranges::end(x.parent_>base_);

friend constexpr bool operator!=(const outer_iterator& x, default_sentinel_t y);
friend constexpr bool operator!=(default_sentinel_t y, const outer_iterator& x);

10 Effects: Equivalent to: return !(x == y);

```

#### 23.7.11.4 Class `split_view::outer_iterator::value_type`

[range.split.outer.value]

```

namespace std::ranges {
 template<class V, class Pattern>
 template<bool Const>
 struct split_view<V, Pattern>::outer_iterator<Const>::value_type {
 private:
 outer_iterator i_ = outer_iterator(); // exposition only
 public:
 value_type() = default;
 constexpr explicit value_type(outer_iterator i);

 constexpr inner_iterator<Const> begin() const;
 constexpr default_sentinel_t end() const;
 };
}

```

```
constexpr explicit value_type(outer_iterator i);
```

1     *Effects:* Initializes `i_` with `i`.

```
constexpr inner_iterator<Const> begin() const;
```

2     *Effects:* Equivalent to: `return inner_iterator<Const>{i_};`

```
constexpr default_sentinel_t end() const;
```

3     *Effects:* Equivalent to: `return default_sentinel;`

### 23.7.11.5 Class template `split_view::inner_iterator`

[range.split.inner]

```
namespace std::ranges {
 template<class V, class Pattern>
 template<bool Const>
 struct split_view<V, Pattern>::inner_iterator { // exposition only
 private:
 using Base =
 conditional_t<Const, const V, V>; // exposition only
 outer_iterator<Const> i_ = outer_iterator<Const>(); // exposition only
 bool incremented_ = false; // exposition only
 public:
 using iterator_concept = typename outer_iterator<Const>::iterator_concept;
 using iterator_category = see below;
 using value_type = iter_value_t<iterator_t<Base>>;
 using difference_type = iter_difference_t<iterator_t<Base>>;

 inner_iterator() = default;
 constexpr explicit inner_iterator(outer_iterator<Const> i);

 constexpr decltype(auto) operator*() const { return *i_.current; }

 constexpr inner_iterator& operator++();
 constexpr decltype(auto) operator++(int) {
 if constexpr (ForwardRange<V>) {
 auto tmp = *this;
 ++*this;
 return tmp;
 } else
 ++*this;
 }

 friend constexpr bool operator==(const inner_iterator& x, const inner_iterator& y)
 requires ForwardRange<Base>;
 friend constexpr bool operator!=(const inner_iterator& x, const inner_iterator& y)
 requires ForwardRange<Base>;

 friend constexpr bool operator==(const inner_iterator& x, default_sentinel_t);
 friend constexpr bool operator==(default_sentinel_t, const inner_iterator& x);
 friend constexpr bool operator!=(const inner_iterator& x, default_sentinel_t y);
 friend constexpr bool operator!=(default_sentinel_t y, const inner_iterator& x);

 friend constexpr decltype(auto) iter_move(const inner_iterator& i)
 noexcept(noexcept(ranges::iter_move(i.i_.current))) {
 return ranges::iter_move(i.i_.current);
 }

 friend constexpr void iter_swap(const inner_iterator& x, const inner_iterator& y)
 noexcept(noexcept(ranges::iter_swap(x.i_.current, y.i_.current)))
 requires IndirectlySwappable<iterator_t<Base>>;
 };
}
```

- <sup>1</sup> The *typedef-name* `iterator_category` denotes `forward_iterator_tag` if `iterator_traits<iterator_t<Base>>::iterator_category` models `DerivedFrom<forward_iterator_tag>`, and `input_iterator_tag` otherwise.

```
constexpr explicit inner_iterator(outer_iterator<Const> i);
```

- <sup>2</sup> *Effects:* Initializes `i_` with `i`.

```
constexpr inner_iterator& operator++() const;
```

- <sup>3</sup> *Effects:* Equivalent to:

```
 incremented_ = true;
 if constexpr (!ForwardRange<Base>) {
 if constexpr (Pattern::size() == 0) {
 return *this;
 }
 }
 ++i_.current;
 return *this;
```

```
friend constexpr bool operator==(const inner_iterator& x, const inner_iterator& y)
 requires ForwardRange<Base>;
```

- <sup>4</sup> *Effects:* Equivalent to: `return x.i_.current_ == y.i_.current_;`

```
friend constexpr bool operator!=(const inner_iterator& x, const inner_iterator& y)
 requires ForwardRange<Base>;
```

- <sup>5</sup> *Effects:* Equivalent to: `return !(x == y);`

```
friend constexpr bool operator==(const inner_iterator& x, default_sentinel_t);
friend constexpr bool operator==(default_sentinel_t, const inner_iterator& x);
```

- <sup>6</sup> *Effects:* Equivalent to:

```
 auto cur = x.i_.current;
 auto end = ranges::end(x.i_.parent_->base_);
 if (cur == end) return true;
 auto [pcur, pend] = subrange{x.i_.parent_->pattern_};
 if (pcur == pend) return x.incremented_;
 do {
 if (*cur != *pcur) return false;
 if (++pcur == pend) return true;
 } while (++cur != end);
 return false;
```

```
friend constexpr bool operator!=(const inner_iterator& x, default_sentinel_t y);
friend constexpr bool operator!=(default_sentinel_t y, const inner_iterator& x);
```

- <sup>7</sup> *Effects:* Equivalent to: `return !(x == y);`

```
friend constexpr void iter_swap(const inner_iterator& x, const inner_iterator& y)
 noexcept(noexcept(ranges::iter_swap(x.i_.current, y.i_.current)))
 requires IndirectlySwappable<iterator_t<Base>>;
```

- <sup>8</sup> *Effects:* Equivalent to `ranges::iter_swap(x.i_.current, y.i_.current)`.

### 23.7.11.6 `view::split`

[range.split.adaptor]

- <sup>1</sup> The name `view::split` denotes a range adaptor object (23.7.1). For some subexpressions `E` and `F`, the expression `view::split(E, F)` is expression-equivalent to `split_view{E, F}`.

### 23.7.12 Counted view

[range.counted]

- <sup>1</sup> The name `view::counted` denotes a customization point object ([customization.point.object]). Let `E` and `F` be expressions, and let `T` be `decay_t<decltype((E))>`. Then the expression `view::counted(E, F)` is expression-equivalent to:

- (1.1) — If `T` models `Iterator` and `decltype((F))` models `ConvertibleTo<iter_difference_t<T>>`,
- (1.1.1) — `subrange{E, E + static_cast<iter_difference_t<T>>(F)}` if `T` models `RandomAccessIterator`.

(1.1.2) — Otherwise, `subrange{counted_iterator{E, F}, default_sentinel}`.

(1.2) — Otherwise, `view::counted(E, F)` is ill-formed. [*Note: This case can result in substitution failure when `view::counted(E, F)` appears in the immediate context of a template instantiation. — end note*]

### 23.7.13 Common view [range.common]

#### 23.7.13.1 Overview [range.common.overview]

<sup>1</sup> `common_view` takes a `View` which has different types for its iterator and sentinel and turns it into a `View` of the same elements with an iterator and sentinel of the same type.

<sup>2</sup> [*Note: `common_view` is useful for calling legacy algorithms that expect a range's iterator and sentinel types to be the same. — end note*]

<sup>3</sup> [*Example:*

```
// Legacy algorithm:
template<class ForwardIterator>
size_t count(ForwardIterator first, ForwardIterator last);

template<ForwardRange R>
void my_algo(R&& r) {
 auto&& common = common_view{r};
 auto cnt = count(common.begin(), common.end());
 // ...
}
```

— end example]

#### 23.7.13.2 Class template `common_view` [range.common.view]

```
namespace std::ranges {
 template<View V>
 requires (!CommonRange<V>)
 class common_view : public view_interface<common_view<V>> {
 private:
 V base_ = V(); // exposition only
 public:
 common_view() = default;

 constexpr explicit common_view(V r);

 template<ViewableRange R>
 requires (!CommonRange<R>) && Constructible<V, all_view<R>>
 constexpr explicit common_view(R&& r);

 constexpr V base() const;

 constexpr auto size() requires SizedRange<V> {
 return ranges::size(base_);
 }
 constexpr auto size() const requires SizedRange<const V> {
 return ranges::size(base_);
 }

 constexpr auto begin() {
 if constexpr (RandomAccessRange<V> && SizedRange<V>)
 return ranges::begin(base_);
 else
 return common_iterator<iterator_t<V>, sentinel_t<V>>(ranges::begin(base_));
 }

 constexpr auto begin() const requires Range<const V> {
 if constexpr (RandomAccessRange<const V> && SizedRange<const V>)
 return ranges::begin(base_);
 else
 return common_iterator<iterator_t<const V>, sentinel_t<const V>>(ranges::begin(base_));
 }
 };
}
```

```

 }

 constexpr auto end() {
 if constexpr (RandomAccessRange<V> && SizedRange<V>)
 return ranges::begin(base_) + ranges::size(base_);
 else
 return common_iterator<iterator_t<V>, sentinel_t<V>>(ranges::end(base_));
 }

 constexpr auto end() const requires Range<const V> {
 if constexpr (RandomAccessRange<const V> && SizedRange<const V>)
 return ranges::begin(base_) + ranges::size(base_);
 else
 return common_iterator<iterator_t<const V>, sentinel_t<const V>>(ranges::end(base_));
 }
};

template<class R>
 common_view(R&&) -> common_view<all_view<R>>;
}

```

```
constexpr explicit common_view(V base_);
```

<sup>1</sup> *Effects:* Initializes `base_` with `std::move(base_)`.

```

template<ViewableRange R>
 requires (!CommonRange<R>) && Constructible<V, all_view<R>>
constexpr explicit common_view(R&& r);

```

<sup>2</sup> *Effects:* Initializes `base_` with `view::all(std::forward<R>(r))`.

```
constexpr V base() const;
```

<sup>3</sup> *Effects:* Equivalent to: `return base_;`

### 23.7.13.3 `view::common`

[range.common.adaptor]

<sup>1</sup> The name `view::common` denotes a range adaptor object (23.7.1). For some subexpression `E`, the expression `view::common(E)` is expression-equivalent to:

- (1.1) — `view::all(E)`, if `decltype((E))` models `CommonRange` and `view::all(E)` is a well-formed expression.
- (1.2) — Otherwise, `common_view{E}`.

## 23.7.14 Reverse view

[range.reverse]

### 23.7.14.1 Overview

[range.reverse.overview]

<sup>1</sup> `reverse_view` takes a bidirectional `View` and produces another `View` that iterates the same elements in reverse order.

<sup>2</sup> [Example:

```

vector<int> is {0,1,2,3,4};
reverse_view rv {is};
for (int i : rv)
 cout << i << ' '; // prints: 4 3 2 1 0

```

— end example]

### 23.7.14.2 Class template `reverse_view`

[range.reverse.view]

```

namespace std::ranges {
 template<View V>
 requires BidirectionalRange<V>
 class reverse_view : public view_interface<reverse_view<V>> {
 private:
 V base_ = V(); // exposition only
 public:
 reverse_view() = default;

```

```

constexpr explicit reverse_view(V r);

template<ViewableRange R>
 requires BidirectionalRange<R> && Constructible<V, all_view<R>>
constexpr explicit reverse_view(R&& r);

constexpr V base() const;

constexpr reverse_iterator<iterator_t<V>> begin();
constexpr reverse_iterator<iterator_t<V>> begin() requires CommonRange<V>;
constexpr reverse_iterator<iterator_t<const V>> begin() const
 requires CommonRange<const V>;

constexpr reverse_iterator<iterator_t<V>> end();
constexpr reverse_iterator<iterator_t<const V>> end() const
 requires CommonRange<const V>;

constexpr auto size() requires SizedRange<V> {
 return ranges::size(base_);
}
constexpr auto size() const requires SizedRange<const V> {
 return ranges::size(base_);
}
};

template<class R>
 reverse_view(R&&) -> reverse_view<all_view<R>>;
}

```

```
constexpr explicit reverse_view(V base);
```

1     *Effects:* Initializes `base_` with `std::move(base)`.

```

template<ViewableRange R>
 requires BidirectionalRange<R> && Constructible<V, all_view<R>>
constexpr explicit reverse_view(R&& r);

```

2     *Effects:* Initializes `base_` with `view::all(std::forward<R>(r))`.

```
constexpr V base() const;
```

3     *Effects:* Equivalent to: `return base_;`

```
constexpr reverse_iterator<iterator_t<V>> begin();
```

4     *Returns:* `make_reverse_iterator(ranges::next(ranges::begin(base_), ranges::end(base_)))`.

5     *Remarks:* In order to provide the amortized constant time complexity required by the `Range` concept, this function caches the result within the `reverse_view` for use on subsequent calls.

```

constexpr reverse_iterator<iterator_t<V>> begin() requires CommonRange<V>;
constexpr reverse_iterator<iterator_t<const V>> begin() const
 requires CommonRange<const V>;

```

6     *Effects:* Equivalent to: `return make_reverse_iterator(ranges::end(base_));`

```

constexpr reverse_iterator<iterator_t<V>> end();
constexpr reverse_iterator<iterator_t<const V>> end() const
 requires CommonRange<const V>;

```

7     *Effects:* Equivalent to: `return make_reverse_iterator(ranges::begin(base_));`

### 23.7.14.3 `view::reverse`

[`range.reverse.adaptor`]

1     The name `view::reverse` denotes a range adaptor object (23.7.1). For some subexpression `E`, the expression `view::reverse(E)` is expression-equivalent to `reverse_view{E}`.

## 24 Algorithms library

[algorithms]

### 24.1 General

[algorithms.general]

[...]

### 24.2 Header <algorithm> synopsis

[algorithm.syn]

```
#include <initializer_list>
```

```
namespace std {
 // 24.5, non-modifying sequence operations
 // 24.5.1, all of
 template<class InputIterator, class Predicate>
 constexpr bool all_of(InputIterator first, InputIterator last, Predicate pred);
 template<class ExecutionPolicy, class ForwardIterator, class Predicate>
 bool all_of(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last, Predicate pred);

 namespace ranges {
 template<InputIterator I, Sentinel<I> S, class Proj = identity,
 IndirectUnaryPredicate<projected<I, Proj>> Pred>
 constexpr bool all_of(I first, S last, Pred pred, Proj proj = {});
 template<InputRange R, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
 constexpr bool all_of(R&& r, Pred pred, Proj proj = {});
 }

 // 24.5.2, any of
 template<class InputIterator, class Predicate>
 constexpr bool any_of(InputIterator first, InputIterator last, Predicate pred);
 template<class ExecutionPolicy, class ForwardIterator, class Predicate>
 bool any_of(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last, Predicate pred);

 namespace ranges {
 template<InputIterator I, Sentinel<I> S, class Proj = identity,
 IndirectUnaryPredicate<projected<I, Proj>> Pred>
 constexpr bool any_of(I first, S last, Pred pred, Proj proj = {});
 template<InputRange R, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
 constexpr bool any_of(R&& r, Pred pred, Proj proj = {});
 }

 // 24.5.3, none of
 template<class InputIterator, class Predicate>
 constexpr bool none_of(InputIterator first, InputIterator last, Predicate pred);
 template<class ExecutionPolicy, class ForwardIterator, class Predicate>
 bool none_of(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last, Predicate pred);

 namespace ranges {
 template<InputIterator I, Sentinel<I> S, class Proj = identity,
 IndirectUnaryPredicate<projected<I, Proj>> Pred>
 constexpr bool none_of(I first, S last, Pred pred, Proj proj = {});
 template<InputRange R, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
 constexpr bool none_of(R&& r, Pred pred, Proj proj = {});
 }
}
```

```

// 24.5.4, for each
template<class InputIterator, class Function>
 constexpr Function for_each(InputIterator first, InputIterator last, Function f);
template<class ExecutionPolicy, class ForwardIterator, class Function>
 void for_each(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last, Function f);

namespace ranges {
 template<class I, class F>
 struct for_each_result {
 I in;
 F fun;
 };

 template<InputIterator I, Sentinel<I> S, class Proj = identity,
 IndirectUnaryInvocable<projected<I, Proj>> Fun>
 constexpr for_each_result<I, Fun>
 for_each(I first, S last, Fun f, Proj proj = {});
 template<InputRange R, class Proj = identity,
 IndirectUnaryInvocable<projected<iterator_t<R>, Proj>> Fun>
 constexpr for_each_result<safe_iterator_t<R>, Fun>
 for_each(R&& r, Fun f, Proj proj = {});
}

template<class InputIterator, class Size, class Function>
 constexpr InputIterator for_each_n(InputIterator first, Size n, Function f);
template<class ExecutionPolicy, class ForwardIterator, class Size, class Function>
 ForwardIterator for_each_n(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, Size n, Function f);

// 24.5.5, find
template<class InputIterator, class T>
 constexpr InputIterator find(InputIterator first, InputIterator last,
 const T& value);
template<class ExecutionPolicy, class ForwardIterator, class T>
 ForwardIterator find(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last,
 const T& value);
template<class InputIterator, class Predicate>
 constexpr InputIterator find_if(InputIterator first, InputIterator last,
 Predicate pred);
template<class ExecutionPolicy, class ForwardIterator, class Predicate>
 ForwardIterator find_if(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last,
 Predicate pred);
template<class InputIterator, class Predicate>
 constexpr InputIterator find_if_not(InputIterator first, InputIterator last,
 Predicate pred);
template<class ExecutionPolicy, class ForwardIterator, class Predicate>
 ForwardIterator find_if_not(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last,
 Predicate pred);

namespace ranges {
 template<InputIterator I, Sentinel<I> S, class T, class Proj = identity>
 requires IndirectRelation<ranges::equal_to<>, projected<I, Proj>, const T*>
 constexpr I find(I first, S last, const T& value, Proj proj = {});
 template<InputRange R, class T, class Proj = identity>
 requires IndirectRelation<ranges::equal_to<>, projected<iterator_t<R>, Proj>, const T*>
 constexpr safe_iterator_t<R>
 find(R&& r, const T& value, Proj proj = {});
 template<InputIterator I, Sentinel<I> S, class Proj = identity,
 IndirectUnaryPredicate<projected<I, Proj>> Pred>
 constexpr I find_if(I first, S last, Pred pred, Proj proj = {});
}

```

```

template<InputRange R, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
constexpr safe_iterator_t<R>
 find_if(R&& r, Pred pred, Proj proj = {});
template<InputIterator I, Sentinel<I> S, class Proj = identity,
 IndirectUnaryPredicate<projected<I, Proj>> Pred>
constexpr I find_if_not(I first, S last, Pred pred, Proj proj = {});
template<InputRange R, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
constexpr safe_iterator_t<R>
 find_if_not(R&& r, Pred pred, Proj proj = {});
}

// 24.5.6, find end
template<class ForwardIterator1, class ForwardIterator2>
constexpr ForwardIterator1
 find_end(ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2);
template<class ForwardIterator1, class ForwardIterator2, class BinaryPredicate>
constexpr ForwardIterator1
 find_end(ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2,
 BinaryPredicate pred);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
ForwardIterator1
 find_end(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2);
template<class ExecutionPolicy, class ForwardIterator1,
 class ForwardIterator2, class BinaryPredicate>
ForwardIterator1
 find_end(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2,
 BinaryPredicate pred);

namespace ranges {
 template<ForwardIterator I1, Sentinel<I1> S1, ForwardIterator I2, Sentinel<I2> S2,
 class Pred = ranges::equal_to<>, class Proj1 = identity, class Proj2 = identity>
 requires IndirectlyComparable<I1, I2, Pred, Proj1, Proj2>
 constexpr subrange<I1>
 find_end(I1 first1, S1 last1, I2 first2, S2 last2, Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
 template<ForwardRange R1, ForwardRange R2,
 class Pred = ranges::equal_to<>, class Proj1 = identity, class Proj2 = identity>
 requires IndirectlyComparable<iterator_t<R1>, iterator_t<R2>, Pred, Proj1, Proj2>
 constexpr safe_subrange_t<R1>
 find_end(R1&& r1, R2&& r2, Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
}

// 24.5.7, find first
template<class InputIterator, class ForwardIterator>
constexpr InputIterator
 find_first_of(InputIterator first1, InputIterator last1,
 ForwardIterator first2, ForwardIterator last2);
template<class InputIterator, class ForwardIterator, class BinaryPredicate>
constexpr InputIterator
 find_first_of(InputIterator first1, InputIterator last1,
 ForwardIterator first2, ForwardIterator last2,
 BinaryPredicate pred);

```

```

template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
 ForwardIterator1
 find_first_of(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2);
template<class ExecutionPolicy, class ForwardIterator1,
 class ForwardIterator2, class BinaryPredicate>
 ForwardIterator1
 find_first_of(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2,
 BinaryPredicate pred);

namespace ranges {
 template<InputIterator I1, Sentinel<I1> S1, ForwardIterator I2, Sentinel<I2> S2,
 class Proj1 = identity, class Proj2 = identity,
 IndirectRelation<projected<I1, Proj1>, projected<I2, Proj2>> Pred = ranges::equal_to<>>
 constexpr I1 find_first_of(I1 first1, S1 last1, I2 first2, S2 last2,
 Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
 template<InputRange R1, ForwardRange R2, class Proj1 = identity,
 class Proj2 = identity,
 IndirectRelation<projected<iterator_t<R1>, Proj1>,
 projected<iterator_t<R2>, Proj2>> Pred = ranges::equal_to<>>
 constexpr safe_iterator_t<R1>
 find_first_of(R1&& r1, R2&& r2,
 Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
}

// 24.5.8, adjacent find
template<class ForwardIterator>
 constexpr ForwardIterator
 adjacent_find(ForwardIterator first, ForwardIterator last);
template<class ForwardIterator, class BinaryPredicate>
 constexpr ForwardIterator
 adjacent_find(ForwardIterator first, ForwardIterator last,
 BinaryPredicate pred);
template<class ExecutionPolicy, class ForwardIterator>
 ForwardIterator
 adjacent_find(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last);
template<class ExecutionPolicy, class ForwardIterator, class BinaryPredicate>
 ForwardIterator
 adjacent_find(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last,
 BinaryPredicate pred);

namespace ranges {
 template<ForwardIterator I, Sentinel<I> S, class Proj = identity,
 IndirectRelation<projected<I, Proj>> Pred = ranges::equal_to<>>
 constexpr I adjacent_find(I first, S last, Pred pred = {},
 Proj proj = {});
 template<ForwardRange R, class Proj = identity,
 IndirectRelation<projected<iterator_t<R>, Proj>> Pred = ranges::equal_to<>>
 constexpr safe_iterator_t<R>
 adjacent_find(R&& r, Pred pred = {}, Proj proj = {});
}

// 24.5.9, count
template<class InputIterator, class T>
 constexpr typename iterator_traits<InputIterator>::difference_type
 count(InputIterator first, InputIterator last, const T& value);

```

```

template<class ExecutionPolicy, class ForwardIterator, class T>
 typename iterator_traits<ForwardIterator>::difference_type
 count(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last, const T& value);
template<class InputIterator, class Predicate>
 constexpr typename iterator_traits<InputIterator>::difference_type
 count_if(InputIterator first, InputIterator last, Predicate pred);
template<class ExecutionPolicy, class ForwardIterator, class Predicate>
 typename iterator_traits<ForwardIterator>::difference_type
 count_if(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last, Predicate pred);

namespace ranges {
 template<InputIterator I, Sentinel<I> S, class T, class Proj = identity>
 requires IndirectRelation<ranges::equal_to<>, projected<I, Proj>, const T*>
 constexpr iter_difference_t<I>
 count(I first, S last, const T& value, Proj proj = {});
 template<InputRange R, class T, class Proj = identity>
 requires IndirectRelation<ranges::equal_to<>, projected<iterator_t<R>, Proj>, const T*>
 constexpr iter_difference_t<iterator_t<R>>
 count(R&& r, const T& value, Proj proj = {});
 template<InputIterator I, Sentinel<I> S, class Proj = identity,
 IndirectUnaryPredicate<projected<I, Proj>> Pred>
 constexpr iter_difference_t<I>
 count_if(I first, S last, Pred pred, Proj proj = {});
 template<InputRange R, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
 constexpr iter_difference_t<iterator_t<R>>
 count_if(R&& r, Pred pred, Proj proj = {});
}

// 24.5.10, mismatch
template<class InputIterator1, class InputIterator2>
 constexpr pair<InputIterator1, InputIterator2>
 mismatch(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2);
template<class InputIterator1, class InputIterator2, class BinaryPredicate>
 constexpr pair<InputIterator1, InputIterator2>
 mismatch(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, BinaryPredicate pred);
template<class InputIterator1, class InputIterator2>
 constexpr pair<InputIterator1, InputIterator2>
 mismatch(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, InputIterator2 last2);
template<class InputIterator1, class InputIterator2, class BinaryPredicate>
 constexpr pair<InputIterator1, InputIterator2>
 mismatch(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, InputIterator2 last2,
 BinaryPredicate pred);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
 pair<ForwardIterator1, ForwardIterator2>
 mismatch(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class BinaryPredicate>
 pair<ForwardIterator1, ForwardIterator2>
 mismatch(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, BinaryPredicate pred);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
 pair<ForwardIterator1, ForwardIterator2>
 mismatch(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,

```

```

 ForwardIterator2 first2, ForwardIterator2 last2);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class BinaryPredicate>
pair<ForwardIterator1, ForwardIterator2>
 mismatch(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2,
 BinaryPredicate pred);

namespace ranges {
 template<class I1, class I2>
 struct mismatch_result {
 I1 in1;
 I2 in2;
 };

 template<InputIterator I1, Sentinel<I1> S1, InputIterator I2, Sentinel<I2> S2,
 class Proj1 = identity, class Proj2 = identity,
 IndirectRelation<projected<I1, Proj1>, projected<I2, Proj2>> Pred = ranges::equal_to<>>
 constexpr mismatch_result<I1, I2>
 mismatch(I1 first1, S1 last1, I2 first2, S2 last2, Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
 template<InputRange R1, InputRange R2,
 class Proj1 = identity, class Proj2 = identity,
 IndirectRelation<projected<iterator_t<R1>, Proj1>,
 projected<iterator_t<R2>, Proj2>> Pred = ranges::equal_to<>>
 constexpr mismatch_result<safe_iterator_t<R1>, safe_iterator_t<R2>>
 mismatch(R1&& r1, R2&& r2, Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
}

// 24.5.11, equal
template<class InputIterator1, class InputIterator2>
 constexpr bool equal(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2);
template<class InputIterator1, class InputIterator2, class BinaryPredicate>
 constexpr bool equal(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, BinaryPredicate pred);
template<class InputIterator1, class InputIterator2>
 constexpr bool equal(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, InputIterator2 last2);
template<class InputIterator1, class InputIterator2, class BinaryPredicate>
 constexpr bool equal(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, InputIterator2 last2,
 BinaryPredicate pred);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
 bool equal(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class BinaryPredicate>
 bool equal(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, BinaryPredicate pred);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
 bool equal(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class BinaryPredicate>
 bool equal(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2,
 BinaryPredicate pred);

```

```

namespace ranges {
 template<InputIterator I1, Sentinel<I1> S1, InputIterator I2, Sentinel<I2> S2,
 class Pred = ranges::equal_to<>, class Proj1 = identity, class Proj2 = identity>
 requires IndirectlyComparable<I1, I2, Pred, Proj1, Proj2>
 constexpr bool equal(I1 first1, S1 last1, I2 first2, S2 last2,
 Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
 template<InputRange R1, InputRange R2, class Pred = ranges::equal_to<>,
 class Proj1 = identity, class Proj2 = identity>
 requires IndirectlyComparable<iterator_t<R1>, iterator_t<R2>, Pred, Proj1, Proj2>
 constexpr bool equal(R1&& r1, R2&& r2, Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
}

// 24.5.12, is_permutation
template<class ForwardIterator1, class ForwardIterator2>
 constexpr bool is_permutation(ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2);
template<class ForwardIterator1, class ForwardIterator2, class BinaryPredicate>
 constexpr bool is_permutation(ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, BinaryPredicate pred);
template<class ForwardIterator1, class ForwardIterator2>
 constexpr bool is_permutation(ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2);
template<class ForwardIterator1, class ForwardIterator2, class BinaryPredicate>
 constexpr bool is_permutation(ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2,
 BinaryPredicate pred);

namespace ranges {
 template<ForwardIterator I1, Sentinel<I1> S1, ForwardIterator I2,
 Sentinel<I2> S2, class Pred = ranges::equal_to<>, class Proj1 = identity,
 class Proj2 = identity>
 requires IndirectlyComparable<I1, I2, Pred, Proj1, Proj2>
 constexpr bool is_permutation(I1 first1, S1 last1, I2 first2, S2 last2,
 Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
 template<ForwardRange R1, ForwardRange R2, class Pred = ranges::equal_to<>,
 class Proj1 = identity, class Proj2 = identity>
 requires IndirectlyComparable<iterator_t<R1>, iterator_t<R2>, Pred, Proj1, Proj2>
 constexpr bool is_permutation(R1&& r1, R2&& r2, Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
}

// 24.5.13, search
template<class ForwardIterator1, class ForwardIterator2>
 constexpr ForwardIterator1
 search(ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2);
template<class ForwardIterator1, class ForwardIterator2, class BinaryPredicate>
 constexpr ForwardIterator1
 search(ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2,
 BinaryPredicate pred);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
 ForwardIterator1
 search(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class BinaryPredicate>
 ForwardIterator1
 search(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,

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```

 ForwardIterator2 first2, ForwardIterator2 last2,
 BinaryPredicate pred);

namespace ranges {
 template<ForwardIterator I1, Sentinel<I1> S1, ForwardIterator I2,
 Sentinel<I2> S2, class Pred = ranges::equal_to<>,
 class Proj1 = identity, class Proj2 = identity>
 requires IndirectlyComparable<I1, I2, Pred, Proj1, Proj2>
 constexpr subrange<I1>
 search(I1 first1, S1 last1, I2 first2, S2 last2, Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
 template<ForwardRange R1, ForwardRange R2, class Pred = ranges::equal_to<>,
 class Proj1 = identity, class Proj2 = identity>
 requires IndirectlyComparable<iterator_t<R1>, iterator_t<R2>, Pred, Proj1, Proj2>
 constexpr safe_subrange_t<R1>
 search(R1&& r1, R2&& r2, Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
}

template<class ForwardIterator, class Size, class T>
 constexpr ForwardIterator
 search_n(ForwardIterator first, ForwardIterator last,
 Size count, const T& value);
template<class ForwardIterator, class Size, class T, class BinaryPredicate>
 constexpr ForwardIterator
 search_n(ForwardIterator first, ForwardIterator last,
 Size count, const T& value,
 BinaryPredicate pred);
template<class ExecutionPolicy, class ForwardIterator, class Size, class T>
 ForwardIterator
 search_n(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last,
 Size count, const T& value);
template<class ExecutionPolicy, class ForwardIterator, class Size, class T,
 class BinaryPredicate>
 ForwardIterator
 search_n(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last,
 Size count, const T& value,
 BinaryPredicate pred);

namespace ranges {
 template<ForwardIterator I, Sentinel<I> S, class T,
 class Pred = ranges::equal_to<>, class Proj = identity>
 requires IndirectlyComparable<I, const T*, Pred, Proj>
 constexpr subrange<I>
 search_n(I first, S last, iter_difference_t<I> count,
 const T& value, Pred pred = {}, Proj proj = {});
 template<ForwardRange R, class T, class Pred = ranges::equal_to<>,
 class Proj = identity>
 requires IndirectlyComparable<iterator_t<R>, const T*, Pred, Proj>
 constexpr safe_subrange_t<R>
 search_n(R&& r, iter_difference_t<iterator_t<R>> count,
 const T& value, Pred pred = {}, Proj proj = {});
}

template<class ForwardIterator, class Searcher>
 constexpr ForwardIterator
 search(ForwardIterator first, ForwardIterator last, const Searcher& searcher);

// 24.6, mutating sequence operations
// 24.6.1, copy
template<class InputIterator, class OutputIterator>
 constexpr OutputIterator copy(InputIterator first, InputIterator last,
 OutputIterator result);

```

```

template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
 ForwardIterator2 copy(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first, ForwardIterator1 last,
 ForwardIterator2 result);

namespace ranges {
 template<class I, class O>
 struct copy_result {
 I in;
 O out;
 };

 template<InputIterator I, Sentinel<I> S, WeaklyIncrementable O>
 requires IndirectlyCopyable<I, O>
 constexpr copy_result<I, O>
 copy(I first, S last, O result);
 template<InputRange R, WeaklyIncrementable O>
 requires IndirectlyCopyable<iterator_t<R>, O>
 constexpr copy_result<safe_iterator_t<R>, O>
 copy(R&& r, O result);
}

template<class InputIterator, class Size, class OutputIterator>
 constexpr OutputIterator copy_n(InputIterator first, Size n,
 OutputIterator result);

template<class ExecutionPolicy, class ForwardIterator1, class Size,
 class ForwardIterator2>
 ForwardIterator2 copy_n(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first, Size n,
 ForwardIterator2 result);

namespace ranges {
 template<class I, class O>
 using copy_n_result = copy_result<I, O>;

 template<InputIterator I, WeaklyIncrementable O>
 requires IndirectlyCopyable<I, O>
 constexpr copy_n_result<I, O>
 copy_n(I first, iter_difference_t<I> n, O result);
}

template<class InputIterator, class OutputIterator, class Predicate>
 constexpr OutputIterator copy_if(InputIterator first, InputIterator last,
 OutputIterator result, Predicate pred);

template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class Predicate>
 ForwardIterator2 copy_if(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first, ForwardIterator1 last,
 ForwardIterator2 result, Predicate pred);

namespace ranges {
 template<class I, class O>
 using copy_if_result = copy_result<I, O>;

 template<InputIterator I, Sentinel<I> S, WeaklyIncrementable O, class Proj = identity,
 IndirectUnaryPredicate<projected<I, Proj>> Pred>
 requires IndirectlyCopyable<I, O>
 constexpr copy_if_result<I, O>
 copy_if(I first, S last, O result, Pred pred, Proj proj = {});
 template<InputRange R, WeaklyIncrementable O, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
 requires IndirectlyCopyable<iterator_t<R>, O>
 constexpr copy_if_result<safe_iterator_t<R>, O>
 copy_if(R&& r, O result, Pred pred, Proj proj = {});
}

```

```

template<class BidirectionalIterator1, class BidirectionalIterator2>
constexpr BidirectionalIterator2
 copy_backward(BidirectionalIterator1 first, BidirectionalIterator1 last,
 BidirectionalIterator2 result);

namespace ranges {
 template<class I1, class I2>
 using copy_backward_result = copy_result<I1, I2>;

 template<BidirectionalIterator I1, Sentinel<I1> S1, BidirectionalIterator I2>
 requires IndirectlyCopyable<I1, I2>
 constexpr copy_backward_result<I1, I2>
 copy_backward(I1 first, S1 last, I2 result);
 template<BidirectionalRange R, BidirectionalIterator I>
 requires IndirectlyCopyable<iterator_t<R>, I>
 constexpr copy_backward_result<safe_iterator_t<R>, I>
 copy_backward(R&& r, I result);
}

// 24.6.2, move
template<class InputIterator, class OutputIterator>
constexpr OutputIterator move(InputIterator first, InputIterator last,
 OutputIterator result);

namespace ranges {
 template<class I, class O>
 using move_result = copy_result<I, O>;

 template<InputIterator I, Sentinel<I> S, WeaklyIncrementable O>
 requires IndirectlyMovable<I, O>
 constexpr move_result<I, O>
 move(I first, S last, O result);
 template<InputRange R, WeaklyIncrementable O>
 requires IndirectlyMovable<iterator_t<R>, O>
 constexpr move_result<safe_iterator_t<R>, O>
 move(R&& r, O result);
}

template<class ExecutionPolicy, class ForwardIterator1,
 class ForwardIterator2>
ForwardIterator2 move(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first, ForwardIterator1 last,
 ForwardIterator2 result);
template<class BidirectionalIterator1, class BidirectionalIterator2>
constexpr BidirectionalIterator2
 move_backward(BidirectionalIterator1 first, BidirectionalIterator1 last,
 BidirectionalIterator2 result);

namespace ranges {
 template<class I1, class I2>
 using move_backward_result = copy_result<I1, I2>;

 template<BidirectionalIterator I1, Sentinel<I1> S1, BidirectionalIterator I2>
 requires IndirectlyMovable<I1, I2>
 constexpr move_backward_result<I1, I2>
 move_backward(I1 first, S1 last, I2 result);
 template<BidirectionalRange R, BidirectionalIterator I>
 requires IndirectlyMovable<iterator_t<R>, I>
 constexpr move_backward_result<safe_iterator_t<R>, I>
 move_backward(R&& r, I result);
}

// 24.6.3, swap
template<class ForwardIterator1, class ForwardIterator2>
constexpr ForwardIterator2

```

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 swap_ranges(ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
ForwardIterator2
 swap_ranges(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2);

namespace ranges {
 template<class I1, class I2>
 using swap_ranges_result = mismatch_result<I1, I2>;

 template<InputIterator I1, Sentinel<I1> S1, InputIterator I2, Sentinel<I2> S2>
 requires IndirectlySwappable<I1, I2>
 constexpr swap_ranges_result<I1, I2>
 swap_ranges(I1 first1, S1 last1, I2 first2, S2 last2);
 template<InputRange R1, InputRange R2>
 requires IndirectlySwappable<iterator_t<R1>, iterator_t<R2>>
 constexpr swap_ranges_result<safe_iterator_t<R1>, safe_iterator_t<R2>>
 swap_ranges(R1&& r1, R2&& r2);
}

template<class ForwardIterator1, class ForwardIterator2>
void iter_swap(ForwardIterator1 a, ForwardIterator2 b);

// 24.6.4, transform
template<class InputIterator, class OutputIterator, class UnaryOperation>
constexpr OutputIterator
 transform(InputIterator first1, InputIterator last1,
 OutputIterator result, UnaryOperation op);
template<class InputIterator1, class InputIterator2, class OutputIterator,
 class BinaryOperation>
constexpr OutputIterator
 transform(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, OutputIterator result,
 BinaryOperation binary_op);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class UnaryOperation>
ForwardIterator2
 transform(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 result, UnaryOperation op);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class ForwardIterator, class BinaryOperation>
ForwardIterator
 transform(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator result,
 BinaryOperation binary_op);

namespace ranges {
 template<class I, class O>
 using unary_transform_result = copy_result<I, O>;

 template<InputIterator I, Sentinel<I> S, WeaklyIncrementable O,
 CopyConstructible F, class Proj = identity>
 requires Writable<O, indirect_result_t<F&, projected<I, Proj>>>
 constexpr unary_transform_result<I, O>
 transform(I first1, S last1, O result, F op, Proj proj = {});
 template<InputRange R, WeaklyIncrementable O, CopyConstructible F,
 class Proj = identity>
 requires Writable<O, indirect_result_t<F&, projected<iterator_t<R>, Proj>>>
 constexpr unary_transform_result<safe_iterator_t<R>, O>
 transform(R&& r, O result, F op, Proj proj = {});
}

```

```

template<class I1, class I2, class O>
struct binary_transform_result {
 I1 in1;
 I2 in2;
 O out;
};

template<InputIterator I1, Sentinel<I1> S1, InputIterator I2, Sentinel<I2> S2,
 WeaklyIncrementable O, CopyConstructible F, class Proj1 = identity,
 class Proj2 = identity>
requires Writable<O, indirect_result_t<F&, projected<I1, Proj1>,
 projected<I2, Proj2>>>
constexpr binary_transform_result<I1, I2, O>
 transform(I1 first1, S1 last1, I2 first2, S2 last2, O result,
 F binary_op, Proj1 proj1 = {}, Proj2 proj2 = {});

template<InputRange R1, InputRange R2, WeaklyIncrementable O,
 CopyConstructible F, class Proj1 = identity, class Proj2 = identity>
requires Writable<O, indirect_result_t<F&,
 projected<iterator_t<R1>, Proj1>, projected<iterator_t<R2>, Proj2>>>
constexpr binary_transform_result<safe_iterator_t<R1>, safe_iterator_t<R2>, O>
 transform(R1&& r1, R2&& r2, O result,
 F binary_op, Proj1 proj1 = {}, Proj2 proj2 = {});
}

// 24.6.5, replace
template<class ForwardIterator, class T>
constexpr void replace(ForwardIterator first, ForwardIterator last,
 const T& old_value, const T& new_value);

template<class ExecutionPolicy, class ForwardIterator, class T>
void replace(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last,
 const T& old_value, const T& new_value);

template<class ForwardIterator, class Predicate, class T>
constexpr void replace_if(ForwardIterator first, ForwardIterator last,
 Predicate pred, const T& new_value);

template<class ExecutionPolicy, class ForwardIterator, class Predicate, class T>
void replace_if(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last,
 Predicate pred, const T& new_value);

namespace ranges {
 template<InputIterator I, Sentinel<I> S, class T1, class T2, class Proj = identity>
 requires Writable<I, const T2&> &&
 IndirectRelation<ranges::equal_to<>, projected<I, Proj>, const T1*>
 constexpr I
 replace(I first, S last, const T1& old_value, const T2& new_value, Proj proj = {});
 template<InputRange R, class T1, class T2, class Proj = identity>
 requires Writable<iterator_t<R>, const T2&> &&
 IndirectRelation<ranges::equal_to<>, projected<iterator_t<R>, Proj>, const T1*>
 constexpr safe_iterator_t<R>
 replace(R&& r, const T1& old_value, const T2& new_value, Proj proj = {});
 template<InputIterator I, Sentinel<I> S, class T, class Proj = identity,
 IndirectUnaryPredicate<projected<I, Proj>> Pred>
 requires Writable<I, const T&>
 constexpr I replace_if(I first, S last, Pred pred, const T& new_value, Proj proj = {});
 template<InputRange R, class T, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
 requires Writable<iterator_t<R>, const T&>
 constexpr safe_iterator_t<R>
 replace_if(R&& r, Pred pred, const T& new_value, Proj proj = {});
}

template<class InputIterator, class OutputIterator, class T>
constexpr OutputIterator replace_copy(InputIterator first, InputIterator last,
 OutputIterator result,

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 const T& old_value, const T& new_value);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2, class T>
 ForwardIterator2 replace_copy(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first, ForwardIterator1 last,
 ForwardIterator2 result,
 const T& old_value, const T& new_value);
template<class InputIterator, class OutputIterator, class Predicate, class T>
 constexpr OutputIterator replace_copy_if(InputIterator first, InputIterator last,
 OutputIterator result,
 Predicate pred, const T& new_value);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class Predicate, class T>
 ForwardIterator2 replace_copy_if(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first, ForwardIterator1 last,
 ForwardIterator2 result,
 Predicate pred, const T& new_value);

namespace ranges {
 template<class I, class O>
 using replace_copy_result = copy_result<I, O>;

 template<InputIterator I, Sentinel<I> S, class T1, class T2, OutputIterator<const T2&> O,
 class Proj = identity>
 requires IndirectlyCopyable<I, O> &&
 IndirectRelation<ranges::equal_to<>, projected<I, Proj>, const T1*>
 constexpr replace_copy_result<I, O>
 replace_copy(I first, S last, O result, const T1& old_value, const T2& new_value,
 Proj proj = {});
 template<InputRange R, class T1, class T2, OutputIterator<const T2&> O,
 class Proj = identity>
 requires IndirectlyCopyable<iterator_t<R>, O> &&
 IndirectRelation<ranges::equal_to<>, projected<iterator_t<R>, Proj>, const T1*>
 constexpr replace_copy_result<safe_iterator_t<R>, O>
 replace_copy(R&& r, O result, const T1& old_value, const T2& new_value,
 Proj proj = {});

 template<class I, class O>
 using replace_copy_if_result = copy_result<I, O>;

 template<InputIterator I, Sentinel<I> S, class T, OutputIterator<const T&> O,
 class Proj = identity, IndirectUnaryPredicate<projected<I, Proj>> Pred>
 requires IndirectlyCopyable<I, O>
 constexpr replace_copy_if_result<I, O>
 replace_copy_if(I first, S last, O result, Pred pred, const T& new_value,
 Proj proj = {});
 template<InputRange R, class T, OutputIterator<const T&> O, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
 requires IndirectlyCopyable<iterator_t<R>, O>
 constexpr replace_copy_if_result<safe_iterator_t<R>, O>
 replace_copy_if(R&& r, O result, Pred pred, const T& new_value,
 Proj proj = {});
}

// 24.6.6, fill
template<class ForwardIterator, class T>
 constexpr void fill(ForwardIterator first, ForwardIterator last, const T& value);
template<class ExecutionPolicy, class ForwardIterator, class T>
 void fill(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last, const T& value);
template<class OutputIterator, class Size, class T>
 constexpr OutputIterator fill_n(OutputIterator first, Size n, const T& value);
template<class ExecutionPolicy, class ForwardIterator,
 class Size, class T>
 ForwardIterator fill_n(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]

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 ForwardIterator first, Size n, const T& value);

namespace ranges {
 template<class T, OutputIterator<const T&> O, Sentinel<O> S>
 constexpr O fill(O first, S last, const T& value);
 template<class T, OutputRange<const T&> R>
 constexpr safe_iterator_t<R> fill(R&& r, const T& value);
 template<class T, OutputIterator<const T&> O>
 constexpr O fill_n(O first, iter_difference_t<O> n, const T& value);
}

// 24.6.7, generate
template<class ForwardIterator, class Generator>
 constexpr void generate(ForwardIterator first, ForwardIterator last,
 Generator gen);
template<class ExecutionPolicy, class ForwardIterator, class Generator>
 void generate(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last,
 Generator gen);
template<class OutputIterator, class Size, class Generator>
 constexpr OutputIterator generate_n(OutputIterator first, Size n, Generator gen);
template<class ExecutionPolicy, class ForwardIterator, class Size, class Generator>
 ForwardIterator generate_n(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, Size n, Generator gen);

namespace ranges {
 template<Iterator O, Sentinel<O> S, CopyConstructible F>
 requires Invocable<F&> && Writable<O, invoke_result_t<F&>>
 constexpr O generate(O first, S last, F gen);
 template<class R, CopyConstructible F>
 requires Invocable<F&> && OutputRange<R, invoke_result_t<F&>>
 constexpr safe_iterator_t<R> generate(R&& r, F gen);
 template<Iterator O, CopyConstructible F>
 requires Invocable<F&> && Writable<O, invoke_result_t<F&>>
 constexpr O generate_n(O first, iter_difference_t<O> n, F gen);
}

// 24.6.8, remove
template<class ForwardIterator, class T>
 constexpr ForwardIterator remove(ForwardIterator first, ForwardIterator last,
 const T& value);
template<class ExecutionPolicy, class ForwardIterator, class T>
 ForwardIterator remove(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last,
 const T& value);
template<class ForwardIterator, class Predicate>
 constexpr ForwardIterator remove_if(ForwardIterator first, ForwardIterator last,
 Predicate pred);
template<class ExecutionPolicy, class ForwardIterator, class Predicate>
 ForwardIterator remove_if(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last,
 Predicate pred);

namespace ranges {
 template<Permutable I, Sentinel<I> S, class T, class Proj = identity>
 requires IndirectRelation<ranges::equal_to<>, projected<I, Proj>, const T*>
 constexpr I remove(I first, S last, const T& value, Proj proj = {});
 template<ForwardRange R, class T, class Proj = identity>
 requires Permutable<iterator_t<R>> &&
 IndirectRelation<ranges::equal_to<>, projected<iterator_t<R>, Proj>, const T*>
 constexpr safe_iterator_t<R>
 remove(R&& r, const T& value, Proj proj = {});
 template<Permutable I, Sentinel<I> S, class Proj = identity,
 IndirectUnaryPredicate<projected<I, Proj>> Pred>
 constexpr I remove_if(I first, S last, Pred pred, Proj proj = {});
}

```

```

template<ForwardRange R, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
requires Permutable<iterator_t<R>>
constexpr safe_iterator_t<R>
 remove_if(R&& r, Pred pred, Proj proj = {});
}

template<class InputIterator, class OutputIterator, class T>
constexpr OutputIterator
 remove_copy(InputIterator first, InputIterator last,
 OutputIterator result, const T& value);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class T>
ForwardIterator2
 remove_copy(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first, ForwardIterator1 last,
 ForwardIterator2 result, const T& value);
template<class InputIterator, class OutputIterator, class Predicate>
constexpr OutputIterator
 remove_copy_if(InputIterator first, InputIterator last,
 OutputIterator result, Predicate pred);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class Predicate>
ForwardIterator2
 remove_copy_if(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first, ForwardIterator1 last,
 ForwardIterator2 result, Predicate pred);

namespace ranges {
 template<class I, class O>
 using remove_copy_result = copy_result<I, O>;

 template<InputIterator I, Sentinel<I> S, WeaklyIncrementable O, class T,
 class Proj = identity>
 requires IndirectlyCopyable<I, O> &&
 IndirectRelation<ranges::equal_to<>, projected<I, Proj>, const T*>
 constexpr remove_copy_result<I, O>
 remove_copy(I first, S last, O result, const T& value, Proj proj = {});
 template<InputRange R, WeaklyIncrementable O, class T, class Proj = identity>
 requires IndirectlyCopyable<iterator_t<R>, O> &&
 IndirectRelation<ranges::equal_to<>, projected<iterator_t<R>, Proj>, const T*>
 constexpr remove_copy_result<safe_iterator_t<R>, O>
 remove_copy(R&& r, O result, const T& value, Proj proj = {});

 template<class I, class O>
 using remove_copy_if_result = copy_result<I, O>;

 template<InputIterator I, Sentinel<I> S, WeaklyIncrementable O,
 class Proj = identity, IndirectUnaryPredicate<projected<I, Proj>> Pred>
 requires IndirectlyCopyable<I, O>
 constexpr remove_copy_if_result<I, O>
 remove_copy_if(I first, S last, O result, Pred pred, Proj proj = {});
 template<InputRange R, WeaklyIncrementable O, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
 requires IndirectlyCopyable<iterator_t<R>, O>
 constexpr remove_copy_if_result<safe_iterator_t<R>, O>
 remove_copy_if(R&& r, O result, Pred pred, Proj proj = {});
}

// 24.6.9, unique
template<class ForwardIterator>
constexpr ForwardIterator unique(ForwardIterator first, ForwardIterator last);
template<class ForwardIterator, class BinaryPredicate>
constexpr ForwardIterator unique(ForwardIterator first, ForwardIterator last,
 BinaryPredicate pred);

```

```

template<class ExecutionPolicy, class ForwardIterator>
 ForwardIterator unique(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last);
template<class ExecutionPolicy, class ForwardIterator, class BinaryPredicate>
 ForwardIterator unique(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last,
 BinaryPredicate pred);

namespace ranges {
 template<Permutable I, Sentinel<I> S, class Proj = identity,
 IndirectRelation<projected<I, Proj>> C = ranges::equal_to<>>
 constexpr I unique(I first, S last, C comp = {}, Proj proj = {});
 template<ForwardRange R, class Proj = identity,
 IndirectRelation<projected<iterator_t<R>, Proj>> C = ranges::equal_to<>>
 requires Permutable<iterator_t<R>>
 constexpr safe_iterator_t<R>
 unique(R&& r, C comp = {}, Proj proj = {});
}

template<class InputIterator, class OutputIterator>
 constexpr OutputIterator
 unique_copy(InputIterator first, InputIterator last,
 OutputIterator result);
template<class InputIterator, class OutputIterator, class BinaryPredicate>
 constexpr OutputIterator
 unique_copy(InputIterator first, InputIterator last,
 OutputIterator result, BinaryPredicate pred);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
 ForwardIterator2
 unique_copy(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first, ForwardIterator1 last,
 ForwardIterator2 result);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class BinaryPredicate>
 ForwardIterator2
 unique_copy(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first, ForwardIterator1 last,
 ForwardIterator2 result, BinaryPredicate pred);

namespace ranges {
 template<class I, class O>
 using unique_copy_result = copy_result<I, O>;

 template<InputIterator I, Sentinel<I> S, WeaklyIncrementable O,
 class Proj = identity, IndirectRelation<projected<I, Proj>> C = ranges::equal_to<>>
 requires IndirectlyCopyable<I, O> &&
 (ForwardIterator<I> ||
 (InputIterator<O> && Same<iter_value_t<I>, iter_value_t<O>>) ||
 IndirectlyCopyableStorable<I, O>)
 constexpr unique_copy_result<I, O>
 unique_copy(I first, S last, O result, C comp = {}, Proj proj = {});
 template<InputRange R, WeaklyIncrementable O, class Proj = identity,
 IndirectRelation<projected<iterator_t<R>, Proj>> C = ranges::equal_to<>>
 requires IndirectlyCopyable<iterator_t<R>, O> &&
 (ForwardIterator<iterator_t<R>> ||
 (InputIterator<O> && Same<iter_value_t<iterator_t<R>>, iter_value_t<O>>) ||
 IndirectlyCopyableStorable<iterator_t<R>, O>)
 constexpr unique_copy_result<safe_iterator_t<R>, O>
 unique_copy(R&& r, O result, C comp = {}, Proj proj = {});
}

// 24.6.10, reverse
template<class BidirectionalIterator>
 void reverse(BidirectionalIterator first, BidirectionalIterator last);

```

```

template<class ExecutionPolicy, class BidirectionalIterator>
 void reverse(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 BidirectionalIterator first, BidirectionalIterator last);

namespace ranges {
 template<BidirectionalIterator I, Sentinel<I> S>
 requires Permutable<I>
 constexpr I reverse(I first, S last);
 template<BidirectionalRange R>
 requires Permutable<iterator_t<R>>
 constexpr safe_iterator_t<R> reverse(R&& r);
}

template<class BidirectionalIterator, class OutputIterator>
 constexpr OutputIterator
 reverse_copy(BidirectionalIterator first, BidirectionalIterator last,
 OutputIterator result);
template<class ExecutionPolicy, class BidirectionalIterator, class ForwardIterator>
 ForwardIterator
 reverse_copy(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 BidirectionalIterator first, BidirectionalIterator last,
 ForwardIterator result);

namespace ranges {
 template<class I, class O>
 using reverse_copy_result = copy_result<I, O>;

 template<BidirectionalIterator I, Sentinel<I> S, WeaklyIncrementable O>
 requires IndirectlyCopyable<I, O>
 constexpr reverse_copy_result<I, O>
 reverse_copy(I first, S last, O result);
 template<BidirectionalRange R, WeaklyIncrementable O>
 requires IndirectlyCopyable<iterator_t<R>, O>
 constexpr reverse_copy_result<safe_iterator_t<R>, O>
 reverse_copy(R&& r, O result);
}

// 24.6.11, rotate
template<class ForwardIterator>
 constexpr ForwardIterator rotate(ForwardIterator first,
 ForwardIterator middle,
 ForwardIterator last);
template<class ExecutionPolicy, class ForwardIterator>
 ForwardIterator rotate(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first,
 ForwardIterator middle,
 ForwardIterator last);

namespace ranges {
 template<Permutable I, Sentinel<I> S>
 constexpr subrange<I> rotate(I first, I middle, S last);
 template<ForwardRange R>
 requires Permutable<iterator_t<R>>
 constexpr safe_subrange_t<R> rotate(R&& r, iterator_t<R> middle);
}

template<class ForwardIterator, class OutputIterator>
 constexpr OutputIterator
 rotate_copy(ForwardIterator first, ForwardIterator middle,
 ForwardIterator last, OutputIterator result);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
 ForwardIterator2
 rotate_copy(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first, ForwardIterator1 middle,
 ForwardIterator1 last, ForwardIterator2 result);

```

```

namespace ranges {
 template<class I, class O>
 using rotate_copy_result = copy_result<I, O>;

 template<ForwardIterator I, Sentinel<I> S, WeaklyIncrementable O>
 requires IndirectlyCopyable<I, O>
 constexpr rotate_copy_result<I, O>
 rotate_copy(I first, I middle, S last, O result);
 template<ForwardRange R, WeaklyIncrementable O>
 requires IndirectlyCopyable<iterator_t<R>, O>
 constexpr rotate_copy_result<safe_iterator_t<R>, O>
 rotate_copy(R&& r, iterator_t<R> middle, O result);
}

// 24.6.12, sample
[...]

// 24.6.13, shuffle
template<class RandomAccessIterator, class UniformRandomBitGenerator>
 void shuffle(RandomAccessIterator first,
 RandomAccessIterator last,
 UniformRandomBitGenerator&& g);

namespace ranges {
 template<RandomAccessIterator I, Sentinel<I> S, class Gen>
 requires Permutable<I> &&
 UniformRandomBitGenerator<remove_reference_t<Gen>> &&
 ConvertibleTo<invoke_result_t<Gen&&, iter_difference_t<I>>>
 I shuffle(I first, S last, Gen&& g);
 template<RandomAccessRange R, class Gen>
 requires Permutable<iterator_t<R>> &&
 UniformRandomBitGenerator<remove_reference_t<Gen>> &&
 ConvertibleTo<invoke_result_t<Gen&&, iter_difference_t<iterator_t<R>>>
 safe_iterator_t<R>
 shuffle(R&& r, Gen&& g);
}

// 24.6.14, shift
[...]

// 24.7, sorting and related operations
// 24.7.1, sorting
template<class RandomAccessIterator>
 constexpr void sort(RandomAccessIterator first, RandomAccessIterator last);
template<class RandomAccessIterator, class Compare>
 constexpr void sort(RandomAccessIterator first, RandomAccessIterator last,
 Compare comp);
template<class ExecutionPolicy, class RandomAccessIterator>
 void sort(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 RandomAccessIterator first, RandomAccessIterator last);
template<class ExecutionPolicy, class RandomAccessIterator, class Compare>
 void sort(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 RandomAccessIterator first, RandomAccessIterator last,
 Compare comp);

namespace ranges {
 template<RandomAccessIterator I, Sentinel<I> S, class Comp = ranges::less<>,
 class Proj = identity>
 requires Sortable<I, Comp, Proj>
 constexpr I
 sort(I first, S last, Comp comp = {}, Proj proj = {});
 template<RandomAccessRange R, class Comp = ranges::less<>, class Proj = identity>
 requires Sortable<iterator_t<R>, Comp, Proj>
 constexpr safe_iterator_t<R>

```

```

 sort(R&& r, Comp comp = {}, Proj proj = {});
 }

template<class RandomAccessIterator>
 void stable_sort(RandomAccessIterator first, RandomAccessIterator last);
template<class RandomAccessIterator, class Compare>
 void stable_sort(RandomAccessIterator first, RandomAccessIterator last,
 Compare comp);
template<class ExecutionPolicy, class RandomAccessIterator>
 void stable_sort(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 RandomAccessIterator first, RandomAccessIterator last);
template<class ExecutionPolicy, class RandomAccessIterator, class Compare>
 void stable_sort(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 RandomAccessIterator first, RandomAccessIterator last,
 Compare comp);

namespace ranges {
 template<RandomAccessIterator I, Sentinel<I> S, class Comp = ranges::less<>,
 class Proj = identity>
 requires Sortable<I, Comp, Proj>
 I stable_sort(I first, S last, Comp comp = {}, Proj proj = {});
 template<RandomAccessRange R, class Comp = ranges::less<>, class Proj = identity>
 requires Sortable<iterator_t<R>, Comp, Proj>
 safe_iterator_t<R>
 stable_sort(R&& r, Comp comp = {}, Proj proj = {});
}

template<class RandomAccessIterator>
 constexpr void partial_sort(RandomAccessIterator first,
 RandomAccessIterator middle,
 RandomAccessIterator last);
template<class RandomAccessIterator, class Compare>
 constexpr void partial_sort(RandomAccessIterator first,
 RandomAccessIterator middle,
 RandomAccessIterator last, Compare comp);
template<class ExecutionPolicy, class RandomAccessIterator>
 void partial_sort(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 RandomAccessIterator first,
 RandomAccessIterator middle,
 RandomAccessIterator last);
template<class ExecutionPolicy, class RandomAccessIterator, class Compare>
 void partial_sort(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 RandomAccessIterator first,
 RandomAccessIterator middle,
 RandomAccessIterator last, Compare comp);

namespace ranges {
 template<RandomAccessIterator I, Sentinel<I> S, class Comp = ranges::less<>,
 class Proj = identity>
 requires Sortable<I, Comp, Proj>
 constexpr I
 partial_sort(I first, I middle, S last, Comp comp = {}, Proj proj = {});
 template<RandomAccessRange R, class Comp = ranges::less<>, class Proj = identity>
 requires Sortable<iterator_t<R>, Comp, Proj>
 constexpr safe_iterator_t<R>
 partial_sort(R&& r, iterator_t<R> middle, Comp comp = {},
 Proj proj = {});
}

template<class InputIterator, class RandomAccessIterator>
 constexpr RandomAccessIterator
 partial_sort_copy(InputIterator first, InputIterator last,
 RandomAccessIterator result_first,
 RandomAccessIterator result_last);

```

```

template<class InputIterator, class RandomAccessIterator, class Compare>
constexpr RandomAccessIterator
 partial_sort_copy(InputIterator first, InputIterator last,
 RandomAccessIterator result_first,
 RandomAccessIterator result_last,
 Compare comp);
template<class ExecutionPolicy, class ForwardIterator, class RandomAccessIterator>
RandomAccessIterator
 partial_sort_copy(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last,
 RandomAccessIterator result_first,
 RandomAccessIterator result_last);
template<class ExecutionPolicy, class ForwardIterator, class RandomAccessIterator,
 class Compare>
RandomAccessIterator
 partial_sort_copy(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last,
 RandomAccessIterator result_first,
 RandomAccessIterator result_last,
 Compare comp);

namespace ranges {
 template<InputIterator I1, Sentinel<I1> S1, RandomAccessIterator I2, Sentinel<I2> S2,
 class Comp = ranges::less<>, class Proj1 = identity, class Proj2 = identity>
 requires IndirectlyCopyable<I1, I2> && Sortable<I2, Comp, Proj2> &&
 IndirectStrictWeakOrder<Comp, projected<I1, Proj1>, projected<I2, Proj2>>
 constexpr I2
 partial_sort_copy(I1 first, S1 last, I2 result_first, S2 result_last,
 Comp comp = {}, Proj1 proj1 = {}, Proj2 proj2 = {});
 template<InputRange R1, RandomAccessRange R2, class Comp = ranges::less<>,
 class Proj1 = identity, class Proj2 = identity>
 requires IndirectlyCopyable<iterator_t<R1>, iterator_t<R2>> &&
 Sortable<iterator_t<R2>, Comp, Proj2> &&
 IndirectStrictWeakOrder<Comp, projected<iterator_t<R1>, Proj1>,
 projected<iterator_t<R2>, Proj2>>
 constexpr safe_iterator_t<R2>
 partial_sort_copy(R1&& r, R2&& result_r, Comp comp = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
}

template<class ForwardIterator>
constexpr bool is_sorted(ForwardIterator first, ForwardIterator last);
template<class ForwardIterator, class Compare>
constexpr bool is_sorted(ForwardIterator first, ForwardIterator last,
 Compare comp);
template<class ExecutionPolicy, class ForwardIterator>
bool is_sorted(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last);
template<class ExecutionPolicy, class ForwardIterator, class Compare>
bool is_sorted(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last,
 Compare comp);

namespace ranges {
 template<ForwardIterator I, Sentinel<I> S, class Proj = identity,
 IndirectStrictWeakOrder<projected<I, Proj>> Comp = ranges::less<>>
 constexpr bool is_sorted(I first, S last, Comp comp = {}, Proj proj = {});
 template<ForwardRange R, class Proj = identity,
 IndirectStrictWeakOrder<projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
 constexpr bool is_sorted(R&& r, Comp comp = {}, Proj proj = {});
}

template<class ForwardIterator>
constexpr ForwardIterator
 is_sorted_until(ForwardIterator first, ForwardIterator last);

```

```

template<class ForwardIterator, class Compare>
constexpr ForwardIterator
 is_sorted_until(ForwardIterator first, ForwardIterator last,
 Compare comp);
template<class ExecutionPolicy, class ForwardIterator>
ForwardIterator
 is_sorted_until(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last);
template<class ExecutionPolicy, class ForwardIterator, class Compare>
ForwardIterator
 is_sorted_until(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last,
 Compare comp);

namespace ranges {
 template<ForwardIterator I, Sentinel<I> S, class Proj = identity,
 IndirectStrictWeakOrder<projected<I, Proj>> Comp = ranges::less<>>
 constexpr I is_sorted_until(I first, S last, Comp comp = {}, Proj proj = {});
 template<ForwardRange R, class Proj = identity,
 IndirectStrictWeakOrder<projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
 constexpr safe_iterator_t<R>
 is_sorted_until(R&& r, Comp comp = {}, Proj proj = {});
}

// 24.7.2, Nth element
template<class RandomAccessIterator>
constexpr void nth_element(RandomAccessIterator first, RandomAccessIterator nth,
 RandomAccessIterator last);
template<class RandomAccessIterator, class Compare>
constexpr void nth_element(RandomAccessIterator first, RandomAccessIterator nth,
 RandomAccessIterator last, Compare comp);
template<class ExecutionPolicy, class RandomAccessIterator>
void nth_element(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 RandomAccessIterator first, RandomAccessIterator nth,
 RandomAccessIterator last);
template<class ExecutionPolicy, class RandomAccessIterator, class Compare>
void nth_element(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 RandomAccessIterator first, RandomAccessIterator nth,
 RandomAccessIterator last, Compare comp);

namespace ranges {
 template<RandomAccessIterator I, Sentinel<I> S, class Comp = ranges::less<>,
 class Proj = identity>
 requires Sortable<I, Comp, Proj>
 constexpr I
 nth_element(I first, I nth, S last, Comp comp = {}, Proj proj = {});
 template<RandomAccessRange R, class Comp = ranges::less<>, class Proj = identity>
 requires Sortable<iterator_t<R>, Comp, Proj>
 constexpr safe_iterator_t<R>
 nth_element(R&& r, iterator_t<R> nth, Comp comp = {}, Proj proj = {});
}

// 24.7.3, binary search
template<class ForwardIterator, class T>
constexpr ForwardIterator
 lower_bound(ForwardIterator first, ForwardIterator last,
 const T& value);
template<class ForwardIterator, class T, class Compare>
constexpr ForwardIterator
 lower_bound(ForwardIterator first, ForwardIterator last,
 const T& value, Compare comp);

namespace ranges {
 template<ForwardIterator I, Sentinel<I> S, class T, class Proj = identity,
 IndirectStrictWeakOrder<const T*, projected<I, Proj>> Comp = ranges::less<>>

```

```

 constexpr I lower_bound(I first, S last, const T& value, Comp comp = {},
 Proj proj = {});
template<ForwardRange R, class T, class Proj = identity,
 IndirectStrictWeakOrder<const T*, projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
 constexpr safe_iterator_t<R>
 lower_bound(R&& r, const T& value, Comp comp = {}, Proj proj = {});
}

template<class ForwardIterator, class T>
constexpr ForwardIterator
 upper_bound(ForwardIterator first, ForwardIterator last,
 const T& value);
template<class ForwardIterator, class T, class Compare>
constexpr ForwardIterator
 upper_bound(ForwardIterator first, ForwardIterator last,
 const T& value, Compare comp);

namespace ranges {
 template<ForwardIterator I, Sentinel<I> S, class T, class Proj = identity,
 IndirectStrictWeakOrder<const T*, projected<I, Proj>> Comp = ranges::less<>>
 constexpr I upper_bound(I first, S last, const T& value, Comp comp = {}, Proj proj = {});
 template<ForwardRange R, class T, class Proj = identity,
 IndirectStrictWeakOrder<const T*, projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
 constexpr safe_iterator_t<R>
 upper_bound(R&& r, const T& value, Comp comp = {}, Proj proj = {});
}

template<class ForwardIterator, class T>
constexpr pair<ForwardIterator, ForwardIterator>
 equal_range(ForwardIterator first, ForwardIterator last,
 const T& value);
template<class ForwardIterator, class T, class Compare>
constexpr pair<ForwardIterator, ForwardIterator>
 equal_range(ForwardIterator first, ForwardIterator last,
 const T& value, Compare comp);

namespace ranges {
 template<ForwardIterator I, Sentinel<I> S, class T, class Proj = identity,
 IndirectStrictWeakOrder<const T*, projected<I, Proj>> Comp = ranges::less<>>
 constexpr subrange<I>
 equal_range(I first, S last, const T& value, Comp comp = {}, Proj proj = {});
 template<ForwardRange R, class T, class Proj = identity,
 IndirectStrictWeakOrder<const T*, projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
 constexpr safe_subrange_t<R>
 equal_range(R&& r, const T& value, Comp comp = {}, Proj proj = {});
}

template<class ForwardIterator, class T>
constexpr bool
 binary_search(ForwardIterator first, ForwardIterator last,
 const T& value);
template<class ForwardIterator, class T, class Compare>
constexpr bool
 binary_search(ForwardIterator first, ForwardIterator last,
 const T& value, Compare comp);

namespace ranges {
 template<ForwardIterator I, Sentinel<I> S, class T, class Proj = identity,
 IndirectStrictWeakOrder<const T*, projected<I, Proj>> Comp = ranges::less<>>
 constexpr bool binary_search(I first, S last, const T& value, Comp comp = {},
 Proj proj = {});
 template<ForwardRange R, class T, class Proj = identity,
 IndirectStrictWeakOrder<const T*, projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
 constexpr bool binary_search(R&& r, const T& value, Comp comp = {},
 Proj proj = {});
}

```

```

 Proj proj = {});
 }

// 24.7.4, partitions
template<class InputIterator, class Predicate>
 constexpr bool is_partitioned(InputIterator first, InputIterator last, Predicate pred);
template<class ExecutionPolicy, class ForwardIterator, class Predicate>
 bool is_partitioned(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last, Predicate pred);

namespace ranges {
 template<InputIterator I, Sentinel<I> S, class Proj = identity,
 IndirectUnaryPredicate<projected<I, Proj>> Pred>
 constexpr bool is_partitioned(I first, S last, Pred pred, Proj proj = {});
 template<InputRange R, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
 constexpr bool is_partitioned(R&& r, Pred pred, Proj proj = {});
}

template<class ForwardIterator, class Predicate>
 constexpr ForwardIterator partition(ForwardIterator first,
 ForwardIterator last,
 Predicate pred);
template<class ExecutionPolicy, class ForwardIterator, class Predicate>
 ForwardIterator partition(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first,
 ForwardIterator last,
 Predicate pred);

namespace ranges {
 template<Permutable I, Sentinel<I> S, class Proj = identity,
 IndirectUnaryPredicate<projected<I, Proj>> Pred>
 constexpr I
 partition(I first, S last, Pred pred, Proj proj = {});
 template<ForwardRange R, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
 requires Permutable<iterator_t<R>>
 constexpr safe_iterator_t<R>
 partition(R&& r, Pred pred, Proj proj = {});
}

template<class BidirectionalIterator, class Predicate>
 BidirectionalIterator stable_partition(BidirectionalIterator first,
 BidirectionalIterator last,
 Predicate pred);
template<class ExecutionPolicy, class BidirectionalIterator, class Predicate>
 BidirectionalIterator stable_partition(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 BidirectionalIterator first,
 BidirectionalIterator last,
 Predicate pred);

namespace ranges {
 template<BidirectionalIterator I, Sentinel<I> S, class Proj = identity,
 IndirectUnaryPredicate<projected<I, Proj>> Pred>
 requires Permutable<I>
 I stable_partition(I first, S last, Pred pred, Proj proj = {});
 template<BidirectionalRange R, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
 requires Permutable<iterator_t<R>>
 safe_iterator_t<R> stable_partition(R&& r, Pred pred, Proj proj = {});
}

template<class InputIterator, class OutputIterator1,
 class OutputIterator2, class Predicate>
 constexpr pair<OutputIterator1, OutputIterator2>
 partition_copy(InputIterator first, InputIterator last,

```

```

 OutputIterator1 out_true, OutputIterator2 out_false,
 Predicate pred);
template<class ExecutionPolicy, class ForwardIterator, class ForwardIterator1,
 class ForwardIterator2, class Predicate>
pair<ForwardIterator1, ForwardIterator2>
partition_copy(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last,
 ForwardIterator1 out_true, ForwardIterator2 out_false,
 Predicate pred);

namespace ranges {
 template<class I, class O1, class O2>
 struct partition_copy_result {
 I in;
 O1 out1;
 O2 out2;
 };

 template<InputIterator I, Sentinel<I> S, WeaklyIncrementable O1, WeaklyIncrementable O2,
 class Proj = identity, IndirectUnaryPredicate<projected<I, Proj>> Pred>
 requires IndirectlyCopyable<I, O1> && IndirectlyCopyable<I, O2>
 constexpr partition_copy_result<I, O1, O2>
 partition_copy(I first, S last, O1 out_true, O2 out_false, Pred pred,
 Proj proj = {});
 template<InputRange R, WeaklyIncrementable O1, WeaklyIncrementable O2,
 class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
 requires IndirectlyCopyable<iterator_t<R>, O1> &&
 IndirectlyCopyable<iterator_t<R>, O2>
 constexpr partition_copy_result<safe_iterator_t<R>, O1, O2>
 partition_copy(R&& r, O1 out_true, O2 out_false, Pred pred, Proj proj = {});
}

template<class ForwardIterator, class Predicate>
constexpr ForwardIterator
partition_point(ForwardIterator first, ForwardIterator last,
 Predicate pred);

namespace ranges {
 template<ForwardIterator I, Sentinel<I> S, class Proj = identity,
 IndirectUnaryPredicate<projected<I, Proj>> Pred>
 constexpr I partition_point(I first, S last, Pred pred, Proj proj = {});
 template<ForwardRange R, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
 constexpr safe_iterator_t<R>
 partition_point(R&& r, Pred pred, Proj proj = {});
}

// 24.7.5, merge
template<class InputIterator1, class InputIterator2, class OutputIterator>
constexpr OutputIterator
merge(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, InputIterator2 last2,
 OutputIterator result);
template<class InputIterator1, class InputIterator2, class OutputIterator,
 class Compare>
constexpr OutputIterator
merge(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, InputIterator2 last2,
 OutputIterator result, Compare comp);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class ForwardIterator>
merge(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,

```

```

 ForwardIterator2 first2, ForwardIterator2 last2,
 ForwardIterator result);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class ForwardIterator, class Compare>
ForwardIterator
merge(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2,
 ForwardIterator result, Compare comp);

namespace ranges {
 template<class I1, class I2, class O>
 using merge_result = binary_transform_result<I1, I2, O>;

 template<InputIterator I1, Sentinel<I1> S1, InputIterator I2, Sentinel<I2> S2,
 WeaklyIncrementable O, class Comp = ranges::less<>, class Proj1 = identity,
 class Proj2 = identity>
 requires Mergeable<I1, I2, O, Comp, Proj1, Proj2>
 constexpr merge_result<I1, I2, O>
 merge(I1 first1, S1 last1, I2 first2, S2 last2, O result,
 Comp comp = {}, Proj1 proj1 = {}, Proj2 proj2 = {});
 template<InputRange R1, InputRange R2, WeaklyIncrementable O, class Comp = ranges::less<>,
 class Proj1 = identity, class Proj2 = identity>
 requires Mergeable<iterator_t<R1>, iterator_t<R2>, O, Comp, Proj1, Proj2>
 constexpr merge_result<safe_iterator_t<R1>, safe_iterator_t<R2>, O>
 merge(R1&& r1, R2&& r2, O result,
 Comp comp = {}, Proj1 proj1 = {}, Proj2 proj2 = {});
}

template<class BidirectionalIterator>
void inplace_merge(BidirectionalIterator first,
 BidirectionalIterator middle,
 BidirectionalIterator last);
template<class BidirectionalIterator, class Compare>
void inplace_merge(BidirectionalIterator first,
 BidirectionalIterator middle,
 BidirectionalIterator last, Compare comp);
template<class ExecutionPolicy, class BidirectionalIterator>
void inplace_merge(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 BidirectionalIterator first,
 BidirectionalIterator middle,
 BidirectionalIterator last);
template<class ExecutionPolicy, class BidirectionalIterator, class Compare>
void inplace_merge(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 BidirectionalIterator first,
 BidirectionalIterator middle,
 BidirectionalIterator last, Compare comp);

namespace ranges {
 template<BidirectionalIterator I, Sentinel<I> S, class Comp = ranges::less<>,
 class Proj = identity>
 requires Sortable<I, Comp, Proj>
 I inplace_merge(I first, I middle, S last, Comp comp = {}, Proj proj = {});
 template<BidirectionalRange R, class Comp = ranges::less<>, class Proj = identity>
 requires Sortable<iterator_t<R>, Comp, Proj>
 safe_iterator_t<R>
 inplace_merge(R&& r, iterator_t<R> middle, Comp comp = {},
 Proj proj = {});
}

// 24.7.6, set operations
template<class InputIterator1, class InputIterator2>
constexpr bool includes(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, InputIterator2 last2);

```

```

template<class InputIterator1, class InputIterator2, class Compare>
 constexpr bool includes(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, InputIterator2 last2,
 Compare comp);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
 bool includes(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class Compare>
 bool includes(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2,
 Compare comp);

namespace ranges {
 template<InputIterator I1, Sentinel<I1> S1, InputIterator I2, Sentinel<I2> S2,
 class Proj1 = identity, class Proj2 = identity,
 IndirectStrictWeakOrder<projected<I1, Proj1>, projected<I2, Proj2>> Comp = ranges::less<>>
 constexpr bool includes(I1 first1, S1 last1, I2 first2, S2 last2, Comp comp = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
 template<InputRange R1, InputRange R2, class Proj1 = identity,
 class Proj2 = identity,
 IndirectStrictWeakOrder<projected<iterator_t<R1>, Proj1>,
 projected<iterator_t<R2>, Proj2>> Comp = ranges::less<>>
 constexpr bool includes(R1&& r1, R2&& r2, Comp comp = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
}

template<class InputIterator1, class InputIterator2, class OutputIterator>
 constexpr OutputIterator
 set_union(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, InputIterator2 last2,
 OutputIterator result);
template<class InputIterator1, class InputIterator2, class OutputIterator, class Compare>
 constexpr OutputIterator
 set_union(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, InputIterator2 last2,
 OutputIterator result, Compare comp);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class ForwardIterator>
 ForwardIterator
 set_union(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2,
 ForwardIterator result);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class ForwardIterator, class Compare>
 ForwardIterator
 set_union(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2,
 ForwardIterator result, Compare comp);

namespace ranges {
 template<class I1, class I2, class O>
 using set_union_result = binary_transform_result<I1, I2, O>;

 template<InputIterator I1, Sentinel<I1> S1, InputIterator I2, Sentinel<I2> S2,
 WeaklyIncrementable O, class Comp = ranges::less<>, class Proj1 = identity, class Proj2 = identity>
 requires Mergeable<I1, I2, O, Comp, Proj1, Proj2>
 constexpr set_union_result<I1, I2, O>
 set_union(I1 first1, S1 last1, I2 first2, S2 last2, O result, Comp comp = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});

```

```

template<InputRange R1, InputRange R2, WeaklyIncrementable O,
 class Comp = ranges::less<>, class Proj1 = identity, class Proj2 = identity>
requires Mergeable<iterator_t<R1>, iterator_t<R2>, O, Comp, Proj1, Proj2>
constexpr set_union_result<safe_iterator_t<R1>, safe_iterator_t<R2>, O>
 set_union(R1&& r1, R2&& r2, O result, Comp comp = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
}

template<class InputIterator1, class InputIterator2, class OutputIterator>
constexpr OutputIterator
 set_intersection(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, InputIterator2 last2,
 OutputIterator result);
template<class InputIterator1, class InputIterator2, class OutputIterator, class Compare>
constexpr OutputIterator
 set_intersection(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, InputIterator2 last2,
 OutputIterator result, Compare comp);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class ForwardIterator>
ForwardIterator
 set_intersection(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2,
 ForwardIterator result);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class ForwardIterator, class Compare>
ForwardIterator
 set_intersection(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2,
 ForwardIterator result, Compare comp);

namespace ranges {
 template<class I1, class I2, class O>
 using set_intersection_result = binary_transform_result<I1, I2, O>;

 template<InputIterator I1, Sentinel<I1> S1, InputIterator I2, Sentinel<I2> S2,
 WeaklyIncrementable O, class Comp = ranges::less<>, class Proj1 = identity, class Proj2 = identity>
 requires Mergeable<I1, I2, O, Comp, Proj1, Proj2>
 constexpr set_intersection_result<I1, I2, O>
 set_intersection(I1 first1, S1 last1, I2 first2, S2 last2, O result,
 Comp comp = {}, Proj1 proj1 = {}, Proj2 proj2 = {});
 template<InputRange R1, InputRange R2, WeaklyIncrementable O,
 class Comp = ranges::less<>, class Proj1 = identity, class Proj2 = identity>
 requires Mergeable<iterator_t<R1>, iterator_t<R2>, O, Comp, Proj1, Proj2>
 constexpr set_intersection_result<safe_iterator_t<R1>, safe_iterator_t<R2>, O>
 set_intersection(R1&& r1, R2&& r2, O result,
 Comp comp = {}, Proj1 proj1 = {}, Proj2 proj2 = {});
}

template<class InputIterator1, class InputIterator2, class OutputIterator>
constexpr OutputIterator
 set_difference(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, InputIterator2 last2,
 OutputIterator result);
template<class InputIterator1, class InputIterator2, class OutputIterator, class Compare>
constexpr OutputIterator
 set_difference(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, InputIterator2 last2,
 OutputIterator result, Compare comp);

```

```

template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class ForwardIterator>
ForwardIterator
 set_difference(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2,
 ForwardIterator result);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class ForwardIterator, class Compare>
ForwardIterator
 set_difference(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2,
 ForwardIterator result, Compare comp);

namespace ranges {
 template<class I, class O>
 using set_difference_result = copy_result<I, O>;

 template<InputIterator I1, Sentinel<I1> S1, InputIterator I2, Sentinel<I2> S2,
 WeaklyIncrementable O, class Comp = ranges::less<>, class Proj1 = identity, class Proj2 = identity>
 requires Mergeable<I1, I2, O, Comp, Proj1, Proj2>
 constexpr set_difference_result<I1, O>
 set_difference(I1 first1, S1 last1, I2 first2, S2 last2, O result,
 Comp comp = {}, Proj1 proj1 = {}, Proj2 proj2 = {});
 template<InputRange R1, InputRange R2, WeaklyIncrementable O,
 class Comp = ranges::less<>, class Proj1 = identity, class Proj2 = identity>
 requires Mergeable<iterator_t<R1>, iterator_t<R2>, O, Comp, Proj1, Proj2>
 constexpr set_difference_result<safe_iterator_t<R1>, O>
 set_difference(R1&& r1, R2&& r2, O result,
 Comp comp = {}, Proj1 proj1 = {}, Proj2 proj2 = {});
}

template<class InputIterator1, class InputIterator2, class OutputIterator>
constexpr OutputIterator
 set_symmetric_difference(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, InputIterator2 last2,
 OutputIterator result);
template<class InputIterator1, class InputIterator2, class OutputIterator, class Compare>
constexpr OutputIterator
 set_symmetric_difference(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, InputIterator2 last2,
 OutputIterator result, Compare comp);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class ForwardIterator>
ForwardIterator
 set_symmetric_difference(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2,
 ForwardIterator result);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class ForwardIterator, class Compare>
ForwardIterator
 set_symmetric_difference(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2,
 ForwardIterator result, Compare comp);

namespace ranges {
 template<class I1, class I2, class O>
 using set_symmetric_difference_result = binary_transform_result<I1, I2, O>;

```

```

template<InputIterator I1, Sentinel<I1> S1, InputIterator I2, Sentinel<I2> S2,
 WeaklyIncrementable O, class Comp = ranges::less<>, class Proj1 = identity, class Proj2 = identity>
requires Mergeable<I1, I2, O, Comp, Proj1, Proj2>
constexpr set_symmetric_difference_result<I1, I2, O>
 set_symmetric_difference(I1 first1, S1 last1, I2 first2, S2 last2, O result,
 Comp comp = {}, Proj1 proj1 = {},
 Proj2 proj2 = {});

template<InputRange R1, InputRange R2, WeaklyIncrementable O,
 class Comp = ranges::less<>, class Proj1 = identity, class Proj2 = identity>
requires Mergeable<iterator_t<R1>, iterator_t<R2>, O, Comp, Proj1, Proj2>
constexpr set_symmetric_difference_result<safe_iterator_t<R1>, safe_iterator_t<R2>, O>
 set_symmetric_difference(R1&& r1, R2&& r2, O result, Comp comp = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
}

// 24.7.7, heap operations
template<class RandomAccessIterator>
constexpr void push_heap(RandomAccessIterator first, RandomAccessIterator last);
template<class RandomAccessIterator, class Compare>
constexpr void push_heap(RandomAccessIterator first, RandomAccessIterator last,
 Compare comp);

namespace ranges {
 template<RandomAccessIterator I, Sentinel<I> S, class Comp = ranges::less<>,
 class Proj = identity>
 requires Sortable<I, Comp, Proj>
 constexpr I
 push_heap(I first, S last, Comp comp = {}, Proj proj = {});
 template<RandomAccessRange R, class Comp = ranges::less<>, class Proj = identity>
 requires Sortable<iterator_t<R>, Comp, Proj>
 constexpr safe_iterator_t<R>
 push_heap(R&& r, Comp comp = {}, Proj proj = {});
}

template<class RandomAccessIterator>
constexpr void pop_heap(RandomAccessIterator first, RandomAccessIterator last);
template<class RandomAccessIterator, class Compare>
constexpr void pop_heap(RandomAccessIterator first, RandomAccessIterator last,
 Compare comp);

namespace ranges {
 template<RandomAccessIterator I, Sentinel<I> S, class Comp = ranges::less<>,
 class Proj = identity>
 requires Sortable<I, Comp, Proj>
 constexpr I
 pop_heap(I first, S last, Comp comp = {}, Proj proj = {});
 template<RandomAccessRange R, class Comp = ranges::less<>, class Proj = identity>
 requires Sortable<iterator_t<R>, Comp, Proj>
 constexpr safe_iterator_t<R>
 pop_heap(R&& r, Comp comp = {}, Proj proj = {});
}

template<class RandomAccessIterator>
constexpr void make_heap(RandomAccessIterator first, RandomAccessIterator last);
template<class RandomAccessIterator, class Compare>
constexpr void make_heap(RandomAccessIterator first, RandomAccessIterator last,
 Compare comp);

namespace ranges {
 template<RandomAccessIterator I, Sentinel<I> S, class Comp = ranges::less<>,
 class Proj = identity>
 requires Sortable<I, Comp, Proj>
 constexpr I
 make_heap(I first, S last, Comp comp = {}, Proj proj = {});
}

```

```

 template<RandomAccessRange R, class Comp = ranges::less<>, class Proj = identity>
 requires Sortable<iterator_t<R>, Comp, Proj>
 constexpr safe_iterator_t<R>
 make_heap(R&& r, Comp comp = {}, Proj proj = {});
}

template<class RandomAccessIterator>
 constexpr void sort_heap(RandomAccessIterator first, RandomAccessIterator last);
template<class RandomAccessIterator, class Compare>
 constexpr void sort_heap(RandomAccessIterator first, RandomAccessIterator last,
 Compare comp);

namespace ranges {
 template<RandomAccessIterator I, Sentinel<I> S, class Comp = ranges::less<>,
 class Proj = identity>
 requires Sortable<I, Comp, Proj>
 constexpr I
 sort_heap(I first, S last, Comp comp = {}, Proj proj = {});
 template<RandomAccessRange R, class Comp = ranges::less<>, class Proj = identity>
 requires Sortable<iterator_t<R>, Comp, Proj>
 constexpr safe_iterator_t<R>
 sort_heap(R&& r, Comp comp = {}, Proj proj = {});
}

template<class RandomAccessIterator>
 constexpr bool is_heap(RandomAccessIterator first, RandomAccessIterator last);
template<class RandomAccessIterator, class Compare>
 constexpr bool is_heap(RandomAccessIterator first, RandomAccessIterator last,
 Compare comp);
template<class ExecutionPolicy, class RandomAccessIterator>
 bool is_heap(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 RandomAccessIterator first, RandomAccessIterator last);
template<class ExecutionPolicy, class RandomAccessIterator, class Compare>
 bool is_heap(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 RandomAccessIterator first, RandomAccessIterator last,
 Compare comp);

namespace ranges {
 template<RandomAccessIterator I, Sentinel<I> S, class Proj = identity,
 IndirectStrictWeakOrder<projected<I, Proj>> Comp = ranges::less<>>
 constexpr bool is_heap(I first, S last, Comp comp = {}, Proj proj = {});
 template<RandomAccessRange R, class Proj = identity,
 IndirectStrictWeakOrder<projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
 constexpr bool is_heap(R&& r, Comp comp = {}, Proj proj = {});
}

template<class RandomAccessIterator>
 constexpr RandomAccessIterator
 is_heap_until(RandomAccessIterator first, RandomAccessIterator last);
template<class RandomAccessIterator, class Compare>
 constexpr RandomAccessIterator
 is_heap_until(RandomAccessIterator first, RandomAccessIterator last,
 Compare comp);
template<class ExecutionPolicy, class RandomAccessIterator>
 RandomAccessIterator
 is_heap_until(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 RandomAccessIterator first, RandomAccessIterator last);
template<class ExecutionPolicy, class RandomAccessIterator, class Compare>
 RandomAccessIterator
 is_heap_until(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 RandomAccessIterator first, RandomAccessIterator last,
 Compare comp);

```

```

namespace ranges {
 template<RandomAccessIterator I, Sentinel<I> S, class Proj = identity,
 IndirectStrictWeakOrder<projected<I, Proj>> Comp = ranges::less<>>
 constexpr I is_heap_until(I first, S last, Comp comp = {}, Proj proj = {});
 template<RandomAccessRange R, class Proj = identity,
 IndirectStrictWeakOrder<projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
 constexpr safe_iterator_t<R>
 is_heap_until(R&& r, Comp comp = {}, Proj proj = {});
}

// 24.7.8, minimum and maximum
template<class T> constexpr const T& min(const T& a, const T& b);
template<class T, class Compare>
 constexpr const T& min(const T& a, const T& b, Compare comp);
template<class T>
 constexpr T min(initializer_list<T> tr);
template<class T, class Compare>
 constexpr T min(initializer_list<T> tr, Compare comp);
namespace ranges {
 template<class T, class Proj = identity,
 IndirectStrictWeakOrder<projected<const T*, Proj>> Comp = ranges::less<>>
 constexpr const T& min(const T& a, const T& b, Comp comp = {}, Proj proj = {});
 template<Copyable T, class Proj = identity,
 IndirectStrictWeakOrder<projected<const T*, Proj>> Comp = ranges::less<>>
 constexpr T min(initializer_list<T> r, Comp comp = {}, Proj proj = {});
 template<InputRange R, class Proj = identity,
 IndirectStrictWeakOrder<projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
 requires IndirectlyCopyableStorable<iterator_t<R>, iter_value_t<iterator_t<R>*>
 constexpr iter_value_t<iterator_t<R>>
 min(R&& r, Comp comp = {}, Proj proj = {});
}

template<class T> constexpr const T& max(const T& a, const T& b);
template<class T, class Compare>
 constexpr const T& max(const T& a, const T& b, Compare comp);
template<class T>
 constexpr T max(initializer_list<T> tr);
template<class T, class Compare>
 constexpr T max(initializer_list<T> tr, Compare comp);
namespace ranges {
 template<class T, class Proj = identity,
 IndirectStrictWeakOrder<projected<const T*, Proj>> Comp = ranges::less<>>
 constexpr const T& max(const T& a, const T& b, Comp comp = {}, Proj proj = {});
 template<Copyable T, class Proj = identity,
 IndirectStrictWeakOrder<projected<const T*, Proj>> Comp = ranges::less<>>
 constexpr T max(initializer_list<T> r, Comp comp = {}, Proj proj = {});
 template<InputRange R, class Proj = identity,
 IndirectStrictWeakOrder<projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
 requires IndirectlyCopyableStorable<iterator_t<R>, iter_value_t<iterator_t<R>*>
 constexpr iter_value_t<iterator_t<R>>
 max(R&& r, Comp comp = {}, Proj proj = {});
}

template<class T> constexpr pair<const T&, const T&> minmax(const T& a, const T& b);
template<class T, class Compare>
 constexpr pair<const T&, const T&> minmax(const T& a, const T& b, Compare comp);
template<class T>
 constexpr pair<T, T> minmax(initializer_list<T> tr);
template<class T, class Compare>
 constexpr pair<T, T> minmax(initializer_list<T> tr, Compare comp);

```

```

namespace ranges {
 template<class T>
 struct minmax_result {
 T min;
 T max;
 };

 template<class T, class Proj = identity,
 IndirectStrictWeakOrder<projected<const T*, Proj>> Comp = ranges::less<>>
 constexpr minmax_result<const T&>
 minmax(const T& a, const T& b, Comp comp = {}, Proj proj = {});
 template<Copyable T, class Proj = identity,
 IndirectStrictWeakOrder<projected<const T*, Proj>> Comp = ranges::less<>>
 constexpr minmax_result<T>
 minmax(initializer_list<T> r, Comp comp = {}, Proj proj = {});
 template<InputRange R, class Proj = identity,
 IndirectStrictWeakOrder<projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
 requires IndirectlyCopyableStorable<iterator_t<R>, iter_value_t<iterator_t<R>>*>
 constexpr minmax_result<iter_value_t<iterator_t<R>>>
 minmax(R&& r, Comp comp = {}, Proj proj = {});
}

template<class ForwardIterator>
 constexpr ForwardIterator min_element(ForwardIterator first, ForwardIterator last);
template<class ForwardIterator, class Compare>
 constexpr ForwardIterator min_element(ForwardIterator first, ForwardIterator last,
 Compare comp);
template<class ExecutionPolicy, class ForwardIterator>
 ForwardIterator min_element(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last);
template<class ExecutionPolicy, class ForwardIterator, class Compare>
 ForwardIterator min_element(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last,
 Compare comp);

namespace ranges {
 template<ForwardIterator I, Sentinel<I> S, class Proj = identity,
 IndirectStrictWeakOrder<projected<I, Proj>> Comp = ranges::less<>>
 constexpr I min_element(I first, S last, Comp comp = {}, Proj proj = {});
 template<ForwardRange R, class Proj = identity,
 IndirectStrictWeakOrder<projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
 constexpr safe_iterator_t<R>
 min_element(R&& r, Comp comp = {}, Proj proj = {});
}

template<class ForwardIterator>
 constexpr ForwardIterator max_element(ForwardIterator first, ForwardIterator last);
template<class ForwardIterator, class Compare>
 constexpr ForwardIterator max_element(ForwardIterator first, ForwardIterator last,
 Compare comp);
template<class ExecutionPolicy, class ForwardIterator>
 ForwardIterator max_element(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last);
template<class ExecutionPolicy, class ForwardIterator, class Compare>
 ForwardIterator max_element(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last,
 Compare comp);

namespace ranges {
 template<ForwardIterator I, Sentinel<I> S, class Proj = identity,
 IndirectStrictWeakOrder<projected<I, Proj>> Comp = ranges::less<>>
 constexpr I max_element(I first, S last, Comp comp = {}, Proj proj = {});
 template<ForwardRange R, class Proj = identity,
 IndirectStrictWeakOrder<projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
 constexpr safe_iterator_t<R>

```

```

 max_element(R&& r, Comp comp = {}, Proj proj = {});
 }

template<class ForwardIterator>
 constexpr pair<ForwardIterator, ForwardIterator>
 minmax_element(ForwardIterator first, ForwardIterator last);
template<class ForwardIterator, class Compare>
 constexpr pair<ForwardIterator, ForwardIterator>
 minmax_element(ForwardIterator first, ForwardIterator last, Compare comp);
template<class ExecutionPolicy, class ForwardIterator>
 pair<ForwardIterator, ForwardIterator>
 minmax_element(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last);
template<class ExecutionPolicy, class ForwardIterator, class Compare>
 pair<ForwardIterator, ForwardIterator>
 minmax_element(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator first, ForwardIterator last, Compare comp);

namespace ranges {
 template<ForwardIterator I, Sentinel<I> S, class Proj = identity,
 IndirectStrictWeakOrder<projected<I, Proj>> Comp = ranges::less<>>
 constexpr minmax_result<I>
 minmax_element(I first, S last, Comp comp = {}, Proj proj = {});
 template<ForwardRange R, class Proj = identity,
 IndirectStrictWeakOrder<projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
 constexpr minmax_result<safe_iterator_t<R>>
 minmax_element(R&& r, Comp comp = {}, Proj proj = {});
}

// 24.7.9, bounded value
[...]

// 24.7.10, lexicographical comparison
template<class InputIterator1, class InputIterator2>
 constexpr bool
 lexicographical_compare(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, InputIterator2 last2);
template<class InputIterator1, class InputIterator2, class Compare>
 constexpr bool
 lexicographical_compare(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, InputIterator2 last2,
 Compare comp);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
 bool
 lexicographical_compare(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class Compare>
 bool
 lexicographical_compare(ExecutionPolicy&& exec, // see [algorithms.parallel.overloads]
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2,
 Compare comp);

namespace ranges {
 template<InputIterator I1, Sentinel<I1> S1, InputIterator I2, Sentinel<I2> S2,
 class Proj1 = identity, class Proj2 = identity,
 IndirectStrictWeakOrder<projected<I1, Proj1>, projected<I2, Proj2>> Comp = ranges::less<>>
 constexpr bool
 lexicographical_compare(I1 first1, S1 last1, I2 first2, S2 last2,
 Comp comp = {}, Proj1 proj1 = {}, Proj2 proj2 = {});
 template<InputRange R1, InputRange R2, class Proj1 = identity,
 class Proj2 = identity,
 IndirectStrictWeakOrder<projected<iterator_t<R1>, Proj1>,

```

```

 projected<iterator_t<R2>, Proj2>> Comp = ranges::less<>>
constexpr bool
 lexicographical_compare(R1&& r1, R2&& r2, Comp comp = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
}

// 24.7.11, three-way comparison algorithms
[...]

// 24.7.12, permutations
template<class BidirectionalIterator>
 constexpr bool next_permutation(BidirectionalIterator first,
 BidirectionalIterator last);
template<class BidirectionalIterator, class Compare>
 constexpr bool next_permutation(BidirectionalIterator first,
 BidirectionalIterator last, Compare comp);

namespace ranges {
 template<BidirectionalIterator I, Sentinel<I> S, class Comp = ranges::less<>,
 class Proj = identity>
 requires Sortable<I, Comp, Proj>
 constexpr bool
 next_permutation(I first, S last, Comp comp = {}, Proj proj = {});
 template<BidirectionalRange R, class Comp = ranges::less<>,
 class Proj = identity>
 requires Sortable<iterator_t<R>, Comp, Proj>
 constexpr bool
 next_permutation(R&& r, Comp comp = {}, Proj proj = {});
}

template<class BidirectionalIterator>
 constexpr bool prev_permutation(BidirectionalIterator first,
 BidirectionalIterator last);
template<class BidirectionalIterator, class Compare>
 constexpr bool prev_permutation(BidirectionalIterator first,
 BidirectionalIterator last, Compare comp);

namespace ranges {
 template<BidirectionalIterator I, Sentinel<I> S, class Comp = ranges::less<>,
 class Proj = identity>
 requires Sortable<I, Comp, Proj>
 constexpr bool
 prev_permutation(I first, S last, Comp comp = {}, Proj proj = {});
 template<BidirectionalRange R, class Comp = ranges::less<>,
 class Proj = identity>
 requires Sortable<iterator_t<R>, Comp, Proj>
 constexpr bool
 prev_permutation(R&& r, Comp comp = {}, Proj proj = {});
}
}

```

### 24.3 Algorithms requirements

[algorithms.requirements]

- <sup>1</sup> All of the algorithms are separated from the particular implementations of data structures and are parameterized by iterator types. Because of this, they can work with program-defined data structures, as long as these data structures have iterator types satisfying the assumptions on the algorithms.
- <sup>2</sup> The entities defined in the `std::ranges` namespace in this Clause are not found by argument-dependent name lookup ([basic.lookup.argdep]). When found by unqualified ([basic.lookup.unqual]) name lookup for the *postfix-expression* in a function call ([expr.call]), they inhibit argument-dependent name lookup.

[Example:

```

void foo() {
 using namespace std::ranges;
 std::vector<int> vec{1,2,3};
}

```

```
 find(begin(vec), end(vec), 2); // #1
}
```

The function call expression at #1 invokes `std::ranges::find`, not `std::find`, despite that (a) the iterator type returned from `begin(vec)` and `end(vec)` may be associated with namespace `std` and (b) `std::find` is more specialized ([`temp.func.order`]) than `std::ranges::find` since the former requires its first two parameters to have the same type. — *end example*

- 3 For purposes of determining the existence of data races, [...]
- 4 Throughout this Clause, where the template parameters are not constrained, the names of template parameters are used to express type requirements. [...]
- 5 If an algorithm's *Effects*: element specifies that a value pointed to by any iterator passed as an argument is modified, then that algorithm has an additional type requirement: The type of that argument shall **Change:** satisfy the requirements of a mutable iterator (22.3). [*Note:* This requirement does not affect arguments that are named `OutputIterator`, `OutputIterator1`, or `OutputIterator2`, because output iterators must always be mutable, nor does it affect arguments that are constrained, for which mutability requirements are expressed explicitly. — *end note*]
- 6 Both in-place and copying versions are provided for certain algorithms. [...]
- 7 When not otherwise constrained, the `Predicate` parameter is used whenever an algorithm expects a function object[`function.objects`] that, when applied to the result of dereferencing the corresponding iterator, [...]
- 8 When not otherwise constrained, the `BinaryPredicate` parameter is used whenever an algorithm expects a function object that [...]
- 11 In the description of the algorithms, operators `+` and `—` are used for some of the iterator categories for which they it does not have to be defined. In these cases the semantics of `a + n` is are the same as that those of

```
Xauto tmp = a;
advance(tmp, n);
for (; n < 0; ++n) --tmp;
for (; n > 0; --n) ++tmp;
return tmp;
```

and that of Similarly, operator `—` is used for some combinations of iterators and sentinel types for which it does not have to be defined. If `[a, b)` denotes a range, the semantics of `b — a` is in these cases are the same as those of

```
iter_difference_t<remove_reference_t<decltype(a)>> n = 0;
for (auto tmp = a; tmp != b; ++tmp) ++n;
return distance(a, b)n;
```

and if `[b, a)` denotes a range, the same as those of

```
iter_difference_t<remove_reference_t<decltype(b)>> n = 0;
for (auto tmp = b; tmp != a; ++tmp) --n;
return n;
```

- 12 In the description of algorithm return values, a sentinel value `s` denoting the end of a range `[i, s)` is sometimes returned where an iterator is expected. In these cases, the semantics are as if the sentinel is converted into an iterator using `ranges::next(i, s)`.
- 13 Overloads of algorithms that take `Range` arguments (23.5.2) behave as if they are implemented by calling `ranges::begin` and `ranges::end` on the `Range(s)` and dispatching to the overload in namespace `ranges` that takes separate iterator and sentinel arguments.
- 14 The number and order of deducible template parameters for algorithm declarations are unspecified, except where explicitly stated otherwise. [*Note:* Consequently, the algorithms may not be called with explicitly-specified template argument lists. — *end note*]
- 15 The class templates `binary_transform_result`, `copy_result`, `for_each_result`, `minmax_result`, `mismatch_result`, and `partition_copy_result` have the template parameters, data members, and special members specified above. They have no base classes or members other than those specified.

## 24.5 Non-modifying sequence operations

[alg.nonmodifying]

### 24.5.1 All of

[alg.all\_of]

```
template<class InputIterator, class Predicate>
constexpr bool all_of(InputIterator first, InputIterator last, Predicate pred);
template<class ExecutionPolicy, class ForwardIterator, class Predicate>
bool all_of(ExecutionPolicy&& exec, ForwardIterator first, ForwardIterator last,
 Predicate pred);
```

```
namespace ranges {
template<InputIterator I, Sentinel<I> S, class Proj = identity,
 IndirectUnaryPredicate<projected<I, Proj>> Pred>
constexpr bool all_of(I first, S last, Pred pred, Proj proj = {});
template<InputRange R, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
constexpr bool all_of(R&& r, Pred pred, Proj proj = {});
}
```

- 1 Let  $E$  be  $\text{pred}(*i)$  and  $\text{invoke}(\text{pred}, \text{invoke}(\text{proj}, *i))$  for the overloads in namespace `std` and `std::ranges`, respectively.
- 2 Returns: ~~true if  $[\text{first}, \text{last})$  is empty or if  $\text{pred}(*i)$  is true for every iterator  $i$  in the range  $[\text{first}, \text{last})$ , and false otherwise.~~ false if  $E$  is false for some iterator  $i$  in the range  $[\text{first}, \text{last})$ , and true otherwise.
- 3 Complexity: At most  $\text{last} - \text{first}$  applications of the predicate and any projection.

### 24.5.2 Any of

[alg.any\_of]

```
template<class InputIterator, class Predicate>
constexpr bool any_of(InputIterator first, InputIterator last, Predicate pred);
template<class ExecutionPolicy, class ForwardIterator, class Predicate>
bool any_of(ExecutionPolicy&& exec, ForwardIterator first, ForwardIterator last,
 Predicate pred);
```

```
namespace ranges {
template<InputIterator I, Sentinel<I> S, class Proj = identity,
 IndirectUnaryPredicate<projected<I, Proj>> Pred>
constexpr bool any_of(I first, S last, Pred pred, Proj proj = {});
template<InputRange R, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
constexpr bool any_of(R&& r, Pred pred, Proj proj = {});
}
```

- 1 Let  $E$  be  $\text{pred}(*i)$  and  $\text{invoke}(\text{pred}, \text{invoke}(\text{proj}, *i))$  for the overloads in namespace `std` and `std::ranges`, respectively.
- 2 Returns: ~~false if  $[\text{first}, \text{last})$  is empty or if there is no iterator  $i$  in the range  $[\text{first}, \text{last})$  such that  $\text{pred}(*i)$  is true, and true otherwise.~~ true if  $E$  is true for some iterator  $i$  in the range  $[\text{first}, \text{last})$ , and false otherwise.
- 3 Complexity: At most  $\text{last} - \text{first}$  applications of the predicate and any projection.

### 24.5.3 None of

[alg.none\_of]

```
template<class InputIterator, class Predicate>
constexpr bool none_of(InputIterator first, InputIterator last, Predicate pred);
template<class ExecutionPolicy, class ForwardIterator, class Predicate>
bool none_of(ExecutionPolicy&& exec, ForwardIterator first, ForwardIterator last,
 Predicate pred);
```

```
namespace ranges {
template<InputIterator I, Sentinel<I> S, class Proj = identity,
 IndirectUnaryPredicate<projected<I, Proj>> Pred>
constexpr bool none_of(I first, S last, Pred pred, Proj proj = {});
}
```

```

template<InputRange R, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
constexpr bool none_of(R&& r, Pred pred, Proj proj = {});
}

```

1     Let  $E$  be `pred(*i)` and `invoke(pred, invoke(proj, *i))` for the overloads in namespace `std` and `std::ranges`, respectively.

2     ~~Returns: true if  $[first, last)$  is empty or if `pred(*i)` is false for every iterator  $i$  in the range  $[first, last)$ , and false otherwise.~~

      Returns: false if  $E$  is true for some iterator  $i$  in the range  $[first, last)$ , and true otherwise.

3     Complexity: At most `last - first` applications of the predicate and any projection.

#### 24.5.4 For each

[**alg.foreach**]

[...]

10     [Note: Does not return a copy of its `Function` parameter, since parallelization may not permit efficient state accumulation. — end note]

```

namespace ranges {
 template<InputIterator I, Sentinel<I> S, class Proj = identity,
 IndirectUnaryInvocable<projected<I, Proj>> Fun>
 constexpr for_each_result<I, Fun>
 for_each(I first, S last, Fun f, Proj proj = {});
 template<InputRange R, class Proj = identity,
 IndirectUnaryInvocable<projected<iterator_t<R>, Proj>> Fun>
 constexpr for_each_result<safe_iterator_t<R>, Fun>
 for_each(R&& r, Fun f, Proj proj = {});
}

```

11     Effects: Calls `invoke(f, invoke(proj, *i))` for every iterator  $i$  in the range  $[first, last)$ , starting from `first` and proceeding to `last - 1`. [Note: If the result of `invoke(proj, *i)` is a mutable reference, `f` may apply non-constant functions. — end note]

12     Returns: `{last, std::move(f)}`.

13     Complexity: Applies `f` and `proj` exactly `last - first` times.

14     Remarks: If `f` returns a result, the result is ignored.

15     [Note: The overloads in namespace `ranges` require `Fun` to model `CopyConstructible`. — end note]

[...]

#### 24.5.5 Find

[**alg.find**]

```

template<class InputIterator, class T>
constexpr InputIterator find(InputIterator first, InputIterator last,
 const T& value);
template<class ExecutionPolicy, class ForwardIterator, class T>
ForwardIterator find(ExecutionPolicy&& exec, ForwardIterator first, ForwardIterator last,
 const T& value);

template<class InputIterator, class Predicate>
constexpr InputIterator find_if(InputIterator first, InputIterator last,
 Predicate pred);
template<class ExecutionPolicy, class ForwardIterator, class Predicate>
ForwardIterator find_if(ExecutionPolicy&& exec, ForwardIterator first, ForwardIterator last,
 Predicate pred);

template<class InputIterator, class Predicate>
constexpr InputIterator find_if_not(InputIterator first, InputIterator last,
 Predicate pred);
template<class ExecutionPolicy, class ForwardIterator, class Predicate>
ForwardIterator find_if_not(ExecutionPolicy&& exec, ForwardIterator first, ForwardIterator last,
 Predicate pred);

```

```

namespace ranges {
 template<InputIterator I, Sentinel<I> S, class T, class Proj = identity>
 requires IndirectRelation<ranges::equal_to<>, projected<I, Proj>, const T*>
 constexpr I find(I first, S last, const T& value, Proj proj = {});
 template<InputRange R, class T, class Proj = identity>
 requires IndirectRelation<ranges::equal_to<>, projected<iterator_t<R>, Proj>, const T*>
 constexpr safe_iterator_t<R>
 find(R&& r, const T& value, Proj proj = {});
 template<InputIterator I, Sentinel<I> S, class Proj = identity,
 IndirectUnaryPredicate<projected<I, Proj>> Pred>
 constexpr I find_if(I first, S last, Pred pred, Proj proj = {});
 template<InputRange R, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
 constexpr safe_iterator_t<R>
 find_if(R&& r, Pred pred, Proj proj = {});
 template<InputIterator I, Sentinel<I> S, class Proj = identity,
 IndirectUnaryPredicate<projected<I, Proj>> Pred>
 constexpr I find_if_not(I first, S last, Pred pred, Proj proj = {});
 template<InputRange R, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
 constexpr safe_iterator_t<R>
 find_if_not(R&& r, Pred pred, Proj proj = {});
}

```

1 Let  $E$  be:

- (1.1) —  $*i == \text{value}$  for `find`,
- (1.2) —  $\text{pred}(*i) != \text{false}$  for `find_if`,
- (1.3) —  $\text{pred}(*i) == \text{false}$  for `find_if_not`,
- (1.4) —  $\text{invoke}(\text{proj}, *i) == \text{value}$  for `ranges::find`,
- (1.5) —  $\text{invoke}(\text{pred}, \text{invoke}(\text{proj}, *i)) != \text{false}$  for `ranges::find_if`,
- (1.6) —  $\text{invoke}(\text{pred}, \text{invoke}(\text{proj}, *i)) == \text{false}$  for `ranges::find_if_not`.

2 *Returns:* The first iterator  $i$  in the range  $[\text{first}, \text{last})$  for which  $E$  is true. ~~the following corresponding conditions hold:  $*i == \text{value}$ ,  $\text{pred}(*i) != \text{false}$ ,  $\text{pred}(*i) == \text{false}$~~ . Returns `last` if no such iterator is found.

3 *Complexity:* At most  $\text{last} - \text{first}$  applications of the corresponding predicate and any projection.

## 24.5.6 Find end

[alg.find.end]

```

template<class ForwardIterator1, class ForwardIterator2>
 constexpr ForwardIterator1
 find_end(ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
 ForwardIterator1
 find_end(ExecutionPolicy&& exec,
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2);

template<class ForwardIterator1, class ForwardIterator2,
 class BinaryPredicate>
 constexpr ForwardIterator1
 find_end(ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2,
 BinaryPredicate pred);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class BinaryPredicate>
 ForwardIterator1
 find_end(ExecutionPolicy&& exec,
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2,

```

```

 BinaryPredicate pred);

namespace ranges {
 template<ForwardIterator I1, Sentinel<I1> S1, ForwardIterator I2, Sentinel<I2> S2,
 class Pred = ranges::equal_to<>, class Proj1 = identity, class Proj2 = identity>
 requires IndirectlyComparable<I1, I2, Pred, Proj1, Proj2>
 constexpr subrange<I1>
 find_end(I1 first1, S1 last1, I2 first2, S2 last2, Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
 template<ForwardRange R1, ForwardRange R2,
 class Pred = ranges::equal_to<>, class Proj1 = identity, class Proj2 = identity>
 requires IndirectlyComparable<iterator_t<R1>, iterator_t<R2>, Pred, Proj1, Proj2>
 constexpr safe_subrange_t<R1>
 find_end(R1&& r1, R2&& r2, Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
}

```

[Editor's note: This wording incorporates the PR for [stl2#180](#) and [stl2#526](#)].

- 1 Let:
- (1.1) — `pred` be `equal_to{}` for the overloads with no parameter `pred`.
- (1.2) —  $E$  be:
  - (1.2.1) — `pred(*(i + n), *(first2 + n))` for the overloads in namespace `std`,
  - (1.2.2) — `invoke(pred, invoke(proj1, *(i + n)), invoke(proj2, *(first2 + n)))` for the overloads in namespace `ranges`.
- (1.3) —  $i$  be `last1` if `[first2, last2)` is empty, or if `(last2 - first2) > (last1 - first1)` is true, or if there is no iterator in the range `[first1, last1 - (last2 - first2))` such that for every non-negative integer  $n < (last2 - first2)$ ,  $E$  is true. Otherwise  $i$  is the last such iterator in `[first1, last1 - (last2 - first2))`.
- 2 *Effects:* Finds a subsequence of equal values in a sequence.
- 3 *Returns:* The last iterator  $i$  in the range `[first1, last1 - (last2 - first2))` such that for every non-negative integer  $n < (last2 - first2)$ , the following corresponding conditions hold: `*(i + n) == *(first2 + n)`, `pred(*(i + n), *(first2 + n)) != false`. Returns `last1` if `[first2, last2)` is empty or if no such iterator is found.
- 4 *Returns:*
  - (4.1) —  $i$  for the overloads in namespace `std`, and
  - (4.2) — `{i, i + (i == last1 ? 0 : last2 - first2)}` for the overloads in namespace `ranges`.
- 5 *Complexity:* At most `(last2 - first2) * (last1 - first1 - (last2 - first2) + 1)` applications of the corresponding predicate [and any projections](#).

## 24.5.7 Find first

[`alg.find.first.of`]

```

template<class InputIterator, class ForwardIterator>
constexpr InputIterator
 find_first_of(InputIterator first1, InputIterator last1,
 ForwardIterator first2, ForwardIterator last2);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
ForwardIterator1
 find_first_of(ExecutionPolicy&& exec,
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2);

template<class InputIterator, class ForwardIterator,
 class BinaryPredicate>
constexpr InputIterator
 find_first_of(InputIterator first1, InputIterator last1,
 ForwardIterator first2, ForwardIterator last2,
 BinaryPredicate pred);

```

```

template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class BinaryPredicate>
ForwardIterator1
find_first_of(ExecutionPolicy&& exec,
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2,
 BinaryPredicate pred);

namespace ranges {
 template<InputIterator I1, Sentinel<I1> S1, ForwardIterator I2, Sentinel<I2> S2,
 class Proj1 = identity, class Proj2 = identity,
 IndirectRelation<projected<I1, Proj1>, projected<I2, Proj2>> Pred = ranges::equal_to<>>
 constexpr I1 find_first_of(I1 first1, S1 last1, I2 first2, S2 last2,
 Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
 template<InputRange R1, ForwardRange R2, class Proj1 = identity,
 class Proj2 = identity,
 IndirectRelation<projected<iterator_t<R1>, Proj1>,
 projected<iterator_t<R2>, Proj2>> Pred = ranges::equal_to<>>
 constexpr safe_iterator_t<R1>
 find_first_of(R1&& r1, R2&& r2,
 Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
}

```

- 1 Let  $E$  be:
- (1.1) —  $*i == *j$  for the overloads with no parameter `pred`,
- (1.2) — `pred(*i, *j) != false` for the overloads with a parameter `pred` and no parameter `proj1`,
- (1.3) — `invoke(pred, invoke(proj1, *i), invoke(proj2, *j)) != false` for the overloads with both parameters `pred` and `proj1`.
- 2 *Effects:* Finds an element that matches one of a set of values.
- 3 *Returns:* The first iterator  $i$  in the range  $[first1, last1)$  such that for some iterator  $j$  in the range  $[first2, last2)$   $E$  holds. ~~the following conditions hold:  $*i == *j$ , `pred(*i, *j) != false`.~~ Returns `last1` if  $[first2, last2)$  is empty or if no such iterator is found.
- 4 *Complexity:* At most  $(last1 - first1) * (last2 - first2)$  applications of the corresponding predicate and any projections.

## 24.5.8 Adjacent find

[alg.adjacent.find]

```

template<class ForwardIterator>
constexpr ForwardIterator
adjacent_find(ForwardIterator first, ForwardIterator last);
template<class ExecutionPolicy, class ForwardIterator>
ForwardIterator
adjacent_find(ExecutionPolicy&& exec,
 ForwardIterator first, ForwardIterator last);

template<class ForwardIterator, class BinaryPredicate>
constexpr ForwardIterator
adjacent_find(ForwardIterator first, ForwardIterator last,
 BinaryPredicate pred);
template<class ExecutionPolicy, class ForwardIterator, class BinaryPredicate>
ForwardIterator
adjacent_find(ExecutionPolicy&& exec,
 ForwardIterator first, ForwardIterator last,
 BinaryPredicate pred);

namespace ranges {
 template<ForwardIterator I, Sentinel<I> S, class Proj = identity,
 IndirectRelation<projected<I, Proj>> Pred = ranges::equal_to<>>
 constexpr I adjacent_find(I first, S last, Pred pred = {},
 Proj proj = {});
}

```

```

template<ForwardRange R, class Proj = identity,
 IndirectRelation<projected<iterator_t<R>, Proj>> Pred = ranges::equal_to<>>
 constexpr safe_iterator_t<R>
 adjacent_find(R&& r, Pred pred = {}, Proj proj = {});
}

```

1 Let  $E$  be:

- (1.1) —  $*i == *(i + 1)$  for the overloads with no parameter `pred`,
- (1.2) —  $\text{pred}(*i, *(i + 1)) \neq \text{false}$  for the overloads with a parameter `pred` and no parameter `proj`,
- (1.3) —  $\text{invoke}(\text{pred}, \text{invoke}(\text{proj}, *i), \text{invoke}(\text{proj}, *(i + 1))) \neq \text{false}$  for the overloads with both parameters `pred` and `proj`.

2 *Returns:* The first iterator  $i$  such that both  $i$  and  $i + 1$  are in the range  $[\text{first}, \text{last})$  for which  $E$  holds. ~~the following corresponding conditions hold:  $*i == *(i + 1)$ ,  $\text{pred}(*i, *(i + 1)) \neq \text{false}$ .~~  
Returns `last` if no such iterator is found.

3 *Complexity:* For the overloads with no `ExecutionPolicy`, exactly  $\min((i - \text{first}) + 1, (\text{last} - \text{first}) - 1)$  applications of the corresponding predicate, where  $i$  is `adjacent_find`'s return value, and no more than twice as many applications of any projection. For the overloads with an `ExecutionPolicy`,  $\mathcal{O}(\text{last} - \text{first})$  applications of the corresponding predicate.

## 24.5.9 Count

[alg.count]

```

template<class InputIterator, class T>
constexpr typename iterator_traits<InputIterator>::difference_type
count(InputIterator first, InputIterator last, const T& value);
template<class ExecutionPolicy, class ForwardIterator, class T>
typename iterator_traits<ForwardIterator>::difference_type
count(ExecutionPolicy&& exec,
 ForwardIterator first, ForwardIterator last, const T& value);

template<class InputIterator, class Predicate>
constexpr typename iterator_traits<InputIterator>::difference_type
count_if(InputIterator first, InputIterator last, Predicate pred);
template<class ExecutionPolicy, class ForwardIterator, class Predicate>
typename iterator_traits<ForwardIterator>::difference_type
count_if(ExecutionPolicy&& exec,
 ForwardIterator first, ForwardIterator last, Predicate pred);

```

```

namespace ranges {
template<InputIterator I, Sentinel<I> S, class T, class Proj = identity>
requires IndirectRelation<ranges::equal_to<>, projected<I, Proj>, const T*>
constexpr iter_difference_t<I>
count(I first, S last, const T& value, Proj proj = {});
template<InputRange R, class T, class Proj = identity>
requires IndirectRelation<ranges::equal_to<>, projected<iterator_t<R>, Proj>, const T*>
constexpr iter_difference_t<iterator_t<R>>
count(R&& r, const T& value, Proj proj = {});
template<InputIterator I, Sentinel<I> S, class Proj = identity,
 IndirectUnaryPredicate<projected<I, Proj>> Pred>
constexpr iter_difference_t<I>
count_if(I first, S last, Pred pred, Proj proj = {});
template<InputRange R, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
constexpr iter_difference_t<iterator_t<R>>
count_if(R&& r, Pred pred, Proj proj = {});
}

```

1 Let  $E$  be:

- (1.1) —  $*i == \text{value}$  for the overloads with no parameter `pred` or `proj`,
- (1.2) —  $\text{pred}(*i) \neq \text{false}$  for the overloads with a parameter `pred` but no parameter `proj`,
- (1.3) —  $\text{invoke}(\text{proj}, *i) == \text{value}$  for the overloads with a parameter `proj` but no parameter `pred`,

- (1.4) — `invoke(pred, invoke(proj, *i)) != false` for the overloads with both parameters `proj` and `pred`.
- 2 *Effects:* Returns the number of iterators `i` in the range `[first, last)` for which `E` holds. ~~the following corresponding conditions hold: `*i == value`, `pred(*i) != false`.~~
- 3 *Complexity:* Exactly `last - first` applications of the corresponding predicate and any projection.

## 24.5.10 Mismatch

[mismatch]

```
template<class InputIterator1, class InputIterator2>
constexpr pair<InputIterator1, InputIterator2>
mismatch(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2);

template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
pair<ForwardIterator1, ForwardIterator2>
mismatch(ExecutionPolicy&& exec,
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2);

template<class InputIterator1, class InputIterator2,
 class BinaryPredicate>
constexpr pair<InputIterator1, InputIterator2>
mismatch(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, BinaryPredicate pred);

template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class BinaryPredicate>
pair<ForwardIterator1, ForwardIterator2>
mismatch(ExecutionPolicy&& exec,
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, BinaryPredicate pred);

template<class InputIterator1, class InputIterator2>
constexpr pair<InputIterator1, InputIterator2>
mismatch(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, InputIterator2 last2);

template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
pair<ForwardIterator1, ForwardIterator2>
mismatch(ExecutionPolicy&& exec,
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2);

template<class InputIterator1, class InputIterator2,
 class BinaryPredicate>
constexpr pair<InputIterator1, InputIterator2>
mismatch(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, InputIterator2 last2,
 BinaryPredicate pred);

template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class BinaryPredicate>
pair<ForwardIterator1, ForwardIterator2>
mismatch(ExecutionPolicy&& exec,
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2,
 BinaryPredicate pred);

namespace ranges {
template<InputIterator I1, Sentinel<I1> S1, InputIterator I2, Sentinel<I2> S2,
 class Proj1 = identity, class Proj2 = identity,
 IndirectRelation<projected<I1, Proj1>, projected<I2, Proj2>> Pred = ranges::equal_to<>>
constexpr mismatch_result<I1, I2>
mismatch(I1 first1, S1 last1, I2 first2, S2 last2, Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
```

```

template<InputRange R1, InputRange R2,
 class Proj1 = identity, class Proj2 = identity,
 IndirectRelation<projected<iterator_t<R1>, Proj1>,
 projected<iterator_t<R2>, Proj2>> Pred = ranges::equal_to<>>
constexpr mismatch_result<safe_iterator_t<R1>, safe_iterator_t<R2>>
 mismatch(R1&& r1, R2&& r2, Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
}

```

- 1 ~~Remarks: If last2 was not given in the argument list, it denotes first2 + (last1 - first1) below.~~
- 2 Let last2 be first2 + (last1 - first1) for the overloads with no parameter last2 or r2.
- 3 Let  $E$  be:
- (3.1) —  $!(*(first1 + n) == *(first2 + n))$  for the overloads with no parameter pred,
- (3.2) —  $pred(*(first1 + n), *(first2 + n)) == false$  for the overloads with a parameter pred and no parameter proj1,
- (3.3) —  $!invoke(pred, invoke(proj1, *(first1 + n)), invoke(proj2, *(first2 + n)))$  for the overloads with both parameters pred and proj1.
- 4 ~~Returns: A pair of iterators first1 + n and first2 + n { first1 + n, first2 + n }, where n is the smallest integer such that  $E$  holds, respectively;~~
- (4.1) —  ~~$!(*(first1 + n) == *(first2 + n))$  or~~
- (4.2) —  ~~$pred(*(first1 + n), *(first2 + n)) == false$ ;~~
- or  $\min(last1 - first1, last2 - first2)$  if no such integer exists.
- 5 *Complexity:* At most  $\min(last1 - first1, last2 - first2)$  applications of the corresponding predicate and any projections.

### 24.5.11 Equal

[alg.equal]

```

template<class InputIterator1, class InputIterator2>
constexpr bool equal(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
bool equal(ExecutionPolicy&& exec,
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2);

template<class InputIterator1, class InputIterator2,
 class BinaryPredicate>
constexpr bool equal(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, BinaryPredicate pred);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class BinaryPredicate>
bool equal(ExecutionPolicy&& exec,
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, BinaryPredicate pred);

template<class InputIterator1, class InputIterator2>
constexpr bool equal(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, InputIterator2 last2);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
bool equal(ExecutionPolicy&& exec,
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2);

template<class InputIterator1, class InputIterator2,
 class BinaryPredicate>
constexpr bool equal(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, InputIterator2 last2,
 BinaryPredicate pred);

```

```

template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class BinaryPredicate>
bool equal(ExecutionPolicy&& exec,
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator2 last2,
 BinaryPredicate pred);

namespace ranges {
 template<InputIterator I1, Sentinel<I1> S1, InputIterator I2, Sentinel<I2> S2,
 class Pred = ranges::equal_to<>, class Proj1 = identity, class Proj2 = identity>
 requires IndirectlyComparable<I1, I2, Pred, Proj1, Proj2>
 constexpr bool equal(I1 first1, S1 last1, I2 first2, S2 last2,
 Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
 template<InputRange R1, InputRange R2, class Pred = ranges::equal_to<>,
 class Proj1 = identity, class Proj2 = identity>
 requires IndirectlyComparable<iterator_t<R1>, iterator_t<R2>, Pred, Proj1, Proj2>
 constexpr bool equal(R1&& r1, R2&& r2, Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
}

```

1 *Remarks:* If `last2` was not given in the argument list, it denotes `first2 + (last1 - first1)` below.

2 *Let:*

(2.1) — `last2` be `first2 + (last1 - first1)` for the overloads with no parameter `last2` or `r2`,

(2.2) — `pred` be `equal_to{}` for the overloads with no parameter `pred`,

(2.3) — *E* be:

(2.3.1) — `pred(*i, *(first2 + (i - first1)))` for the overloads with no parameter `proj1`,

(2.3.2) — `invoke(pred, invoke(proj1, *i), invoke(proj2, *(first2 + (i - first1))))` for the overloads with parameter `proj1`.

3 *Returns:* If `last1 - first1 != last2 - first2`, return `false`. Otherwise return `true` if *E* holds for every iterator *i* in the range `[first1, last1)` ~~the following corresponding conditions hold: `*i == *(first2 + (i - first1))`, `pred(*i, *(first2 + (i - first1))) != false`~~. Otherwise, returns `false`.

4 *Complexity:* If the types of `first1`, `last1`, `first2`, and `last2`:

(4.1) — meet the *Cpp17RandomAccessIterator* requirements (22.3.5.6) for the overloads in namespace `std`, or

(4.2) — pairwise model *SizedSentinel* (22.3.4.8) for the overloads in namespace `ranges`,

and `last1 - first1 != last2 - first2`, then no applications of the corresponding predicate and each projection; otherwise,

(4.3) — For the overloads with no *ExecutionPolicy*, at most `min(last1 - first1, last2 - first2)` applications of the corresponding predicate and any projections.

(4.3.1) — ~~if *InputIterator1* and *InputIterator2* meet the *Cpp17RandomAccessIterator* requirements (22.3.5.6) and `last1 - first1 != last2 - first2`, then no applications of the corresponding predicate; otherwise,~~

(4.4) — For the overloads with an *ExecutionPolicy*,  $\mathcal{O}(\min(\text{last1} - \text{first1}, \text{last2} - \text{first2}))$  applications of the corresponding predicate.

(4.4.1) — ~~if *ForwardIterator1* and *ForwardIterator2* meet the *Cpp17RandomAccessIterator* requirements and `last1 - first1 != last2 - first2`, then no applications of the corresponding predicate; otherwise,~~

## 24.5.12 Is permutation

[`alg.is_permutation`]

[...]

```

namespace ranges {
 template<ForwardIterator I1, Sentinel<I1> S1, ForwardIterator I2,
 Sentinel<I2> S2, class Pred = ranges::equal_to<>, class Proj1 = identity,

```

```

 class Proj2 = identity>
requires IndirectlyComparable<I1, I2, Pred, Proj1, Proj2>
constexpr bool is_permutation(I1 first1, S1 last1, I2 first2, S2 last2,
 Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
template<ForwardRange R1, ForwardRange R2, class Pred = ranges::equal_to<>,
 class Proj1 = identity, class Proj2 = identity>
requires IndirectlyComparable<iterator_t<R1>, iterator_t<R2>, Pred, Proj1, Proj2>
constexpr bool is_permutation(R1&& r1, R2&& r2, Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
}

```

5 *Returns:* If `last1 - first1 != last2 - first2`, return false. Otherwise return true if there exists a permutation of the elements in the range `[first2, last2)`, bounded by `[pfirst, plast)`, such that `ranges::equal(first1, last1, pfirst, plast, pred, proj1, proj2)` returns true; otherwise, returns false.

6 *Complexity:* No applications of the corresponding predicate and projections if:

- (6.1) — `S1` and `I1` model `SizedSentinel`,
- (6.2) — `S2` and `I2` model `SizedSentinel`, and
- (6.3) — `last1 - first1 != last2 - first2`.

Otherwise, exactly `last1 - first1` applications of the corresponding predicate and projections if `ranges::equal(first1, last1, first2, last2, pred, proj1, proj2)` would return true; otherwise, at worst  $\mathcal{O}(N^2)$ , where  $N$  has the value `last1 - first1`.

## 24.5.13 Search

[alg.search]

[...]

3 *Complexity:* At most `(last1 - first1) * (last2 - first2)` applications of the corresponding predicate.

```

namespace ranges {
template<ForwardIterator I1, Sentinel<I1> S1, ForwardIterator I2,
 Sentinel<I2> S2, class Pred = ranges::equal_to<>,
 class Proj1 = identity, class Proj2 = identity>
requires IndirectlyComparable<I1, I2, Pred, Proj1, Proj2>
constexpr subrange<I1>
search(I1 first1, S1 last1, I2 first2, S2 last2, Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
template<ForwardRange R1, ForwardRange R2, class Pred = ranges::equal_to<>,
 class Proj1 = identity, class Proj2 = identity>
requires IndirectlyComparable<iterator_t<R1>, iterator_t<R2>, Pred, Proj1, Proj2>
constexpr safe_subrange_t<R1>
search(R1&& r1, R2&& r2, Pred pred = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
}

```

[Editor's note: This wording incorporates the PR for [stl2#180](#) and [stl2#526](#).]

4 *Effects:* Finds a subsequence of equal values in a sequence.

5 *Returns:*

- (5.1) — `{i, i + (last2 - first2)}`, where `i` is the first iterator in the range `[first1, last1 - (last2 - first2))` such that for every non-negative integer `n` less than `last2 - first2` the condition `bool(invoke(pred, invoke(proj1, *(i + n)), invoke(proj2, *(first2 + n))))` is true;
- (5.2) — Returns `{last1, last1}` if no such iterator exists.

6 *Complexity:* At most `(last1 - first1) * (last2 - first2)` applications of the corresponding predicate and projections.

```
template<class ForwardIterator, class Size, class T>
constexpr ForwardIterator
search_n(ForwardIterator first, ForwardIterator last,
 Size count, const T& value);
```

[...]

10 *Complexity:* At most `last - first` applications of the corresponding predicate.

```
namespace ranges {
 template<ForwardIterator I, Sentinel<I> S, class T,
 class Pred = ranges::equal_to<>, class Proj = identity>
 requires IndirectlyComparable<I, const T*, Pred, Proj>
 constexpr subrange<I>
 search_n(I first, S last, iter_difference_t<I> count,
 const T& value, Pred pred = {}, Proj proj = {});
 template<ForwardRange R, class T, class Pred = ranges::equal_to<>,
 class Proj = identity>
 requires IndirectlyComparable<iterator_t<R>, const T*, Pred, Proj>
 constexpr safe_subrange_t<R>
 search_n(R&& r, iter_difference_t<iterator_t<R>> count,
 const T& value, Pred pred = {}, Proj proj = {});
}
```

[Editor's note: This wording incorporates the PR for [stl2#180](#) and [stl2#526](#).]

11 *Effects:* Finds a subsequence of equal values in a sequence.

12 *Returns:* `{i, i + count}` where `i` is the first iterator in the range `[first, last - count)` such that for every non-negative integer `n` less than `count`, the following condition holds: `invoke(pred, invoke(proj, *(i + n)), value)`. Returns `{last, last}` if no such iterator is found.

13 *Complexity:* At most `last - first` applications of the corresponding predicate and projection.

[...]

## 24.6 Mutating sequence operations

[alg.modifying.operations]

### 24.6.1 Copy

[alg.copy]

```
template<class InputIterator, class OutputIterator>
constexpr OutputIterator copy(InputIterator first, InputIterator last,
 OutputIterator result);
```

```
namespace ranges {
 template<InputIterator I, Sentinel<I> S, WeaklyIncrementable O>
 requires IndirectlyCopyable<I, O>
 constexpr copy_result<I, O>
 copy(I first, S last, O result);
 template<InputRange R, WeaklyIncrementable O>
 requires IndirectlyCopyable<iterator_t<R>, O>
 constexpr copy_result<safe_iterator_t<R>, O>
 copy(R&& r, O result);
}
```

1 *Requires:* `result` shall not be in the range `[first, last)`.

2 *Effects:* Copies elements in the range `[first, last)` into the range `[result, result + (last - first))` starting from `first` and proceeding to `last`. For each non-negative integer `n` ~~`< (last - first)`~~, performs `*(result + n) = *(first + n)`.

3 *Returns:*

- (3.1) — `result + (last - first)` for the overload in namespace `std`, or
- (3.2) — `{last, result + (last - first)}` for the overloads in namespace `ranges`.

4 *Complexity:* Exactly `last - first` assignments.

```
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
ForwardIterator2 copy(ExecutionPolicy&& policy,
 ForwardIterator1 first, ForwardIterator1 last,
 ForwardIterator2 result);
```

5 *Requires:* The ranges  $[first, last)$  and  $[result, result + (last - first))$  shall not overlap.

6 *Effects:* Copies elements in the range  $[first, last)$  into the range  $[result, result + (last - first))$ . For each non-negative integer  $n < (last - first)$ , performs  $*(result + n) = *(first + n)$ .

7 *Returns:*  $result + (last - first)$ .

8 *Complexity:* Exactly  $last - first$  assignments.

```
template<class InputIterator, class Size, class OutputIterator>
constexpr OutputIterator copy_n(InputIterator first, Size n,
 OutputIterator result);
template<class ExecutionPolicy, class ForwardIterator1, class Size, class ForwardIterator2>
ForwardIterator2 copy_n(ExecutionPolicy&& exec,
 ForwardIterator1 first, Size n,
 ForwardIterator2 result);
```

```
namespace ranges {
 template<InputIterator I, WeaklyIncrementable O>
 requires IndirectlyCopyable<I, O>
 constexpr copy_n_result<I, O>
 copy_n(I first, iter_difference_t<I> n, O result);
}
```

[Editor's note: This wording incorporates the PR for [stl2#498](#).]

9 Let  $M$  be  $\max(n, 0)$ .

10 *Effects:* For each non-negative integer  $i < n$ , performs  $*(result + i) = *(first + i)$ .

11 *Returns:*

(11.1) —  $result + n$  for the overloads in namespace `std`, or

(11.2) —  $\{first + M, result + M\}$  for the overload in namespace `ranges`.

12 *Complexity:* Exactly  $n$  assignments.

```
template<class InputIterator, class OutputIterator, class Predicate>
constexpr OutputIterator copy_if(InputIterator first, InputIterator last,
 OutputIterator result, Predicate pred);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class Predicate>
ForwardIterator2 copy_if(ExecutionPolicy&& exec,
 ForwardIterator1 first, ForwardIterator1 last,
 ForwardIterator2 result, Predicate pred);

namespace ranges {
 template<InputIterator I, Sentinel<I> S, WeaklyIncrementable O, class Proj = identity,
 IndirectUnaryPredicate<projected<I, Proj>> Pred>
 requires IndirectlyCopyable<I, O>
 constexpr copy_if_result<I, O>
 copy_if(I first, S last, O result, Pred pred, Proj proj = {});
 template<InputRange R, WeaklyIncrementable O, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
 requires IndirectlyCopyable<iterator_t<R>, O>
 constexpr copy_if_result<safe_iterator_t<R>, O>
 copy_if(R&& r, O result, Pred pred, Proj proj = {});
}
```

13 *Let  $E$  be:*

(13.1) —  $\text{bool}(\text{pred}(*i))$  for the overloads in namespace `std`, or

(13.2) —  $\text{bool}(\text{invoke}(\text{pred}, \text{invoke}(\text{proj}, *i)))$  for the overloads in namespace `ranges`.

and  $N$  be the number of iterators  $i$  in the range  $[first, last)$  for which the condition  $E$  holds.

*Requires:* The ranges  $[first, last)$  and  $[result, result + (last - first))$  shall not overlap.  
*[Note:* For the overload with an `ExecutionPolicy`, there may be a performance cost if `iterator_traits<ForwardIterator1>::value_type` is not *Cpp17MoveConstructible* ([\[tab:moveconstructible\]](#)).  
— end note]

*Effects:* Copies all of the elements referred to by the iterator  $i$  in the range  $[first, last)$  for which `pred(*i)E` is true.

*Returns:* ~~The end of the resulting range.~~

— `result + N` for the overloads in namespace `std`, or

— `{last, result + N}` for the overloads in namespace `ranges`.

*Complexity:* Exactly  $last - first$  applications of the corresponding predicate and any projection.

*Remarks:* Stable ([\[algorithm.stable\]](#)).

```
template<class BidirectionalIterator1, class BidirectionalIterator2>
constexpr BidirectionalIterator2
copy_backward(BidirectionalIterator1 first,
 BidirectionalIterator1 last,
 BidirectionalIterator2 result);
```

```
namespace ranges {
 template<BidirectionalIterator I1, Sentinel<I1> S1, BidirectionalIterator I2>
 requires IndirectlyCopyable<I1, I2>
 constexpr copy_backward_result<I1, I2>
 copy_backward(I1 first, S1 last, I2 result);
 template<BidirectionalRange R, BidirectionalIterator I>
 requires IndirectlyCopyable<iterator_t<R>, I>
 constexpr copy_backward_result<safe_iterator_t<R>, I>
 copy_backward(R&& r, I result);
}
```

*Requires:* `result` shall not be in the range  $(first, last]$ .

*Effects:* Copies elements in the range  $[first, last)$  into the range  $[result - (last - first), result)$  starting from  $last - 1$  and proceeding to `first`.<sup>5</sup> For each positive integer  ~~$n \leq (last - first)$~~   $n \leq (last - first)$ , performs  $*(result - \underline{nn}) = *(last - \underline{nn})$ .

*Returns:*

— `result - (last - first)` for the overload in namespace `std`, or

— `{last, result - (last - first)}` for the overloads in namespace `ranges`.

*Complexity:* Exactly  $last - first$  assignments.

## 24.6.2 Move

[alg.move]

```
template<class InputIterator, class OutputIterator>
constexpr OutputIterator move(InputIterator first, InputIterator last,
 OutputIterator result);
```

```
namespace ranges {
 template<InputIterator I, Sentinel<I> S, WeaklyIncrementable O>
 requires IndirectlyMovable<I, O>
 constexpr move_result<I, O>
 move(I first, S last, O result);
 template<InputRange R, WeaklyIncrementable O>
 requires IndirectlyMovable<iterator_t<R>, O>
 constexpr move_result<safe_iterator_t<R>, O>
 move(R&& r, O result);
}
```

Let  $E$  be

<sup>5</sup>) `copy_backward` should be used instead of `copy` when `last` is in the range  $[result - (last - first), result)$ .

- (1.1) — `std::move(*(first + n))` for the overload in namespace `std`, or
- (1.2) — `ranges::iter_move(first + n)` for the overloads in namespace `ranges`.
- 2 *Requires:* `result` shall not be in the range `[first, last)`.
- 3 *Effects:* Moves elements in the range `[first, last)` into the range `[result, result + (last - first))` starting from `first` and proceeding to `last`. For each non-negative integer  $n < (last - first)$ , performs ~~`*(result + n) = std::move(*(first + n))`~~ `*(result + n) = E`.
- 4 *Returns:*
- (4.1) — `result + (last - first)` for the overload in namespace `std`, or
- (4.2) — `{last, result + (last - first)}` for the overloads in namespace `ranges`.
- 5 *Complexity:* Exactly `last - first` ~~move~~ assignments.

```
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
ForwardIterator2 move(ExecutionPolicy&& policy,
 ForwardIterator1 first, ForwardIterator1 last,
 ForwardIterator2 result);
```

- 6 *Requires:* The ranges `[first, last)` and `[result, result + (last - first))` shall not overlap.
- 7 *Effects:* Moves elements in the range `[first, last)` into the range `[result, result + (last - first))`. For each non-negative integer  $n < (last - first)$ , performs `*(result + n) = std::move(*(first + n))`.
- 8 *Returns:* `result + (last - first)`.
- 9 *Complexity:* Exactly `last - first` assignments.

```
template<class BidirectionalIterator1, class BidirectionalIterator2>
constexpr BidirectionalIterator2
move_backward(BidirectionalIterator1 first, BidirectionalIterator1 last,
 BidirectionalIterator2 result);
```

```
namespace ranges {
template<BidirectionalIterator I1, Sentinel<I1> S1, BidirectionalIterator I2>
requires IndirectlyMovable<I1, I2>
constexpr move_backward_result<I1, I2>
move_backward(I1 first, S1 last, I2 result);
template<BidirectionalRange R, BidirectionalIterator I>
requires IndirectlyMovable<iterator_t<R>, I>
constexpr move_backward_result<safe_iterator_t<R>, I>
move_backward(R&& r, I result);
}
```

- 10 Let  $E$  be
- (10.1) — `std::move(*(last - n))` for the overload in namespace `std`, or
- (10.2) — `ranges::iter_move(last - n)` for the overloads in namespace `ranges`.
- 11 *Requires:* `result` shall not be in the range `(first, last]`.
- 12 *Effects:* Moves elements in the range `[first, last)` into the range `[result - (last - first), result)` starting from `last - 1` and proceeding to `first`.<sup>6</sup> For each positive integer  $n < (last - first)$ , performs ~~`*(result - n) = std::move(*(last - n))`~~ `*(result - n) = E`.
- 13 *Returns:*
- (13.1) — `result - (last - first)` for the overload in namespace `std`, or
- (13.2) — `{last, result - (last - first)}` for the overloads in namespace `ranges`.
- 14 *Complexity:* Exactly `last - first` assignments.

### 24.6.3 Swap

[alg.swap]

[Editor's note: This wording integrates the PR for [stl2#415](#).]

<sup>6</sup> `move_backward` should be used instead of ~~`move`~~ `move` when `last` is in the range `[result - (last - first), result)`.

```

template<class ForwardIterator1, class ForwardIterator2>
constexpr ForwardIterator2
 swap_ranges(ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
ForwardIterator2
 swap_ranges(ExecutionPolicy&& exec,
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2);

namespace ranges {
 template<InputIterator I1, Sentinel<I1> S1, InputIterator I2, Sentinel<I2> S2>
 requires IndirectlySwappable<I1, I2>
 constexpr swap_ranges_result<I1, I2>
 swap_ranges(I1 first1, S1 last1, I2 first2, S2 last2);
 template<InputRange R1, InputRange R2>
 requires IndirectlySwappable<iterator_t<R1>, iterator_t<R2>>
 constexpr swap_ranges_result<safe_iterator_t<R1>, safe_iterator_t<R2>>
 swap_ranges(R1&& r1, R2&& r2);
}

```

1 Let:

- (1.1) —  $last2$  be  $first2 + (last1 - first1)$  for the overloads with no parameter named  $last2$ , and
- (1.2) —  $M$  be  $\min(last1 - first1, last2 - first2)$ .

2 *Requires:* The two ranges  $[first1, last1)$  and  $[first2, ~~first2 + (last1 - first1)~~ last2)$  shall not overlap. For the overloads in namespace `std`,  $*(first1 + n)$  shall be swappable with  $([swappable.requirements]) * (first2 + n)$ .

3 *Effects:* For each non-negative integer  $n < (last1 - first1)$   $n < M$  performs:  ~~$swap(*(first1 + n), *(first2 + n))$~~

- (3.1) —  $swap(*(first1 + n), *(first2 + n))$  for the overloads in namespace `std`, or
- (3.2) — `ranges::iter_swap(first1 + n, first2 + n)` for the overloads in namespace `ranges`.

4 *Returns:*  ~~$first2 + (last1 - first1)$~~

- (4.1) —  $last2$  for the overloads in namespace `std`, or
- (4.2) —  $\{first1 + M, first2 + M\}$  for the overloads in namespace `ranges`.

5 *Complexity:* Exactly  ~~$last1 - first1$~~  $M$  swaps.

```

template<class ForwardIterator1, class ForwardIterator2>
constexpr void iter_swap(ForwardIterator1 a, ForwardIterator2 b);
[...]
```

## 24.6.4 Transform

[alg.transform]

```

template<class InputIterator, class OutputIterator,
 class UnaryOperation>
constexpr OutputIterator
 transform(InputIterator first1, InputIterator last1,
 OutputIterator result, UnaryOperation op);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class UnaryOperation>
ForwardIterator2
 transform(ExecutionPolicy&& exec,
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 result, UnaryOperation op);

template<class InputIterator1, class InputIterator2,
 class OutputIterator, class BinaryOperation>
constexpr OutputIterator
 transform(InputIterator1 first1, InputIterator1 last1,
 InputIterator2 first2, OutputIterator result,
 BinaryOperation binary_op);

```

```

template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class ForwardIterator, class BinaryOperation>
ForwardIterator
transform(ExecutionPolicy&& exec,
 ForwardIterator1 first1, ForwardIterator1 last1,
 ForwardIterator2 first2, ForwardIterator result,
 BinaryOperation binary_op);

namespace ranges {
 template<InputIterator I, Sentinel<I> S, WeaklyIncrementable O,
 CopyConstructible F, class Proj = identity>
 requires Writable<O, indirect_result_t<F&, projected<I, Proj>>>
 constexpr unary_transform_result<I, O>
 transform(I first1, S last1, O result, F op, Proj proj = {});
 template<InputRange R, WeaklyIncrementable O, CopyConstructible F,
 class Proj = identity>
 requires Writable<O, indirect_result_t<F&,
 projected<iterator_t<R>, Proj>>>
 constexpr unary_transform_result<safe_iterator_t<R>, O>
 transform(R&& r, O result, F op, Proj proj = {});
 template<InputIterator I1, Sentinel<I1> S1, InputIterator I2, Sentinel<I2> S2,
 WeaklyIncrementable O, CopyConstructible F, class Proj1 = identity,
 class Proj2 = identity>
 requires Writable<O, indirect_result_t<F&, projected<I1, Proj1>,
 projected<I2, Proj2>>>
 constexpr binary_transform_result<I1, I2, O>
 transform(I1 first1, S1 last1, I2 first2, S2 last2, O result,
 F binary_op, Proj1 proj1 = {}, Proj2 proj2 = {});
 template<InputRange R1, InputRange R2, WeaklyIncrementable O,
 CopyConstructible F, class Proj1 = identity, class Proj2 = identity>
 requires Writable<O, indirect_result_t<F&,
 projected<iterator_t<R1>, Proj1>, projected<iterator_t<R2>, Proj2>>>
 constexpr binary_transform_result<safe_iterator_t<R1>, safe_iterator_t<R2>, O>
 transform(R1&& r1, R2&& r2, O result,
 F binary_op, Proj1 proj1 = {}, Proj2 proj2 = {});
}

```

1 Let:

- (1.1) — `last2` be `first2 + (last1 - first1)` for the overloads with parameter `first2` but no parameter `last2`,
- (1.2) —  $N$  be `last1 - first1` for unary transforms, or  $\min(\text{last1} - \text{first1}, \text{last2} - \text{first2})$  for binary transforms, and
- (1.3) —  $E$  be
  - (1.3.1) — `op(*(first1 + (i - result)))` for unary transforms defined in namespace `std`,
  - (1.3.2) — `binary_op(*(first1 + (i - result)), *(first2 + (i - result)))` for binary transforms defined in namespace `std`,
  - (1.3.3) — `invoke(op, invoke(proj, *(first1 + (i - result))))` for unary transforms defined in namespace `ranges`, or
  - (1.3.4) — `invoke(binary_op, invoke(proj1, *(first1 + (i - result))), invoke(proj2, *(first2 + (i - result))))` for binary transforms defined in namespace `ranges`.

2 *Requires:* `op` and `binary_op` shall not invalidate iterators or subranges, or modify elements in the ranges

- (2.1) — `[first1, last1first1 +  $N$ ]`,
- (2.2) — `[first2, first2 + (last1 - first1) $N$ ]`, and
- (2.3) — `[result, result + (last1 - first1) $N$ ]`.<sup>7</sup>

<sup>7</sup> The use of fully closed ranges is intentional.

3 *Effects:* Assigns through every iterator  $i$  in the range  $[\text{result}, \text{result} + (\text{last1} - \text{first1})N)$  a new corresponding value equal to  $E \text{ op}(*(\text{first1} + (i - \text{result})))$  or  $\text{binary\_op}(*(\text{first1} + (i - \text{result})), *(\text{first2} + (i - \text{result})))$ .

4 *Returns:*

(4.1) —  $\text{result} + (\text{last1} - \text{first1})N$  for the overloads defined in namespace `std`,

(4.2) —  $\{\text{first1} + N, \text{result} + N\}$  for unary transforms defined in namespace `ranges`, or

(4.3) —  $\{\text{first1} + N, \text{first2} + N, \text{result} + N\}$  for binary transforms defined in namespace `ranges`.

5 *Complexity:* Exactly  $\text{last1} - \text{first1}N$  applications of `op` or `binary_op`, and any projections. This requirement also applies to the overload with an `ExecutionPolicy`.

6 *Remarks:* `result` may be equal to `first1` or `first2`.

## 24.6.5 Replace

[alg.replace]

```
template<class ForwardIterator, class T>
constexpr void replace(ForwardIterator first, ForwardIterator last,
 const T& old_value, const T& new_value);
template<class ExecutionPolicy, class ForwardIterator, class T>
void replace(ExecutionPolicy&& exec,
 ForwardIterator first, ForwardIterator last,
 const T& old_value, const T& new_value);

template<class ForwardIterator, class Predicate, class T>
constexpr void replace_if(ForwardIterator first, ForwardIterator last,
 Predicate pred, const T& new_value);
template<class ExecutionPolicy, class ForwardIterator, class Predicate, class T>
void replace_if(ExecutionPolicy&& exec,
 ForwardIterator first, ForwardIterator last,
 Predicate pred, const T& new_value);

namespace ranges {
template<InputIterator I, Sentinel<I> S, class T1, class T2, class Proj = identity>
requires Writable<I, const T2&> &&
IndirectRelation<ranges::equal_to<>, projected<I, Proj>, const T1*>
constexpr I
replace(I first, S last, const T1& old_value, const T2& new_value, Proj proj = {});
template<InputRange R, class T1, class T2, class Proj = identity>
requires Writable<iterator_t<R>, const T2&> &&
IndirectRelation<ranges::equal_to<>, projected<iterator_t<R>, Proj>, const T1*>
constexpr safe_iterator_t<R>
replace(R&& r, const T1& old_value, const T2& new_value, Proj proj = {});
template<InputIterator I, Sentinel<I> S, class T, class Proj = identity,
 IndirectUnaryPredicate<projected<I, Proj>> Pred>
requires Writable<I, const T&>
constexpr I replace_if(I first, S last, Pred pred, const T& new_value, Proj proj = {});
template<InputRange R, class T, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
requires Writable<iterator_t<R>, const T&>
constexpr safe_iterator_t<R>
replace_if(R&& r, Pred pred, const T& new_value, Proj proj = {});
}
```

1 Let  $E$  be

(1.1) —  $\text{bool}(*i == \text{old\_value})$  for `replace`,

(1.2) —  $\text{bool}(\text{pred}(*i))$  for `replace_if`,

(1.3) —  $\text{bool}(\text{invoke}(\text{proj}, *i) == \text{old\_value})$  for `ranges::replace`, or

(1.4) —  $\text{bool}(\text{invoke}(\text{pred}, \text{invoke}(\text{proj}, *i)))$  for `ranges::replace_if`.

- 2 *Requires:* The expression `*first = new_value` shall be valid.
- 3 *Effects:* Substitutes elements referred by the iterator `i` in the range `[first, last)` with `new_value`,  
when *E* is true ~~the following corresponding conditions hold: `*i == old_value`, `pred(*i) != false`.~~
- 4 *Returns:* `last` for the overloads in namespace `ranges`.
- 5 *Complexity:* Exactly `last - first` applications of the corresponding predicate and any projection.

```
template<class InputIterator, class OutputIterator, class T>
constexpr OutputIterator
replace_copy(InputIterator first, InputIterator last,
 OutputIterator result,
 const T& old_value, const T& new_value);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2, class T>
ForwardIterator2
replace_copy(ExecutionPolicy&& exec,
 ForwardIterator1 first, ForwardIterator1 last,
 ForwardIterator2 result,
 const T& old_value, const T& new_value);

template<class InputIterator, class OutputIterator, class Predicate, class T>
constexpr OutputIterator
replace_copy_if(InputIterator first, InputIterator last,
 OutputIterator result,
 Predicate pred, const T& new_value);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class Predicate, class T>
ForwardIterator2
replace_copy_if(ExecutionPolicy&& exec,
 ForwardIterator1 first, ForwardIterator1 last,
 ForwardIterator2 result,
 Predicate pred, const T& new_value);

namespace ranges {
template<InputIterator I, Sentinel<I> S, class T1, class T2, OutputIterator<const T2&> O,
 class Proj = identity>
requires IndirectlyCopyable<I, O> &&
IndirectRelation<ranges::equal_to<>, projected<I, Proj>, const T1*>
constexpr replace_copy_result<I, O>
replace_copy(I first, S last, O result, const T1& old_value, const T2& new_value,
 Proj proj = {});
template<InputRange R, class T1, class T2, OutputIterator<const T2&> O,
 class Proj = identity>
requires IndirectlyCopyable<iterator_t<R>, O> &&
IndirectRelation<ranges::equal_to<>, projected<iterator_t<R>, Proj>, const T1*>
constexpr replace_copy_result<safe_iterator_t<R>, O>
replace_copy(R&& r, O result, const T1& old_value, const T2& new_value,
 Proj proj = {});

template<InputIterator I, Sentinel<I> S, class T, OutputIterator<const T&> O,
 class Proj = identity, IndirectUnaryPredicate<projected<I, Proj>> Pred>
requires IndirectlyCopyable<I, O>
constexpr replace_copy_if_result<I, O>
replace_copy_if(I first, S last, O result, Pred pred, const T& new_value,
 Proj proj = {});
template<InputRange R, class T, OutputIterator<const T&> O, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
requires IndirectlyCopyable<iterator_t<R>, O>
constexpr replace_copy_if_result<safe_iterator_t<R>, O>
replace_copy_if(R&& r, O result, Pred pred, const T& new_value,
 Proj proj = {});
}

```

- 6 Let *E* be
- (6.1) — `bool(*(first + (i - result)) == old_value)` for `replace_copy`,

- (6.2) — `bool(pred(*(first + (i - result))))` for `replace_copy_if`,  
 (6.3) — `bool(invoker(proj, *(first + (i - result))) == old_value)` for `ranges::replace_copy`,  
 (6.4) — `bool(invoker(pred, invoker(proj, *(first + (i - result))))` for `ranges::replace_copy_if`.

7 *Requires:* The results of the expressions `*first` and `new_value` shall be writable (22.3.1) to the `result` output iterator. The ranges `[first, last)` and `[result, result + (last - first))` shall not overlap.

8 *Effects:* Assigns to every iterator `i` in the range `[result, result + (last - first))` either `new_value` or `*(first + (i - result))` depending on whether *E* holds. ~~the following corresponding conditions hold:~~

```
*(first + (i - result)) == old_value
pred(*(first + (i - result))) != false
```

9 *Returns:*

- (9.1) — `result + (last - first)` for the overloads in namespace `std`, or  
 (9.2) — `{last, result + (last - first)}` for the overloads in namespace `ranges`.

10 *Complexity:* Exactly `last - first` applications of the corresponding predicate and any projection.

## 24.6.6 Fill

[alg.fill]

```
template<class ForwardIterator, class T>
constexpr void fill(ForwardIterator first, ForwardIterator last, const T& value);
template<class ExecutionPolicy, class ForwardIterator, class T>
void fill(ExecutionPolicy&& exec,
 ForwardIterator first, ForwardIterator last, const T& value);

template<class OutputIterator, class Size, class T>
constexpr OutputIterator fill_n(OutputIterator first, Size n, const T& value);
template<class ExecutionPolicy, class ForwardIterator, class Size, class T>
ForwardIterator fill_n(ExecutionPolicy&& exec,
 ForwardIterator first, Size n, const T& value);

namespace ranges {
 template<class T, OutputIterator<const T> O, Sentinel<O> S>
 constexpr O fill(O first, S last, const T& value);
 template<class T, OutputRange<const T> R>
 constexpr safe_iterator_t<R> fill(R&& r, const T& value);
 template<class T, OutputIterator<const T> O>
 constexpr O fill_n(O first, iter_difference_t<O> n, const T& value);
}
```

1 Let  $N$  be  $\max(0, n)$  for the `fill_n` algorithms, and `last - first` for the `fill` algorithms.

2 *Requires:* The expression `value` shall be writable (22.3.1) to the output iterator. The type `Size` shall be convertible to an integral type ([conv.integral], [class.conv]).

3 *Effects:* Assigns `value` through all the iterators in the range `[first, first + N)`. ~~The `fill` algorithms assign `value` through all the iterators in the range `[first, last)`. The `fill_n` algorithms assign `value` through all the iterators in the range `[first, first + n)` if `n` is positive, otherwise they do nothing.~~

4 *Returns:* `first + N`. `fill_n` returns `first + n` for non-negative values of `n` and `first` for negative values.

5 *Complexity:* Exactly  $N$  ~~`last - first`, `n`, or `0`~~ assignments, ~~respectively~~.

## 24.6.7 Generate

[alg.generate]

```
template<class ForwardIterator, class Generator>
constexpr void generate(ForwardIterator first, ForwardIterator last,
 Generator gen);
```

```

template<class ExecutionPolicy, class ForwardIterator, class Generator>
 void generate(ExecutionPolicy&& exec,
 ForwardIterator first, ForwardIterator last,
 Generator gen);

template<class OutputIterator, class Size, class Generator>
 constexpr OutputIterator generate_n(OutputIterator first, Size n, Generator gen);
template<class ExecutionPolicy, class ForwardIterator, class Size, class Generator>
 ForwardIterator generate_n(ExecutionPolicy&& exec,
 ForwardIterator first, Size n, Generator gen);

```

```

namespace ranges {
 template<Iterator O, Sentinel<O> S, CopyConstructible F>
 requires Invocable<F&> && Writable<O, invoke_result_t<F&>>
 constexpr O generate(O first, S last, F gen);
 template<class R, CopyConstructible F>
 requires Invocable<F&> && OutputRange<R, invoke_result_t<F&>>
 constexpr safe_iterator_t<R> generate(R&& r, F gen);
 template<Iterator O, CopyConstructible F>
 requires Invocable<F&> && Writable<O, invoke_result_t<F&>>
 constexpr O generate_n(O first, iter_difference_t<O> n, F gen);
}

```

- 1 Let  $N$  be  $\max(0, n)$  for the `generate_n` algorithms, and  $\text{last} - \text{first}$  for the `generate` algorithms.
- 2 *Requires:* ~~`gen` takes no arguments~~; `Size` shall be convertible to an integral type (`[conv.integral]`, `[class.conv]`).
- 3 *Effects:* Assigns the result of successive evaluations of `gen()` through each iterator in the range  $[\text{first}, \text{first} + N)$ .  
~~The `generate` algorithms invoke the function object `gen` and assign the return value of `gen` through all the iterators in the range  $[\text{first}, \text{last})$ . The `generate_n` algorithms invoke the function object `gen` and assign the return value of `gen` through all the iterators in the range  $[\text{first}, \text{first} + n)$  if  $n$  is positive, otherwise they do nothing.~~
- 4 *Returns:*  $\text{first} + N$ .  
~~`generate_n` returns  $\text{first} + n$  for non-negative values of  $n$  and  $\text{first}$  for negative values.~~
- 5 *Complexity:* Exactly  $N$  last — first,  $n$ , or 0 invocationsevaluations of `gen()` and assignments; ~~respectively.~~

## 24.6.8 Remove

[alg.remove]

```

template<class ForwardIterator, class T>
 constexpr ForwardIterator remove(ForwardIterator first, ForwardIterator last,
 const T& value);
template<class ExecutionPolicy, class ForwardIterator, class T>
 ForwardIterator remove(ExecutionPolicy&& exec,
 ForwardIterator first, ForwardIterator last,
 const T& value);

template<class ForwardIterator, class Predicate>
 constexpr ForwardIterator remove_if(ForwardIterator first, ForwardIterator last,
 Predicate pred);
template<class ExecutionPolicy, class ForwardIterator, class Predicate>
 ForwardIterator remove_if(ExecutionPolicy&& exec,
 ForwardIterator first, ForwardIterator last,
 Predicate pred);

namespace ranges {
 template<Permutable I, Sentinel<I> S, class T, class Proj = identity>
 requires IndirectRelation<ranges::equal_to<>, projected<I, Proj>, const T*>
 constexpr I remove(I first, S last, const T& value, Proj proj = {});
}

```

```

template<ForwardRange R, class T, class Proj = identity>
requires Permutable<iterator_t<R>> &&
 IndirectRelation<ranges::equal_to<>, projected<iterator_t<R>, Proj>, const T*>
constexpr safe_iterator_t<R>
 remove(R&& r, const T& value, Proj proj = {});
template<Permutable I, Sentinel<I> S, class Proj = identity,
 IndirectUnaryPredicate<projected<I, Proj>> Pred>
constexpr I remove_if(I first, S last, Pred pred, Proj proj = {});
template<ForwardRange R, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
requires Permutable<iterator_t<R>>
constexpr safe_iterator_t<R>
 remove_if(R&& r, Pred pred, Proj proj = {});
}

```

- 1     Let *E* be
- (1.1)     — `bool(*i == value)` for `remove`,
- (1.2)     — `bool(pred(*i))` for `remove_if`,
- (1.3)     — `bool(invoker(proj, *i) == value)` for `ranges::remove`, or
- (1.4)     — `bool(invoker(pred, invoker(proj, *i)))` for `ranges::remove_if`.
- 2     *Requires:* ~~The~~For the algorithms in namespace `std`, the type of `*first` shall ~~satisfy~~meet the *Cpp17MoveAssignable* requirements ([[tab:moveassignable](#)]).
- 3     *Effects:* Eliminates all the elements referred to by iterator *i* in the range `[first, last)` for which *E* holds ~~the following corresponding conditions hold: `*i == value`, `pred(*i) != false`.~~
- 4     *Returns:* The end of the resulting range.
- 5     *Remarks:* Stable ([[algorithm.stable](#)]).
- 6     *Complexity:* Exactly `last - first` applications of the corresponding predicate and any projection.
- 7     [*Note:* Each element in the range `[ret, last)`, where `ret` is the returned value, has a valid but unspecified state, because the algorithms can eliminate elements by moving from elements that were originally in that range. — *end note*]

```

template<class InputIterator, class OutputIterator, class T>
constexpr OutputIterator
 remove_copy(InputIterator first, InputIterator last,
 OutputIterator result, const T& value);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class T>
ForwardIterator2
 remove_copy(ExecutionPolicy&& exec,
 ForwardIterator1 first, ForwardIterator1 last,
 ForwardIterator2 result, const T& value);

template<class InputIterator, class OutputIterator, class Predicate>
constexpr OutputIterator
 remove_copy_if(InputIterator first, InputIterator last,
 OutputIterator result, Predicate pred);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class Predicate>
ForwardIterator2
 remove_copy_if(ExecutionPolicy&& exec,
 ForwardIterator1 first, ForwardIterator1 last,
 ForwardIterator2 result, Predicate pred);

namespace ranges {
 template<InputIterator I, Sentinel<I> S, WeaklyIncrementable O, class T,
 class Proj = identity>
 requires IndirectlyCopyable<I, O> &&
 IndirectRelation<ranges::equal_to<>, projected<I, Proj>, const T*>
 constexpr remove_copy_result<I, O>
 remove_copy(I first, S last, O result, const T& value, Proj proj = {});
}

```

```

template<InputRange R, WeaklyIncrementable O, class T, class Proj = identity>
 requires IndirectlyCopyable<iterator_t<R>, O> &&
 IndirectRelation<ranges::equal_to<>, projected<iterator_t<R>, Proj>, const T*>
 constexpr remove_copy_result<safe_iterator_t<R>, O>
 remove_copy(R&& r, O result, const T& value, Proj proj = {});
template<InputIterator I, Sentinel<I> S, WeaklyIncrementable O,
 class Proj = identity, IndirectUnaryPredicate<projected<I, Proj>> Pred>
 requires IndirectlyCopyable<I, O>
 constexpr remove_copy_if_result<I, O>
 remove_copy_if(I first, S last, O result, Pred pred, Proj proj = {});
template<InputRange R, WeaklyIncrementable O, class Proj = identity,
 IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
 requires IndirectlyCopyable<iterator_t<R>, O>
 constexpr remove_copy_if_result<safe_iterator_t<R>, O>
 remove_copy_if(R&& r, O result, Pred pred, Proj proj = {});
}

```

8 Let  $E$  be

- (8.1) — `bool(*i == value)` for `remove_copy`,
- (8.2) — `bool(pred(*i))` for `remove_copy_if`,
- (8.3) — `bool(invoker(proj, *i) == value)` for `ranges::remove_copy`, or
- (8.4) — `bool(invoker(pred, invoker(proj, *i)))` for `ranges::remove_copy_if`.

9 Let  $N$  be the number of elements in `[first, last)` for which  $E$  is false.

10 *Requires:* The ranges `[first, last)` and `[result, result + (last - first))` shall not overlap. The expression `*result = *first` shall be valid. [Note: For the overloads with an `ExecutionPolicy`, there may be a performance cost if `iterator_traits<ForwardIterator1>::value_type` ~~is~~ does not meet the *Cpp17MoveConstructible* ([\[tab:moveconstructible\]](#)) requirements. — end note]

11 *Effects:* Copies all the elements referred to by the iterator  $i$  in the range `[first, last)` for which  $E$  is false ~~the following corresponding conditions do not hold:  $*i == value$ ,  $pred(*i) != false$ .~~

12 *Returns:* ~~The end of the resulting range.~~

(12.1) — `result + N`, for the algorithms in namespace `std`, or

(12.2) — `{last, result + N}`, for the algorithms in namespace `ranges`.

13 *Complexity:* Exactly `last - first` applications of the corresponding predicate and any projection.

14 *Remarks:* Stable ([\[algorithm.stable\]](#)).

## 24.6.9 Unique

[alg.unique]

```

template<class ForwardIterator>
 constexpr ForwardIterator unique(ForwardIterator first, ForwardIterator last);
template<class ExecutionPolicy, class ForwardIterator>
 ForwardIterator unique(ExecutionPolicy&& exec,
 ForwardIterator first, ForwardIterator last);

template<class ForwardIterator, class BinaryPredicate>
 constexpr ForwardIterator unique(ForwardIterator first, ForwardIterator last,
 BinaryPredicate pred);
template<class ExecutionPolicy, class ForwardIterator, class BinaryPredicate>
 ForwardIterator unique(ExecutionPolicy&& exec,
 ForwardIterator first, ForwardIterator last,
 BinaryPredicate pred);

namespace ranges {
 template<Permutable I, Sentinel<I> S, class Proj = identity,
 IndirectRelation<projected<I, Proj>> C = ranges::equal_to<>>
 constexpr I unique(I first, S last, C comp = {}, Proj proj = {});
 template<ForwardRange R, class Proj = identity,
 IndirectRelation<projected<iterator_t<R>, Proj>> C = ranges::equal_to<>>
 requires Permutable<iterator_t<R>>
 constexpr safe_iterator_t<R>

```

```

 unique(R&& r, C comp = {}, Proj proj = {});
}

```

1 Let `pred` be `equal_to{}` for the overloads with no parameter `pred`, and let  $E$  be

(1.1) — `bool(pred(*(i - 1), *i))` for the overloads in namespace `std`, or

(1.2) — `bool(invoke(comp, invoke(proj, *(i - 1)), invoke(proj, *i)))` for the overloads in namespace `ranges`.

2 *Requires:* ~~The comparison function~~ For the overloads in namespace `std`, `pred` shall be an equivalence relation. ~~The and the~~ type of `*first` shall ~~satisfy~~meet the *Cpp17MoveAssignable* requirements ([[tab:moveassignable](#)]).

3 *Effects:* For a nonempty range, eliminates all but the first element from every consecutive group of equivalent elements referred to by the iterator `i` in the range `[first + 1, last)` for which  $E$  is true. ~~the following conditions hold:  $*(i - 1) == *i$  or  $pred(*(i - 1), *i) != false$ .~~

4 *Returns:* The end of the resulting range.

5 *Complexity:* ~~For nonempty ranges, e~~Exactly  $(last - first) - 1$  applications of the corresponding predicate and no more than twice as many applications of any projection.

```

template<class InputIterator, class OutputIterator>
constexpr OutputIterator
 unique_copy(InputIterator first, InputIterator last,
 OutputIterator result);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
ForwardIterator2
 unique_copy(ExecutionPolicy&& exec,
 ForwardIterator1 first, ForwardIterator1 last,
 ForwardIterator2 result);

template<class InputIterator, class OutputIterator,
 class BinaryPredicate>
constexpr OutputIterator
 unique_copy(InputIterator first, InputIterator last,
 OutputIterator result, BinaryPredicate pred);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
 class BinaryPredicate>
ForwardIterator2
 unique_copy(ExecutionPolicy&& exec,
 ForwardIterator1 first, ForwardIterator1 last,
 ForwardIterator2 result, BinaryPredicate pred);

namespace ranges {
 template<InputIterator I, Sentinel<I> S, WeaklyIncrementable O,
 class Proj = identity, IndirectRelation<projected<I, Proj>> C = ranges::equal_to<>>
 requires IndirectlyCopyable<I, O> &&
 (ForwardIterator<I> ||
 (InputIterator<O> && Same<iter_value_t<I>, iter_value_t<O>>) ||
 IndirectlyCopyableStorable<I, O>)
 constexpr unique_copy_result<I, O>
 unique_copy(I first, S last, O result, C comp = {}, Proj proj = {});
 template<InputRange R, WeaklyIncrementable O, class Proj = identity,
 IndirectRelation<projected<iterator_t<R>, Proj>> C = ranges::equal_to<>>
 requires IndirectlyCopyable<iterator_t<R>, O> &&
 (ForwardIterator<iterator_t<R>> ||
 (InputIterator<O> && Same<iter_value_t<iterator_t<R>>, iter_value_t<O>>) ||
 IndirectlyCopyableStorable<iterator_t<R>, O>)
 constexpr unique_copy_result<safe_iterator_t<R>, O>
 unique_copy(R&& r, O result, C comp = {}, Proj proj = {});
}

```

6 Let `pred` be `equal_to{}` for the overloads in namespace `std` with no parameter `pred`, and let  $E$  be

(6.1) — `bool(pred(*i, *(i - 1)))` for the overloads in namespace `std`, or

- (6.2) — `bool(invoker(comp, invoker(proj, *i), invoker(proj, *(i - 1))))` for the overloads in namespace `ranges`.

7 *Requires:*

- (7.1) — ~~The comparison function shall be an equivalence relation.~~
- (7.2) — The ranges `[first, last)` and `[result, result + (last - first))` shall not overlap.
- (7.3) — For the overloads in namespace `std`:
- (7.3.1) — The comparison function shall be an equivalence relation.
- (7.3.2) — The expression `*result = *first` shall be valid.
- (7.3.3) — For the overloads with no `ExecutionPolicy`, let `T` be the value type of `InputIterator`. If `InputIterator` meets the ~~forward iterator~~ `Cpp17ForwardIterator` requirements, then there are no additional requirements for `T`. Otherwise, if `OutputIterator` meets the ~~forward iterator~~ `Cpp17ForwardIterator` requirements and its value type is the same as `T`, then `T` shall ~~be~~ meet the `Cpp17CopyAssignable` ([tab:copyassignable]) requirements. Otherwise, `T` shall ~~be both~~ meet the `Cpp17CopyConstructible` ([tab:copyconstructible]) and `Cpp17CopyAssignable` requirements. [Note: For the overloads with an `ExecutionPolicy`, there may be a performance cost if the value type of `ForwardIterator1` ~~is not both~~ does not meet the `Cpp17CopyConstructible` and `Cpp17CopyAssignable` requirements. — end note]

8 *Effects:* Copies only the first element from every consecutive group of equal elements referred to by the iterator `i` in the range `[first, last)` for which `E` holds. ~~the following corresponding conditions hold: `*i == *(i - 1)` or `pred(*i, *(i - 1)) != false`.~~

9 *Returns:* ~~The end of the resulting range.~~

- (9.1) — `result + N` for the overloads in namespace `std`, or
- (9.2) — `{last, result + N}` for the overloads in namespace `ranges`

where `N` is the number of groups of equal elements.

10 *Complexity:* ~~For nonempty ranges, e~~ Exactly `last - first - 1` applications of the corresponding predicate and no more than twice as many applications of any projection.

## 24.6.10 Reverse

[alg.reverse]

```
template<class BidirectionalIterator>
constexpr void reverse(BidirectionalIterator first, BidirectionalIterator last);
template<class ExecutionPolicy, class BidirectionalIterator>
void reverse(ExecutionPolicy&& exec,
 BidirectionalIterator first, BidirectionalIterator last);

namespace ranges {
 template<BidirectionalIterator I, Sentinel<I> S>
 requires Permutable<I>
 constexpr I reverse(I first, S last);
 template<BidirectionalRange R>
 requires Permutable<iterator_t<R>>
 constexpr safe_iterator_t<R> reverse(R&& r);
}
```

1 *Requires:* For the overloads in namespace `std`, `BidirectionalIterator` shall ~~satisfy~~ meet the `Cpp17ValueSwappable` requirements ([swappable.requirements]).

2 *Effects:* For each non-negative integer `i < (last - first) / 2`, applies `std::iter_swap`, or `ranges::iter_swap` for the overloads in namespace `ranges`, to all pairs of iterators `first + i`, `(last - i) - 1`.

3 *Returns:* `last` for the overloads in namespace `ranges`.

4 *Complexity:* Exactly `(last - first) / 2` swaps.

```
template<class BidirectionalIterator, class OutputIterator>
constexpr OutputIterator
reverse_copy(BidirectionalIterator first, BidirectionalIterator last,
 OutputIterator result);
```

```
template<class ExecutionPolicy, class BidirectionalIterator, class ForwardIterator>
ForwardIterator
reverse_copy(ExecutionPolicy&& exec,
 BidirectionalIterator first, BidirectionalIterator last,
 ForwardIterator result);
```

```
namespace ranges {
template<BidirectionalIterator I, Sentinel<I> S, WeaklyIncrementable O>
requires IndirectlyCopyable<I, O>
constexpr reverse_copy_result<I, O>
reverse_copy(I first, S last, O result);
template<BidirectionalRange R, WeaklyIncrementable O>
requires IndirectlyCopyable<iterator_t<R>, O>
constexpr reverse_copy_result<safe_iterator_t<R>, O>
reverse_copy(R&& r, O result);
}
```

5 Let  $N$  be  $\text{last} - \text{first}$ .

6 *Requires:* The ranges  $[\text{first}, \text{last})$  and  $[\text{result}, \text{result} + \text{N})$  shall not overlap.

7 *Effects:* Copies the range  $[\text{first}, \text{last})$  to the range  $[\text{result}, \text{result} + \text{N})$  such that for every non-negative integer  $i < \text{N}$  the following assignment takes place:  
 $*(\text{result} + \text{N} - 1 - i) = *(\text{first} + i)$ .

8 *Returns:*

(8.1) —  $\text{result} + \text{N}$  for the overloads in namespace `std`, or

(8.2) —  $\{\text{last}, \text{result} + \text{N}\}$  for the overloads in namespace `ranges`.

9 *Complexity:* Exactly  $\text{last} - \text{first}$  assignments.

## 24.6.11 Rotate

[alg.rotate]

[Editor's note: This wording incorporates the PR for [stl2#526](#).]

```
template<class ForwardIterator>
constexpr ForwardIterator
rotate(ForwardIterator first, ForwardIterator middle, ForwardIterator last);
template<class ExecutionPolicy, class ForwardIterator>
ForwardIterator
rotate(ExecutionPolicy&& exec,
 ForwardIterator first, ForwardIterator middle, ForwardIterator last);
```

```
namespace ranges {
template<Permutable I, Sentinel<I> S>
constexpr subrange<I> rotate(I first, I middle, S last);
}
```

1 *Requires:*  $[\text{first}, \text{middle})$  and  $[\text{middle}, \text{last})$  shall be valid ranges. For the overloads in namespace `std`, `ForwardIterator` shall [satisfy meet](#) the *Cpp17ValueSwappable* requirements ([\[swappable.requirements\]](#)). The, and the type of  $*\text{first}$  shall [satisfy meet](#) the *Cpp17MoveConstructible* ([\[tab:moveconstructible\]](#)) and *Cpp17MoveAssignable* ([\[tab:moveassignable\]](#)) requirements.

2 *Effects:* For each non-negative integer  $i < (\text{last} - \text{first})$ , places the element from the position  $\text{first} + i$  into position  $\text{first} + (i + (\text{last} - \text{middle})) \% (\text{last} - \text{first})$ . [Note: This is a left rotate. — end note]

3 *Returns:*

(3.1) —  $\text{first} + (\text{last} - \text{middle})$  for the overloads in namespace `std`, or

(3.2) —  $\{\text{first} + (\text{last} - \text{middle}), \text{last}\}$  for the overload in namespace `ranges`.

4 ~~*Remarks:* This is a left rotate.~~

5 *Complexity:* At most  $\text{last} - \text{first}$  swaps.

```

namespace ranges {
 template<ForwardRange R>
 requires Permutable<iterator_t<R>>
 constexpr safe_subrange_t<R> rotate(R&& r, iterator_t<R> middle);
}

```

6 *Effects:* Equivalent to: `return ranges::rotate(ranges::begin(r), middle, ranges::end(r));`

```

template<class ForwardIterator, class OutputIterator>
constexpr OutputIterator
 rotate_copy(ForwardIterator first, ForwardIterator middle, ForwardIterator last,
 OutputIterator result);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
ForwardIterator2
 rotate_copy(ExecutionPolicy&& exec,
 ForwardIterator1 first, ForwardIterator1 middle, ForwardIterator1 last,
 ForwardIterator2 result);

```

```

namespace ranges {
 template<ForwardIterator I, Sentinel<I> S, WeaklyIncrementable O>
 requires IndirectlyCopyable<I, O>
 constexpr rotate_copy_result<I, O>
 rotate_copy(I first, I middle, S last, O result);
}

```

7 Let  $N$  be `last - first`.

8 *Requires:* `[first, middle)` and `[middle, last)` shall be valid ranges. The ranges `[first, last)` and `[result, result + (last - first) $N$ )` shall not overlap.

9 *Effects:* Copies the range `[first, last)` to the range `[result, result + (last - first) $N$ )` such that for each non-negative integer  $i < \text{~~(last - first)~~ $N$ }$  the following assignment takes place: `*(result + i) = *(first + (i + (middle - first)) % (last - first) $N$ )`.

10 *Returns:*

(10.1) — `result + (last - first) $N$`  for the overloads in namespace `std`, or

(10.2) — `{last, result +  $N$ }` for the overload in namespace `ranges`.

11 *Complexity:* Exactly  ~~$\text{last} - \text{first}$~~  $N$  assignments.

```

namespace ranges {
 template<ForwardRange R, WeaklyIncrementable O>
 requires IndirectlyCopyable<iterator_t<R>, O>
 constexpr rotate_copy_result<safe_iterator_t<R>, O>
 rotate_copy(R&& r, iterator_t<R> middle, O result);
}

```

12 *Effects:* Equivalent to:

```
return ranges::rotate_copy(ranges::begin(r), middle, ranges::end(r), result);
```

## 24.6.12 Sample

[`alg.random.sample`]

[...]

## 24.6.13 Shuffle

[`alg.random.shuffle`]

```

template<class RandomAccessIterator, class UniformRandomBitGenerator>
void shuffle(RandomAccessIterator first,
 RandomAccessIterator last,
 UniformRandomBitGenerator&& g);

namespace ranges {
 template<RandomAccessIterator I, Sentinel<I> S, class Gen>
 requires Permutable<I> &&
 UniformRandomBitGenerator<remove_reference_t<Gen>>
 I shuffle(I first, S last, Gen&& g);
}

```

```

template<RandomAccessRange R, class Gen>
requires Permutable<iterator_t<R>> &&
 UniformRandomBitGenerator<remove_reference_t<Gen>>
safe_iterator_t<R> shuffle(R&& r, Gen&& g);
}

```

- 1 *Requires:* For the overload in namespace `std`:
- (1.1) — `RandomAccessIterator` shall ~~satisfy~~meet the *Cpp17ValueSwappable* requirements ([swappable.requirements]).
- (1.2) — The type `remove_reference_t<UniformRandomBitGenerator>` shall ~~satisfy the requirements of~~meet the uniform random bit generator ([rand.req.urng]) ~~requirements~~type whose return type  
is convertible to `iterator_traits<RandomAccessIterator>::difference_type`.
- 2 *Effects:* Permutes the elements in the range `[first, last)` such that each possible permutation of those elements has equal probability of appearance.
- 3 *Returns:* `last` for the overloads in namespace `ranges`.
- 4 *Complexity:* Exactly `(last - first) - 1` swaps.
- 5 *Remarks:* To the extent that the implementation of this function makes use of random numbers, the object referenced by `g` shall serve as the implementation's source of randomness.

#### 24.6.14 Shift

[`alg.shift`]

[...]

### 24.7 Sorting and related operations

[`alg.sorting`]

- 1 ~~All the~~The operations in 24.7 defined directly in namespace `std` have two versions: one that takes a function object of type `Compare` and one that uses an `operator<`.
- 2 `Compare` is a function object type[function.objects]. The return value of the function call operation applied to an object of type `Compare`, when contextually converted to `bool`[conv], yields `true` if the first argument of the call is less than the second, and `false` otherwise. `Compare` `comp` is used throughout for algorithms assuming an ordering relation. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator.
- 3 For all algorithms that take `Compare`, there is a version that uses `operator<` instead. That is, `comp(*i, *j) != false` defaults to `*i < *j != false`. For algorithms other than those described in 24.7.3, `comp` shall induce a strict weak ordering on the values.
- 4 The term *strict* refers to the requirement of an irreflexive relation (`!comp(x, x)` for all `x`), and the term *weak* to requirements that are not as strong as those for a total ordering, but stronger than those for a partial ordering. If we define `equiv(a, b)` as `!comp(a, b) && !comp(b, a)`, then the requirements are that `comp` and `equiv` both be transitive relations:
- (4.1) — `comp(a, b) && comp(b, c)` implies `comp(a, c)`
- (4.2) — `equiv(a, b) && equiv(b, c)` implies `equiv(a, c)`
- [*Note:* Under these conditions, it can be shown that
- (4.3) — `equiv` is an equivalence relation
- (4.4) — `comp` induces a well-defined relation on the equivalence classes determined by `equiv`
- (4.5) — The induced relation is a strict total ordering.
- *end note*]
- 5 A sequence is ~~sorted with respect to a comparator `comp`~~ sorted with respect to `comp` and `proj` for a comparator and projection `comp` and `proj` if for every iterator `i` pointing to the sequence and every non-negative integer `n` such that `i + n` is a valid iterator pointing to an element of the sequence, ~~`comp(*(i + n), *i) == false`~~  
`bool(invoke(comp, invoke(proj, *(i + n)), invoke(proj, *i)))`  
is false.
- 6 A sequence `[start, finish)` is *partitioned with respect to an expression `f(e)`* if there exists an integer `n` such that for all `0 <= i < (finish - start)`, `f(*(start + i))` is `true` if and only if `i < n`.
- 7 In the descriptions of the functions that deal with ordering relationships we frequently use a notion of equivalence to describe concepts such as stability. The equivalence to which we refer is not necessarily an

`operator==`, but an equivalence relation induced by the strict weak ordering. That is, two elements `a` and `b` are considered equivalent if and only if `!(a < b) && !(b < a)`.

## 24.7.1 Sorting

[alg.sort]

### 24.7.1.1 sort

[sort]

```
template<class RandomAccessIterator>
constexpr void sort(RandomAccessIterator first, RandomAccessIterator last);
template<class ExecutionPolicy, class RandomAccessIterator>
void sort(ExecutionPolicy&& exec,
 RandomAccessIterator first, RandomAccessIterator last);

template<class RandomAccessIterator, class Compare>
constexpr void sort(RandomAccessIterator first, RandomAccessIterator last,
 Compare comp);
template<class ExecutionPolicy, class RandomAccessIterator, class Compare>
void sort(ExecutionPolicy&& exec,
 RandomAccessIterator first, RandomAccessIterator last,
 Compare comp);

namespace ranges {
 template<RandomAccessIterator I, Sentinel<I> S, class Comp = ranges::less<>,
 class Proj = identity>
 requires Sortable<I, Comp, Proj>
 constexpr I
 sort(I first, S last, Comp comp = {}, Proj proj = {});
 template<RandomAccessRange R, class Comp = ranges::less<>, class Proj = identity>
 requires Sortable<iterator_t<R>, Comp, Proj>
 constexpr safe_iterator_t<R>
 sort(R&& r, Comp comp = {}, Proj proj = {});
}
```

- 1 Let `comp` be `less{}` and `proj` be `identity{}` for the overloads with no parameters by those names.
- 2 *Requires:* For the overloads in namespace `std`, `RandomAccessIterator` shall ~~satisfy~~meet the *Cpp17ValueSwappable* requirements ([swappable.requirements]). ~~The and the~~ type of `*first` shall ~~satisfy~~meet the *Cpp17MoveConstructible* ([tab:moveconstructible]) and *Cpp17MoveAssignable* ([tab:moveassignable]) requirements.
- 3 *Effects:* Sorts the elements in the range `[first, last)` with respect to `comp` and `proj`.
- 4 *Returns:* `last`, for the overloads in namespace `ranges`.
- 5 *Complexity:* Let  $N$  be `last - first`.  $\mathcal{O}(N \log N)$  comparisons and projections. ~~where  $N = last - first$ .~~

### 24.7.1.2 stable\_sort

[stable.sort]

```
template<class RandomAccessIterator>
void stable_sort(RandomAccessIterator first, RandomAccessIterator last);
template<class ExecutionPolicy, class RandomAccessIterator>
void stable_sort(ExecutionPolicy&& exec,
 RandomAccessIterator first, RandomAccessIterator last);

template<class RandomAccessIterator, class Compare>
void stable_sort(RandomAccessIterator first, RandomAccessIterator last,
 Compare comp);
template<class ExecutionPolicy, class RandomAccessIterator, class Compare>
void stable_sort(ExecutionPolicy&& exec,
 RandomAccessIterator first, RandomAccessIterator last,
 Compare comp);

namespace ranges {
 template<RandomAccessIterator I, Sentinel<I> S, class Comp = ranges::less<>,
 class Proj = identity>
 requires Sortable<I, Comp, Proj>
 I stable_sort(I first, S last, Comp comp = {}, Proj proj = {});
}
```

```

template<RandomAccessRange R, class Comp = ranges::less<>, class Proj = identity>
 requires Sortable<iterator_t<R>, Comp, Proj>
 safe_iterator_t<R>
 stable_sort(R&& r, Comp comp = {}, Proj proj = {});
}

```

- 1 Let `comp` be `less{}` and `proj` be `identity{}` for the overloads with no parameters by those names.
- 2 *Requires:* For the overloads in namespace `std`, `RandomAccessIterator` shall **satisfy**meet the *Cpp17ValueSwappable* requirements ([swappable.requirements]). ~~The and the~~ type of `*first` shall **satisfy**meet the *Cpp17MoveConstructible* ([tab:moveconstructible]) and *Cpp17MoveAssignable* ([tab:moveassignable]) requirements.
- 3 *Effects:* Sorts the elements in the range `[first, last)` with respect to `comp` and `proj`.
- 4 *Returns:* `last` for the overloads in namespace `ranges`.
- 5 *Complexity:* Let  $N$  be `last - first`. If enough extra memory is available,  $N \log(N)$  comparisons. Otherwise, at most  $N \log^2(N)$  comparisons. In either case, twice as many projections as the number of comparisons. ~~At most  $N \log^2(N)$  comparisons, where  $N = last - first$ , but only  $N \log N$  comparisons if there is enough extra memory.~~
- 6 *Remarks:* Stable ([algorithm.stable]).

### 24.7.1.3 partial\_sort

[partial.sort]

```

template<class RandomAccessIterator>
 constexpr void partial_sort(RandomAccessIterator first,
 RandomAccessIterator middle,
 RandomAccessIterator last);

template<class ExecutionPolicy, class RandomAccessIterator>
 void partial_sort(ExecutionPolicy&& exec,
 RandomAccessIterator first,
 RandomAccessIterator middle,
 RandomAccessIterator last);

template<class RandomAccessIterator, class Compare>
 constexpr void partial_sort(RandomAccessIterator first,
 RandomAccessIterator middle,
 RandomAccessIterator last,
 Compare comp);

template<class ExecutionPolicy, class RandomAccessIterator, class Compare>
 void partial_sort(ExecutionPolicy&& exec,
 RandomAccessIterator first,
 RandomAccessIterator middle,
 RandomAccessIterator last,
 Compare comp);

namespace ranges {
 template<RandomAccessIterator I, Sentinel<I> S, class Comp = ranges::less<>,
 class Proj = identity>
 requires Sortable<I, Comp, Proj>
 constexpr I
 partial_sort(I first, I middle, S last, Comp comp = {}, Proj proj = {});
}

```

- 1 Let `comp` be `less{}` and `proj` be `identity{}` for the overloads with no parameters by those names.
- 2 *Requires:* [`first, middle`) and [`middle, last`) shall be valid ranges. For the overloads in namespace `std`, `RandomAccessIterator` shall **satisfy**meet the *Cpp17ValueSwappable* requirements ([swappable.requirements]). ~~The and the~~ type of `*first` shall **satisfy**meet the *Cpp17MoveConstructible* ([tab:moveconstructible]) and *Cpp17MoveAssignable* ([tab:moveassignable]) requirements.
- 3 *Effects:* Places the `first middle - first` ~~sorted~~ elements from the range `[first, last)` as sorted with respect to `comp` and `proj`, into the range `[first, middle)`. The rest of the elements in the range `[middle, last)` are placed in an unspecified order.
- 4 *Returns:* `last` for the overload in namespace `ranges`.

- 5 *Complexity:* Approximately  $(\text{last} - \text{first}) * \log(\text{middle} - \text{first})$  comparisons, and twice as many projections.

```
namespace ranges {
 template<RandomAccessRange R, class Comp = ranges::less<>, class Proj = identity>
 requires Sortable<iterator_t<R>, Comp, Proj>
 constexpr safe_iterator_t<R>
 partial_sort(R&& r, iterator_t<R> middle, Comp comp = {}, Proj proj = {});
}
```

- 6 *Effects:* Equivalent to:

```
return ranges::partial_sort(ranges::begin(r), middle, ranges::end(r), comp, proj);
```

#### 24.7.1.4 partial\_sort\_copy

[partial.sort.copy]

```
template<class InputIterator, class RandomAccessIterator>
 constexpr RandomAccessIterator
 partial_sort_copy(InputIterator first, InputIterator last,
 RandomAccessIterator result_first,
 RandomAccessIterator result_last);

template<class ExecutionPolicy, class ForwardIterator, class RandomAccessIterator>
 RandomAccessIterator
 partial_sort_copy(ExecutionPolicy&& exec,
 ForwardIterator first, ForwardIterator last,
 RandomAccessIterator result_first,
 RandomAccessIterator result_last);

template<class InputIterator, class RandomAccessIterator,
 class Compare>
 constexpr RandomAccessIterator
 partial_sort_copy(InputIterator first, InputIterator last,
 RandomAccessIterator result_first,
 RandomAccessIterator result_last,
 Compare comp);

template<class ExecutionPolicy, class ForwardIterator, class RandomAccessIterator,
 class Compare>
 RandomAccessIterator
 partial_sort_copy(ExecutionPolicy&& exec,
 ForwardIterator first, ForwardIterator last,
 RandomAccessIterator result_first,
 RandomAccessIterator result_last,
 Compare comp);
```

```
namespace ranges {
 template<InputIterator I1, Sentinel<I1> S1, RandomAccessIterator I2, Sentinel<I2> S2,
 class Comp = ranges::less<>, class Proj1 = identity, class Proj2 = identity>
 requires IndirectlyCopyable<I1, I2> && Sortable<I2, Comp, Proj2> &&
 IndirectStrictWeakOrder<Comp, projected<I1, Proj1>, projected<I2, Proj2>>
 constexpr I2
 partial_sort_copy(I1 first, S1 last, I2 result_first, S2 result_last,
 Comp comp = {}, Proj1 proj1 = {}, Proj2 proj2 = {});

 template<InputRange R1, RandomAccessRange R2, class Comp = ranges::less<>,
 class Proj1 = identity, class Proj2 = identity>
 requires IndirectlyCopyable<iterator_t<R1>, iterator_t<R2>> &&
 Sortable<iterator_t<R2>, Comp, Proj2> &&
 IndirectStrictWeakOrder<Comp, projected<iterator_t<R1>, Proj1>,
 projected<iterator_t<R2>, Proj2>>
 constexpr safe_iterator_t<R2>
 partial_sort_copy(R1&& r, R2&& result_r, Comp comp = {},
 Proj1 proj1 = {}, Proj2 proj2 = {});
}
```

- 1 Let  $N$  be  $\min(\text{last} - \text{first}, \text{result\_last} - \text{result\_first})$ . Let  $\text{comp}$  be  $\text{less}\{\}$ , and  $\text{proj1}$  and  $\text{proj2}$  be  $\text{identity}\{\}$  for the overloads with no parameters by those names.

- 2 *Requires:* For the overloads in namespace `std`, `RandomAccessIterator` shall ~~satisfy~~meet the *Cpp17ValueSwappable* requirements ([swappable.requirements]). ~~The~~, the type of `*result_first` shall ~~satisfy~~meet the *Cpp17MoveConstructible* ([tab:moveconstructible]) and *Cpp17MoveAssignable* ([tab:moveassignable]) requirements, and the expression `*first` shall be writable (22.3.1) to `result_first`.
- 3 *Expects:* For iterators `a1` and `b1` in `[first, last)`, and iterators `x2` and `y2` in `[result_first, result_last)`, after evaluating the assignment `*y2 = *b1`, let *E* be the value of  
`bool(invoker(comp, invoker(proj1, *a1), invoker(proj2, *y2)))`.  
Then after evaluating the assignment `*x2 = *a1`, *E* shall be equal to  
`bool(invoker(comp, invoker(proj2, *x2), invoker(proj2, *y2)))`.  
[Note: Writing a value from the input range into the output range does not affect how it is ordered by `comp` and `proj1` or `proj2`. — end note]
- 4 *Effects:* Places the first ~~min(last — first, result\_last — result\_first)~~*N* sorted elements as sorted with respect to `comp` and `proj2` into the range `[result_first, result_first + min(last — first, result_last — result_first)N).  
Returns: The smaller of: result_last or result_first + (last — first)N.`
- 5 *Complexity:* Approximately  $(last - first) * \log(\min(last - first, result\_last - result\_first))$  *N* comparisons, and twice as many projections.

#### 24.7.1.5 `is_sorted`

[is.sorted]

- ```
template<class ForwardIterator>
constexpr bool is_sorted(ForwardIterator first, ForwardIterator last);
```
- 1 ~~Returns:~~ *Effects:* Equivalent to: `return is_sorted_until(first, last) == last;`
- ```
template<class ExecutionPolicy, class ForwardIterator>
bool is_sorted(ExecutionPolicy&& exec,
 ForwardIterator first, ForwardIterator last);
```
- 2 ~~Returns:~~ *Effects:* Equivalent to:  
`return is_sorted_until(std::forward<ExecutionPolicy>(exec), first, last) == last;`
- ```
template<class ForwardIterator, class Compare>
constexpr bool is_sorted(ForwardIterator first, ForwardIterator last,
                        Compare comp);
```
- 3 ~~Returns:~~ *Effects:* Equivalent to: `return is_sorted_until(first, last, comp) == last;`
- ```
template<class ExecutionPolicy, class ForwardIterator, class Compare>
bool is_sorted(ExecutionPolicy&& exec,
 ForwardIterator first, ForwardIterator last,
 Compare comp);
```
- 4 ~~Returns:~~ *Effects:* Equivalent to:  
`return is_sorted_until(std::forward<ExecutionPolicy>(exec), first, last, comp) == last;`
- ```
namespace ranges {
    template<ForwardIterator I, Sentinel<I> S, class Proj = identity,
            IndirectStrictWeakOrder<projected<I, Proj>> Comp = ranges::less<>>
    constexpr bool is_sorted(I first, S last, Comp comp = {}, Proj proj = {});
    template<ForwardRange R, class Proj = identity,
            IndirectStrictWeakOrder<projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
    constexpr bool is_sorted(R&& r, Comp comp = {}, Proj proj = {});
}
```
- 5 *Effects:* Equivalent to: `return ranges::is_sorted_until(first, last, comp, proj) == last;`
- ```
template<class ForwardIterator>
constexpr ForwardIterator
is_sorted_until(ForwardIterator first, ForwardIterator last);
```

```

template<class ExecutionPolicy, class ForwardIterator>
 ForwardIterator
 is_sorted_until(ExecutionPolicy&& exec,
 ForwardIterator first, ForwardIterator last);

template<class ForwardIterator, class Compare>
 constexpr ForwardIterator
 is_sorted_until(ForwardIterator first, ForwardIterator last,
 Compare comp);
template<class ExecutionPolicy, class ForwardIterator, class Compare>
 ForwardIterator
 is_sorted_until(ExecutionPolicy&& exec,
 ForwardIterator first, ForwardIterator last,
 Compare comp);

namespace ranges {
 template<ForwardIterator I, Sentinel<I> S, class Proj = identity,
 IndirectStrictWeakOrder<projected<I, Proj>> Comp = ranges::less<>>
 constexpr I is_sorted_until(I first, S last, Comp comp = {}, Proj proj = {});
 template<ForwardRange R, class Proj = identity,
 IndirectStrictWeakOrder<projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
 constexpr safe_iterator_t<R>
 is_sorted_until(R&& r, Comp comp = {}, Proj proj = {});
}

```

- 6     Let `comp` be `less{}` and `proj` be `identity{}` for the overloads with no parameters by those names.
- 7     Returns: ~~If  $(last - first) < 2$ , returns `last`. Otherwise, returns `t`~~The last iterator `i` in `[first, last]` for which the range `[first, i)` is sorted with respect to `comp` and `proj`.
- 8     Complexity: Linear.

## 24.7.2 Nth element

[alg.nth.element]

```

template<class RandomAccessIterator>
 constexpr void nth_element(RandomAccessIterator first, RandomAccessIterator nth,
 RandomAccessIterator last);
template<class ExecutionPolicy, class RandomAccessIterator>
 void nth_element(ExecutionPolicy&& exec,
 RandomAccessIterator first, RandomAccessIterator nth,
 RandomAccessIterator last);

template<class RandomAccessIterator, class Compare>
 constexpr void nth_element(RandomAccessIterator first, RandomAccessIterator nth,
 RandomAccessIterator last, Compare comp);
template<class ExecutionPolicy, class RandomAccessIterator, class Compare>
 void nth_element(ExecutionPolicy&& exec,
 RandomAccessIterator first, RandomAccessIterator nth,
 RandomAccessIterator last, Compare comp);

namespace ranges {
 template<RandomAccessIterator I, Sentinel<I> S, class Comp = ranges::less<>,
 class Proj = identity>
 requires Sortable<I, Comp, Proj>
 constexpr I
 nth_element(I first, I nth, S last, Comp comp = {}, Proj proj = {});
}

```

- 1     Let `comp` be `less{}` and `proj` be `identity{}` for the overloads with no parameters by those names.
- 2     Requires: `[first, nth)` and `[nth, last)` shall be valid ranges. For the overloads in namespace `std`, `RandomAccessIterator` shall ~~satisfy~~meet the `Cpp17ValueSwappable` requirements ([swappable.requirements])-  
~~The~~and the type of `*first` shall ~~satisfy~~meet the `Cpp17MoveConstructible` ([tab:moveconstructible])  
 and `Cpp17MoveAssignable` ([tab:moveassignable]) requirements.
- 3     Effects: After `nth_element` the element in the position pointed to by `nth` is the element that would be in that position if the whole range were sorted with respect to `comp` and `proj`, unless `nth == last`. Also for

every iterator *i* in the range `[first, nth)` and every iterator *j* in the range `[nth, last)` it holds that:  
~~!(\*j < \*i) or comp(\*j, \*i) == false;~~ `bool(invoker(comp, invoker(proj, *j), invoker(proj, *i)))`  
is false.

4 *Returns:* `last` for the overload in namespace `ranges`.

5 *Complexity:* For the overloads with no `ExecutionPolicy`, linear on average. For the overloads with an `ExecutionPolicy`,  $\mathcal{O}(N)$  applications of the predicate, and  $\mathcal{O}(N \log N)$  swaps, where  $N = \text{last} - \text{first}$ .

```
namespace ranges {
 template<RandomAccessRange R, class Comp = ranges::less<>, class Proj = identity>
 requires Sortable<iterator_t<R>, Comp, Proj>
 constexpr safe_iterator_t<R>
 nth_element(R&& r, iterator_t<R> nth, Comp comp = {}, Proj proj = {});
}
```

6 *Effects:* Equivalent to:

```
return ranges::nth_element(ranges::begin(r), nth, ranges::end(r), comp, proj);
```

### 24.7.3 Binary search

[alg.binary.search]

1 All of the algorithms in this subclause are versions of binary search and assume that the sequence being searched is partitioned with respect to an expression formed by binding the search key to an argument of the ~~implied-or-explicit~~ comparison function. They work on non-random access iterators minimizing the number of comparisons, which will be logarithmic for all types of iterators. They are especially appropriate for random access iterators, because these algorithms do a logarithmic number of steps through the data structure. For non-random access iterators they execute a linear number of steps.

#### 24.7.3.1 lower\_bound

[lower.bound]

```
template<class ForwardIterator, class T>
 constexpr ForwardIterator
 lower_bound(ForwardIterator first, ForwardIterator last,
 const T& value);

template<class ForwardIterator, class T, class Compare>
 constexpr ForwardIterator
 lower_bound(ForwardIterator first, ForwardIterator last,
 const T& value, Compare comp);

namespace ranges {
 template<ForwardIterator I, Sentinel<I> S, class T, class Proj = identity,
 IndirectStrictWeakOrder<const T*, projected<I, Proj>> Comp = ranges::less<>>
 constexpr I lower_bound(I first, S last, const T& value, Comp comp = {},
 Proj proj = {});
 template<ForwardRange R, class T, class Proj = identity,
 IndirectStrictWeakOrder<const T*, projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
 constexpr safe_iterator_t<R>
 lower_bound(R&& r, const T& value, Comp comp = {}, Proj proj = {});
}
```

1 Let `comp` be `less{}` and `proj` be `identity{}` for overloads with no parameters by those names.

2 *Requires:* The elements *e* of `[first, last)` shall be partitioned with respect to the expression  
~~`e < value` or~~ `bool(invoker(comp, invoker(proj, e), value))`.

3 *Returns:* The furthestmost iterator *i* in the range `[first, last]` such that for every iterator *j* in the range `[first, i)`, ~~the following corresponding conditions hold: `*j < value` or~~ `bool(invoker(comp, invoker(proj, *j), value))` is true. ~~`!= false`.~~

4 *Complexity:* At most  $\log_2(\text{last} - \text{first}) + \mathcal{O}(1)$  comparisons and projections.

#### 24.7.3.2 upper\_bound

[upper.bound]

```
template<class ForwardIterator, class T>
 constexpr ForwardIterator
 upper_bound(ForwardIterator first, ForwardIterator last,
```

```

 const T& value);

template<class ForwardIterator, class T, class Compare>
constexpr ForwardIterator
 upper_bound(ForwardIterator first, ForwardIterator last,
 const T& value, Compare comp);

namespace ranges {
 template<ForwardIterator I, Sentinel<I> S, class T, class Proj = identity,
 IndirectStrictWeakOrder<const T*, projected<I, Proj>> Comp = ranges::less<>>
 constexpr I upper_bound(I first, S last, const T& value, Comp comp = {}, Proj proj = {});
 template<ForwardRange R, class T, class Proj = identity,
 IndirectStrictWeakOrder<const T*, projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
 constexpr safe_iterator_t<R>
 upper_bound(R&& r, const T& value, Comp comp = {}, Proj proj = {});
}

```

- 1 Let `comp` be `less{}` and `proj` be `identity{}` for overloads with no parameters by those names.
- 2 *Requires:* The elements `e` of `[first, last)` shall be partitioned with respect to the expression ~~`!(value < e)`~~ or `!bool(invoke(comp, value, invoke(proj, e)))`.
- 3 *Returns:* The furthestmost iterator `i` in the range `[first, last]` such that for every iterator `j` in the range `[first, i)`, ~~the following corresponding conditions hold: `!(value < *j)` or `!bool(invoke(comp, value, invoke(proj, *j)))` is true. `== false`.~~
- 4 *Complexity:* At most  $\log_2(\text{last} - \text{first}) + \mathcal{O}(1)$  comparisons and projections.

### 24.7.3.3 equal\_range

[`equal.range`]

[Editor's note: This wording incorporates the PR for [stl2#526](#).]

```

template<class ForwardIterator, class T>
constexpr pair<ForwardIterator, ForwardIterator>
 equal_range(ForwardIterator first,
 ForwardIterator last, const T& value);

template<class ForwardIterator, class T, class Compare>
constexpr pair<ForwardIterator, ForwardIterator>
 equal_range(ForwardIterator first,
 ForwardIterator last, const T& value,
 Compare comp);

namespace ranges {
 template<ForwardIterator I, Sentinel<I> S, class T, class Proj = identity,
 IndirectStrictWeakOrder<const T*, projected<I, Proj>> Comp = ranges::less<>>
 constexpr subrange<I>
 equal_range(I first, S last, const T& value, Comp comp = {}, Proj proj = {});
 template<ForwardRange R, class T, class Proj = identity,
 IndirectStrictWeakOrder<const T*, projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
 constexpr safe_subrange_t<R>
 equal_range(R&& r, const T& value, Comp comp = {}, Proj proj = {});
}

```

- 1 Let `comp` be `less{}` and `proj` be `identity{}` for overloads with no parameters by those names.
- 2 *Requires:* The elements `e` of `[first, last)` shall be partitioned with respect to the expressions ~~`e < value`~~ and ~~`!(value < e)`~~ or `bool(invoke(comp, invoke(proj, e), value))` and `!bool(invoke(comp, value, invoke(proj, e)))`. Also, for all elements `e` of `[first, last)`, ~~`e < value` shall imply~~ `!(value < e)` or `bool(comp(e, value))` shall imply `!bool(comp(value, e))` for the overloads in namespace `std`.
- 3 *Returns:*
- (3.1) — For the overloads in namespace `std`:
- ```

        make_pair(lower_bound(first, last, value),
                  upper_bound(first, last, value))
    or

```

```

make_pair(lower_bound(first, last, value, comp),
            upper_bound(first, last, value, comp))

```

(3.2) — For the overloads in namespace `ranges`:

```

{ranges::lower_bound(first, last, value, comp, proj),
 ranges::upper_bound(first, last, value, comp, proj)}

```

4 *Complexity:* At most $2 * \log_2(\text{last} - \text{first}) + \mathcal{O}(1)$ comparisons and projections.

24.7.3.4 binary_search

[binary.search]

```

template<class ForwardIterator, class T>

```

```

constexpr bool

```

```

    binary_search(ForwardIterator first, ForwardIterator last,
                  const T& value);

```

```

template<class ForwardIterator, class T, class Compare>

```

```

constexpr bool

```

```

    binary_search(ForwardIterator first, ForwardIterator last,
                  const T& value, Compare comp);

```

```

namespace ranges {

```

```

    template<ForwardIterator I, Sentinel<I> S, class T, class Proj = identity,
            IndirectStrictWeakOrder<const T*, projected<I, Proj>> Comp = ranges::less<>>
    constexpr bool binary_search(I first, S last, const T& value, Comp comp = {},
                                Proj proj = {});

```

```

    template<ForwardRange R, class T, class Proj = identity,
            IndirectStrictWeakOrder<const T*, projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
    constexpr bool binary_search(R&& r, const T& value, Comp comp = {},
                                Proj proj = {});
}

```

1 Let `comp` be `less{}` and `proj` be `identity{}` for overloads with no parameters by those names.

2 *Requires:* The elements `e` of `[first, last)` shall be partitioned with respect to the expressions ~~`e < value`~~ and ~~`!(value < e)`~~ or `bool(invoker(comp, invoker(proj, e), value))` and `!bool(invoker(comp, value, invoker(proj, e)))`. Also, for all elements `e` of `[first, last)`, ~~`e < value`~~ shall imply ~~`!(value < e)`~~ or `bool(comp(e, value))` shall imply `!bool(comp(value, e))` for the overloads in namespace `std`.

3 *Returns:* `true` if ~~there is an~~ and only if for some iterator `i` in the range `[first, last)`, ~~that satisfies the corresponding conditions:~~ ~~`!(i < value) && !(value < *i)`~~ or `!bool(invoker(comp, invoker(proj, *i), value))` ~~`== false`~~ `&& !bool(invoker(comp, value, invoker(proj, *i)))` ~~`== false`~~ is `true`.

4 *Complexity:* At most $\log_2(\text{last} - \text{first}) + \mathcal{O}(1)$ comparisons and projections.

24.7.4 Partitions

[alg.partitions]

```

template<class InputIterator, class Predicate>

```

```

constexpr bool is_partitioned(InputIterator first, InputIterator last, Predicate pred);

```

```

template<class ExecutionPolicy, class ForwardIterator, class Predicate>

```

```

    bool is_partitioned(ExecutionPolicy&& exec,
                        ForwardIterator first, ForwardIterator last, Predicate pred);

```

```

namespace ranges {

```

```

    template<InputIterator I, Sentinel<I> S, class Proj = identity,
            IndirectUnaryPredicate<projected<I, Proj>> Pred>
    constexpr bool is_partitioned(I first, S last, Pred pred, Proj proj = {});
    template<InputRange R, class Proj = identity,
            IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
    constexpr bool is_partitioned(R&& r, Pred pred, Proj proj = {});
}

```

1 Let `proj` be `identity{}` for the overloads with no parameter named `proj`.

2 *Requires:* For the overload with no `ExecutionPolicy`, `InputIterator`'s value type shall be convertible to `Predicate`'s argument type. For the overload with an `ExecutionPolicy`, `ForwardIterator`'s value type shall be convertible to `Predicate`'s argument type.

3 *Returns:* `true` if ~~`[first, last)` is empty or~~ and only if the elements `e` of `[first, last)` are partitioned with respect to the expression `bool(invoker(pred, invoke(proj, e)))`.

4 *Complexity:* Linear. At most `last - first` applications of `pred` and proj.

```
template<class ForwardIterator, class Predicate>
constexpr ForwardIterator
    partition(ForwardIterator first, ForwardIterator last, Predicate pred);
template<class ExecutionPolicy, class ForwardIterator, class Predicate>
ForwardIterator
    partition(ExecutionPolicy&& exec,
              ForwardIterator first, ForwardIterator last, Predicate pred);
```

```
namespace ranges {
    template<Permutable I, Sentinel<I> S, class Proj = identity,
            IndirectUnaryPredicate<projected<I, Proj>> Pred>
    constexpr I
        partition(I first, S last, Pred pred, Proj proj = {});
    template<ForwardRange R, class Proj = identity,
            IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
    requires Permutable<iterator_t<R>>
    constexpr safe_iterator_t<R>
        partition(R&& r, Pred pred, Proj proj = {});
}
```

5 Let `proj` be `identity{}` for the overloads with no parameter named `proj` and let $E(x)$ be `bool(invoker(pred, invoke(proj, x)))`.

6 *Requires:* For the overloads in namespace `std`, `ForwardIterator` shall ~~satisfy~~ meet the `Cpp17ValueSwappable` requirements ([swappable.requirements]).

7 *Effects:* Places all the elements `e` in the range `[first, last)` that satisfy ~~`pred`~~ $E(e)$ before all the elements that do not ~~satisfy it~~.

8 *Returns:* An iterator `i` such that $E(*j)$ is `true` for every iterator `j` in ~~the range~~ `[first, i)` ~~`pred(*j) != false`~~, and `false` for every iterator ~~`kj`~~ `kj` in ~~the range~~ `[i, last)`; ~~`pred(*k) == false` is `false`~~.

9 *Complexity:* Let $N = \text{last} - \text{first}$:

(9.1) — For the overloads with no `ExecutionPolicy`, exactly N applications of the predicate and projection. At most $N/2$ swaps if ~~`ForwardIterator`~~ the type of `first` meets the `Cpp17BidirectionalIterator` requirements for the overloads in namespace `std` or models `BidirectionalIterator` for the overloads in namespace `ranges`, and at most N swaps otherwise.

(9.2) — For the overload with an `ExecutionPolicy`, $\mathcal{O}(N \log N)$ swaps and $\mathcal{O}(N)$ applications of the predicate.

```
template<class BidirectionalIterator, class Predicate>
BidirectionalIterator
    stable_partition(BidirectionalIterator first, BidirectionalIterator last, Predicate pred);
template<class ExecutionPolicy, class BidirectionalIterator, class Predicate>
BidirectionalIterator
    stable_partition(ExecutionPolicy&& exec,
                    BidirectionalIterator first, BidirectionalIterator last, Predicate pred);
```

```
namespace ranges {
    template<BidirectionalIterator I, Sentinel<I> S, class Proj = identity,
            IndirectUnaryPredicate<projected<I, Proj>> Pred>
    requires Permutable<I>
    I stable_partition(I first, S last, Pred pred, Proj proj = {});
    template<BidirectionalRange R, class Proj = identity,
            IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
    requires Permutable<iterator_t<R>>
    requires Permutable<iterator_t<R>>
```

```

    safe_iterator_t<R> stable_partition(R&& r, Pred pred, Proj proj = {});
}

```

10 Let `proj` be `identity{}` for the overloads with no parameter named `proj` and let $E(x)$ be `bool(invoker(pred, invoke(proj, x)))`.

11 *Requires:* For the overloads in namespace `std`, `BidirectionalIterator` shall ~~satisfy~~ meet the *Cpp17ValueSwappable* requirements (`[swappable.requirements]`). ~~The and the~~ type of `*first` shall ~~satisfy~~ meet the *Cpp17MoveConstructible* (`[tab:moveconstructible]`) and *Cpp17MoveAssignable* (`[tab:moveassignable]`) requirements.

12 *Effects:* Places all the elements `e` in ~~the range~~ `[first, last)` that satisfy ~~`pred E(e)`~~ before all the elements that do not ~~satisfy it~~. The relative order of the elements in both groups is preserved.

13 *Returns:* An iterator `i` such that for every iterator `j` in ~~the range~~ `[first, i)`, ~~`pred(*j) != false`~~ `E(*j) is true`, and for every iterator ~~`kj`~~ `kj` in ~~the range~~ `[i, last)`, ~~`pred(*k) == false`~~ `E(*j) is false`. ~~The relative order of the elements in both groups is preserved.~~

14 *Complexity:* Let $N = \text{last} - \text{first}$:

- (14.1) — For the overloads with no `ExecutionPolicy`, at most $N \log N$ swaps, but only $\mathcal{O}(N)$ swaps if there is enough extra memory. Exactly N applications of the predicate and projection.
- (14.2) — For the overload with an `ExecutionPolicy`, $\mathcal{O}(N \log N)$ swaps and $\mathcal{O}(N)$ applications of the predicate.

```

template<class InputIterator, class OutputIterator1,
        class OutputIterator2, class Predicate>
constexpr pair<OutputIterator1, OutputIterator2>
partition_copy(InputIterator first, InputIterator last,
              OutputIterator1 out_true, OutputIterator2 out_false, Predicate pred);
template<class ExecutionPolicy, class ForwardIterator, class ForwardIterator1,
        class ForwardIterator2, class Predicate>
pair<ForwardIterator1, ForwardIterator2>
partition_copy(ExecutionPolicy&& exec,
              ForwardIterator first, ForwardIterator last,
              ForwardIterator1 out_true, ForwardIterator2 out_false, Predicate pred);

namespace ranges {
    template<InputIterator I, Sentinel<I> S, WeaklyIncrementable O1, WeaklyIncrementable O2,
            class Proj = identity, IndirectUnaryPredicate<projected<I, Proj>> Pred>
            requires IndirectlyCopyable<I, O1> && IndirectlyCopyable<I, O2>
            constexpr partition_copy_result<I, O1, O2>
            partition_copy(I first, S last, O1 out_true, O2 out_false, Pred pred,
                          Proj proj = {});
    template<InputRange R, WeaklyIncrementable O1, WeaklyIncrementable O2,
            class Proj = identity,
            IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
            requires IndirectlyCopyable<iterator_t<R>, O1> &&
            IndirectlyCopyable<iterator_t<R>, O2>
            constexpr partition_copy_result<safe_iterator_t<R>, O1, O2>
            partition_copy(R&& r, O1 out_true, O2 out_false, Pred pred, Proj proj = {});
}

```

15 Let `proj` be `identity{}` for the overloads with no parameter named `proj` and let $E(x)$ be `bool(invoker(pred, invoke(proj, x)))`.

16 *Requires:* The input range and output ranges shall not overlap ~~with either of the output ranges~~. For the overloads in namespace `std`, the expression `*first` shall be writable (22.3.1) to `out_true` and `out_false`. [Note: For the overload with an `ExecutionPolicy`, there may be a performance cost if `first`'s value type ~~is not~~ does not meet the *Cpp17CopyConstructible* requirements. — end note]

- (16.1) — For the overload with no `ExecutionPolicy`, `InputIterator`'s value type shall be *Cpp17CopyAssignable* (`[tab:copyassignable]`), and shall be writable (22.3.1) to the `out_true` and `out_false` `OutputIterators`, and shall be convertible to `Predicate`'s argument type.
- (16.2) — For the overload with an `ExecutionPolicy`, `ForwardIterator`'s value type shall be *Cpp17CopyAssignable*, and shall be writable to the `out_true` and `out_false` `ForwardIterators`, and shall be convertible

to `Predicate`'s argument type. [Note: There may be a performance cost if `ForwardIterator`'s value type is not *Cpp17CopyConstructible*. — end note]

(16.3) — For both overloads, the input range shall not overlap with either of the output ranges.

17 *Effects:* For each iterator `i` in `[first, last)`, copies `*i` to the output range beginning with `out_true` if ~~`pred(*i)`~~ `E(*i)` is true, or to the output range beginning with `out_false` otherwise.

18 *Returns:* ~~A pair `p` such that `p.first` is the end of the output range beginning at `out_true` and `p.second` is the end of the output range beginning at `out_false`.~~ Let `o1` be the end of the output range beginning at `out_true`, and `o2` the end of the output range beginning at `out_false`. Returns

(18.1) — `{o1, o2}` for the overloads in namespace `std`, or

(18.2) — `{last, o1, o2}` for the overloads in namespace `ranges`.

19 *Complexity:* Exactly `last - first` applications of `pred` and `proj`.

```
template<class ForwardIterator, class Predicate>
constexpr ForwardIterator
partition_point(ForwardIterator first, ForwardIterator last, Predicate pred);
```

```
namespace ranges {
    template<ForwardIterator I, Sentinel<I> S, class Proj = identity,
            IndirectUnaryPredicate<projected<I, Proj>> Pred>
        constexpr I partition_point(I first, S last, Pred pred, Proj proj = {});
    template<ForwardRange R, class Proj = identity,
            IndirectUnaryPredicate<projected<iterator_t<R>, Proj>> Pred>
        constexpr safe_iterator_t<R>
            partition_point(R&& r, Pred pred, Proj proj = {});
}
```

20 Let `proj` be `identity{}` for the overloads with no parameter named `proj` and let $E(x)$ be `bool(invoker(pred, invoker(proj, x)))`.

21 *Requires:* ~~`ForwardIterator`'s value type shall be convertible to `Predicate`'s argument type.~~ The elements `e` of `[first, last)` shall be partitioned with respect to ~~the expression `pred(e)`~~ `E(e)`.

22 *Returns:* An iterator `mid` such that ~~`all_of(first, mid, pred)` and `none_of(mid, last, pred)` are both true~~ `E(*i)` is true for all iterators `i` in `[first, mid)`, and false for all iterators `i` in `[mid, last)`.

23 *Complexity:* $\mathcal{O}(\log(\text{last} - \text{first}))$ applications of `pred` and `proj`.

24.7.5 Merge

[alg.merge]

```
template<class InputIterator1, class InputIterator2,
        class OutputIterator>
constexpr OutputIterator
merge(InputIterator1 first1, InputIterator1 last1,
      InputIterator2 first2, InputIterator2 last2,
      OutputIterator result);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
        class ForwardIterator>
ForwardIterator
merge(ExecutionPolicy&& exec,
      ForwardIterator1 first1, ForwardIterator1 last1,
      ForwardIterator2 first2, ForwardIterator2 last2,
      ForwardIterator result);

template<class InputIterator1, class InputIterator2,
        class OutputIterator, class Compare>
constexpr OutputIterator
merge(InputIterator1 first1, InputIterator1 last1,
      InputIterator2 first2, InputIterator2 last2,
      OutputIterator result, Compare comp);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
        class ForwardIterator, class Compare>
ForwardIterator
merge(ExecutionPolicy&& exec,
```

```

    ForwardIterator1 first1, ForwardIterator1 last1,
    ForwardIterator2 first2, ForwardIterator2 last2,
    ForwardIterator result, Compare comp);

```

```

namespace ranges {
    template<InputIterator I1, Sentinel<I1> S1, InputIterator I2, Sentinel<I2> S2,
            WeaklyIncrementable O, class Comp = ranges::less<>, class Proj1 = identity,
            class Proj2 = identity>
    requires Mergeable<I1, I2, O, Comp, Proj1, Proj2>
    constexpr merge_result<I1, I2, O>
        merge(I1 first1, S1 last1, I2 first2, S2 last2, O result,
              Comp comp = {}, Proj1 proj1 = {}, Proj2 proj2 = {});
    template<InputRange R1, InputRange R2, WeaklyIncrementable O, class Comp = ranges::less<>,
            class Proj1 = identity, class Proj2 = identity>
    requires Mergeable<iterator_t<R1>, iterator_t<R2>, O, Comp, Proj1, Proj2>
    constexpr merge_result<safe_iterator_t<R1>, safe_iterator_t<R2>, O>
        merge(R1&& r1, R2&& r2, O result,
              Comp comp = {}, Proj1 proj1 = {}, Proj2 proj2 = {});
}

```

- 1 Let N be $(last1 - first1) + (last2 - first2)$. Let `comp` be `less{}`, `proj1` be `identity{}`, and `proj2` be `identity{}`, for the overloads with no parameters by those names.
- 2 *Requires:* The ranges $[first1, last1)$ and $[first2, last2)$ shall be sorted with respect to **`operator<`** ~~or `comp`~~ and `proj1` or `proj2`, respectively. The resulting range shall not overlap with either of the original ranges.
- 3 *Effects:* Copies all the elements of the two ranges $[first1, last1)$ and $[first2, last2)$ into the range $[result, result_last)$, where `result\_last` is `result +  $N$` ~~$(last1 - first1) + (last2 - first2)$~~ , such that the resulting range satisfies is_sorted(result, result_last) or is_sorted(result, result_last, comp), respectively. If an element a precedes b in an input range, a is copied into the output range before b . If $e1$ is an element of $[first1, last1)$ and $e2$ of $[first2, last2)$, $e2$ is copied into the output range before $e1$ if and only if `bool(invoker(comp, invoker(proj2, e2), invoker(proj1, e1)))` is true.
- 4 *Returns:* ~~`result + (last1 - first1) + (last2 - first2)`~~.
 - (4.1) — `result\_last` for the overloads in namespace `std`, or
 - (4.2) — `{last1, last2, result\_last}` for the overloads in namespace `ranges`.
- 5 *Complexity:* ~~Let $N = (last1 - first1) + (last2 - first2)$:~~
 - (5.1) — For the overloads with no `ExecutionPolicy`, at most $N - 1$ comparisons and applications of each projection.
 - (5.2) — For the overloads with an `ExecutionPolicy`, $\mathcal{O}(N)$ comparisons.
- 6 *Remarks:* Stable ([\[algorithm.stable\]](#)).

```

template<class BidirectionalIterator>
    void inplace_merge(BidirectionalIterator first,
                      BidirectionalIterator middle,
                      BidirectionalIterator last);
template<class ExecutionPolicy, class BidirectionalIterator>
    void inplace_merge(ExecutionPolicy&& exec,
                      BidirectionalIterator first,
                      BidirectionalIterator middle,
                      BidirectionalIterator last);

template<class BidirectionalIterator, class Compare>
    void inplace_merge(BidirectionalIterator first,
                      BidirectionalIterator middle,
                      BidirectionalIterator last, Compare comp);
template<class ExecutionPolicy, class BidirectionalIterator, class Compare>
    void inplace_merge(ExecutionPolicy&& exec,
                      BidirectionalIterator first,
                      BidirectionalIterator middle,
                      BidirectionalIterator last, Compare comp);

```

```

namespace ranges {
    template<BidirectionalIterator I, Sentinel<I> S, class Comp = ranges::less<>,
            class Proj = identity>
    requires Sortable<I, Comp, Proj>
    I inplace_merge(I first, I middle, S last, Comp comp = {}, Proj proj = {});
}

```

Let `comp` be `less{}` and `proj` be `identity{}` for the overloads with no parameters by those names.

Requires: ~~The~~ `ranges` `[first, middle)` and `[middle, last)` shall be valid ranges sorted with respect to ~~operator<~~ ~~or~~ `comp` and `proj`. For the overloads in namespace `std`, `BidirectionalIterator` shall ~~satisfy~~ meet the *Cpp17ValueSwappable* requirements ([swappable.requirements]) ~~—The and the~~ type of `*first` shall ~~satisfy~~ meet the *Cpp17MoveConstructible* ([tab:moveconstructible]) and *Cpp17MoveAssignable* ([tab:moveassignable]) requirements.

Effects: Merges two sorted consecutive ranges `[first, middle)` and `[middle, last)`, putting the result of the merge into the range `[first, last)`. The resulting range is sorted with respect to `comp` and `proj`. ~~will be in non-decreasing order; that is, for every iterator `i` in `[first, last)` other than `first`, the condition `*i < *(i - 1)` or, respectively, `comp(*i, *(i - 1))` will be false.~~

Returns: `last` for the overload in namespace `ranges`.

Complexity: Let $N = \text{last} - \text{first}$:

— For the overloads with no `ExecutionPolicy`, and if enough additional memory is available, exactly $N - 1$ comparisons.

~~For the overloads with no `ExecutionPolicy` if no additional memory is available, $\mathcal{O}(N \log N)$ comparisons.~~

~~Otherwise~~ For the overloads with an `ExecutionPolicy`, $\mathcal{O}(N \log N)$ comparisons.

In either case, twice as many projections as comparisons.

Remarks: Stable ([algorithm.stable]).

```

namespace ranges {
    template<BidirectionalRange R, class Comp = ranges::less<>, class Proj = identity>
    requires Sortable<iterator_t<R>, Comp, Proj>
    safe_iterator_t<R>
    inplace_merge(R&& r, iterator_t<R> middle, Comp comp = {}, Proj proj = {});
}

```

Effects: Equivalent to:

```
return ranges::inplace_merge(ranges::begin(r), middle, ranges::end(r), comp, proj);
```

24.7.6 Set operations on sorted structures

[alg.set.operations]

- 1 This subclause defines all the basic set operations on sorted structures. They also work with `multisets` ([multiset]) containing multiple copies of equivalent elements. The semantics of the set operations are generalized to `multisets` in a standard way by defining `set_union()` to contain the maximum number of occurrences of every element, `set_intersection()` to contain the minimum, and so on.

24.7.6.1 includes

[includes]

[Editor's note: This wording includes (pun intended) the proposed resolution for LWG 3115.]

```

template<class InputIterator1, class InputIterator2>
constexpr bool includes(InputIterator1 first1, InputIterator1 last1,
                       InputIterator2 first2, InputIterator2 last2);

template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
bool includes(ExecutionPolicy&& exec,
             ForwardIterator1 first1, ForwardIterator1 last1,
             ForwardIterator2 first2, ForwardIterator2 last2);

template<class InputIterator1, class InputIterator2, class Compare>
constexpr bool includes(InputIterator1 first1, InputIterator1 last1,
                       InputIterator2 first2, InputIterator2 last2,
                       Compare comp);

```

```

template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2, class Compare>
    bool includes(ExecutionPolicy&& exec,
                  ForwardIterator1 first1, ForwardIterator1 last1,
                  ForwardIterator2 first2, ForwardIterator2 last2,
                  Compare comp);

namespace ranges {
    template<InputIterator I1, Sentinel<I1> S1, InputIterator I2, Sentinel<I2> S2,
             class Proj1 = identity, class Proj2 = identity,
             IndirectStrictWeakOrder<projected<I1, Proj1>, projected<I2, Proj2>> Comp = ranges::less<>>
             constexpr bool includes(I1 first1, S1 last1, I2 first2, S2 last2, Comp comp = {},
                                     Proj1 proj1 = {}, Proj2 proj2 = {});
    template<InputRange R1, InputRange R2, class Proj1 = identity,
             class Proj2 = identity,
             IndirectStrictWeakOrder<projected<iterator_t<R1>, Proj1>,
             projected<iterator_t<R2>, Proj2>> Comp = ranges::less<>>
             constexpr bool includes(R1&& r1, R2&& r2, Comp comp = {},
                                     Proj1 proj1 = {}, Proj2 proj2 = {});
}

```

- 1 Let `comp` be `less{}`, `proj1` be `identity{}`, and `proj2` be `identity{}`, for the overloads with no parameters by those names.
- 2 *Requires:* The ranges `[first1, last1)` and `[first2, last2)` shall be sorted with respect to `comp` and `proj1` or `proj2`, respectively.
- 3 *Returns:* `true` if and only if `[first2, last2)` is ~~empty or if every element in the range `[first2, last2)` is contained in the range `[first1, last1)`. Returns `false` otherwise.~~ a subsequence of `[first1, last1)`. [Note: A sequence *S* is a subsequence of another sequence *T* if *S* can be obtained from *T* by removing some, all, or none of *T*'s elements and keeping the remaining elements in the same order. — end note]
- 4 *Complexity:* At most $2 * (last1 - first1)$ comparisons and applications of each projection.

24.7.6.2 set_union

[set.union]

```

template<class InputIterator1, class InputIterator2,
         class OutputIterator>
    constexpr OutputIterator
        set_union(InputIterator1 first1, InputIterator1 last1,
                  InputIterator2 first2, InputIterator2 last2,
                  OutputIterator result);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
         class ForwardIterator>
    ForwardIterator
        set_union(ExecutionPolicy&& exec,
                  ForwardIterator1 first1, ForwardIterator1 last1,
                  ForwardIterator2 first2, ForwardIterator2 last2,
                  ForwardIterator result);

template<class InputIterator1, class InputIterator2,
         class OutputIterator, class Compare>
    constexpr OutputIterator
        set_union(InputIterator1 first1, InputIterator1 last1,
                  InputIterator2 first2, InputIterator2 last2,
                  OutputIterator result, Compare comp);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
         class ForwardIterator, class Compare>
    ForwardIterator
        set_union(ExecutionPolicy&& exec,
                  ForwardIterator1 first1, ForwardIterator1 last1,
                  ForwardIterator2 first2, ForwardIterator2 last2,
                  ForwardIterator result, Compare comp);

```

```

namespace ranges {
    template<InputIterator I1, Sentinel<I1> S1, InputIterator I2, Sentinel<I2> S2,
            WeaklyIncrementable O, class Comp = ranges::less<>, class Proj1 = identity, class Proj2 = identity>
    requires Mergeable<I1, I2, O, Comp, Proj1, Proj2>
    constexpr set_union_result<I1, I2, O>
        set_union(I1 first1, S1 last1, I2 first2, S2 last2, O result, Comp comp = {},
            Proj1 proj1 = {}, Proj2 proj2 = {});
    template<InputRange R1, InputRange R2, WeaklyIncrementable O,
            class Comp = ranges::less<>, class Proj1 = identity, class Proj2 = identity>
    requires Mergeable<iterator_t<R1>, iterator_t<R2>, O, Comp, Proj1, Proj2>
    constexpr set_union_result<safe_iterator_t<R1>, safe_iterator_t<R2>, O>
        set_union(R1&& r1, R2&& r2, O result, Comp comp = {},
            Proj1 proj1 = {}, Proj2 proj2 = {});
}

```

1 Let `comp` be `less{}`, and `proj1` and `proj2` be `identity{}` for the overloads with no parameters by those names.

2 *Requires:* [The ranges `\[first1, last1\)` and `\[first2, last2\)` shall be sorted with respect to `comp` and `proj1` or `proj2`, respectively.](#) The resulting range shall not overlap with either of the original ranges.

3 *Effects:* Constructs a sorted union of the elements from the two ranges; that is, the set of elements that are present in one or both of the ranges.

4 *Returns:* [Let `result\_last` be](#) ~~The~~ [the end of the constructed range.](#) [Returns](#)

(4.1) — `result_last` for the overloads in namespace `std`, or

(4.2) — `{last1, last2, result_last}` for the overloads in namespace `ranges`.

5 *Complexity:* At most $2 * ((last1 - first1) + (last2 - first2)) - 1$ comparisons [and applications of each projection](#).

6 *Remarks:* [Stable \(\[algorithm.stable\]\).](#) If `[first1, last1)` contains m elements that are equivalent to each other and `[first2, last2)` contains n elements that are equivalent to them, then all m elements from the first range ~~shall be~~ copied to the output range, in order, and then [the final](#) $\max(n - m, 0)$ elements from the second range ~~shall be~~ copied to the output range, in order.

24.7.6.3 set_intersection

[set.intersection]

```

template<class InputIterator1, class InputIterator2,
        class OutputIterator>
constexpr OutputIterator
    set_intersection(InputIterator1 first1, InputIterator1 last1,
                    InputIterator2 first2, InputIterator2 last2,
                    OutputIterator result);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
        class ForwardIterator>
ForwardIterator
    set_intersection(ExecutionPolicy&& exec,
                    ForwardIterator1 first1, ForwardIterator1 last1,
                    ForwardIterator2 first2, ForwardIterator2 last2,
                    ForwardIterator result);

template<class InputIterator1, class InputIterator2,
        class OutputIterator, class Compare>
constexpr OutputIterator
    set_intersection(InputIterator1 first1, InputIterator1 last1,
                    InputIterator2 first2, InputIterator2 last2,
                    OutputIterator result, Compare comp);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
        class ForwardIterator, class Compare>
ForwardIterator
    set_intersection(ExecutionPolicy&& exec,
                    ForwardIterator1 first1, ForwardIterator1 last1,
                    ForwardIterator2 first2, ForwardIterator2 last2,
                    ForwardIterator result, Compare comp);

```

```

namespace ranges {
    template<InputIterator I1, Sentinel<I1> S1, InputIterator I2, Sentinel<I2> S2,
            WeaklyIncrementable O, class Comp = ranges::less<>, class Proj1 = identity, class Proj2 = identity>
    requires Mergeable<I1, I2, O, Comp, Proj1, Proj2>
    constexpr set_intersection_result<I1, I2, O>
        set_intersection(I1 first1, S1 last1, I2 first2, S2 last2, O result,
                        Comp comp = {}, Proj1 proj1 = {}, Proj2 proj2 = {});
    template<InputRange R1, InputRange R2, WeaklyIncrementable O,
            class Comp = ranges::less<>, class Proj1 = identity, class Proj2 = identity>
    requires Mergeable<iterator_t<R1>, iterator_t<R2>, O, Comp, Proj1, Proj2>
    constexpr set_intersection_result<safe_iterator_t<R1>, safe_iterator_t<R2>, O>
        set_intersection(R1&& r1, R2&& r2, O result,
                        Comp comp = {}, Proj1 proj1 = {}, Proj2 proj2 = {});
}

```

1 Let `comp` be `less{}`, and `proj1` and `proj2` be `identity{}` for the overloads with no parameters by those names.

2 *Requires:* [The ranges `\[first1, last1\)` and `\[first2, last2\)` shall be sorted with respect to `comp` and `proj1` or `proj2`, respectively.](#) The resulting range shall not overlap with either of the original ranges.

3 *Effects:* Constructs a sorted intersection of the elements from the two ranges; that is, the set of elements that are present in both of the ranges.

4 *Returns:* [Let `result\_last` be](#) ~~The~~ [the end of the constructed range.](#) [Returns](#)

(4.1) — `result_last` for the overloads in namespace `std`, or

(4.2) — `{last1, last2, result_last}` for the overloads in namespace `ranges`.

5 *Complexity:* At most $2 * ((last1 - first1) + (last2 - first2)) - 1$ comparisons [and applications of each projection](#).

6 *Remarks:* [Stable](#) ([\[algorithm.stable\]](#)). If `[first1, last1)` contains m elements that are equivalent to each other and `[first2, last2)` contains n elements that are equivalent to them, the first $\min(m, n)$ elements ~~shall be~~ copied from the first range to the output range, in order.

24.7.6.4 `set_difference`

[`set.difference`]

```

template<class InputIterator1, class InputIterator2,
        class OutputIterator>
constexpr OutputIterator
    set_difference(InputIterator1 first1, InputIterator1 last1,
                  InputIterator2 first2, InputIterator2 last2,
                  OutputIterator result);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
        class ForwardIterator>
ForwardIterator
    set_difference(ExecutionPolicy&& exec,
                  ForwardIterator1 first1, ForwardIterator1 last1,
                  ForwardIterator2 first2, ForwardIterator2 last2,
                  ForwardIterator result);

template<class InputIterator1, class InputIterator2,
        class OutputIterator, class Compare>
constexpr OutputIterator
    set_difference(InputIterator1 first1, InputIterator1 last1,
                  InputIterator2 first2, InputIterator2 last2,
                  OutputIterator result, Compare comp);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
        class ForwardIterator, class Compare>
ForwardIterator
    set_difference(ExecutionPolicy&& exec,
                  ForwardIterator1 first1, ForwardIterator1 last1,
                  ForwardIterator2 first2, ForwardIterator2 last2,
                  ForwardIterator result, Compare comp);

```

```

namespace ranges {
    template<InputIterator I1, Sentinel<I1> S1, InputIterator I2, Sentinel<I2> S2,
            WeaklyIncrementable O, class Comp = ranges::less<>, class Proj1 = identity, class Proj2 = identity>
    requires Mergeable<I1, I2, O, Comp, Proj1, Proj2>
    constexpr set_difference_result<I1, O>
        set_difference(I1 first1, S1 last1, I2 first2, S2 last2, O result,
            Comp comp = {}, Proj1 proj1 = {}, Proj2 proj2 = {});
    template<InputRange R1, InputRange R2, WeaklyIncrementable O,
            class Comp = ranges::less<>, class Proj1 = identity, class Proj2 = identity>
    requires Mergeable<iterator_t<R1>, iterator_t<R2>, O, Comp, Proj1, Proj2>
    constexpr set_difference_result<safe_iterator_t<R1>, O>
        set_difference(R1&& r1, R2&& r2, O result,
            Comp comp = {}, Proj1 proj1 = {}, Proj2 proj2 = {});
}

```

1 Let `comp` be `less{}`, and `proj1` and `proj2` be `identity{}` for the overloads with no parameters by those names.

2 *Requires:* [The ranges `\[first1, last1\)` and `\[first2, last2\)` shall be sorted with respect to `comp` and `proj1` or `proj2`, respectively.](#) The resulting range shall not overlap with either of the original ranges.

3 *Effects:* Copies the elements of the range `[first1, last1)` which are not present in the range `[first2, last2)` to the range beginning at `result`. The elements in the constructed range are sorted.

4 *Returns:* [Let `result\_last` be](#) ~~The~~ [the end of the constructed range.](#) [Returns](#)

(4.1) — `result_last` for the overloads in namespace `std`, or

(4.2) — `{last1, result_last}` for the overloads in namespace `ranges`.

5 *Complexity:* At most $2 * ((last1 - first1) + (last2 - first2)) - 1$ comparisons [and applications of each projection](#).

6 *Remarks:* [Stable](#) ([algorithm.stable](#)). If `[first1, last1)` contains m elements that are equivalent to each other and `[first2, last2)` contains n elements that are equivalent to them, the last $\max(m - n, 0)$ elements from `[first1, last1)` ~~shall be~~ copied to the output range, [in order](#).

24.7.6.5 set_symmetric_difference

[set.symmetric.difference]

```

template<class InputIterator1, class InputIterator2,
        class OutputIterator>
constexpr OutputIterator
    set_symmetric_difference(InputIterator1 first1, InputIterator1 last1,
                            InputIterator2 first2, InputIterator2 last2,
                            OutputIterator result);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
        class ForwardIterator>
ForwardIterator
    set_symmetric_difference(ExecutionPolicy&& exec,
                            ForwardIterator1 first1, ForwardIterator1 last1,
                            ForwardIterator2 first2, ForwardIterator2 last2,
                            ForwardIterator result);

template<class InputIterator1, class InputIterator2,
        class OutputIterator, class Compare>
constexpr OutputIterator
    set_symmetric_difference(InputIterator1 first1, InputIterator1 last1,
                            InputIterator2 first2, InputIterator2 last2,
                            OutputIterator result, Compare comp);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
        class ForwardIterator, class Compare>
ForwardIterator
    set_symmetric_difference(ExecutionPolicy&& exec,
                            ForwardIterator1 first1, ForwardIterator1 last1,
                            ForwardIterator2 first2, ForwardIterator2 last2,
                            ForwardIterator result, Compare comp);

```

```

namespace ranges {
    template<InputIterator I1, Sentinel<I1> S1, InputIterator I2, Sentinel<I2> S2,
            WeaklyIncrementable O, class Comp = ranges::less<>, class Proj1 = identity, class Proj2 = identity>
    requires Mergeable<I1, I2, O, Comp, Proj1, Proj2>
    constexpr set_symmetric_difference_result<I1, I2, O>
        set_symmetric_difference(I1 first1, S1 last1, I2 first2, S2 last2, O result,
                                Comp comp = {}, Proj1 proj1 = {},
                                Proj2 proj2 = {});

    template<InputRange R1, InputRange R2, WeaklyIncrementable O,
            class Comp = ranges::less<>, class Proj1 = identity, class Proj2 = identity>
    requires Mergeable<iterator_t<R1>, iterator_t<R2>, O, Comp, Proj1, Proj2>
    constexpr set_symmetric_difference_result<safe_iterator_t<R1>, safe_iterator_t<R2>, O>
        set_symmetric_difference(R1&& r1, R2&& r2, O result, Comp comp = {},
                                Proj1 proj1 = {}, Proj2 proj2 = {});
}

```

1 Let `comp` be `less{}`, and `proj1` and `proj2` be `identity{}` for the overloads with no parameters by those names.

2 *Requires:* [The ranges `\[first1, last1\)` and `\[first2, last2\)` shall be sorted with respect to `comp` and `proj1` or `proj2`, respectively.](#) The resulting range shall not overlap with either of the original ranges.

3 *Effects:* Copies the elements of the range `[first1, last1)` that are not present in the range `[first2, last2)`, and the elements of the range `[first2, last2)` that are not present in the range `[first1, last1)` to the range beginning at `result`. The elements in the constructed range are sorted.

4 *Returns:* [Let `result\_last` be](#) ~~The end of the constructed range.~~ [Returns](#)

(4.1) — `result_last` for the overloads in namespace `std`, or

(4.2) — `{last1, last2, result_last}` for the overloads in namespace `ranges`.

5 *Complexity:* At most $2 * ((last1 - first1) + (last2 - first2)) - 1$ comparisons [and applications of each projection](#).

6 *Remarks:* [Stable](#) ([\[algorithm.stable\]](#)). If `[first1, last1)` contains m elements that are equivalent to each other and `[first2, last2)` contains n elements that are equivalent to them, then $|m - n|$ of those elements ~~shall be~~[are](#) copied to the output range: the last $m - n$ of these elements from `[first1, last1)` if $m > n$, and the last $n - m$ of these elements from `[first2, last2)` if $m < n$. [In either case, the elements are copied in order.](#)

24.7.7 Heap operations

[alg.heap.operations]

1 A *heap* is a particular organization of elements in a range between two random access iterators `[a, b)` such that:

2 A random access range `[a, b)` is a *heap with respect to `comp` and `proj`* for a comparator and projection `comp` and `proj` if its elements are organized such that:

(2.1) — With $N = b - a$, for all $i, 0 < i < N$, [bool](#)([invoke](#)(`comp`, [invoke](#)(`proj`, `a`[$\lfloor \frac{i-1}{2} \rfloor$]), [invoke](#)(`proj`, `a`[i])) is false.

(2.2) — `*a` may be removed by `pop_heap`[\(\)](#), or a new element added by `push_heap`[\(\)](#), in $\mathcal{O}(\log N)$ time.

3 These properties make heaps useful as priority queues.

4 `make_heap`[\(\)](#) converts a range into a heap and `sort_heap`[\(\)](#) turns a heap into a sorted sequence.

24.7.7.1 push_heap

[push.heap]

```

template<class RandomAccessIterator>
constexpr void push_heap(RandomAccessIterator first, RandomAccessIterator last);

template<class RandomAccessIterator, class Compare>
constexpr void push_heap(RandomAccessIterator first, RandomAccessIterator last,
                        Compare comp);

```

```

namespace ranges {
    template<RandomAccessIterator I, Sentinel<I> S, class Comp = ranges::less<>,
            class Proj = identity>
    requires Sortable<I, Comp, Proj>

```

```

constexpr I
    push_heap(I first, S last, Comp comp = {}, Proj proj = {});
template<RandomAccessRange R, class Comp = ranges::less<>, class Proj = identity>
    requires Sortable<iterator_t<R>, Comp, Proj>
constexpr safe_iterator_t<R>
    push_heap(R&& r, Comp comp = {}, Proj proj = {});
}

```

- 1 Let `comp` be `less{}` and `proj` be `identity{}` for the overloads with no parameters by those names.
- 2 *Requires:* The range `[first, last - 1)` shall be a valid heap with respect to `comp` and `proj`. ~~For the~~ overloads in namespace `std`, the type of `*first` shall satisfy ~~meet~~ the *Cpp17MoveConstructible* requirements ([[tab:moveconstructible](#)]) and the *Cpp17MoveAssignable* requirements ([[tab:moveassignable](#)]).
- 3 *Effects:* Places the value in the location `last - 1` into the resulting heap `[first, last)`.
- 4 *Returns:* `last` for the overloads in namespace `ranges`.
- 5 *Complexity:* At most $\log(\text{last} - \text{first})$ comparisons and twice as many projections.

24.7.7.2 pop_heap

[pop.heap]

```

template<class RandomAccessIterator>
    constexpr void pop_heap(RandomAccessIterator first, RandomAccessIterator last);

template<class RandomAccessIterator, class Compare>
    constexpr void pop_heap(RandomAccessIterator first, RandomAccessIterator last,
                           Compare comp);

namespace ranges {
    template<RandomAccessIterator I, Sentinel<I> S, class Comp = ranges::less<>,
            class Proj = identity>
        requires Sortable<I, Comp, Proj>
        constexpr I
            pop_heap(I first, S last, Comp comp = {}, Proj proj = {});
    template<RandomAccessRange R, class Comp = ranges::less<>, class Proj = identity>
        requires Sortable<iterator_t<R>, Comp, Proj>
        constexpr safe_iterator_t<R>
            pop_heap(R&& r, Comp comp = {}, Proj proj = {});
}

```

- 1 Let `comp` be `less{}` and `proj` be `identity{}` for the overloads with no parameters by those names.
- 2 *Requires:* The range `[first, last)` shall be a valid non-empty heap with respect to `comp` and `proj`. For the overloads in namespace `std`, `RandomAccessIterator` shall satisfy ~~meet~~ the *Cpp17ValueSwappable* requirements ([[swappable.requirements](#)]). ~~The~~ and the type of `*first` shall satisfy ~~meet~~ the *Cpp17MoveConstructible* ([[tab:moveconstructible](#)]) and *Cpp17MoveAssignable* ([[tab:moveassignable](#)]) requirements.
- 3 *Effects:* Swaps the value in the location `first` with the value in the location `last - 1` and makes `[first, last - 1)` into a heap with respect to `comp` and `proj`.
- 4 *Returns:* `last` for the overloads in namespace `ranges`.
- 5 *Complexity:* At most $2 \log(\text{last} - \text{first})$ comparisons and twice as many projections.

24.7.7.3 make_heap

[make.heap]

```

template<class RandomAccessIterator>
    constexpr void make_heap(RandomAccessIterator first, RandomAccessIterator last);

template<class RandomAccessIterator, class Compare>
    constexpr void make_heap(RandomAccessIterator first, RandomAccessIterator last,
                           Compare comp);

namespace ranges {
    template<RandomAccessIterator I, Sentinel<I> S, class Comp = ranges::less<>,
            class Proj = identity>
        requires Sortable<I, Comp, Proj>
        constexpr I
            make_heap(I first, S last, Comp comp = {}, Proj proj = {});
}

```

```

template<RandomAccessRange R, class Comp = ranges::less<>, class Proj = identity>
requires Sortable<iterator_t<R>, Comp, Proj>
constexpr safe_iterator_t<R>
    make_heap(R&& r, Comp comp = {}, Proj proj = {});
}

```

1 Let `comp` be `less{}` and `proj` be `identity{}` for the overloads with no parameters by those names.

2 *Requires:* For the overloads in namespace `std`, the type of `*first` shall ~~satisfy~~meet the *Cpp17MoveConstructible* requirements ([[tab:moveconstructible](#)]) and the *Cpp17MoveAssignable* requirements ([[tab:moveassignable](#)]).

3 *Effects:* Constructs a heap with respect to `comp` and `proj` out of the range `[first, last)`.

4 *Returns:* `last` for the overloads in namespace `ranges`.

5 *Complexity:* At most $3(\text{last} - \text{first})$ comparisons and twice as many projections.

24.7.7.4 `sort_heap`

[[sort.heap](#)]

```

template<class RandomAccessIterator>
constexpr void sort_heap(RandomAccessIterator first, RandomAccessIterator last);

template<class RandomAccessIterator, class Compare>
constexpr void sort_heap(RandomAccessIterator first, RandomAccessIterator last,
    Compare comp);

```

```

namespace ranges {
template<RandomAccessIterator I, Sentinel<I> S, class Comp = ranges::less<>,
    class Proj = identity>
requires Sortable<I, Comp, Proj>
constexpr I
    sort_heap(I first, S last, Comp comp = {}, Proj proj = {});
template<RandomAccessRange R, class Comp = ranges::less<>, class Proj = identity>
requires Sortable<iterator_t<R>, Comp, Proj>
constexpr safe_iterator_t<R>
    sort_heap(R&& r, Comp comp = {}, Proj proj = {});
}

```

1 Let `comp` be `less{}` and `proj` be `identity{}` for the overloads with no parameters by those names.

2 *Requires:* The range `[first, last)` shall be a valid heap with respect to `comp` and `proj`. For the overloads in namespace `std`, `RandomAccessIterator` shall ~~satisfy~~meet the *Cpp17ValueSwappable* requirements ([[swappable.requirements](#)]). ~~The~~ and the type of `*first` shall ~~satisfy~~meet the *Cpp17MoveConstructible* ([[tab:moveconstructible](#)]) and *Cpp17MoveAssignable* ([[tab:moveassignable](#)]) requirements.

3 *Effects:* Sorts elements in the heap `[first, last)` with respect to `comp` and `proj`.

4 *Returns:* `last` for the overloads in namespace `ranges`.

5 *Complexity:* At most $2N \log N$ comparisons, where $N = \text{last} - \text{first}$, and twice as many projections.

24.7.7.5 `is_heap`

[[is.heap](#)]

```

template<class RandomAccessIterator>
constexpr bool is_heap(RandomAccessIterator first, RandomAccessIterator last);

```

1 ~~Returns:~~ Effects: Equivalent to: `return is_heap_until(first, last) == last;`

```

template<class ExecutionPolicy, class RandomAccessIterator>
bool is_heap(ExecutionPolicy&& exec,
    RandomAccessIterator first, RandomAccessIterator last);

```

2 ~~Returns:~~ Effects: Equivalent to:

`return is_heap_until(std::forward<ExecutionPolicy>(exec), first, last) == last;`

```

template<class RandomAccessIterator, class Compare>
constexpr bool is_heap(RandomAccessIterator first, RandomAccessIterator last,
    Compare comp);

```

3 ~~Returns:~~ Effects: Equivalent to: `return is_heap_until(first, last, comp) == last;`

```
template<class ExecutionPolicy, class RandomAccessIterator, class Compare>
bool is_heap(ExecutionPolicy&& exec,
             RandomAccessIterator first, RandomAccessIterator last,
             Compare comp);
```

4 ~~Returns:~~ Effects: Equivalent to:

```
    return is_heap_until(std::forward<ExecutionPolicy>(exec), first, last, comp) == last;
```

```
namespace ranges {
    template<RandomAccessIterator I, Sentinel<I> S, class Proj = identity,
            IndirectStrictWeakOrder<projected<I, Proj>> Comp = ranges::less<>>
        constexpr bool is_heap(I first, S last, Comp comp = {}, Proj proj = {});
    template<RandomAccessRange R, class Proj = identity,
            IndirectStrictWeakOrder<projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
        constexpr bool is_heap(R&& r, Comp comp = {}, Proj proj = {});
```

5 Effects: Equivalent to: `return ranges::is_heap_until(first, last, comp, proj) == last;`

```
template<class RandomAccessIterator>
constexpr RandomAccessIterator
    is_heap_until(RandomAccessIterator first, RandomAccessIterator last);
template<class ExecutionPolicy, class RandomAccessIterator>
RandomAccessIterator
    is_heap_until(ExecutionPolicy&& exec,
                  RandomAccessIterator first, RandomAccessIterator last);

template<class RandomAccessIterator, class Compare>
constexpr RandomAccessIterator
    is_heap_until(RandomAccessIterator first, RandomAccessIterator last,
                  Compare comp);
template<class ExecutionPolicy, class RandomAccessIterator, class Compare>
RandomAccessIterator
    is_heap_until(ExecutionPolicy&& exec,
                  RandomAccessIterator first, RandomAccessIterator last,
                  Compare comp);
```

```
namespace ranges {
    template<RandomAccessIterator I, Sentinel<I> S, class Proj = identity,
            IndirectStrictWeakOrder<projected<I, Proj>> Comp = ranges::less<>>
        constexpr I is_heap_until(I first, S last, Comp comp = {}, Proj proj = {});
    template<RandomAccessRange R, class Proj = identity,
            IndirectStrictWeakOrder<projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
        constexpr safe_iterator_t<R>
            is_heap_until(R&& r, Comp comp = {}, Proj proj = {});
}
```

6 Let `comp` be `less{}` and `proj` be `identity{}` for the overloads with no parameters by those names.

7 ~~Returns: If $(last - first) < 2$, returns `last`. Otherwise, returns~~ the last iterator `i` in `[first, last]` for which the range `[first, i)` is a heap with respect to `comp` and `proj`.

8 Complexity: Linear.

24.7.8 Minimum and maximum

[alg.min.max]

```
template<class T> constexpr const T& min(const T& a, const T& b);
template<class T, class Compare>
constexpr const T& min(const T& a, const T& b, Compare comp);

namespace ranges {
    template<class T, class Proj = identity,
            IndirectStrictWeakOrder<projected<const T*, Proj>> Comp = ranges::less<>>
        constexpr const T& min(const T& a, const T& b, Comp comp = {}, Proj proj = {});
}
```

1 Requires: For the first form, type `T` shall be *Cpp17LessThanComparable* ([tab:lessthancomparable]).

2 Returns: The smaller value.

3 *Remarks:* Returns the first argument when the arguments are equivalent. [An invocation may explicitly specify an argument for the template parameter T of the overloads in namespace std.](#)

4 *Complexity:* Exactly one comparison [and two applications of the projection, if any.](#)

```
template<class T>
constexpr T min(initializer_list<T> tr);
template<class T, class Compare>
constexpr T min(initializer_list<T> tr, Compare comp);

namespace ranges {
    template<Copyable T, class Proj = identity,
            IndirectStrictWeakOrder<projected<const T*, Proj>> Comp = ranges::less<>>
    constexpr T min(initializer_list<T> r, Comp comp = {}, Proj proj = {});
    template<InputRange R, class Proj = identity,
            IndirectStrictWeakOrder<projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
    requires IndirectlyCopyableStorable<iterator_t<R>, iter_value_t<iterator_t<R>>*>
    constexpr iter_value_t<iterator_t<R>>
        min(R&& r, Comp comp = {}, Proj proj = {});
}
```

5 *Requires:* [ranges::distance\(r\) > 0](#). For the overloads in namespace `std`, T shall be *Cpp17CopyConstructible* and ~~t.size()~~→0. For the first form, type T shall be *Cpp17LessThanComparable*.

6 *Returns:* The smallest value in the [initializer-list](#) [input range](#).

7 *Remarks:* Returns a copy of the leftmost [argumentelement](#) when several [argumentelements](#) are equivalent to the smallest. [An invocation may explicitly specify an argument for the template parameter T of the overloads in namespace std.](#)

8 *Complexity:* Exactly ~~t.size()~~[ranges::distance\(r\)](#) - 1 comparisons [and twice as many applications of the projection, if any.](#)

```
template<class T> constexpr const T& max(const T& a, const T& b);
template<class T, class Compare>
constexpr const T& max(const T& a, const T& b, Compare comp);

namespace ranges {
    template<class T, class Proj = identity,
            IndirectStrictWeakOrder<projected<const T*, Proj>> Comp = ranges::less<>>
    constexpr const T& max(const T& a, const T& b, Comp comp = {}, Proj proj = {});
}
```

9 *Requires:* For the first form, type T shall be *Cpp17LessThanComparable* ([\[tab:lessthancomparable\]](#)).

10 *Returns:* The larger value.

11 *Remarks:* Returns the first argument when the arguments are equivalent. [An invocation may explicitly specify an argument for the template parameter T of the overloads in namespace std.](#)

12 *Complexity:* Exactly one comparison [and two applications of the projection, if any.](#)

```
template<class T>
constexpr T max(initializer_list<T> tr);
template<class T, class Compare>
constexpr T max(initializer_list<T> tr, Compare comp);

namespace ranges {
    template<Copyable T, class Proj = identity,
            IndirectStrictWeakOrder<projected<const T*, Proj>> Comp = ranges::less<>>
    constexpr T max(initializer_list<T> r, Comp comp = {}, Proj proj = {});
    template<InputRange R, class Proj = identity,
            IndirectStrictWeakOrder<projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
    requires IndirectlyCopyableStorable<iterator_t<R>, iter_value_t<iterator_t<R>>*>
    constexpr iter_value_t<iterator_t<R>>
        max(R&& r, Comp comp = {}, Proj proj = {});
}
```

13 *Requires:* [ranges::distance\(r\) > 0](#). For the overloads in namespace `std`, T shall be *Cpp17CopyConstructible* and ~~t.size()~~→0. For the first form, type T shall be *Cpp17LessThanComparable*.

- 14 *Returns:* The largest value in the ~~initializer-list~~ input range.
- 15 *Remarks:* Returns a copy of the leftmost ~~argumentelement~~ argumentelement when several ~~argumentelements~~ argumentelements are equivalent to the largest. An invocation may explicitly specify an argument for the template parameter T of the overloads in namespace std.
- 16 *Complexity:* Exactly ~~t.size()~~ ranges::distance(r) - 1 comparisons and twice as many applications of the projection, if any.

```
template<class T> constexpr pair<const T&, const T&> minmax(const T& a, const T& b);
template<class T, class Compare>
    constexpr pair<const T&, const T&> minmax(const T& a, const T& b, Compare comp);
```

```
namespace ranges {
    template<class T, class Proj = identity,
        IndirectStrictWeakOrder<projected<const T*, Proj>> Comp = ranges::less<>>
        constexpr minmax_result<const T&>
            minmax(const T& a, const T& b, Comp comp = {}, Proj proj = {});
}
```

- 17 *Requires:* For the first form, type T shall be *Cpp17LessThanComparable* ([~~tab:lessthancomparable~~]).
- 18 *Returns:* ~~pair<const T&, const T&>({b, a})~~ if b is smaller than a, and ~~pair<const T&, const T&>({a, b})~~ otherwise.
- 19 *Remarks:* ~~Returns pair<const T&, const T&>(a, b) when the arguments are equivalent.~~ An invocation may explicitly specify an argument for the template parameter T of the overloads in namespace std.
- 20 *Complexity:* Exactly one comparison and two applications of the projection, if any.

```
template<class T>
    constexpr pair<T, T> minmax(initializer_list<T> tr);
template<class T, class Compare>
    constexpr pair<T, T> minmax(initializer_list<T> tr, Compare comp);

namespace ranges {
    template<Copyable T, class Proj = identity,
        IndirectStrictWeakOrder<projected<const T*, Proj>> Comp = ranges::less<>>
        constexpr minmax_result<T>
            minmax(initializer_list<T> r, Comp comp = {}, Proj proj = {});
    template<InputRange R, class Proj = identity,
        IndirectStrictWeakOrder<projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
        requires IndirectlyCopyableStorable<iterator_t<R>, iter_value_t<iterator_t<R>>*>
        constexpr minmax_result<iter_value_t<iterator_t<R>>>
            minmax(R&& r, Comp comp = {}, Proj proj = {});
}
```

- 21 *Requires:* ranges::distance(r) > 0. For the overloads in namespace std, T shall be *Cpp17CopyConstructible* ~~and t.size() > 0.~~ For the first form, type T shall be *Cpp17LessThanComparable*.
- 22 *Returns:* Let X be the return type. Returns ~~pair<T, T>({x, y})~~, where x ~~has~~ is a copy of the leftmost element with the smallest value and y ~~has~~ a copy of the rightmost element with the largest value in the initializer-list input range.
- 23 *Remarks:* ~~x is a copy of the leftmost argument when several arguments are equivalent to the smallest. y is a copy of the rightmost argument when several arguments are equivalent to the largest.~~ An invocation may explicitly specify an argument for the template parameter T of the overloads in namespace std.
- 24 *Complexity:* At most $(3/2)t.size()$ ranges::distance(r) applications of the corresponding predicate and twice as many applications of the projection, if any.

```
template<class ForwardIterator>
    constexpr ForwardIterator min_element(ForwardIterator first, ForwardIterator last);

template<class ExecutionPolicy, class ForwardIterator>
    ForwardIterator min_element(ExecutionPolicy&& exec,
        ForwardIterator first, ForwardIterator last);
```

```

template<class ForwardIterator, class Compare>
constexpr ForwardIterator min_element(ForwardIterator first, ForwardIterator last,
                                     Compare comp);
template<class ExecutionPolicy, class ForwardIterator, class Compare>
ForwardIterator min_element(ExecutionPolicy&& exec,
                           ForwardIterator first, ForwardIterator last,
                           Compare comp);

```

```

namespace ranges {
template<ForwardIterator I, Sentinel<I> S, class Proj = identity,
        IndirectStrictWeakOrder<projected<I, Proj>> Comp = ranges::less<>>
constexpr I min_element(I first, S last, Comp comp = {}, Proj proj = {});
template<ForwardRange R, class Proj = identity,
        IndirectStrictWeakOrder<projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
constexpr safe_iterator_t<R>
min_element(R&& r, Comp comp = {}, Proj proj = {});
}

```

25 Let `comp` be `less{}` and `proj` be `identity{}` for the overloads with no parameters by those names.

26 *Returns:* The first iterator `i` in the range `[first, last)` such that for every iterator `j` in the range `[first, last)`,

`bool(invoke(comp, invoke(proj, *j), invoke(proj, *i)))`
is false. ~~the following corresponding conditions hold: `!(*j < *i)` or `comp(*j, *i) == false`.~~ Re-
turns last if `first == last`.

27 *Complexity:* Exactly $\max(\text{last} - \text{first} - 1, 0)$ ~~applications of the corresponding~~ comparisons and
twice as many projections.

```

template<class ForwardIterator>
constexpr ForwardIterator max_element(ForwardIterator first, ForwardIterator last);
template<class ExecutionPolicy, class ForwardIterator>
ForwardIterator max_element(ExecutionPolicy&& exec,
                           ForwardIterator first, ForwardIterator last);

```

```

template<class ForwardIterator, class Compare>
constexpr ForwardIterator max_element(ForwardIterator first, ForwardIterator last,
                                     Compare comp);
template<class ExecutionPolicy, class ForwardIterator, class Compare>
ForwardIterator max_element(ExecutionPolicy&& exec,
                           ForwardIterator first, ForwardIterator last,
                           Compare comp);

```

```

namespace ranges {
template<ForwardIterator I, Sentinel<I> S, class Proj = identity,
        IndirectStrictWeakOrder<projected<I, Proj>> Comp = ranges::less<>>
constexpr I max_element(I first, S last, Comp comp = {}, Proj proj = {});
template<ForwardRange R, class Proj = identity,
        IndirectStrictWeakOrder<projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
constexpr safe_iterator_t<R>
max_element(R&& r, Comp comp = {}, Proj proj = {});
}

```

28 Let `comp` be `less{}` and `proj` be `identity{}` for the overloads with no parameters by those names.

29 *Returns:* The first iterator `i` in the range `[first, last)` such that for every iterator `j` in the range `[first, last)`,

`bool(invoke(comp, invoke(proj, *i), invoke(proj, *j)))`
is false. ~~the following corresponding conditions hold: `!(*i < *j)` or `comp(*i, *j) == false`.~~ Re-
turns last if `first == last`.

30 *Complexity:* Exactly $\max(\text{last} - \text{first} - 1, 0)$ ~~applications of the corresponding~~ comparisons and
twice as many projections.

```

template<class ForwardIterator>
constexpr pair<ForwardIterator, ForwardIterator>
minmax_element(ForwardIterator first, ForwardIterator last);
template<class ExecutionPolicy, class ForwardIterator>
pair<ForwardIterator, ForwardIterator>
minmax_element(ExecutionPolicy&& exec,
               ForwardIterator first, ForwardIterator last);

template<class ForwardIterator, class Compare>
constexpr pair<ForwardIterator, ForwardIterator>
minmax_element(ForwardIterator first, ForwardIterator last, Compare comp);
template<class ExecutionPolicy, class ForwardIterator, class Compare>
pair<ForwardIterator, ForwardIterator>
minmax_element(ExecutionPolicy&& exec,
               ForwardIterator first, ForwardIterator last, Compare comp);

namespace ranges {
template<ForwardIterator I, Sentinel<I> S, class Proj = identity,
        IndirectStrictWeakOrder<projected<I, Proj>> Comp = ranges::less<>>
constexpr minmax_result<I>
minmax_element(I first, S last, Comp comp = {}, Proj proj = {});
template<ForwardRange R, class Proj = identity,
        IndirectStrictWeakOrder<projected<iterator_t<R>, Proj>> Comp = ranges::less<>>
constexpr minmax_result<safe_iterator_t<R>>
minmax_element(R&& r, Comp comp = {}, Proj proj = {});
}

```

31 *Returns:* `make_pair({first, first})` if `[first, last)` is empty, otherwise `make_pair({m, M})`, where `m` is the first iterator in `[first, last)` such that no iterator in the range refers to a smaller element, and where `M` is the last iterator⁸ in `[first, last)` such that no iterator in the range refers to a larger element.

32 *Complexity:* Let N be `last - first`. At most $\max(\lfloor \frac{3}{2}(N - 1) \rfloor, 0)$ comparisons and twice as many applications of the projection, if any. ~~applications of the corresponding predicate, where N is `last - first`.~~

24.7.9 Bounded value

[alg.clamp]

[...]

24.7.10 Lexicographical comparison

[alg.lex.comparison]

```

template<class InputIterator1, class InputIterator2>
constexpr bool
lexicographical_compare(InputIterator1 first1, InputIterator1 last1,
                       InputIterator2 first2, InputIterator2 last2);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2>
bool
lexicographical_compare(ExecutionPolicy&& exec,
                       ForwardIterator1 first1, ForwardIterator1 last1,
                       ForwardIterator2 first2, ForwardIterator2 last2);

template<class InputIterator1, class InputIterator2, class Compare>
constexpr bool
lexicographical_compare(InputIterator1 first1, InputIterator1 last1,
                       InputIterator2 first2, InputIterator2 last2,
                       Compare comp);
template<class ExecutionPolicy, class ForwardIterator1, class ForwardIterator2,
        class Compare>
bool
lexicographical_compare(ExecutionPolicy&& exec,
                       ForwardIterator1 first1, ForwardIterator1 last1,
                       ForwardIterator2 first2, ForwardIterator2 last2,
                       Compare comp);

```

⁸) This behavior intentionally differs from `max_element()`.

```

namespace ranges {
    template<InputIterator I1, Sentinel<I1> S1, InputIterator I2, Sentinel<I2> S2,
            class Proj1 = identity, class Proj2 = identity,
            IndirectStrictWeakOrder<projected<I1, Proj1>, projected<I2, Proj2>> Comp = ranges::less<>>
    constexpr bool
        lexicographical_compare(I1 first1, S1 last1, I2 first2, S2 last2,
                                Comp comp = {}, Proj1 proj1 = {}, Proj2 proj2 = {});
    template<InputRange R1, InputRange R2, class Proj1 = identity,
            class Proj2 = identity,
            IndirectStrictWeakOrder<projected<iterator_t<R1>, Proj1>,
            projected<iterator_t<R2>, Proj2>> Comp = ranges::less<>>
    constexpr bool
        lexicographical_compare(R1&& r1, R2&& r2, Comp comp = {},
                                Proj1 proj1 = {}, Proj2 proj2 = {});
}

```

- 1 *Returns:* true if and only if the sequence of elements defined by the range [first1, last1) is lexicographically less than the sequence of elements defined by the range [first2, last2) ~~and false otherwise.~~
- 2 *Complexity:* At most 2min(last1 - first1, last2 - first2) applications of the corresponding comparison and each projection, if any.
- 3 *Remarks:* If two sequences have the same number of elements and their corresponding elements (if any) are equivalent, then neither sequence is lexicographically less than the other. If one sequence is a proper prefix of the other, then the shorter sequence is lexicographically less than the longer sequence. Otherwise, the lexicographical comparison of the sequences yields the same result as the comparison of the first corresponding pair of elements that are not equivalent.
- 4 *[Example:* ~~The following sample implementation satisfies these requirements:~~
ranges::lexicographical_compare(I1, S1, I2, S2, Comp, Proj1, Proj2) could be implemented as:

```

for ( ; first1 != last1 && first2 != last2; ++first1, (void) ++first2) {
    if (*first1 < *first2) return true;
    if (*first2 < *first1) return false;
    if (invoke(comp, invoke(proj1, *first1), invoke(proj2, *first2))) return true;
    if (invoke(comp, invoke(proj2, *first2), invoke(proj1, *first1))) return false;
}
return first1 == last1 && first2 != last2;

```

— end example]
- 5 *[Note:* An empty sequence is lexicographically less than any non-empty sequence, but not less than any empty sequence. *— end note]*

24.7.11 Three-way comparison algorithms

[alg.3way]

[...]

24.7.12 Permutation generators

[alg.permutation.generators]

```

template<class BidirectionalIterator>
    constexpr bool next_permutation(BidirectionalIterator first,
                                    BidirectionalIterator last);

template<class BidirectionalIterator, class Compare>
    constexpr bool next_permutation(BidirectionalIterator first,
                                    BidirectionalIterator last, Compare comp);

namespace ranges {
    template<BidirectionalIterator I, Sentinel<I> S, class Comp = ranges::less<>,
            class Proj = identity>
    requires Sortable<I, Comp, Proj>
    constexpr bool
        next_permutation(I first, S last, Comp comp = {}, Proj proj = {});
}

```

```

template<BidirectionalRange R, class Comp = ranges::less<>,
        class Proj = identity>
requires Sortable<iterator_t<R>, Comp, Proj>
constexpr bool
    next_permutation(R&& r, Comp comp = {}, Proj proj = {});
}

```

1 Let `comp` be `less{}` and `proj` be `identity{}` for overloads with no parameters by those names.

2 *Requires:* For the overloads in namespace `std`, `BidirectionalIterator` shall ~~satisfy~~meet the `Cpp17ValueSwappable` requirements ([swappable.requirements]).

3 *Effects:* Takes a sequence defined by the range `[first, last)` and transforms it into the next permutation. The next permutation is found by assuming that the set of all permutations is lexicographically sorted with respect to ~~`operator<` or `comp`~~ and `proj`. If no such permutation exists, transforms the sequence into the first permutation; that is, the ascendingly-sorted one.

4 *Returns:* ~~`true` if such a permutation exists. Otherwise, it transforms the sequence into the smallest permutation, that is, the ascendingly sorted one, and returns `false`.~~ and only if a next permutation was found.

5 *Complexity:* At most $(\text{last} - \text{first}) / 2$ swaps.

```

template<class BidirectionalIterator>
constexpr bool prev_permutation(BidirectionalIterator first,
                                BidirectionalIterator last);

template<class BidirectionalIterator, class Compare>
constexpr bool prev_permutation(BidirectionalIterator first,
                                BidirectionalIterator last, Compare comp);

```

```

namespace ranges {
    template<BidirectionalIterator I, Sentinel<I> S, class Comp = ranges::less<>,
            class Proj = identity>
    requires Sortable<I, Comp, Proj>
    constexpr bool
        prev_permutation(I first, S last, Comp comp = {}, Proj proj = {});
    template<BidirectionalRange R, class Comp = ranges::less<>,
            class Proj = identity>
    requires Sortable<iterator_t<R>, Comp, Proj>
    constexpr bool
        prev_permutation(R&& r, Comp comp = {}, Proj proj = {});
}

```

6 Let `comp` be `less{}` and `proj` be `identity{}` for overloads with no parameters by those names.

7 *Requires:* For the overloads in namespace `std`, `BidirectionalIterator` shall ~~satisfy~~meet the `Cpp17ValueSwappable` requirements ([swappable.requirements]).

8 *Effects:* Takes a sequence defined by the range `[first, last)` and transforms it into the previous permutation. The previous permutation is found by assuming that the set of all permutations is lexicographically sorted with respect to ~~`operator<` or `comp`~~ and `proj`. If no such permutation exists, transforms the sequence into the last permutation; that is, the descendingly-sorted one.

9 *Returns:* ~~`true` if such a permutation exists. Otherwise, it transforms the sequence into the largest permutation, that is, the descendingly sorted one, and returns `false`.~~ and only if a previous permutation was found.

10 *Complexity:* At most $(\text{last} - \text{first}) / 2$ swaps.

[...]

25 Numerics library

[numerics]

[...]

25.8 Numeric arrays

[numarray]

[...]

25.8.10 valarray range access

[valarray.range]

- ¹ In the `begin` and `end` function templates that follow, *unspecified1* is a type that meets the requirements of a mutable ~~random-access iterator~~ [Cpp17RandomAccessIterator](#) (22.3.5.6) ~~and of a contiguous iterator (22.3.1) and models ContiguousIterator (22.3.4.14)~~, whose `value_type` is the template parameter `T` and whose `reference` type is `T&`. *unspecified2* is a type that meets the requirements of a constant ~~random-access iterator (22.3.5.6)~~ [Cpp17RandomAccessIterator](#) ~~and of a contiguous iterator (22.3.1) and models ContiguousIterator~~, whose `value_type` is the template parameter `T` and whose `reference` type is `const T&`.

[...]

31 Thread support library

[thread]

[...]

31.2 Requirements

[thread.req]

[...]

31.2.6 decay_copy

[thread.decaycopy]

- ¹ In several places in this Clause the operation *DECAY_COPY*(*x*) is used. All such uses mean call the function *decay_copy*(*x*) and use the result, where *decay_copy* is defined as follows:

```
template<class T>
  constexpr decay_t<T> decay_copy(T&& v)
    noexcept(is_nothrow_convertible_v<T, decay_t<T>>)
  { return std::forward<T>(v); }
```

[...]

Annex C (informative)

Compatibility

[diff]

C.1 C++ and ISO C

[diff.iso]

[...]

C.5 C++ and ISO C++ 2017

[diff.cpp17]

[...]

C.5.8 [Clause 21](#): containers library

[diff.cpp17.containers]

[...]

C.5.9 [Clause 24](#): algorithms library

[diff.cpp17.alg.reqs]

¹ Affected subclause: [24.3](#)

Change: The number and order of deducible template parameters for algorithm declarations is now unspecified, instead of being as-declared.

Rationale: Increase implementor freedom and allow some function templates to be implemented as function objects with templated call operators.

Effect on original feature: A valid C++ 2017 program that passes explicit template arguments to algorithms not explicitly specified to allow such in this version of C++ may fail to compile or have undefined behavior.

C.5.10 Annex D: compatibility features

[diff.cpp17.depr]

[...]

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